THEORETICAL AND COMPUTATIONAL STUDIES OF COLLECTIVE CELL BEHAVIOR AND CELL SENSING

by
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A dissertation submitted to The Johns Hopkins University in conformity with the requirements for the degree of Doctor of Philosophy

Baltimore, Maryland
June 2026

Abstract

Groups of cells, including clusters of cancer cells, developing organs, and whole organisms, can both grow and break apart. What controls when cells detach from a group, and what sets the size of the pieces that break off? How accurately can a cell sense that it has touched a neighbor? This dissertation develops simple theoretical and computational models for two problems: the mechanics of cell-cell detachment and tissue fracture, and the physical limits on cell-cell sensing.

I first study how cancer cells dissociate from a collectively invading strand in confinement, an *in vitro* mimic of cancer metastasis. Using a phase-field model, I find that single cells break off most often, that fast "leader" cells explain this, and that stronger cell-cell adhesion produces fewer but larger ruptures. Unjamming the tissue is necessary but not sufficient for rupture. Building on this, I ask what sets the size of a growing tissue, where division competes with motility-driven fracture. Modeling the tissue as self-propelled particles joined by breakable junctions, I find that its size depends only on the ratio of the junction break rate to the cell division rate, so motility can effectively regulate organ size.

To follow rupture into larger tissues, such as *Trichoplax adhaerens*, I need a model that scales to many cells while keeping tissue fluidity. I turn to the finite Voronoi model, a variant of the standard Voronoi model for nonconfluent tissue, but find that its detachment forces diverge as cells lose contact. This makes the simulated fracture timescale depend on the numerical time step rather than the physics, which I fix with a regularization calibrated against a deformable polygon model.

Finally, I turn from mechanics to sensing. In contact inhibition of locomotion, a cell repolarizes away from a neighbor upon contact. Modeling this as a cell reading ephrin bound to Eph receptors against nonspecific interfering ligands, I find that a cell can reliably decide whether a neighbor is present, even when the interfering ligand concentration is ten times that of the correct ligands. Together, these models show how cell groups hold together, break apart, and sense their neighbors.

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This thesis is dedicated to my parents.

Acknowledgement

First and foremost, I would like to express my heartfelt gratitude to my advisor, Professor Brian A. Camley. The first year of my Ph.D. was tough. Back then, I was taking classes and doing research remotely from China, with day and night reversed, because of the pandemic. I joined his group in the middle of that year, transferring from hard condensed matter physics into soft matter and biophysics. I had only a little background in this field, and his patience, encouragement, and clear guidance made the transition not only possible but genuinely enjoyable. His scientific insight and his dedication to his students have shaped me as a researcher, and I am also grateful that he supported me as a research assistant throughout, which let me focus on my research without worrying about teaching duties. Most of all, he showed me, as a physicist, how to find the simple principles behind complex biology. I could not have asked for a better mentor.

I am also grateful to Professor Daniel H. Reich and Professor Konstantinos Konstantopoulos for their kind support when I was applying for postdoc positions. I would further like to thank Professor Reich and Professor Yaojun Zhang for serving on my thesis advisory committee and for their helpful advice on my academic career. I also want to thank Ms. Kelley Key, who has been very helpful with the administrative side of my studies.

I thank all the members of the Camley group—Dr. Emiliano Perez Ipiña, Dr. Indrajit Badvaram, Dr. Ifunanya Nwogbaga, Dr. Pedrom Zadeh, Dr. Kurmanbek Kaiyrbekov, Dr. Yongtian Luo, Daiyue Sun, Mariia Kryvoruchko, and Yuntong Zhu—for the many helpful discussions and for making the group such a welcoming place to work.

I am lucky to have friends here, also in the Department of Physics and Astronomy at Johns Hopkins University—especially Dr. Xueheng Luo, Xiaofan Wu, Boran Zhou, and Dr. Chih-Chieh Chiang—for the lively dinners full of conversation and the workouts in the gym, which helped me keep a healthy work-life balance. I would also like to thank my first roommate, Dr. Yuanyuan Xu, for his help, especially when I first arrived in the United States.

Finally, I want to thank my parents and my beloved, Yue Wu, for their unconditional love and support, and for always believing in me even from so far away. Their encouragement, understanding, and patience have been a constant source of strength, and I could not have come this far without them.

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Chapter 1
Introduction

The ability of cells to move is central to life. During embryonic development, cells migrate over long distances to build tissues and organs [1, 2]; in the adult, they crawl to close wounds [3] and patrol the body as part of the immune response [4]; and in disease, the same migratory ability allows cancer cells to escape a primary tumor and seed metastases [5, 6]. Many of these processes are not carried out by isolated cells but by groups that move, grow, and remodel together [7–9]. A recurring theme is that the behavior of such a group is more than the sum of its cells: collective motion, mechanical properties, and the ability to sense the environment all emerge from interactions between neighboring cells [10, 11].

This dissertation studies two faces of this collective behavior. The first is mechanical: how a group of cells holds together and, just as importantly, how it comes apart. Tissues must be cohesive enough to maintain their shape, yet many biological processes depend on cells detaching in a controlled way [7, 8]. A primary tumor extends a strand of cells into the surrounding tissue and then sheds single cells and small clusters that can seed metastases [5, 6]. Germline cysts in the mouse reach different characteristic sizes during embryonic development through a balance between cell division, which grows the cluster, and fracture of intercellular bridges, which breaks it into smaller groups [12]. The metazoan *Trichoplax adhaerens*, one of the simplest known animals, reproduces by a motility-induced fracture [13]. In each case the outcome is set by mechanics: cells adhere to one another, crawl, generate forces, and eventually detach. This raises a pair of physical questions that run through the first half of this dissertation: what controls when a group of cells breaks apart, and what sets the size of the pieces that result?

The second face is informational: to move sensibly, a cell must read its surroundings. Cells follow chemical gradients [14, 15], mechanical cues [16], and electromagnetic signals [17, 18], and they do so with surprising accuracy despite the noise inherent in counting a finite number of molecules [19]. A particularly basic version of this problem is sensing other cells. When two crawling cells collide, each often repolarizes and the two move apart, a behavior known as contact inhibition of locomotion (CIL) [20, 21], which can be triggered by a contact only micron-scale in size [22]. This motivates the question taken up in the second half of the dissertation: how accurately can a cell sense that it has touched a neighbor?

This dissertation develops theoretical and computational models for these two problems: the mechanics of how cell groups detach and fracture, and the physical limits on how a cell senses its neighbors. My aim throughout is to use simple models to expose the physical mechanisms behind these observed behaviors.

1.1 Insights from experiments

Collective cell migration appears in many biological processes, from embryonic development to wound healing to cancer [7, 8]. The border cells of the *Drosophila* egg chamber move as a small cluster of six to eight cells through the surrounding nurse cells [23]; the vertebrate neural crest migrates as multicellular streams that disperse across the embryo to form cartilage, peripheral nerves, and pigment cells [1]; and during wound healing a single epithelial sheet advances over the wound bed to close it [3]. In cancer, invasion of the surrounding tissue is often collective, with strands and buds of tumor cells extending from the primary tumor mass [7, 8]. In each case the cells stay attached to their neighbors through cadherin junctions and coordinate their motion, so that the group migrates as one mechanical unit [8].

A group of cells can also display collective properties not seen in single cells [7]. A confluent monolayer can jam into a solid-like state, in which each cell is caged by its neighbors, or unjam into a fluid-like state, in which cells rearrange and flow past one another, with the transition controlled by cell shape, motility, and adhesion [24–26], and this has been proposed as a route by which tumor cells escape a primary tumor and invade [27]. A migrating group can also be polarized supracellularly, as a whole rather than cell by cell: the cells at the front lead with forward protrusions while the rear is contractile, so the group has a single front-to-back axis, much like one large cell [7, 8]. For instance, in the cranial neural crest of *Xenopus*, the rear cells build a shared actomyosin cable that runs along the back of the cohort and helps propel it forward [7].

A question that recurs across these systems is how a group stays cohesive and yet still lets cells leave in a controlled way [7, 8]. One route is the epithelial-mesenchymal transition (EMT), in which epithelial cells loosen their junctions and become individually migratory, a program that is reactivated during cancer invasion [28, 29]. But cells need not complete that transition to leave: in many epithelial cancers the disseminating cells retain E-cadherin and their cell-cell junctions, so cohesive clusters break away and travel together [7]. What sets when a group fragments, and how large the resulting pieces are, is then a question of mechanics, of how cells adhere, crawl, and pull on one another.

In vivo, collective migration is often confined by adjacent tissues, by the dense fibers of the extracellular matrix (ECM), and by the vasculature, the walls of blood and lymphatic vessels [30–32]. During cancer invasion, the primary tumor extends a strand of cells into the surrounding tissue, and intravital microscopy shows that tumor cells preferentially move along channel-like tracks created by these anatomical structures, with widths from a few microns up to wider than a single cell [33–35]. Whether cells leave the strand one at a time or as clusters matters for the disease: clusters of circulating tumor cells are rare, but they seed metastases far more efficiently than single cells [6, 36, 37]. Moreover, it has been reported that circulating tumor cells (CTCs) may even cluster with white blood cells or platelets to escape immune surveillance [38]. To re-create this confinement *in vitro*, Law *et al.* induced human A431 carcinoma cells by a serum gradient to invade microchannels of different widths, from 6 to 50 μm, and tracked single cells and clusters detaching from the invading front [39]. Most detachment events involve a single cell, and larger clusters (up to roughly twenty cells in the widest channels) detach more often in wider channels. Surprisingly, the time a strand takes to rupture does not depend on the channel width. These observations raise the question of what mechanical features of the strand and the channel set the rupture statistics, and motivate the model developed in Chapter 2.

Recent experiments on germline cysts in the mouse have shown that two competing processes can set the size of a small cell cluster: cell division builds the cluster, while random cell motility fractures the intercellular bridges that connect its cells [12]. A related mechanism drives reproduction in the metazoan *Trichoplax adhaerens*, one of the simplest known animals. *Trichoplax* has no nervous system, no basement membrane, and no visible extracellular matrix; its body is a flat sheet of two epithelial layers only about 25 μm thick but several millimeters wide [13, 40, 41]. This minimal anatomy makes it simple enough to model and yet still a complete animal, and it reproduces by motility-induced fracture as it crawls [13]. A similar fracture-limited size control occurs in multicellular snowflake yeast, though there it is driven by crowding stress rather than by motility [42]. Tissue and organ sizes are more commonly thought to be set by regulation of cell proliferation [43], but in these systems they instead emerge from a balance of growth and fracture. Moreover, both *Trichoplax* and germline cysts show significant variability in sizes despite the presence of a characteristic size scale [12, 13, 44, 45]. This combination of a well-defined typical size with substantial individual-to-individual variability raises the question of how reliably motility-driven fracture can regulate size at all. This question is taken up in Chapter 3.

Contact between cells does more than hold a group together: through contact inhibition of locomotion (CIL), it also helps steer and organize collective migration. When two crawling cells collide, each retracts its protrusions at the point of contact, repolarizes, and the two move apart, a behavior first seen in fibroblasts by Abercrombie more than half a century ago [20, 46] and since found in many healthy and cancerous cell types [21, 47]. Because each cell turns away from the neighbors it touches, contacts in the interior of a group suppress protrusions there and bias them toward the free edge, so that the group spreads outward or streams in a common direction; in the vertebrate neural crest, this contact-driven repolarization controls the directional migration of the cell streams [1, 21]. CIL also sets how cells space themselves out, and *Drosophila* macrophages rely on it to disperse evenly through the embryo [48, 49]. At the molecular level the response can be triggered by the Eph-ephrin pathway, in which Eph receptors on one cell bind ephrin ligands anchored to the surface of the cell it touches [50, 51], and it is precise enough to fire even when two cells crawling on nanofibers touch over a patch only a micron wide [22]. Detecting a neighbor this way can therefore be cast as a sensing problem, namely how reliably a cell can tell that a neighbor is present, which we study in Chapter 5.

1.2 Physical models of collective cell migration

1.2.1 Agent-based models of tissues

The molecular biology of cell motility is rich: integrins coupling the actin cytoskeleton to the extracellular matrix, E-cadherins coupling cells to each other, Rho-family GTPases setting the polarity, and branched actin networks pushing the cell forward [53, 55–58]. A faithful model of this molecular detail would itself be complex. For the tissue-scale questions of this dissertation, however, such a faithful description is usually not necessary. What matters is whether a few simple physical ingredients, mechanical contact, an active velocity, a few rate constants, are enough to reproduce a given experimental observation. Physical models of motile cells therefore vary widely in their level of detail, from cells as single self-propelled points to fields that resolve each cell's shape [Fig. 1.1(a)], and each chapter of this dissertation uses a model from this hierarchy chosen to match its question. A more complete survey of the modeling toolbox is given by the review [52].

The simplest models treat each cell as a self-propelled particle (SPP) [59–62]. In the active Brownian particle (ABP) model the velocity has fixed magnitude and a direction that rotates diffusively, while in the active Ornstein-Uhlenbeck particle (AOUP) model the velocity itself is an Ornstein-Uhlenbeck (OU) process with fluctuating magnitude and direction [59, 63, 64]. These models are cheap enough to simulate many particles and are often analytically tractable, which makes them well suited to statistical questions about clustering, fracture rates, and steady-state distributions [64–66]. Chapter 3 uses the active Ornstein-Uhlenbeck model to study tissue size control by motility-driven fracture. What such models give up is cell shape. Moreover, dense packings of identical particles tend to crystallize into ordered lattices [13], which is a problem for modeling real tissues, where the cells are disordered.

Adding cell shape gives vertex and Voronoi models for confluent tissue, in which cells tile space without gaps [24, 25, 65, 67]. In a vertex model, each cell is a polygon defined by its vertices, and the energy penalizes departures from a preferred area and perimeter [68]. Because the vertices outnumber the cells, the number of degrees of freedom is larger than in a particle model, and the model predicts

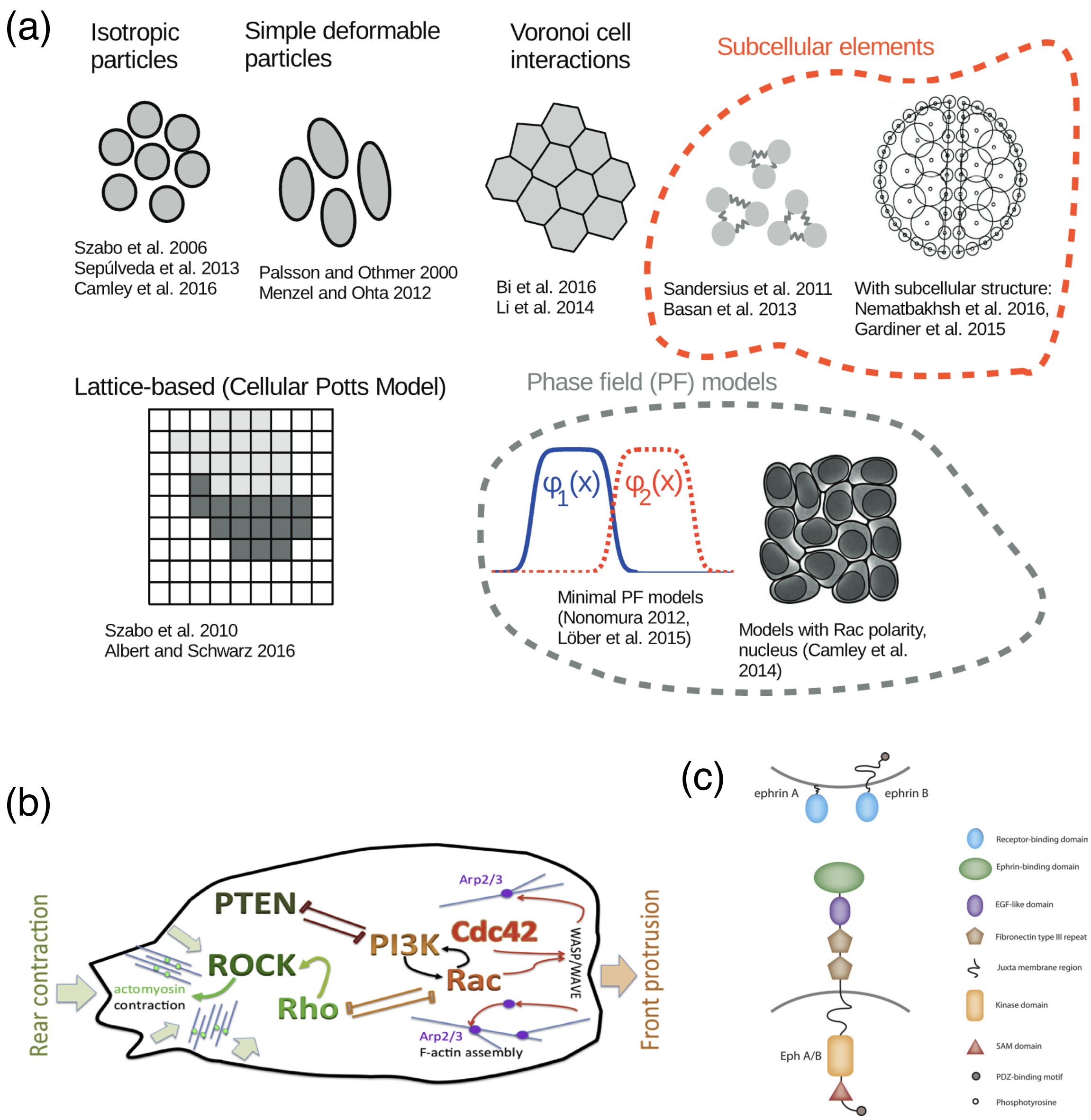

Figure 1.1: (a) Examples of common models of collective cell migration. Reprinted from Fig. 2 of Ref. [52], with permission from IOP Publishing (Order License ID: 1739114-1). (b) A schematic sketch of proteins associated with front (in shades of red) and rear (green) of a polarized cell. Reprinted from Fig. 1D of Ref. [53], with permission from Elsevier (License Number: 6285460585190). (c) The structure of the Eph receptors and their ephrin ligands. Reprinted from Fig. 1 of Ref. [54], under the terms of the CC BY-NC-ND 4.0 license.

a rigidity transition tuned by the cell shape index, with floppy modes near the transition [24, 67, 69]. The active Voronoi model is more economical: each cell is the Voronoi region of its center, so the only degrees of freedom are the N cell-center positions, the same count as a particle model. This geometric constraint matters. At zero temperature the static Voronoi model lacks the shape-index rigidity transition seen in vertex models because each cell's shape is fixed by the Voronoi tessellation of the cell centers, leaving no independent shape degrees of freedom to tune [70]; the active Voronoi model with self-propulsion does, however, exhibit a motility-driven glass transition [25]. However, because cells in either model tile space, neither allows cells to pull apart.

Phase-field models go further, representing each cell as a smooth field on a grid that is one inside the cell and zero outside [71–76]. The dynamics of these fields are motivated by classical phase-separation equations. The Cahn-Hilliard equation describes a strictly conserved order parameter and is more commonly applied to systems such as liquid droplets with fixed total volume [77, 78]. The Allen-Cahn equation describes an unconserved order parameter and is the more natural starting point for cells, since the cell-interior indicator is not itself a conserved quantity; cell area is instead held near a preferred value by a soft energetic penalty, in the same spirit as the area term in vertex and Voronoi models [73, 78]. Because the cell boundary is free to take any shape, phase-field models resolve deformation, allow cells to separate, and reproduce the unjamming transition of confluent tissues [73, 76]. The cost is computational: representing every cell as a field on a grid limits phase-field simulations to relatively small numbers of cells. Chapter 2 uses a phase-field model for cancer cell dissociation, where resolving cell shape and detachment matters more than reaching large system sizes.

For larger nonconfluent tissues, such as the metazoan *Trichoplax adhaerens* (an animal composed of up to $\sim 10^5$ cells), neither family of models is ideal: phase-field simulations do not scale to that many cells, and the standard vertex and Voronoi models tile space without gaps and so cannot represent a nonconfluent tissue. Some intermediate approaches have been proposed to bridge the two regimes. The finite Voronoi model truncates each cell's Voronoi region at a maximum radius, so that cells keep a polygonal shape where they touch but can also pull apart and leave gaps between cells [79–81]. Deformable polygon models instead represent each cell by an explicit polygon with many vertices, without imposing a Voronoi geometry [82, 83]. Subcellular element models go further by representing each cell as a collection of interacting subunits [84]. Chapter 4 studies the finite Voronoi model in detail.

1.2.2 Modelling interactions and polarity dynamics

A crawling cell anchors itself to its environment through transmembrane integrin receptors, which couple the extracellular matrix (ECM) to the actin cytoskeleton at focal adhesions and generate the traction force needed to crawl [55, 85]. Two cells in contact form a different kind of adhesion: cadherins on the membrane of one cell bind cadherins on the other, and the cytoplasmic tails of these cadherins are linked through catenins to the actin cortex, mechanically coupling the two cells [58, 86]. A third class of contacts uses receptor tyrosine kinases such as Eph receptors that bind ephrin ligands on the neighbor, triggering downstream signaling rather than a long-lived mechanical attachment [50, 51].

Physical models capture these interactions through effective potentials and energies that summarize the underlying molecular and mechanical detail. For cell-cell coupling, particle models use a short-range pairwise potential with soft repulsion and an attractive tail [61, 62], or explicit breakable spring junctions as in the chain of Chapter 3; vertex and Voronoi models encode adhesion in a line tension on cell-cell interfaces, where a lower tension corresponds to stronger adhesion and a longer preferred contact [25, 67, 81]; and phase-field models use an interaction energy between the fields of neighboring cells, typically a surface-overlap term that lowers the energy when two cell boundaries coincide [71, 75]. Cell-environment coupling enters analogously, as a soft-wall potential or an attraction to a substrate field. For numerical stability at strong adhesion, Chapter 2 adapts the phase-field framework by introducing an advection term that handles both the cell-cell and cell-wall adhesion.

The direction a cell wants to go is set by its polarity, an asymmetry between leading and trailing edges that is maintained by the Rho family of small GTPases [53]. As sketched in Fig. 1.1(b), Rac1 and Cdc42 are active at the leading edge, where they activate the Arp2/3 complex to nucleate the branched actin networks that push out the lamellipodium [57, 85]. RhoA is active at the rear, where it promotes the actomyosin contractility that pulls the cell body [53, 85]. External cues, including chemical gradients, electromagnetic fields, the mechanical stiffness of the substrate, and physical contact with a neighbor, reshape this polarity by changing the spatial distribution of these GTPases [18, 53, 57, 87].

Physical models usually reduce this machinery to a phenomenological dynamics for the cell's polarity vector, the direction in which it tends to move. In the simplest case the polarity diffuses freely, giving the persistent random walks of active Brownian and active Ornstein-Uhlenbeck particles introduced above. The Vicsek model is the canonical example of velocity alignment: each particle rotates its velocity toward the average velocity of its neighbors, originally for flocks of self-propelled particles and later for cells [88]. A common way to model chemotaxis is to bias the polarity along an external chemical gradient, while in models of CIL, the polarity is most often rotated away from contacting neighbors, taken as the vector sum of unit vectors pointing away from each contact [14, 15, 61, 62]. The phase-field model of Chapter 2 includes guided cells whose polarity vector points along the serum gradient imposed in the experiments.

1.3 Physical limits on cell sensing

To migrate purposefully, adopt the correct developmental fate, and respond to neighboring cells, cells must sense and interpret their surroundings despite noisy molecular signals [89]. These functions can all be cast as problems of information sensing and processing: chemotaxing cells follow shallow chemical gradients [90], cells in a developing tissue interpret morphogen concentrations to decide their fate [91], and immune cells discriminate self from non-self at a cell-cell contact [92]. A central question in cell biophysics is how accurately a cell can perform such tasks given that the underlying signals, the arrival of individual molecules at individual receptors, are inherently stochastic [19]. Answering it sets a benchmark: If an observed behavior approaches the physical limit, the cell is using its information nearly optimally, whereas a large gap points to additional noise or unused information.

A cell measures a chemical signal by counting molecules that bind to receptors on its membrane. Berg and Purcell showed that this counting is itself a fundamental source of noise: because molecules arrive at random, a receptor monitored for a finite time returns only an estimate of the concentration, with an error that shrinks slowly as the observation time and the number of receptors grow [19]. This argument sets a physical limit that no downstream processing can beat, and it has become the starting point for a large body of theory on the physics of cellular information processing [93–95].

The Berg-Purcell argument has been extended to the harder tasks that real cells perform. Cells do not only measure a uniform concentration; they need to estimate signals that change in space and time. They read spatial gradients to decide which way to crawl [90, 96], they estimate the position of a spatially localized signal [97], they sense gradients collectively as a group [14, 15], and they track signals that change in time [98]. Even for the single task of finding a gradient, a cell faces a choice of strategy: it can compare concentrations across its body at one instant (spatial sensing) or compare them at different points along its path over time (temporal sensing), and which is more accurate depends on the cell's size, speed, and persistence [99]. The cell's shape matters too: because gradient information is limited by how widely the receptors are dispersed, an elongated or branched shape can sense a shallow gradient more accurately than a round one [100]. A useful way to treat all of these problems is as statistical estimation: the cell is assumed to extract as much information as it can from its receptors using maximum-likelihood estimation (MLE), and the best accuracy any such estimate can reach is bounded by the Cramér-Rao bound [101]. This framework gives a clean way to ask what limits a given sensing task, separate from the details of any particular signaling network.

Several broad questions have guided this field. A major line of research concerns the fundamental limits and asks how cells can beat the naive Berg-Purcell limit: Time-averaging, spatial averaging over many receptors, and comparisons across the cell body can all reduce sensing error, although each strategy has its own trade-offs and ultimate limits. [94]. A related line of work asks how groups of cells sense better than individuals, sharing information across cell-cell contacts to detect gradients that are too shallow for a single cell to resolve [14, 15]. Some other questions concern adaptation and dynamic range, namely how a cell can stay sensitive to relative changes in signal across many orders of magnitude of background concentration [102, 103].

The second half of this dissertation focuses on a different but closely related problem: specificity. Receptors are not perfectly specific. A receptor tuned to one ligand will also bind other molecules that resemble it, and these spurious binding events add noise that is distinct from counting noise [104, 105]. When the correct ligand is rare and the interfering ligands are common, telling them apart becomes its own limit on sensing accuracy. This problem arises whenever a cell must read a specific molecular cue against a noisy chemical background, from immune recognition to the detection of a neighboring cell.

CIL can be viewed in exactly these terms. Research has shown that CIL is regulated by the Eph-ephrin pathway on the cell membrane [Fig. 1.1(c)]: Eph receptors on one cell bind ephrin ligands on the other and initiate the contact response, modulating the Rho GTPases of Sec. 1.2.2 to increase actomyosin contractility and inhibit lamellipodium formation at the contact, so that the cell repolarizes and moves away [87, 106–108]. For this response to fire reliably, a cell must first detect the presence and position of the cell-cell contact, a spatially localized signal made of ephrin ligands bound to Eph receptors [50, 51]. The detection is made harder by interfering ligands that bind the same receptors [109–111], and harder still when the contact is only micron-scale [22]. Chapter 5 treats CIL as a sensing problem of this kind, and asks how accurately a cell can locate a contact and how reliably it can decide that a neighbor is present at all.

1.4 Overview of dissertation

The remaining chapters are organized around the two main themes of my Ph.D. research. The first three develop models of tissue rupture; the last develops a model of cell sensing. Each is adapted from a published paper or a preprint, cited at the start of the chapter.

Chapter 2 studies how cancer cells dissociate from a collectively invading strand under confinement. With a phase-field model of cells crawling through microchannels, I recapitulate the dissociation statistics measured by Law *et al.* and find that single cells dissociate most often, that fast leader cells explain this predominance, and that stronger cell-cell adhesion produces fewer but larger ruptures. Unjamming the tissue turns out to be necessary but not sufficient to create ruptures.

Building on this picture of fracture in a confined strand, Chapter 3 asks whether motility-driven fracture can also regulate the size of a growing tissue. Modeling a cluster as active particles joined by breakable junctions, I derive the exact steady-state size distribution and show that it depends only on the ratio of the junction break rate to the cell division rate. Size control improves if division is restricted to the cluster edge or if fracture is localized to its middle.

To follow these results into larger nonconfluent tissues, such as *Trichoplax adhaerens*, a more scalable model than phase-field is needed, one that still allows the tissue to remain fluid. Chapter 4 examines the finite Voronoi model as a candidate, and shows that its detachment forces diverge as cells lose contact, so that the simulated time of fracture depends on the numerical time step rather than on the physics. I then propose a regularization, calibrated against a deformable polygon model that shares the same energy function but does not impose a Voronoi shape.

Chapter 5 turns from mechanics to sensing and treats CIL as a problem of cell sensing. Modeling a cell that detects a neighbor by binding ephrin to its Eph receptors against nonspecific interfering ligands, I use maximum-likelihood estimation and the Cramér-Rao bound to find the best possible sensing accuracy. I find that a cell can decide whether a neighbor is present with high reliability even when the concentration of interfering ligands is ten times that of the correct ligand, and that the accuracy of locating the contact depends only weakly on its width.

Chapter 6 provides a summary of the dissertation and outlines potential directions for future research.

Chapter 2
Confinement, jamming, and adhesion in cancer cell dissociations

This chapter is adapted from the following article published by the American Physical Society:

- Wang, W., Law, R. A., Perez Ipiña, E., Konstantopoulos, K. & Camley, B. A. Confinement, jamming, and adhesion in cancer cells dissociating from a collectively invading strand. *PRX Life* **3,** 013012. DOI: 10.1103/PRXLife.3.013012 (2025).

2.1 Background

Collective cell migration shows up in many developmental processes, like wound healing [3], embryogenesis [2, 113], and tumor growth and invasion [5, 6]. Compared to single-cell migration, collective migration is often more persistent since cells are more coordinated due to their mutual communications and interactions [6, 36]. The ability of cells to rearrange fluidly in collective migration may be controlled by an unjamming transition [25, 73, 76, 114] shown in dense biological tissues, which is similar to the well-known solid-liquid phase transition of materials [73]. This jamming-unjamming transition has been extensively studied using different theoretical models [25, 73, 76, 82, 114], which predict that cell tissues can be fluidized by increasing cell motility and deformability [25, 73, 76].

Collective cell migration *in vivo* often occurs within confined spaces [30–32]. For example, during cancer invasion, the primary tumor extends a strand of cells into surrounding tissues, resulting in cells being confined, followed by the detachment of a single cell or a group of cells from the strand. Cancer metastasis is a complex process that requires cancer cells first to detach from the invading strand of the primary tumor either individually or as small clusters, and then migrate through adjacent tissue, circulate in the vasculature, and finally survive and proliferate in distant organs [6]. Intravital microscopy studies reveal that tumor cells preferentially migrate along channel-like tracks created by various anatomical structures *in vivo* [33]. The widths of these tracks range from highly confining ($\leqslant 3\,\mu$m) to larger than the individual tumor cells [34, 35]. Cancer cells that break away from the original tumor can migrate through interstitial tissues, enter the bloodstream, attach to the vascular endothelium, and cross it to infiltrate the target tissues [37]. Whether cells detach as single cells or clusters is a key controlling factor for cancer progression. Though clusters of tumor cells (usually 2–50 cells) circulating in the bloodstream are rather rare, they can drastically increase metastatic potential compared to single-cell seeding [6, 36], dramatically increasing the harms from the cancer.

In this work, we study how single cells and clusters of cells dissociate from an invading front of a tumor under different degrees of geometric confinement. We use the approach developed in [39] to study invasion and dissociation *in vitro*, with cells induced to invade into a narrow microchannel by a gradient of serum [Fig. 2.1]. The narrow microchannel is designed to mimic the confinement that cells encounter *in vivo* as they migrate through the extracellular matrix (ECM), adjacent tissues, or the vasculature mentioned above. In these experiments, we see one or several cells as a cluster detach from an invading stream in a microchannel of controllable width [39]. We have observed that the majority of dissociation events ("ruptures") are single-cell ruptures, but in wide channels rupture of larger clusters is more common. Interestingly, the timescale of rupture appears to be independent of channel width in the experiments. We construct a model of this dissociation process by describing the cells as deformable crawling objects using a phase field model [73, 74]. This model allows us to describe the shape of each individual cell, which is key to unjamming [25, 73, 76, 114], while allowing us to model cells detaching from one another. By incorporating different cell states and cell-wall adhesion into the standard phase field model, we find that our model recapitulates the cluster-size distribution and survival probability observed in experiments, but this requires the presence of leader cells and a decrease in intercellular adhesion under confinement. Subsequently, we investigate the impact of four critical parameters in our model—cell-cell adhesion strength, cell-wall adhesion strength, chemotaxis strength, and the number of leader cells—on the rupture behavior of invading monolayers in confinement. In particular, we find that increasing cell-cell adhesion makes dissociation slower, but also increases the average size of the dissociating cluster. Leader cell formation probability controls the balance between single-cell and multiple-cell dissociations. Furthermore, we explore the relationship between cell dissociation behavior and the unjamming transition, finding that unjamming is necessary but not sufficient to create ruptures.

2.2 Experimental results

We begin with a focus on our recent experiments using human A431 epidermoid carcinoma cells in microchannels [39], where confluent cell monolayers follow a gradient of chemoattractant (fetal bovine serum, FBS) to enter the microchannels. These experiments revealed details of the biochemical mechanism of cell dissociation from collectively migrating strands in confinement [39]. Figure 2.1(a) shows experimental snapshots for different microchannel widths. We can observe that cells are entering microchannels from the seeding region (the wide region with cells at the bottom) and following the chemical gradient. As the cells crawl through the microchannels toward the exit region on the opposite side, we observe that single cells or clusters of several cells can dissociate from the main bulk of the cell collective inside the microchannels. The detached cells then leave the bulk, crawling independently, and enter the exit region. We show an example of a single-cell dissociation in a 6-micron channel and a dissociation of a large cluster in Fig. 2.1(b).

We re-analyze experiments of invasion in different channel widths first presented in Ref. [39], measuring the sizes of clusters breaking off and the times of cluster ruptures. We plot the distributions of cluster size as a function of microchannel width in Fig. 2.2(a). The mean cluster size A, plotted as dashed lines in Fig. 2.2(a), increases with channel width W. We see that across all the channel widths, though cluster size increases, the most common outcome is a single-cell dissociation. The probability of having an n-cell rupture decreases sharply with increasing n. However, in wide channels (20 μm and 50 μm), we do see a good number of larger ruptures [Fig. 2.2(a)]. In fact, in the 50-micron channel, observing clusters of >10 cells is the second most probable category. For full histograms without categorization, see Sec. 2.6.5.

How quickly do these ruptures happen? We characterize the timescale for the first rupture to happen by measuring the survival probability $S(t)$, defined as the probability that the monolayer in a microchannel has not ruptured at time t. By comparing $S(t)$ curves for different channel widths, we can see how fast ruptures happen in these channels, and the probability of any eventual rupture. Figure 2.2(b) shows the survival curves (probability of no rupture in a channel) and the corresponding error intervals obtained by standard Kaplan-Meier survival analysis [115]. By intuition, we would

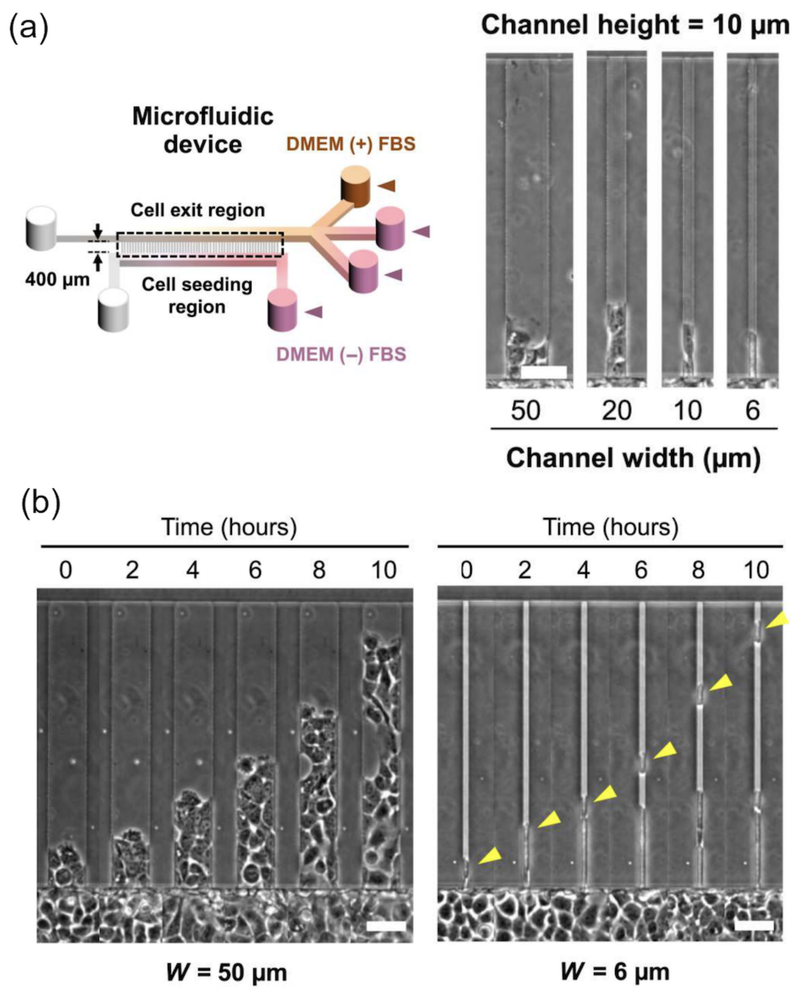

Figure 2.1: Experimental setup. (a) Schematic of microfluidic device with microchannels of prescribed dimensions and phase-contrast images of microchannels of different widths. Cells were cultured in Dulbecco's modified Eagle's medium (DMEM). (b) Representative phase-contrast snapshots of invading human A431 epidermoid carcinoma cells confined in 50 μm and 6 μm microchannels. Yellow arrowheads indicate an ongoing single-cell dissociation. Cells are exposed to a gradient of fetal bovine serum (FBS); FBS concentration increases in the $+y$ direction. Gradients are established between the wells with FBS, labeled DMEM (+) FBS, which have serum-containing DMEM (20% FBS with 1% penicillin-streptomycin) and the wells without FBS, labeled DMEM (−) FBS, which have serum-free DMEM (containing only 1% penicillin-streptomycin). Scale bars, 50 μm. Reprinted from Ref. [39]. Channel height from the substrate is 10 microns.

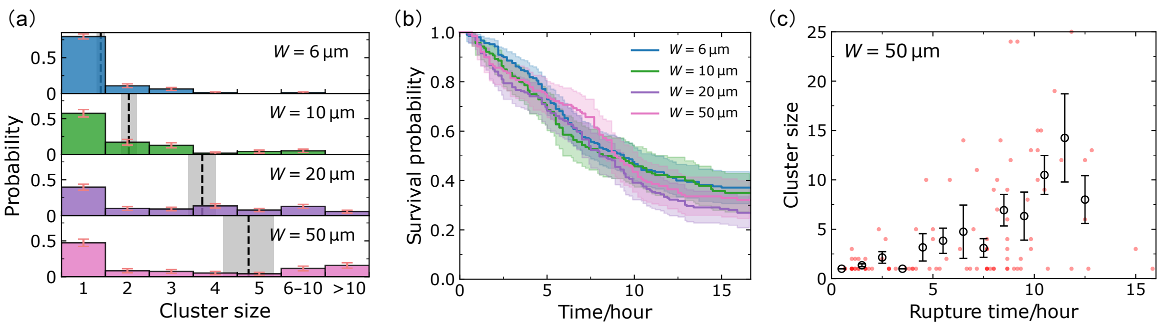

Figure 2.2: Analysis of experimental dissociation events. (a) Dissociation cluster size distributions for $W = 6$, 10, 20, and $50\,\mu$m microchannels, based on 137, 92, 138, and 95 dissociation events, respectively. The dashed lines and gray areas denote the average cluster sizes $A = 1.4$, 2.0, 3.7, 4.8, and the associated standard errors of the mean (mean $\pm$ SE). Note that we are using cluster size categories 6–10 and >10 to group together similar-sized large clusters; full histograms are shown in Fig. 2.12(a). Panel (b) shows the survival curves for different channel widths extracted by Kaplan-Meier survival analysis [115]. The shaded areas represent the 95% confidence intervals; the final survival probabilities are $K_s = 73/210\,(35\%)$, $48/140\,(34\%)$, $51/189\,(27\%)$, $45/140\,(32\%)$, respectively; the denominators of K_s represent the number of assays in the experiment. (c) Scatter plot in the (rupture time, cluster size) phase plane. Data points (red) are plotted with transparency; darker points indicate ruptures occurring in more than one assay with the same cluster size and rupture time. The circles represent the average cluster sizes within each time bin (1 hour), and the error bars indicate the corresponding standard errors. Figures for all channel widths are given in Fig. 2.12(b).

expect that larger channels would have more ruptures and ruptures occur sooner because more cells enter the larger channels—meaning more possible cells to dissociate. Surprisingly, the survival curves are—to the level of our statistical uncertainties—independent of channel width. We also see that the survival probability saturates to $\sim 30\%$ at long times, i.e., roughly $\sim 70\%$ of channels have a dissociation event within the experimental measurement.

To see the correlation between cluster size and rupture time, we make a scatter plot of cluster size as a function of the rupture time for the 50-micron channels; each dot in Fig. 2.2(c) corresponds to one dissociation event [analogous plots for smaller channel sizes shown in Fig. 2.12(b)]. Roughly 50% of the dissociations are single-cell ruptures; the dots corresponding to these are primarily in the bottom-left corner of Fig. 2.2(c), showing these single-cell events often occur at early times. On the other hand, larger ruptures are relatively infrequent and tend to occur at later times. This trend is further supported by binning the data into small time ranges, where we can see that the average cluster size—black circles in Fig. 2.2(c)—increases with rupture time. We believe the average cluster size increases with time in part simply because to have a rupture of n cells, we must wait for n cells to enter into the channel.

2.3 Model

2.3.1 Phase field approach

To capture the complex shapes of cells, we use a multi-cell phase field approach [71, 72, 116, 117] to track the cell boundaries. We can model a cell interface by introducing an auxiliary phase field $\phi_i(\mathbf{r})$ for each cell i. This field smoothly varies from $\phi_i(\mathbf{r}) = 1$ inside the cell to $\phi_i(\mathbf{r}) = 0$ outside cell i. Consequently, the cell interface can be easily tracked at $\phi_i(\mathbf{r}) = 1/2$ [72].

To model the dynamics of the cell, we adopt a Newtonian-like advection scheme appropriate for an overdamped environment [73, 118]:

$$\frac{\partial \phi_i(\mathbf{r}, t)}{\partial t} + \mathbf{P}_i \cdot \boldsymbol{\nabla} \phi_i + \left.\frac{\partial \phi_i}{\partial t}\right|_{\mathrm{adh}} = -\frac{1}{\gamma}\frac{\delta \mathcal{H}}{\delta \phi_i}, \tag{2.1}$$

where the first two terms on the left-hand side describe cell i as being advected with a constant velocity $\mathbf{P}_i$. We call $\mathbf{P}_i$ the "polarity" of cell i—the velocity it would have in the absence of other cells pushing on it or forces arising from deformation of its boundary. The third term describes the evolution of ϕ_i due to cell-cell and cell-wall adhesion, which we will describe in detail later. On the right-hand side we introduce the functional derivative of a Hamiltonian $\mathcal{H}$ [114, 118, 119]. This term tends to minimize $\mathcal{H}$ [72]. γ is a friction coefficient so the whole Eq. (2.1) can be thought of as a force balance in an overdamped environment [120]—though see note [1]. We include a Cahn-Hilliard term, an area constraint, and cell-cell and cell-wall exclusion in the Hamiltonian [73, 118]:

$$\mathcal{H} = \mathcal{H}_{\mathrm{CH}} + \mathcal{H}_{\mathrm{area}} + \mathcal{H}_{\mathrm{exclusion}}. \tag{2.2}$$

The first term in $\mathcal{H}$ is the Cahn-Hilliard energy [114, 118]

$$\mathcal{H}_{\mathrm{CH}} = \sum_i \int \mathrm{d}^2 r \left[\frac{\alpha}{4} \phi_i^2 (\phi_i - 1)^2 + \frac{K}{2} (\boldsymbol{\nabla} \phi_i)^2 \right],$$

where $\phi_i^2(\phi_i - 1)^2$ in the integrand is a double-well potential ensuring there exist two minima corresponding to the two phases $\phi(\mathbf{r}) = 0$ and $\phi(\mathbf{r}) = 1$; the second term $(\boldsymbol{\nabla}\phi_i)^2$ tells the energy cost of forming a domain wall between the two phases. In general, the Cahn-Hilliard energy describes the interfacial tension [121] and controls the interfacial width $d = \sqrt{2K/\alpha}$ (see Sec. 2.6.3). Also, we assume our cells resist changes in area, introducing a term [114, 118]

$$\mathcal{H}_{\mathrm{area}} = \sum_i \lambda \left(1 - \int \mathrm{d}^2 r \frac{\phi_i^2}{\pi R^2} \right)^2,$$

which penalizes any deviation from the preferred area πR^2. We next introduce an energetic penalty to prevent cells from overlapping with other cells [118] or the walls of the microchannel,

$$\mathcal{H}_{\mathrm{exclusion}} = \sum_{i>j} \int \mathrm{d}^2 r \; g \phi_i^2 \phi_j^2 + \sum_i \int \mathrm{d}^2 r \; g_{\mathrm{wall}} \phi_i^2 \phi_{\mathrm{wall}}^2.$$

[1]We note this force balance interpretation is not quite correct for our model, because the terms $\partial\phi_i/\partial t|_{\mathrm{adh}}$ are not derived from a Hamiltonian.

The first term in $\mathcal{H}_{\text{exclusion}}$ describes cell-cell exclusion, which is a penalty for any overlapping between cells, since the integrand is only non-zero when both ϕ_i and ϕ_j are non-zero; the second term describes cell-wall exclusion in the same way as cell-cell exclusion—$\phi_{\text{wall}}(\mathbf{r})$ is another auxiliary field introduced to depict the geometric confinement—$\phi_{\text{wall}} = 0$ inside the channel so cells can exist (no exclusion), and $\phi_{\text{wall}} = 1$ for regions where cells are not allowed to enter.

The term $\partial\phi_i/\partial t|_{\text{adh}}$ in Eq. (2.1) models the effect of cell-cell and cell-wall adhesion,

$$\left.\frac{\partial \phi_i}{\partial t}\right|_{\text{adh}} = \sum_j \omega f(\boldsymbol{\nabla}\phi_j) \cdot \boldsymbol{\nabla}\phi_i + \kappa f(\boldsymbol{\nabla}\phi_{\text{wall}}) \cdot \boldsymbol{\nabla}\phi_i.$$

The first term treats cell-cell adhesion with the advection-like term of [117], and describes the attractive interactions between interfaces of different cells. Note that $\boldsymbol{\nabla}\phi$ is a vector pointing normally into the cell described by ϕ. $f(\boldsymbol{\zeta}) = \boldsymbol{\zeta}/\sqrt{1+\epsilon|\boldsymbol{\zeta}|^2}$ is a function that saturates at large $|\boldsymbol{\zeta}|$ to avoid numerical instability [117]. We have chosen this form over alternate implementations [71, 122], as we have found it possible to resolve high cell-cell adhesions without numerical instability. The second term in $\partial\phi_i/\partial t|_{\text{adh}}$ models the cell-wall adhesion in the same way, with a strength κ. Cell-wall adhesion arises because all the surfaces of the channel are coated with adhesive fibronectin [39]; we do not treat cell-substrate adhesion explicitly here because all cells have roughly the same contact area with the bottom substrate.

Each cell in our model is assigned a polarity $\mathbf{P}_i = p_i(\cos\theta_i, \sin\theta_i)$. It is important to note that $\mathbf{P}_i$ does not represent the actual velocity of the cell. As mentioned earlier, polarity $\mathbf{P}_i$ corresponds to the velocity that the cell would exhibit in the absence of any interactions with other cells or obstacles, i.e., considering only the first two terms in Eq. (2.1). The actual velocity of each cell is the velocity once all the other effects have been applied. We have set the default value of the magnitude p_i to $p_0 = 13.3\,\mu$m/h—at the typical experimental cell velocity scale, though slower than cells that typically break off [39]. For the direction θ_i, we assume that cells tend to align their polarity to the direction θ_0 of the chemical gradient but there are still fluctuations that would lead to rotational diffusion,

$$\frac{\partial \theta_i}{\partial t} = -k_\theta(\theta_i - \theta_0) + \sqrt{2D_r}\xi_i(t), \tag{2.3}$$

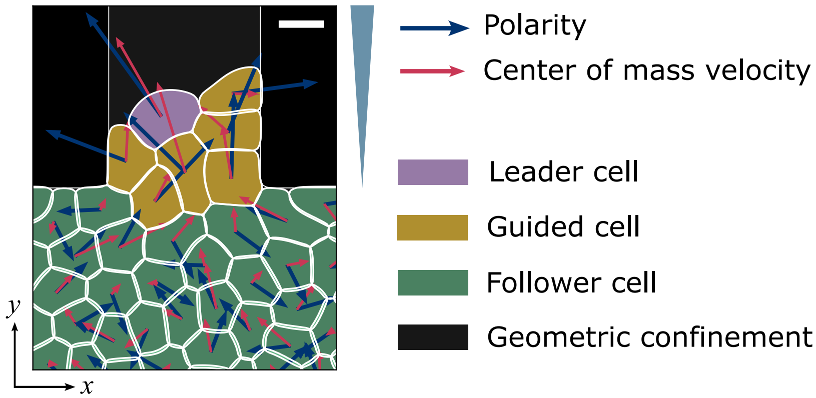

Figure 2.3: Schematic of the model. For each cell, the blue arrow represents its polarity $\mathbf{P}_i$, and the red arrow shows its center of mass velocity $\mathbf{v}_i^{\text{c.m.}}$; chemoattractant concentration increases in the $+y$ direction. Scale bar, 15 μm.

where $\xi_i(t)$ is a Gaussian white noise satisfying $\langle \xi_i(t) \rangle = 0$, and $\langle \xi_i(t)\xi_j(t') \rangle = \delta_{ij}\delta(t-t')$, and the preferred direction θ_0 is determined by the chemotactic cue. The first term on the right-hand side drives θ_i to relax to θ_0 on a timescale of $\sim 1/k_\theta$. In the absence of chemotaxis ($k_\theta = 0$), the angle θ_i will simply diffuse with angular diffusion coefficient D_r. This representation of cell polarity is a very simplified one, and could be extended by models of Rho GTPase dynamics [74, 123] or stochastic protrusion dynamics [75].

2.3.2 Cell states—chemotaxis and leader cells

Above, we described cells as following a chemoattractant gradient. However, not all of the cells are likely to be able to sense the chemical signal—since the chemoattractant concentration will be diluted as it reaches the bottom of the microchannel. Rather than explicitly modeling chemoattractant dynamics, we make a simple assumption and distinguish the cells that are in the seeding region (reservoir at the bottom of Fig. 2.1), and cells that are in the microchannel. We assume cells located in the seeding region are not guided by the chemical gradient—cells in the seeding region are just followers of those guided cells moving along the chemical gradient in the channel.

In addition, there is well-established evidence that in collective migrations like wound healing [3], and cancer metastasis [5, 6, 124], cells invade the free space under the apparent guidance of some highly motile "leader" at their leading edge [30, 125, 126], which can be crucial in collective invasion [1, 127]. These leader cells can invade the free surface more easily than other cells and coordinate their motion with their followers. Within cancerous collectives, leader cells may be keratin-14 positive cells which can either activate at or move to the leading edge and guide collective migration in tumor invasion [128–130]. Many effects of cell leadership have been proposed, including creating a path or coordinating with follower cells [124–126]. Instead, we take a relatively simplified assumption, which is that the only phenotypic difference between leader cells and other cells is that leader cells will have a larger self-propulsion strength p_i than other cells.

We illustrate the three cell states—follower (green), guided (yellow), and leader (purple)—described in our model in Fig. 2.3. The blue arrows represent the polarity $\mathbf{P}_i$ and the red arrows indicate the center of mass velocity $\mathbf{v}_i^{\text{c.m.}}$ (see Sec. 2.6.4). Followers (green) are cells with chemotaxis strength $k_\theta = 0$ and polarity magnitude $p_{\text{follower}} = p_0$. Guided cells (yellow) sense the chemoattractants and typically exhibit higher velocities than followers—modeling chemokinesis and chemotaxis [89]—i.e., guided cells have a non-zero k_θ and larger magnitude of polarity $p_{\text{guided}} = 2p_0$. At the time a cell reaches the front, it has a probability of becoming a leader cell that is dependent on its contact length with other cells:

$$P_i^{\text{leader}} = \max(1 - L_i/L_c,\, 0), \tag{2.4}$$

where L_i is the contact length of the ith cell with its neighbors, and L_c is a characteristic length. In practice, we count the number of cell-cell contact points n_i for each cell, hence $P_i^{\text{leader}} = \max(1 - n_i/n_c,\, 0)$, where n_c is the characteristic number of contact points (see Sec. 2.6.4 for details). Equation (2.4) reflects the idea that cell-cell contact inhibits leadership—similar to the proposal of [30]. This differs slightly from other models that choose leader cells randomly near the front [125, 126].

The only distinction between leader cells and guided cells is that leader cells have stronger self-propulsion $p_{\text{leader}} = 3p_0$ (see Sec. 2.6.4 for details of how we define reaching the front in the model). With this value, leader cells that dissociate will have speeds of up to $\sim 3p_0 \approx 40\,\mu\text{m/h}$ while non-leader cells that dissociate will have speeds of up to $\sim 2p_0 \approx 26\,\mu\text{m/h}$. These are both within the broad range of typical dissociating cell speeds (up to $\sim 50\,\mu$m/h) observed in 6-micron channels [39]. Experimentally in 6-micron channels, the difference between the mean speed of a cell that dissociates and a cell that is left behind after dissociating is roughly $\sim 20\,\mu$m/h (Fig. 1D of [39]). Our model results are also roughly consistent with this observation, with typical y velocities of cells inside the invading front of 0–20 μm/h, and escaping single cells able to reach up to $\sim$ 26–40 μm/h, depending on whether the dissociating cell is a leader.

Currently, the best model we find that can fit the experimental data (as shown in Fig. 2.2) is composed of these three types of cells: cells near the channel entrance and inside the channel are identified as guided cells (labeled yellow in Fig. 2.3), and cells in the chamber are the followers (green) following the guided cells, while leader cells (purple) are tip cells at the invading front. Alternate models without leader cells are shown in Sec. 2.6.1.

2.4 Model results

2.4.1 Matching with experimental results

We simulate our phase field model for a maximum duration of 18 hours, after an initial equilibration time where cells relax to more physical shapes (see Sec. 2.6.4). Since our focus is on the first dissociation events and their sizes, the simulations are terminated as soon as the first detachment occurs—except for those we show for illustration. Figure 2.4 shows the simulation snapshots for $W = 6$, 10, 20, 50 μm microchannels. We observe both single-cell and cluster ruptures. Figure 2.5 shows statistics on rupture size and times for our simulations, analogous to our experimental results in Fig. 2.2. We have been able to recapitulate several key factors of the experiment. Specifically, our model reproduces (i) the predominance of single-cell ruptures, (ii) the occurrence of larger ruptures in wider channels, (iii) the independence of survival curves from microchannel width, and (iv) the increase of average cluster size with rupture time.

When varying the channel width, we include the possibility that cell tension and adhesion dynamics change in confinement. Earlier experiments showed that in a narrow confinement, contractility

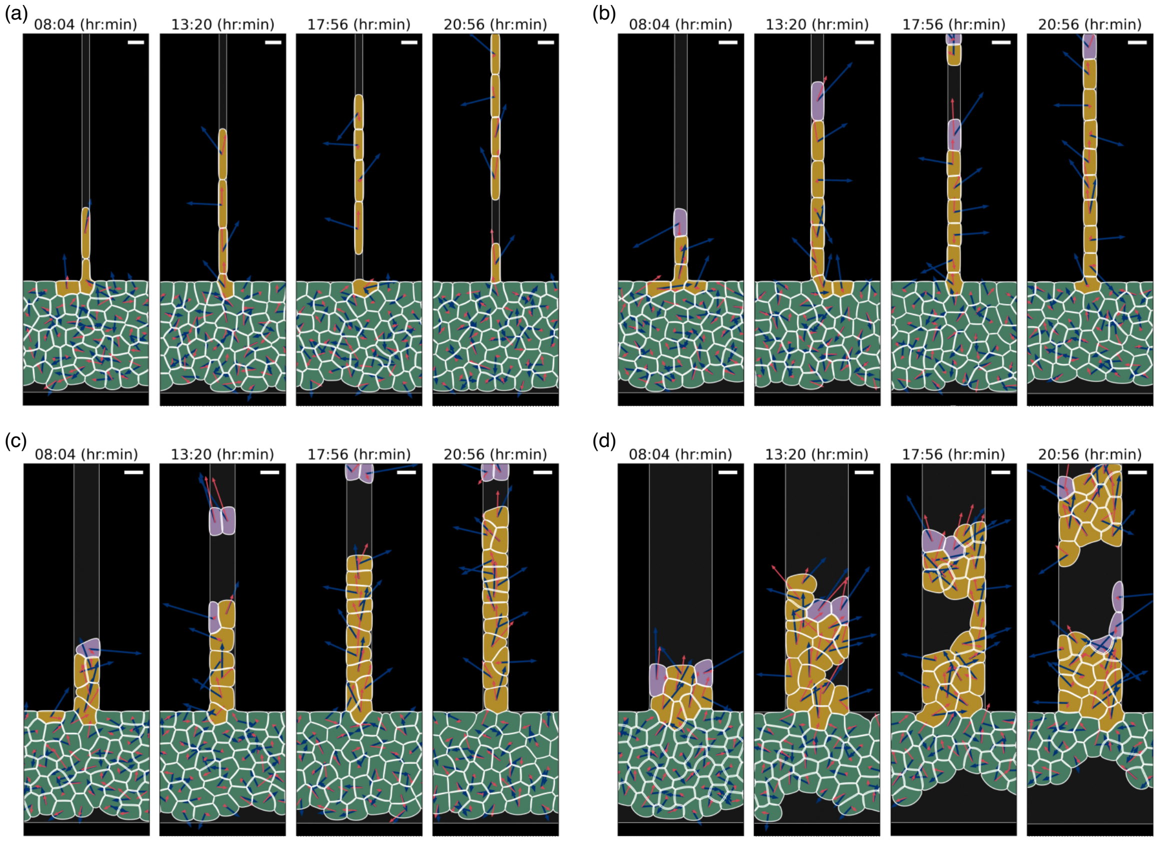

Figure 2.4: Evolution of rupture for simulations of cells in microchannels with different widths. Panels (a)–(d) correspond to $W = 6$, 10, 20, and 50 μm, respectively. There are chemical gradients along the microchannels ($+y$ direction) driving cells to move upwards. Scale bars, 15 μm.

increases, which leads to E-cadherin disengagement. In addition, cell-cell adhesions are more dynamic in narrow confinement [39, 131, 132]. We thus expect that the effective cell-cell adhesion should decrease as cells are placed into narrow confinement—but there is no direct quantitative measurement of adhesion strength as a function of confinement. This unavoidably requires some degree of tuning of parameters as a function of channel width. We take the simplest route we think is reasonable and change only the cell-cell adhesion parameter. To match the survival probability with experiments in Fig. 2.2(b), we have to increase cell-cell adhesion strength ω from $0.7\omega_0$ to ω_0 for channel widths from 6–50 μm, where ω_0 is the reference value of adhesion strength in our model (see Table 2.1 for details). This is a relatively small tuning of adhesion strength, but is sufficient to ensure that the survival curves do not differ strongly as a function of channel width [Fig. 2.5(b)]. If we do not make

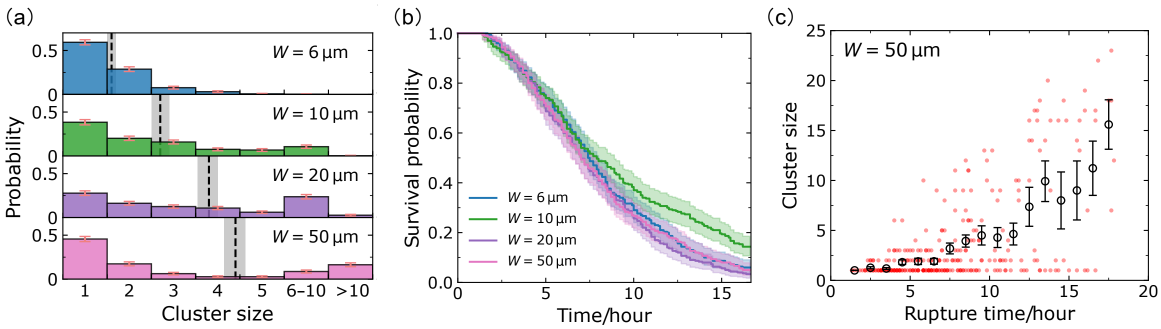

Figure 2.5: Simulation results to match experimental data. We adjust $\omega = 0.7\omega_0$ for the two narrowest channels, $\omega = 0.8\omega_0$ for $20\,\mu$m channels, and $\omega = \omega_0$ for $50\,\mu$m channels. (a) Cluster-size distributions for $W = 6$, 10, 20, $50\,\mu$m microchannels, based on 284, 257, 292, 287 dissociation events, respectively. The corresponding average cluster sizes (dashed lines; mean $\pm$ SE) are $A = 1.6$, 2.7, 3.8, 4.4. The complete histograms are shown in Fig. 2.13(a). Panel (b) shows the survival probabilities for different channel widths; the final survival probabilities are $K_s = 16/300$, $43/300$, $8/300$, $13/300$, respectively. Panel (c) shows the (rupture time, cluster size) phase plane scatter plot. The circles represent the average cluster sizes within each time bin (1 hour), and the error bars indicate the corresponding standard errors. Figures for all channel widths are given in Fig. 2.13(b). We conduct 300 independent simulations for each microchannel width W with $k_\theta = 1.4k_\theta^0$ in all channels; ω_0 and k_θ^0 represent the reference values for adhesion and chemotaxis strength in the simulation.

this assumption, and instead assume that all channels have the same value of cell-cell adhesion strength ω_0, we find that the 10-micron channel has slower rupture than other channels [Figs. 2.10(b) and 2.10(f)]. This is because cells in the 10-micron channel have a broader cell-cell contact line than the 6-micron channel—but in the 20-micron channel, multiple cells can enter at once, so ruptures do not require breaking a larger contact line (see Sec. 2.6.2 and Fig. 2.11).

The experimental cluster-size distributions [Fig. 2.2(a)] are well captured by our model [Fig. 2.5(a)]. We observe that single-cell ruptures are always predominant, though there is a population of larger ruptures in wide channels, leading to a mean rupture size that increases with channel width. We note that the single-cell ruptures being predominant depends on the presence of leader cells that are phenotypically different from other cells; we show how cluster statistics depend on the leader cell number in Sec. 2.4.6, and show results without leader cells in Figs. 2.10(c) and 2.10(d).

We plot rupture cluster size as a function of rupture time in Figs. 2.5(c) and 2.13(b). Similar to Fig. 2.2(c), we find that small ruptures dominate and usually happen at early times, and large ruptures are less probable and tend to happen at later times. Moreover, we see—consistent with the experiment—that the average cluster size increases with rupture time.

We note that the survival curves in Fig. 2.5(b) drop nearly to zero, which the experimental curves do not. We think this may reflect the evolution of parameters like the chemoattractant strength and cell-cell adhesion over time in our experiments, and address this further in Sec. 2.5.

Next, we systematically adjust several key parameters in the model to investigate their impact on the rupture behavior of the monolayers, including the cell-cell adhesion strength ω, cell-wall adhesion strength κ, chemotaxis strength k_θ, and the probability for tip cells to become leader cells.

2.4.2 Rough intuitive picture: active escape over a barrier

Thinking about a chain of invading cells, such as shown in Fig. 2.4, we expect that rupture is not likely to happen if all cells are moving with consistent velocities. Differences in the self-propulsion from one cell to another lead to the cell centers moving apart from one another—increased strain. This difference in self-propulsion could arise because cell polarities point in different directions, e.g., due to the rotational noise D_r, or have different magnitudes, due to one cell having a different state from its neighbors. The difference in motilities between cells tends to work against forces keeping cells together—largely cell-cell adhesion. In this way, we think of rupture as being initiated by fluctuations of cell motility leading to the cells crossing an energy barrier [133, 134]. For channels wider than a single cell can span, a dissociation event will start with the breakage of a single cell-cell junction, but this fracture will then have to evolve to cross the channel, as seen in, e.g., Refs. [135, 136] and in Fig. 2.1(b) for the 50-micron channel. Our picture here is an oversimplification because there is not a single energy barrier, but a complex process involving re-arrangement of cells and cell boundary deformations. Nonetheless, this rough picture makes some clear predictions. We would expect faster rupture if the cell motion is noisier, and more rupture if there is more variability in cell speeds—both increasing the tendency for cells to be pulled apart. We would expect less rupture if there is high cell-cell adhesion (increasing the barrier height)—and if the process of rupture is nucleated at the walls, then cell-wall adhesion could also influence the barrier. We test all of these ideas below.

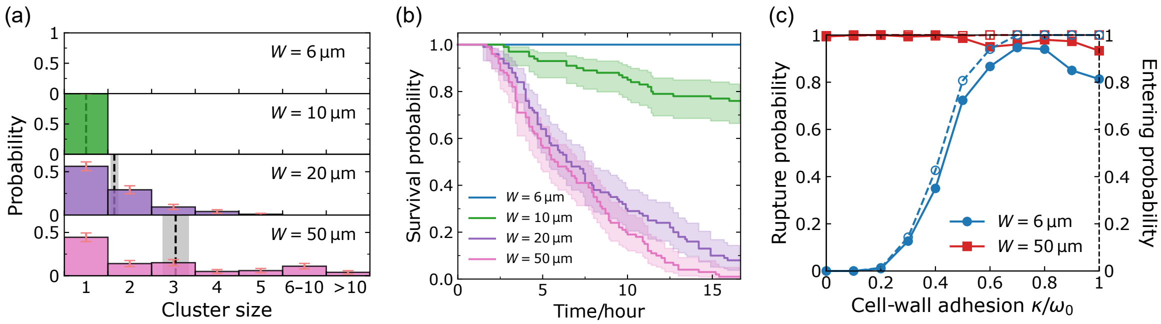

Figure 2.6: Cell-wall adhesion helps cells invade into narrow channels. Panels (a) and (b) are simulation results (100 independent simulations) when there is no cell-wall adhesion term. The corresponding average cluster sizes (dashed lines; mean ± SE) are $A = 0$, 1.0, 1.6, 3.1, and the final survival probabilities are $K_s = 100/100$, 76/100, 8/100, 1/100. The complete histograms are shown in Fig. 2.14(a). Panel (c) shows the rupture probability $1 - K_s$ (solid lines) and the probability of cells entering the channel (dashed lines) as a function of cell-wall adhesion strength κ for narrow ($6\,\mu$m) and wide ($50\,\mu$m) microchannels; 300 independent simulations for each point.

2.4.3 Cell-wall adhesion allows cells to invade into narrow channels

Cell-wall adhesion is vital to observe ruptures in narrow channels. Figures 2.6(a) and 2.6(b) show the cluster-size distributions and survival curves when we turn this term off. The rupture behavior in wide channels (like $50\,\mu$m channels) is not immediately different in the absence of cell-wall adhesion, but ruptures are strongly suppressed in narrow ($6\,\mu$m or $10\,\mu$m) channels—most dramatically, no ruptures occur in $6\,\mu$m microchannels. This is because, in this scenario, cells fail to enter the narrow channels. We change the cell-wall adhesion strength κ and see how this influences rupture in narrow (6-micron) and wide (50-micron) channels. As shown in Fig. 2.6(c), κ has little effect on $50\,\mu$m channels, but it can control the rupture probability ($1-$final survival probability) in narrow channels. Rupture fraction in small channels increases rapidly with cell-wall adhesion and then saturates, consistent with the idea that cell-wall adhesion is helping cells overcome an energy barrier at the channel entrance [137]. The rupture fraction is low at small cell-wall adhesion because in these cases, few cells manage to enter the channel. We plot the probability of any cells entering the channel in Fig. 2.6(c) using dashed lines, and it closely tracks the rupture fraction—at small κ, if any cell enters the channel, there will almost always be a dissociation. Interestingly, rupture probability in both 6- and 50-micron channels eventually decreases as cell-wall adhesion is increased. We believe this is because some ruptures require detachment from the wall as part of their dissociation process—e.g.,

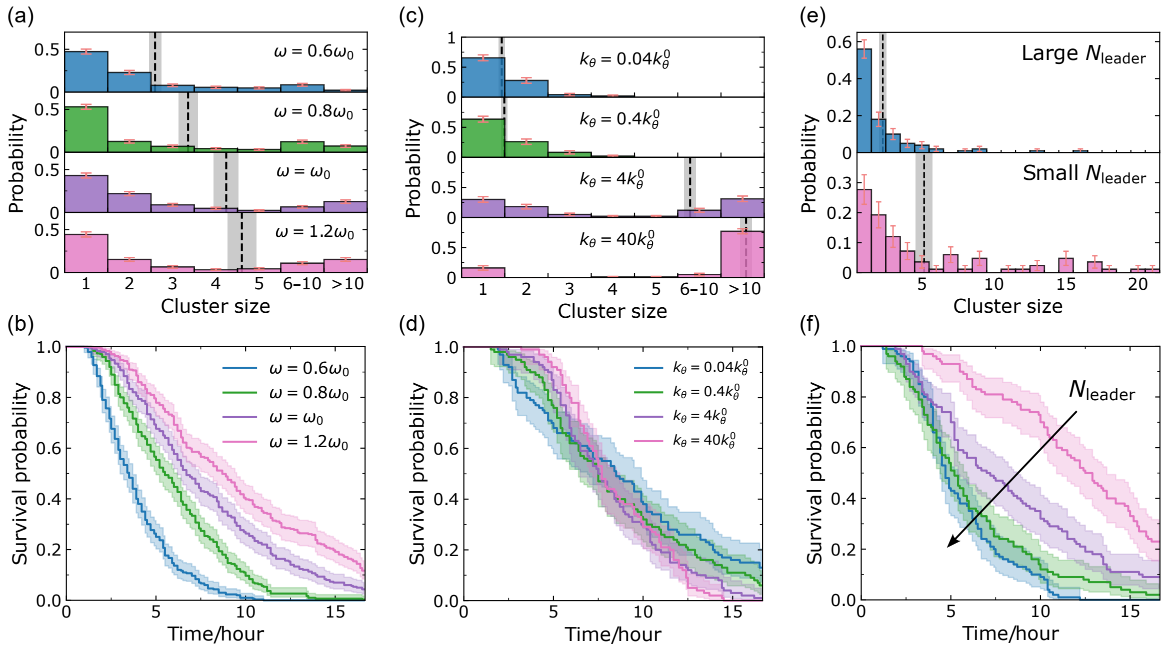

Figure 2.7: How intercellular adhesion, chemotaxis, and leader cells affect the dissociation behaviors. Panels (a), (b) are simulation results for different cell-cell adhesion strength ω in $W = 50\,\mu$m channels. The corresponding average cluster sizes (dashed lines; mean $\pm$ SE) are $A = 2.6$, 3.4, 4.2, 4.6, and the final survival probabilities are $K_s = 0$, 1/100, 4/100, 11/100. Panels (c), (d) are simulation results for different chemotaxis strength k_θ in $W = 50\,\mu$m microchannels. The corresponding average cluster sizes (dashed lines; mean $\pm$ SE) are $A = 1.4$, 1.5, 6.9, 13.4, and the final survival probabilities are $K_s = 12/100$, 6/100, 1/100, 0/100. Panels (e), (f) are simulation results when the probability of leader cell formation [Eq. (2.4)] is altered by setting the characteristic number of contact points n_c to $0.5n_c^0$, n_c^0, $2n_c^0$, and $10n_c^0$, where $n_c^0 = 256$ (see Sec. 2.6.4). Larger n_c corresponds to larger N_{leader}. The corresponding average cluster sizes (dashed lines; mean $\pm$ SE) are $A = 2.3$, 5.2. The complete histograms for panels (a), (c) are shown in Figs. 2.14(b), (c).

the formation of the "notch" in the 50-micron channel simulation in Fig. 2.4(d). Cell-wall adhesion thus has two effects: some amount of cell-wall adhesion helps cells enter channels, but large cell-wall adhesion helps prevent cells from dissociating.

2.4.4 Strong cell-cell adhesion leads to rarer but larger ruptures

Ruptures occur at cell-cell junctions for epithelial monolayers, and the intercellular adhesion complexes, such as different members of the cadherin family, can control the strength of tissues [135]. In our model, cell-cell adhesion strength ω is one of the most important parameters that prescribe the rupture behavior. Our intuition is that increasing cell-cell adhesion strength ω should increase the barrier to rupture and reduce the frequency of rupture events. Figures 2.7(a) and 2.7(b) show the simulation results for $50\,\mu$m microchannels. With the increase of cell-cell adhesion, the survival curve drops much more slowly, and the final survival rate rises a little—rupture is less common due to the strong connection between cells. This influence of cell-cell adhesion is consistent with our earlier experimental work, which found that in 6-micron channels, increasing the cell-cell adhesion by overexpressing E-cadherin led to suppressed rupture, and downregulation by shRNA led to increased rupture [39].

From the cluster-size distribution, we find there are more ruptures of larger clusters (>10 cells) happening in systems with strong cell-cell adhesion, and mean cluster sizes increase with increasing adhesion ω. In general, strong cell-cell adhesion leads to less rupture, but if we get a rupture, it is more likely to be a large rupture. Our results show that the influence of changes in cell-cell adhesion on metastasis may be complex, and we would expect its effect on overall risk to depend on the relative importance of the rate and size of metastases, even neglecting any effect of cadherins on cell proliferation and survival.

2.4.5 Strong chemotaxis strength leads to larger and faster ruptures

We expect that the rate of rupture could potentially be increased if the cells are noisier in their motility—i.e., if they are less uniformly guided by the presence of the chemoattractant gradient. This is in part controlled by the chemotaxis strength k_θ.

To better understand the magnitude of k_θ, we think about the dynamics of cell orientation θ for a single isolated cell, which follows Eq. (2.3). We take the preferred direction $\theta_0 = 0$ then by the properties of Ornstein-Uhlenbeck processes [138], the average is $\langle\theta\rangle = 0$, and the variance is $\langle\theta^2\rangle = D_r/k_\theta$. Cell chemotactic accuracy is often characterized by a "chemotactic index" $\mathrm{CI} = \langle\cos\theta\rangle$ (or related definitions) [139]; $\mathrm{CI} = -1$ would be perfect chemorepulsion, $\mathrm{CI} = 0$ is undirected migration, and $\mathrm{CI} = 1$ is perfect, deterministic motion up the gradient. For our model, we can compute for an isolated cell $\mathrm{CI} = \langle\cos\theta\rangle = \exp(-\langle\theta^2\rangle/2) = \exp(-D_r/2k_\theta)$, using the Gaussian distribution of θ. This gives us a way to interpret k_θ. Our default value of $k_\theta = 1.4k_\theta^0 = 0.014\,\mathrm{min}^{-1}$ corresponds to a chemotactic index of $\langle \cos\theta \rangle \approx 0.45$. However, this estimate—which essentially assumes an isolated cell not in confinement—may not always reflect the chemotactic accuracy of cells in our full simulation. First, in our system, we have many cells—so the motion of their center of mass will be more accurate than any individual cell [139]. Secondly, cells in narrow channels are forced to have their velocity in the direction of the channel, so they will be more directional in confinement [31, 140].

Figures 2.7(c) and 2.7(d) show the cluster-size distributions and the survival curves for different chemotaxis strengths k_θ. These simulations are in 50 μm microchannels, and cell-cell adhesion strength is set to the default $\omega = \omega_0$. We vary k_θ across a large range ($0.04k_\theta^0$–$40k_\theta^0$, which corresponds to a CI range of roughly 0–1). From Fig. 2.7(c), we see that more ruptures of larger size happen at large k_θ when cell chemotaxis is more deterministic—cells move together as a large pack. In Fig. 2.7(d), when there is only very weak guidance (small k_θ), when the cells can be polarized in almost any direction, not only toward the channel, we can see more early ruptures (survival curve drops earlier) but also see more channels that never have ruptures. By contrast, at very strong guidance (large k_θ), the survival curves become steeper, indicating that a large chunk of cells move together and faster, and at $k_\theta = 40k_\theta^0$ large clusters are the majority outcome.

2.4.6 Leader cells drive single-cell rupture

Within our model, the key factor that controls whether cells have variable states is the probability for cells that reach the leading edge to become "metabolically active" leaders—with a high degree of self-propulsion, given by Eq. (2.4). We can increase the probability of cells becoming leader cells—and thus the average number of leader cells N_{leader}—by increasing the characteristic number of contact sites n_c, and vice versa.

Figure 2.7(e) shows the cluster-size distributions for large N_{leader} and small N_{leader}, and Fig. 2.7(f) shows the survival curves for different N_{leader} in 50 μm channels, with $\omega = \omega_0$ and $k_\theta = 1.4k_\theta^0$. The survival curves decrease faster with larger N_{leader}, which tells us that more leader cells lead to faster

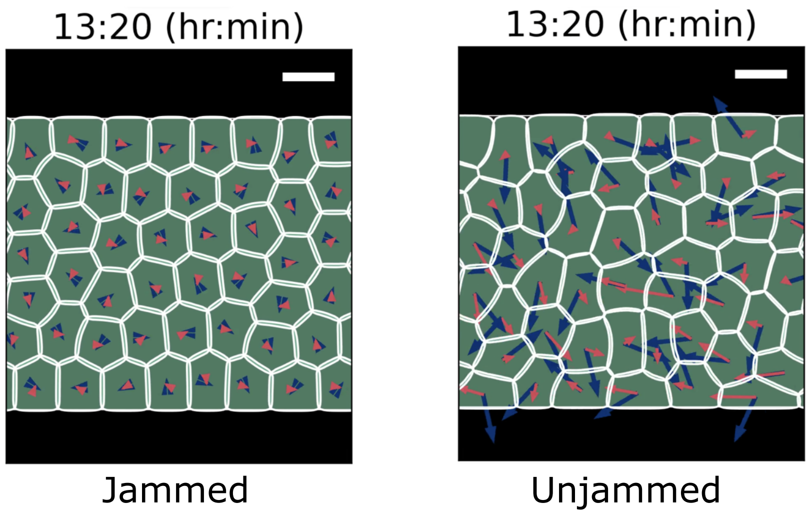

Figure 2.8: Simulation snapshots for typical systems in jammed (Pe $= 0.14$) and unjammed (Pe $= 1.43$) states at the same simulation time. Blue arrows represent the polarity $\mathbf{P}_i$ and red arrows indicate the center of mass velocity $\mathbf{v}_i^{\text{c.m.}}$. Scale bars, $15\,\mu$m.

ruptures. Also, from Fig. 2.7(e) we can see that the leader cells increase the ratio of single-cell ruptures to all ruptures, which ensures the absolute predominance of single-cell ruptures shown in experiments. These leader cells, which have the largest velocity in the system and are located at the leading edge of the expanding monolayer, can dissociate from the main bulk more easily. Since leaders are relatively rare in our default parameters, this may explain the universal most probable outcome of single-cell rupture in all cluster-size distributions.

2.4.7 Unjamming is necessary but not sufficient to create ruptures

A confluent monolayer can go from a jammed state to an unjammed state, allowing cells within the monolayer to more freely rearrange [24, 141]. Experiments have shown that jamming is influenced by factors such as cell shape, cell motility, and cell-cell adhesion [24, 69, 142], and the spatiotemporal control of tissue states may represent a generic physical mechanism of embryonic morphogenesis [143]. Reference [73] shows that several different order parameters can capture this transition behavior. For instance, if a dimensionless quantity, called shape index, $q = \text{cell perimeter}/\sqrt{\text{cell area}}$, is larger than some critical value q_c (Ref. [25] indicates $q_c \approx 3.81$), then the system is in a fluid-like unjammed state, where cells can squeeze between neighbors; if q is smaller than q_c, the system is in a more

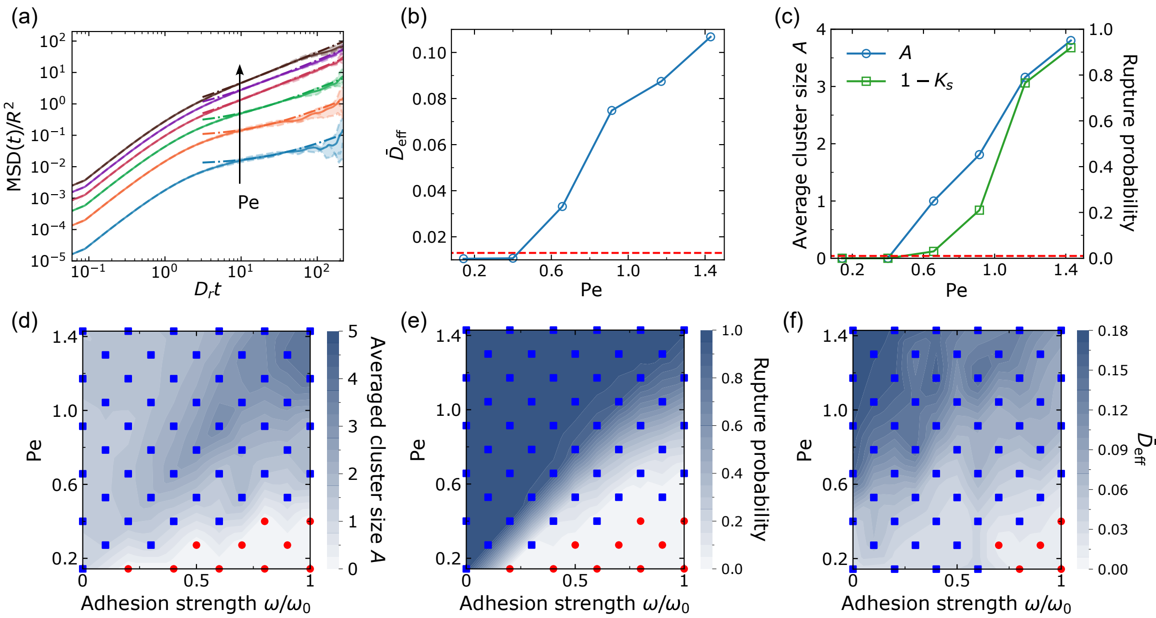

Figure 2.9: Unjamming and rupture. (a) Mean squared displacement $\mathrm{MSD}(t)$ for different cell motility quantified by a dimensionless Péclet number $\mathrm{Pe} := p_0/(RD_r)$. The different Pe shown in this panel are 0.14, 0.40, 0.66, 0.91, 1.17, and 1.43 (default); we have rescaled MSD and t to make them dimensionless; for each Péclet number, the dashed lines are results from three independent simulations, and the solid line is their average, and the dash-dotted lines are linear fit at late times where the effective diffusivity $\bar{D}_{\mathrm{eff}}$ is extracted. (b) The effective diffusivity $\bar{D}_{\mathrm{eff}}$ as an order parameter for solid-liquid transition. (c) Corresponding rupture transition of average cluster size (blue) and rupture probability (green). Panels (d) and (e) are phase diagrams for rupture transition. Red points indicate values equal to 0 and blue squares represent values greater than 0. (f) Phase diagram for jamming transition. Red points indicate values below the threshold value 0.012 and blue squares represent values above the threshold value. 300 independent simulations for each point in panels (c)–(f).

solid-like jammed state, where cells are somewhat caged by their nearest neighbors. References [73] and [25] also show that a system in the solid state can be fluidized by increasing cell motility, cell deformability, or the preferred shape index of the tissue.

Does the jamming transition influence the presence of rupture? To characterize the extent of jamming, we follow [73] and measure the effective diffusivity of cells as the order parameter, $\bar{D}_{\text{eff}} = \lim_{t\to\infty} \text{MSD}(t)/4D_0 t$, where $\text{MSD}(t)$ is the mean squared displacement of the cells (see Sec. 2.6.4). This is similar to the experimental quantification of jamming by, e.g., [24]. Here, $D_0 = p_0^2/2D_r$ is the diffusion coefficient for an isolated follower cell [25, 73]. We control jamming by varying cell-cell adhesion and cell speed p_0. To compare with previous work, we characterize the cell's speed by computing the dimensionless Péclet number $\text{Pe} = p_0/(RD_r)$ [73], which is roughly speaking the distance an isolated cell would travel in the time the cell takes to reorient $(1/D_r)$, scaled by its size R. We measure the effective diffusivity of cells in the rectangular confinement (with periodic boundary condition in the x direction, see Fig. 2.8) of our previous simulations—but do not open the channel entrance, so all cells are confined in the original reservoir and there is no invasion. As shown in Fig. 2.9(a), we see a transition between near-ballistic motion at short time scales to more-diffusive motion at long times $(t > 1/D_r)$. Mean squared displacements are naturally larger for cells with larger motility p_0 (larger Pe). Fitting to a linear function to extrapolate to measure $\bar{D}_{\text{eff}}$, we can see that there is a clear transition in $\bar{D}_{\text{eff}}$ representing a fluid-solid transition [Fig. 2.9(b)]. Cells with low motility (the first two data points below the red dashed line) are in the solid-like jammed state, and cells with high motility (the rest points above the red dashed line) are in the unjammed state, which is consistent with Refs. [25, 73]. The red dashed line is the threshold value 0.012, which we use to indicate the presence of a solid-fluid transition. This value is chosen both based on the sudden transition in Fig. 2.9(b), but also on the appearance of simulation movies showing the degree of cell rearrangement. Simulation snapshots for a typical jammed state and unjammed state are given in Fig. 2.8. We see that cells in the jammed state are highly ordered and near-crystalline, while the unjammed state shows more elongated, disordered cell shapes, as in [73].

To study the relationship between this jamming transition and our dissociation statistics, we run simulations of ruptures in 50-micron channels using the same procedure we used for Fig. 2.5, but varying the Péclet number. While we are varying the motility Pe, we keep R and D_r constant and change the basal polarity magnitude to $p_0 = \mathrm{Pe} \cdot RD_r$. However, we still maintain the relationships $p_{\mathrm{guided}} = 2p_0$ and $p_{\mathrm{leader}} = 3p_0$, i.e., we are changing the self-propulsion of all cells in the system. Figure 2.9(c) shows there is no rupture for a system with p_0 at its two lowest values—when our jamming simulations indicate the tissue is in a solid state—and ruptures start to happen after the system is fluidized by increasing cell motility.

Is this a coincidence, or is the jamming transition always aligned with the rupture transition? We sweep the (ω, Pe) parameter plane to construct several phase diagrams, as shown in Figs. 2.9(d)–2.9(f). Both Figs. 2.9(d) and 2.9(e) show the rupture transition and that ruptures are completely abolished when the tissue has strong cell-cell adhesion but lower motility (red-point regions in the bottom right). We also notice—consistently with our earlier results—that larger ruptures tend to happen in a system of cells with strong cell-cell adhesion and high motility [Fig. 2.9(d), upper right corner].

We also measure jamming as a function of adhesion and motility in Fig. 2.9(f). Consistent with prior research findings [25, 73], we have observed that a jammed state can be transitioned to an unjammed state by elevating cell motility. Furthermore, we also find the reinforcement of cell-cell adhesion will suppress unjamming, which is supported by experiments [32, 144].

Upon comparing Fig. 2.9(f) to Figs. 2.9(d) and 2.9(e), it is evident that the red-point region indicating the solid states in Fig. 2.9(f) is a subset of the red-point region corresponding to no rupture in Figs. 2.9(d) and 2.9(e). In other words, unjamming alone does not ensure rupture—it is possible to have cells that are unable to invade a channel and rupture cell-cell adhesions (e.g., owing to low motility) but still have sufficient motility to fluidize the tissue. This includes, e.g., the points located in the lower center region of the (ω, Pe) plane. Therefore, we conclude that unjamming is necessary but not sufficient to create ruptures.

2.5 Discussion

To investigate the mechanism of how cells dissociate from collectively migrating strands in confinement, we re-analyzed data from our previous experiments [39], and used these results to motivate a phase-field simulation of invasion. We found in our experimental data that while larger channels led to larger cluster sizes, the most common outcome was always a single-cell dissociation event. To reproduce this effect within our model, we had to introduce the possibility that cells at the invasive front might spontaneously develop a leader state with a stronger self-propulsion. Experimental measurements found no key difference between the time to rupture (survival time curves) for different channel sizes—opposite to the naive expectation that ruptures should be more common in channels with more cells. We were able to explain this feature of the data by assuming the cell adhesion weakens slightly in narrower channels relative to wider channels—a decrease of 30% in the narrowest channels. This was motivated by earlier results which find that contractility increases (leading to E-cadherin disengagement) and adhesions are more dynamic in confinement [39, 131, 132], but our results show that these effects need not be quantitatively dramatic to explain relatively large changes in rupture rates. Because explaining the survival curve in particular required this tuning of adhesion strength, we view this tuning as a prediction that could be validated by simultaneous measurement of cell-cell adhesion strength and rupture rates—though measuring cell-cell adhesion strength in the context of a narrow channel will likely be very experimentally difficult. We also identify key control factors within our model, showing that increases in cell-cell adhesion strength suppress rupture (as seen experimentally [39]), that the probability of cells becoming leaders changes the distribution of cluster sizes, and that increasing cell-wall adhesion can both permit rupture by permitting cells to invade into narrow channels, as well as suppressing it, by preventing cells from detaching from the wall.

For simplicity, in our current model, we have not included cell division and cell death. We think this is reasonable, given the timescale of our study (18 hours)—we believe cells are primarily being driven by migration. However, division could quantitatively change the jamming phase diagram we find here, e.g., via changing fluidity [145, 146], or by effects on cell polarity, speed, and cell-cell adhesion [22].

Earlier work has broadly argued that the degree of jamming and the ability of the tumor cells to rearrange may reflect whether cells from that tumor will metastasize [132, 147, 148]. We study, within our model, both the jamming transition and the rupture transition, and find that unjamming of the tissue is *necessary*, but not sufficient to predict rupture—there are many parameter sets where tissues are fluid but cancer cells are unable to dissociate. The details of this result may depend on assumptions of our model about cell-cell adhesion and motility. Researchers have employed various models to investigate the unjamming transition in tissues as a function of different control variables [24, 25, 114, 117]. For example, studies using the vertex model [24] and the Voronoi model [25] have explored how cell shape index and motility influence the jamming transition. These models make predictions about the influence of cell-cell adhesion—but adhesion enters only as an effective interfacial tension term [25, 149], changing the ability of cells to deform, but not immediately penalizing cell-cell separation or sliding of cell-cell junctions. Our focus on non-confluent tissues forces us to include cell-cell adhesion in a different form. We find that, in our phase field model, increasing cell-cell adhesion leads to less rearrangement and increased jamming. However, even within phase field models, there is no universal form of the cell-cell adhesion model, with Refs. [71, 114, 117, 122] all including adhesion in slightly different forms, and [73] disregarding it entirely, while [76] do not include a cell-cell adhesive force but do include the effect of a cell-cell friction, which also changes dynamics of rearrangement. Our model could also be extended to include more explicit dynamics of bond rupture [135, 150, 151]. Broadly, our results that increasing cell-cell adhesion leads to jamming is supported by experiments, but with some contradictory results. Many experiments support the idea that decreasing cell-cell adhesion can lead to tissue fluidization [32, 144, 152, 153] consistent with our observation. However, unjamming may also co-occur with the amplification of local adhesive intercellular stresses in human bronchial epithelial cells [24].

Our work shows that increasing cell-cell adhesion strength decreases the rate of rupture, but—if that rupture happens—increases the average size of the cluster that dissociates. This result may be relevant to the larger re-evaluation of the role of E-cadherin in cancer. The loss of intercellular adhesion molecule E-cadherin has long been considered a hallmark when tumor cells transition into an invasive phenotype resulting in cancer metastasis [154]. However, recent experiments have shown that E-cadherin expression is paradoxically correlated with cancer metastasis—while loss of E-cadherin increases invasion, it also reduces cancer cell proliferation and survival rates [154, 155]. Larger dissociation events may also be an aspect of this puzzle, because large-cluster dissociations may be more successful metastases [6].

We note that in our simulations at default parameters, all the survival probabilities drop to almost 0 [Fig. 2.5(b)], while the experimental survival curves saturate to approximately 30% [Fig. 2.2(b)]. The saturation of the experimental survival curves means either that there is something special about 30% of experiments (e.g., high degree of variability between channels), or that the system is aging—i.e., the probability of dissociation must decrease sharply as a function of time. There are two known sources of aging. First, the chemical gradient in the experiments of [39] is created by simple diffusion and will become more shallow over time, potentially decreasing the strength of chemotaxis, though we do not expect this to be relevant in the most narrow channels. Decreasing chemotactic strength leads to the flattening of survival curves in our model [Fig. 2.7(d)]. However, this is a weaker effect, and does not completely suppress dissociation. We view it as more likely that cell-cell adhesion strength increases over the experiment time, as has been previously observed in epithelial jamming [156]. Increasing cell-cell adhesion can completely suppress dissociations (Fig. 2.9). The survival of a small percentage of channels could also potentially reflect channel-to-channel variability in the experiment, e.g., in geometry or fibronectin coating. In a trivial example, if 30% of channels were not adherent, there would be no invasion in these channels—and hence no dissociations. While we do not observe anything this dramatic, variability of this sort (or day-to-day variability [157]) may represent additional potential confounders.

Our model simply assumes the polarity of cells follows a random reorientation, biased by the chemoattractant gradient [Eq. (2.3)]. This neglects any possible coordination of cell polarities between neighboring cells, e.g., alignment of cell polarity to neighbors [88], or contact inhibition of locomotion (CIL) leading to repolarization [71, 123, 158], or many others reviewed in Ref. [52]. We view our assumptions here as a first theoretical approach, but careful tests of cell-cell interaction may require us to extend our model. For instance, Ref. [127] shows that relatively rare populations of invasive leader cells can drive the invasion of non-leader cells, which could potentially be captured by this sort of cell-cell guidance [30, 159]. We also can address the possible roles of chemotaxis—directed cell migration—vs. chemokinesis—chemically-dependent increase in speed—in our simulations and experiments. We expect that the chemoattractant FBS is important both in terms of a chemokinetic factor and a chemoattractant. In the experiment, the absence of FBS results in significantly fewer dissociation events in 6-micron channels (see Fig. S3c from [39]), but changing gradient strength does not change dissociation. In our simulations, if there is no chemokinetic effect—i.e., if we reduce the cell velocities so that leader cells have the base self-propulsion strength p_0, as a model for the no-FBS condition—rupture is strongly suppressed. This no-FBS condition is similar to the point $\mathrm{Pe} \approx 0.5$ in the phase diagram of Fig. 2.9. The absence of chemotaxis but the presence of a chemokinetic factor (modeling constant but high FBS concentration) can be modeled by setting $k_\theta = 0$ in the model—removing any directed migration. Our results in Figs. 2.7(c) and 2.7(d) for very small k_θ indicate that cells are less coordinated, which leads to smaller cluster sizes. However, the survival probability curves are less strongly affected by the absence of chemotaxis.

Our work has focused on the rupture and dissociation in collective cancer invasion, but it may have interesting connections to fracture events in other biological systems. Reproduction by fission in the metazoan *Trichoplax adhaerens* occurs by a motility-driven ductile-to-brittle transition of their epithelial layers [13], which might also reflect some of the control parameters we identify here. Fracture of epithelial monolayers under strain has been shown to depend on both keratin organization within the cell and cell-cell adhesive bond rupture [135]. Epithelial fracture has also been modeled within a vertex model with a detachment energy [136], predicting that the transition between the ability of the tissue to rearrange under strain and its tendency to break depends on the effective interfacial tension and contractility of the cells. Our model may also reflect this difference—but in our case, ruptures are not induced by a strain applied to the tissue but self-generated by the cells' noisy motility. Formation of holes and voids can also occur in the spongy mesophyll, resulting in stable porous cell networks [160], or driven by cell contractility in epithelioid tissues [83]. An exciting future line of research is to understand to what extent and at what rate fracture events of all these various types can occur spontaneously, driven by randomness in cell motility and by chemical

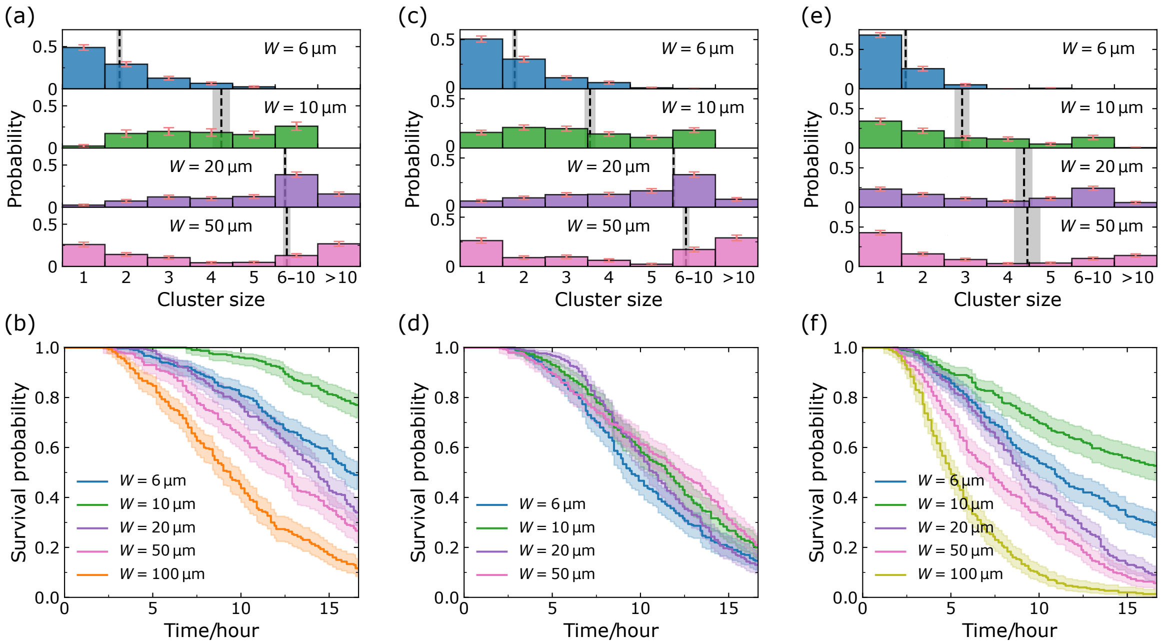

Figure 2.10: Simulation results for alternate models. Panels (a), (b) are simulation results with neither leader cells nor tuning of cell-cell adhesion ($\omega = \omega_0$ in all microchannels). Panels (c), (d) are simulation results after we tune down cell-cell adhesion in narrow microchannels (but no leader cells). Panels (e), (f) are simulation results after we introduce the leader cells but still keep cell-cell adhesion $\omega = \omega_0$ in all microchannels. 300 independent simulations; chemotaxis strength $k_\theta = 1.4k_\theta^0$. For the simulation results of 100 μm channels shown in panels (b) and (f), the total number of cells N is increased to 72.

gradients, as observed in our work.

Data and code to reproduce this chapter have been deposited at Zenodo, doi: 10.5281/zenodo.14890129.

2.6 Appendices

2.6.1 Alternate models

In this section, we present various models with different assumptions we have tested and show the process of how we reach our final model given in the main text.

2.6.1.1 Initial baseline model

As a starting point, we implement the simplest model with fixed intercellular adhesion $\omega = \omega_0$ and only two cell states. We designate cells inside the channel or near the channel entrance as guided cells, while all other cells are classified as follower cells. The statistical results of our simulations are shown in Figs. 2.10(a) and 2.10(b). We can see that the average cluster sizes are larger than those observed in the experiments, and single-cell ruptures are not predominant in all channels. Additionally, the survival curves decline much more slowly compared to the experiments and do not collapse. By intuition, we would expect the survival probability to decrease with increasing channel width. Surprisingly, we notice that the 10 μm curve is higher than the other curves. We will discuss this fact later in Sec. 2.6.2.

2.6.1.2 Downregulating intercellular adhesion

In the experiments, we observe a good collapse of all survival curves for different channel widths on top of each other. However, the survival curves of our initial model in Fig. 2.10(b) do not collapse. We attempt to address this issue by reducing intercellular adhesion in narrow channels, as motivated by earlier experimental work [39]. With a reduction of only 30% in 6- and 10-micron channels and 20% in 20-micron channels, we achieve the collapse of all survival curves [see Fig. 2.10(d)], but the cluster-size distribution and the average cluster size [Fig. 2.10(c)] are still qualitatively different from the experimental data, with only rare single-cell ruptures.

2.6.1.3 Introducing leader cells

Compared to the initial model, here we introduce the high-metabolic-activity leader cells at the leading edge into the model, but do not tune cell-cell adhesion as a function of channel width. We notice that single-cell ruptures are predominant now [see Fig. 2.10(e)]. As for the survival probability, they decrease more rapidly with time, but still do not collapse onto a single curve [Fig. 2.10(f)]. Again, we see that the 10-micron channel has the longest survival curves, which we address in Sec. 2.6.2.

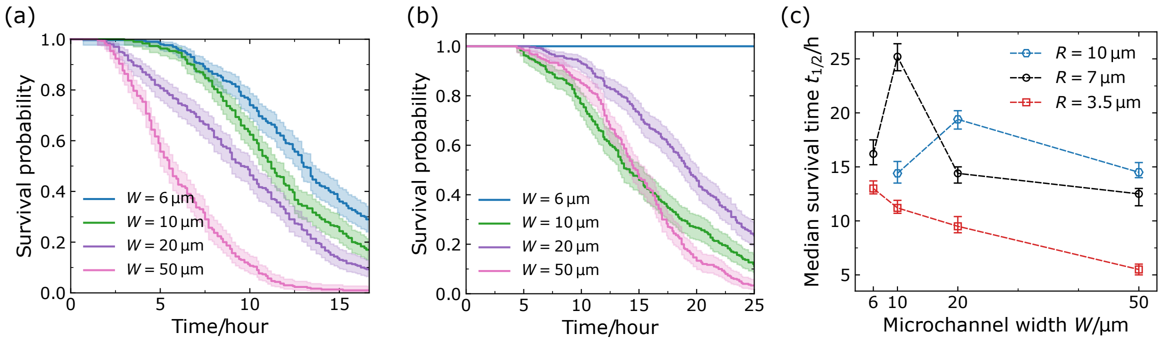

Figure 2.11: Simulation results for different cell sizes without special tunings (300 independent simulations, $\omega = \omega_0$ in all microchannels, $k_\theta = 1.4k_\theta^0$, no leader cells). Panels (a) and (b) show the survival curves for different channel widths when the cell radius $R = 3.5\,\mu$m and $10\,\mu$m, respectively. We notice that the 6-micron channels are too narrow for cells with a radius of $10\,\mu$m to enter, resulting in no ruptures occurring [blue curve in Panel (b)]. Panel (c) shows the median survival time $t_{1/2}$ as a function of microchannel width W for different cell sizes; the results for $R = 7\,\mu$m (black) are extracted from Fig. 2.10(b); the error bars indicate the 95% confidence intervals. When changing the cell radius R to 3.5 and $10\,\mu$m, we proportionally adjust the interfacial thickness $d = \sqrt{2K/\alpha}$ by varying α to fix d/R, and we also modify the number of cells N to 96 and 30 accordingly.

2.6.1.4 Final model

Our final model featured in the main text combines the adjustments made in the two models above: downregulating intercellular adhesion in narrow channels and introducing leader cells, which recapitulates both the cluster-size distribution and the collapse of survival curves observed in experiments.

2.6.2 How does cell size affect ruptures?

By intuition, we would expect the survival probability to decrease with increasing channel width. However, as we can see in Figs. 2.10(b) and 2.10(f), the 10-micron channel has slower ruptures than other channels. The reasoning behind the slower ruptures in the 10-micron channel lies in the proximity of its size to the default cell size (preferred radius $R = 7\,\mu$m) in our model. In the 10-micron channel, cells exhibit a broader cell-cell contact line compared to the 6-micron channel, making it harder to have ruptures—but in the 20-micron or wider channels, multiple cells can enter at once, so ruptures do not require breaking a larger contact line. Within our rough guiding hypothesis, we think of the 10-micron channel as having a higher initial energy barrier.

To verify our hypothesis, we first define the median survival time as the "half-life" of a survival curve $S(t)$, i.e., the time when $S(t) = 0.5$, or more explicitly,

$$t_{1/2} = S^{-1}(0.5). \tag{2.5}$$

For the default $R = 7\,\mu$m cells, we observe a clear peak of $t_{1/2}$ [extracted from Fig. 2.10(b)] at $W = 10\,\mu$m in Fig. 2.11(c). If our hypothesis holds true, the position of the peak should always be around the cell diameter $2R$. Therefore, we next vary the preferred cell radius R to see how the peak of median survival time $t_{1/2}$ changes. Figures 2.11(a) and 2.11(b) present the survival probabilities of the initial model for $R = 3.5\,\mu$m and $R = 10\,\mu$m, respectively. We notice that in Fig. 2.11(b), the 6-micron channels are too narrow for the cells with a diameter of $\sim 20\,\mu$m to enter, resulting in the survival probability remaining at 1. After computing the corresponding median survival times, we find that the channel width where $t_{1/2}$ peaks is indeed around the cell diameter and increases with cell size [Fig. 2.11(c)], which confirms our thoughts.

2.6.3 Initial conditions and domain wall profile

We first consider the static solution to the equation of motion [Eq. (2.1)] for an isolated cell in one dimension $\phi(x)$, so the active polarity vector $\mathbf{P} = 0$, $\mathcal{H}_{\text{exclusion}} = 0$, and all adhesion terms vanish. If the area of the cell is fixed, then the area deviation term can also be ignored. Thus Eq. (2.1) degenerates to $\delta\mathcal{H}_{\text{CH}}/\delta\phi = 0$ now, and the corresponding Euler-Lagrange equation of $\mathcal{H}_{\text{CH}}$ is [118]

$$\frac{\mathrm{d}^2\phi(x)}{\mathrm{d}x^2} = \frac{\alpha}{K}\left[\phi^3(x) - \frac{3}{2}\phi^2(x) + \frac{1}{2}\phi(x)\right]. \tag{2.6}$$

A domain wall can be introduced by forcing the two sides of the field to be in different phases, e.g., requiring $\phi(x \to -\infty) = 0$ and $\phi(x \to \infty) = 1$ as boundary conditions. By using the identity $\mathrm{d}^2\tanh(ax)/\mathrm{d}x^2 = -2a^2\mathrm{sech}^2(ax)\tanh(ax)$, it can be readily verified that the profile

$$\phi(x) = \frac{1}{2} + \frac{1}{2}\tanh\left(\frac{x - x_0}{2d}\right), \tag{2.7}$$

is a solution to the above nonlinear differential equation [Eq. (2.6)] adhering to the boundary conditions, provided $d = \sqrt{2K/\alpha}$ and x_0 represent the width and position of the domain wall. Since our phase field model is in two dimensions, we use

$$\phi(\mathbf{r}) = \frac{1}{2} + \frac{1}{2}\tanh\left(\frac{R-r}{2d}\right), \tag{2.8}$$

as the initial configuration, where $r = |\mathbf{r} - \mathbf{r}_0|$ is the distance to the center of the cell $\mathbf{r}_0$. The geometric-confinement field $\phi_{\text{wall}}(\mathbf{r})$ is defined by a series of logistic sigmoid functions in a similar manner.

2.6.4 Numerical details

By evaluating the functional derivative of the total Hamiltonian [Eq. (2.2)], the equation of motion [Eq. (2.1)] can be written as

$$\begin{aligned} \frac{\partial \phi_i}{\partial t} = & -\mathbf{P}_i \cdot \boldsymbol{\nabla}\phi_i - \frac{1}{\gamma}\left[\alpha\left(\phi_i^3 - \frac{3}{2}\phi_i^2 + \frac{1}{2}\phi_i\right) - K\boldsymbol{\nabla}^2\phi_i - \frac{4\lambda\phi_i}{\pi R^2}\delta V[\phi_i]\right. \\ & \left. + 2g\phi_i\left(h(\mathbf{r}) - \phi_i^2\right) + 2g_{\text{wall}}\phi_i\phi_{\text{wall}}^2\right] - \sum_{j\neq i}\omega f(\boldsymbol{\nabla}\phi_j)\cdot\boldsymbol{\nabla}\phi_i - \kappa f(\boldsymbol{\nabla}\phi_{\text{wall}})\cdot\boldsymbol{\nabla}\phi_i, \end{aligned} \tag{2.9}$$

where we have introduced another auxiliary field $h(\mathbf{r}) = \sum_i \phi_i^2(\mathbf{r})$ for parallel computing purpose (the simulation `C++` code is parallelized on multiprocessors by `Open MPI`) [73]. The area deviation term is $\delta V[\phi_i] = 1 - \int \mathrm{d}^2 r\ \phi_i^2/(\pi R^2)$, and we normalize the gradient via a saturation function $f(\boldsymbol{\zeta}) = \boldsymbol{\zeta}/\sqrt{1+\epsilon|\boldsymbol{\zeta}|^2}$ to prevent numerical instability resulting from very steep gradients [117]. Then, we employ forward differences to evolve the phase field equation for each cell, i.e.,

$$\phi_i(t+\Delta t) = \phi_i(t) + \text{R.H.S.}(t)\times\Delta t, \tag{2.10a}$$

$$\theta_i(t+\Delta t) = \theta_i(t) - k_\theta[\theta_i(t) - \theta_0]\times\Delta t + \Theta, \tag{2.10b}$$

where R.H.S.(t) is the right-hand side of Eq. (2.9), the noise $\Theta \sim \mathcal{N}(0,\, 2D_r\Delta t)$, and the preferred direction θ_0 points toward or along channels ($\theta_0 = \pi/2$). When computing the gradient $\boldsymbol{\nabla}\phi$ and Laplacian $\boldsymbol{\nabla}^2\phi$, we apply first- and second-order central differences, respectively. We have matched

Table 2.1: Table of simulation parameters. These parameters are used throughout the chapter; any deviation from them is explicitly noted.

Parameter	Description	Dimension	Dimensionless	Real value
δx	Lattice constant	L	1	$1\,\mu$m
δt	Simulation time unit	T	1	15 s
N	Number of cells	1	48	48
Δt	Simulation time step	T	0.4	6 s
R	Target cell radius	L	7	$7\,\mu$m
p_0	Advection velocity (polarity)	L/T	5.556×10^{-2}	$13.3\,\mu$m/h
D_r^{-1}	Reciprocal of rotational diffusion coefficient	T	180	45 min
d	Interfacial thickness	L	2	$2\,\mu$m
γ	Friction coefficient	ET/L^2	12	
K	Cell tension coefficient	E	3.5	
α	Cell interfacial coefficient ($2K/d^2$)	E/L^2	1.75	
λ	Cell area coefficient	E	250	
g	Cell-cell repulsion coefficient	E/L^2	4	
g_{wall}	Cell-wall repulsion coefficient	E/L^2	5	
ω	Cell-cell adhesion coefficient	L^2/T	3.5–5	
ω_0	Reference adhesion coefficient	L^2/T	5	$\sim 0.3\,\mu\text{m}^2/\text{s}$
κ	Cell-wall adhesion coefficient	L^2/T	3.5	$0.23\,\mu\text{m}^2/\text{s}$
ϵ	Regularization coefficient	L^2	37.25	$37.25\,\mu\text{m}^2$
k_θ	Chemotaxis strength coefficient	$1/T$	10^{-3}–0.05	0.004–0.2 min^{-1}
k_θ^0	Reference chemotaxis strength coefficient	$1/T$	0.0025	0.01 min^{-1}
n_c^0	Default characteristic number of contact points n_c	1	256	

all default parameter values in our model with the experiments, which are given in Table 2.1.

During the simulation, we first implement a "thermalization" process for $T_{\text{therm}} = 5$ hours which allows all cells to evolve from the initial conditions we imposed and reach more natural shapes. After this process, we open the channels and keep on watching for $T_{\text{tot}} = 18$ hours, consistent with the experiments. Our study focuses on the first dissociation event within the channel. Therefore, we terminate the simulations once the first dissociation occurs for each channel, recording the rupture times and rupture sizes for statistical analysis. (The only exception to this is the simulations in Fig. 2.4, which we run to show the separation dynamics of the clusters as an illustration.) As a result, reattachment following dissociation is not considered in our analysis. Note that while we implement periodic boundary conditions in both the x and y directions, there is also a bottom wall, making the channels effectively closed at the top. We expect that the presence of this wall does not influence our core statistical results on dissociations. In our simulations, ruptures generally occur before cells hit the wall, and all channels generally have a rupture (Fig. 2.5)—except when motility is slowed (low Pe), in which case cells do not even reach the end of the channel. Once the first cells of the invading strand hit the wall, this suppresses dissociation—similar to if we had ended the simulation when cells impact the wall or removed them from the system, the other plausible boundary conditions. However, the presence of a wall (like the presence of a finite channel size in the experiments of [39]) could impact the very long-time dynamics in cases when dissociation is slow. Our simulated system size (channel length of 200 microns) is slightly smaller than the experimental channels (~ 400 microns); this effect, like our finite number of cells, could in principle reduce the likelihood of extremely large clusters being found, though we have seen no evidence of this in the experiment.

The initial magnitudes of polarity for all cells are set to p_0. Cells become guided once they get close enough to the channel. We define a box of $(W + 24\,\mu\text{m}) \times 6\,\mu\text{m}$ at the channel entrance in the simulation. Once the center of mass of a cell crosses into this region or inside the channel, the cell is considered a guided cell ($k_\theta \neq 0$), which can sense the chemical gradient and move with a larger velocity (magnitude of polarity set to $2p_0$). We also introduced leader cells at the invading front in our model—cells have a probability of becoming leader cells when they reach the front (become a

Algorithm 1: How to identify cell states

```
Input: Cell states at previous time t − Δt
Output: Cell states at current time t

foreach cell within the monolayer do
    if cell is inside the channel then
        if cell is at the leading edge then
            if cell is at the leading edge at t − Δt then
                keep the cell state;
            else if RandomUniform(0, 1) < P_i^leader then
                /* P_i^leader is given by Eq. (2.4)                 */
                cell state ← leader;
            else cell state ← guided;
        else
            cell state ← guided;
        end
    else
        cell state ← follower;
    end
end
```

tip cell). First, we have to define what a tip cell is: if there is any point of the cell surface where there is not a cell in the monolayer immediately above it, it is at the tip. In our model, whenever a guided cell becomes a tip cell, there is a probability for it to transition into a leader (magnitude of polarity set to $3p_0$). We give a sketch of the algorithm for this process in Algorithm 1. As discussed in the main text, the probability of becoming a leader cell inversely scales with its contact length with other cells in our model—cells are more likely to become leader cells if they have a lot of free space. Based on the form of the adhesion term in the model, we count a grid point at position $\mathbf{r}$ as a contact point of cell i when $|\boldsymbol{\nabla}\phi_i(\mathbf{r}) \cdot \sum_{j\neq i} \boldsymbol{\nabla}\phi_j(\mathbf{r})| = |\boldsymbol{\nabla}\phi_i \cdot (\boldsymbol{\nabla}h - \boldsymbol{\nabla}\phi_i)| > 0.01\,\mu\mathrm{m}^{-2}$. Therefore, Eq. (2.4) is calculated as $P_i^{\mathrm{leader}} = \max(1 - n_i/n_c,\, 0)$ in practice, where n_i is the number of contact points of cell i, and n_c is set to 256 in our simulation by default. Also, we note that in Algorithm 1 when a leader cell is no longer at the tip, it will revert to a guided cell state.

To improve computational efficiency, we also implement domain decomposition [71, 73, 116, 120], where we solve for each ϕ_i only within a smaller subdomain that shifts with the center of mass $\mathbf{R}_i^{\mathrm{c.m.}}$

of each cell. The center of mass position for each cell is computed using the following equation:

$$\mathbf{R}_i^{\text{c.m.}} = \frac{1}{M_i} \int \mathrm{d}^2 r \; \mathbf{r}\, \phi_i(\mathbf{r}), \tag{2.11}$$

where $M_i := \int \mathrm{d}^2 r \; \phi_i(\mathbf{r})$ is the total "mass" of the cell. Thus the center of mass velocity is readily given by $\mathbf{v}_i^{\text{c.m.}} = \mathrm{d}\mathbf{R}_i^{\text{c.m.}}/\mathrm{d}t$. The mean squared displacement mentioned in Sec. 2.4.7 is computed by [73]

$$\mathrm{MSD}(t) = \left\langle \frac{1}{N} \sum_{i=1}^{N} \left[\mathbf{R}_i^{\text{c.m.}}(t+\tau) - \mathbf{R}_i^{\text{c.m.}}(\tau)\right]^2 \right\rangle_\tau , \tag{2.12}$$

where the average $\langle \cdots \rangle$ is taken over time $\tau \in [0, T_{\text{tot}} - t]$.

We apply standard Kaplan-Meier survival analysis [115] to handle the data and generate survival probability plots using the `lifelines.KaplanMeierFitter` class provided by the `Python` survival-analysis package `lifelines` [161]. The 95% confidence intervals are computed using the exponential Greenwood confidence interval method [162].

2.6.5 Additional figures

Within the main text of this chapter (e.g., Fig. 2.2 and Fig. 2.5), we show the cluster size histograms with some cluster sizes grouped together, for legibility and clarity. In this section, we provide the full histograms. We also show the scatter plots of cluster size vs. rupture time for different channel widths.

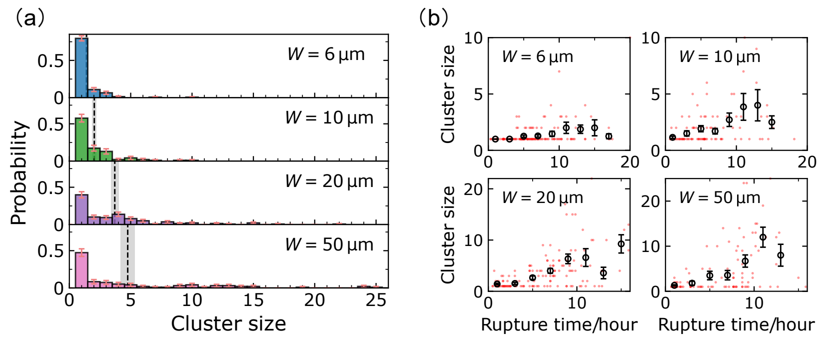

Figure 2.12: Experimental results. (a) Cluster-size distributions for microchannels of different widths. The dashed lines and gray areas denote the average cluster sizes A and their corresponding standard errors of the mean (SEM, or just SE). The error bars of the probabilities are determined by the binomial interval formula $\pm\sqrt{p(1-p)/n}$. (b) Scatter plots exhibit the (rupture time, cluster size) phase plane. The circles represent the average cluster sizes within each time bin (2 hours), and the error bars indicate the corresponding SE. The standard error of the mean (SE) is calculated as $\mathrm{SE} = \mathrm{SD}/\sqrt{n}$, where SD is the standard deviation and n is the sample size.

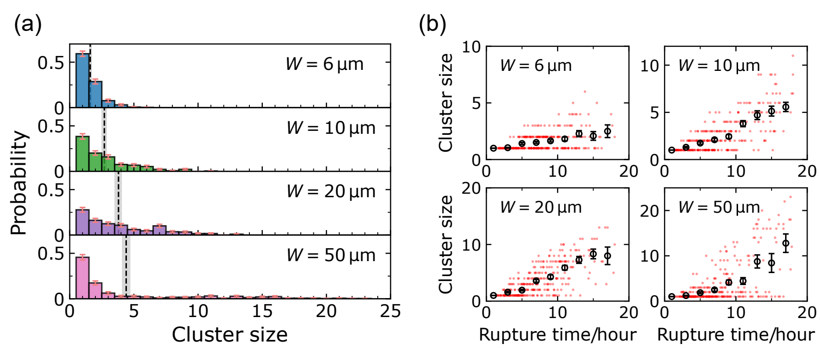

Figure 2.13: Simulation results with $\omega = 0.7\omega_0$ for narrow channels, $\omega = 0.8\omega_0$ for $20\,\mu m$ channels, and $\omega = \omega_0$ for $50\,\mu m$ channels. (a) Cluster-size distributions for different channel widths. The dashed lines and gray areas denote the average cluster sizes A and their corresponding standard errors of the mean (SE). Panel (b) displays the (rupture time, cluster size) phase plane scatter plots. The circles represent the average cluster sizes within each time bin (2 hours), and the error bars indicate the corresponding SE.

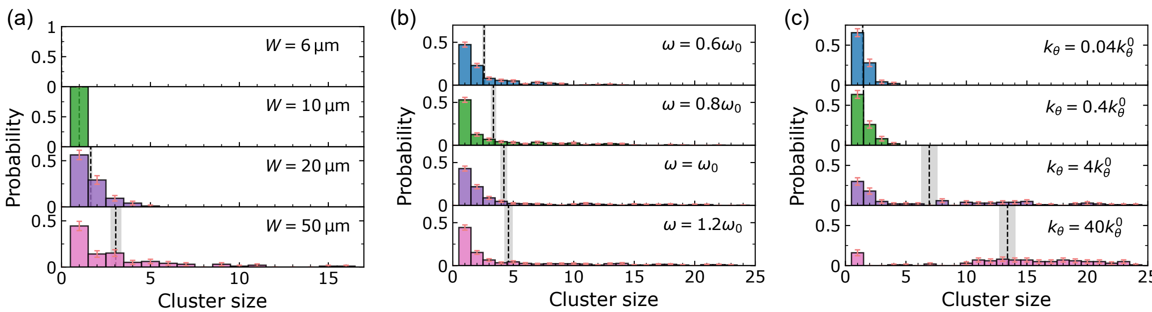

Figure 2.14: Simulation results. (a) Histograms when cell-wall adhesion κ is set to 0. (b) Histograms for different adhesion strength ω in $50\,\mu$m channels. (c) Histograms for different chemotaxis strength k_θ in $50\,\mu$m channels. The dashed lines and gray areas in (a)–(c) denote the average cluster sizes A and their corresponding SE.

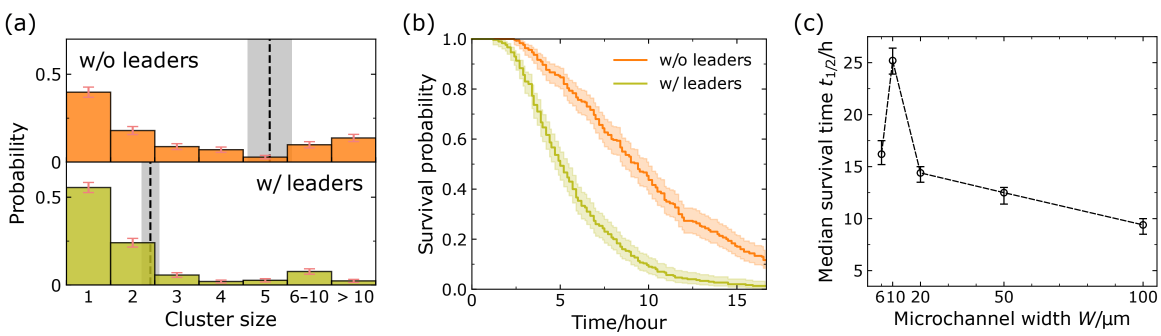

Figure 2.15: Simulation results for $W = 100\,\mu$m channels with cell-cell adhesion fixed at $\omega = \omega_0$. (a) Histograms for Model 1 (without leader cells) and Model 3 (with leader cells). (b) Survival curves for Model 1 and Model 3, which are also presented in Figs. 2.10(b) and 2.10(f). (c) Median survival time $t_{1/2}$ for $R = 7\,\mu$m, extended to $100\,\mu$m based on Fig. 2.11(c). Results are based on 300 independent simulations with the chemotaxis strength of $k_\theta = 1.4k_\theta^0$, and the number of cells N increased to 72.

Chapter 3
Controlling tissue size by active fracture

This chapter is adapted from the following article with permission from the American Physical Society:

- Wang, W. & Camley, B. A. Controlling tissue size by active fracture. *Phys. Rev. E* **113,** 034405. DOI: 10.1103/dk15-hwzg (2026).

3.1 Introduction

How does an organ or an organism control its size? Size is thought to be tightly regulated by feedbacks controlling cell division [43]. However, two recent experiments suggest a different possibility—that the size of a group of cells can arise from a competition between growth, which tends to make the group larger, and random cell motility, which can make the group fracture into multiple pieces, reducing group size. In the metazoan *Trichoplax adhaerens*, asexual reproduction by fission is driven by motility-induced fractures [13]. Similarly, germline cysts in mice are formed by a combination of cell division and fracture of intercellular bridges by random cell motility [12], as shown in Fig. 3.1. These mechanisms, which depend on material failure, are reminiscent of how cancerous cells break from an invading front [39, 112, 163] (see also Chapter 2)—which seems very different than a well-regulated organism attempting to keep a fixed size. Analogous fracture-limited size control also occurs in multicellular snowflake yeast clusters, though driven by crowding-induced mechanical stress, not motility [42]. We want to know: can motility-driven fracture alone reliably regulate the size of a group of cells? Both *T. adhaerens* and germline cysts show significant variability in size [12, 13, 44, 45]. What physical factors control the group size and its variability?

We argue that the size of cell clusters controlled by fracture is set by competition between the break rate of cell-cell junctions k_b and the cell division rate k_d [164]. We model break rate k_b from a mechanical perspective, using a simple one-dimensional model of cells as active particles connected by springs. The break rate depends on typical cell speed, cell-cell junction strength, and the cell's persistence time. We then develop models of cluster growth and fracture. We derive the exact

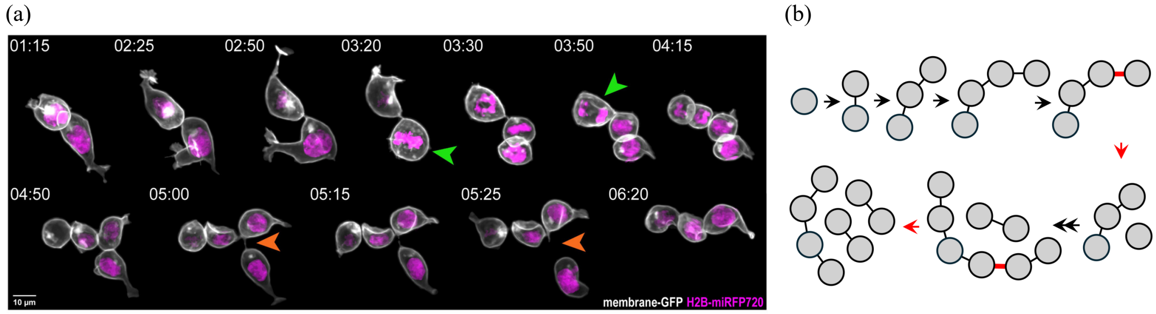

Figure 3.1: (a) Live image of an E12.5 ovary shows a germline cyst undergoing both cell division and fracture. Green arrows indicate cells that are about to divide, and orange arrows show the fracture of an intercellular bridge. Reprinted from Ref. [12], with permission from Elsevier (License Number: 6105550597121). (b) Schematic illustration of cell division and fracture. Red lines denote intercellular bridges that undergo fracture.

steady-state cluster size distribution, finding that cluster sizes depend solely on the ratio k_b/k_d, allowing us to link cluster size to cell adhesion and motility. The quality of cluster size control can be improved if only cells on the edge of the cluster divide or if only cell-cell junctions near the cluster middle can fracture.

In this chapter, we first develop a one-dimensional theory for motility-driven fracture and size regulation in cell clusters in Sec. 3.2. We start, in Sec. 3.2.1, by deriving an analytic expression for the junction break rate from a minimal mechanical model. Sec. 3.2.2 subsequently analyzes three biologically motivated growth scenarios and obtains exact steady-state size distributions. Sec. 3.2.3 shows how the model parameters are constrained by available experimental data, and Sec. 3.2.4 demonstrates that the survival probability of an intact cluster where all cells divide follows the universal decay $S(t) = \mathrm{e}^{-k_d t}$, governed solely by the cell division rate. Finally, we test the robustness of the theory by comparing with two-dimensional active-particle simulations in Sec. 3.3.

3.2 Results for one-dimensional chain

3.2.1 Mechanics

We initially model our cell cluster as a one-dimensional chain of cells. This geometry is natural for germline cysts, which are composed of potentially-branched chains of cells that can be fragmented by a single junction break (Fig. 3.1), and linear geometries of strings of cells are often found in cells confined in extracellular matrix or experiments performed in microchannels or micropatterns [39, 112, 165, 166]. One dimension may be appropriate for *T. adhaerens* when it takes on an elongated string-like shape [13], though a higher-dimensional model is necessary for a full study of *T. adhaerens*.

We treat our chain of cells as self-propelled active particles with positions $\{x_n\}$ connected by springs [Fig. 3.2(a)], assuming an overdamped environment:

$$\dot{x}_n = -\mu \nabla_n \Phi + v_n(t), \tag{3.1}$$

where μ is the particle mobility and $\Phi = \frac{1}{2}k \sum_{\langle i,j \rangle} \left(|x_i - x_j| - \ell_0\right)^2$ is the cell-cell interaction energy. Here, k is the spring constant, ℓ_0 is the natural length of the springs, and $\langle i,j \rangle$ denotes the summation over nearest neighbors. The active velocities v_n—the velocities the cell would have in the absence of cell-cell interactions—are Ornstein-Uhlenbeck (OU) processes [63]:

$$\tau \dot{v}_n = -v_n + \sqrt{2D}\xi_n(t), \tag{3.2}$$

where $\xi_n(t)$ are independent, zero-mean, unit-variance Gaussian white noises. τ is the persistence time of the cell, while D controls the typical cell speeds—v_n have mean zero and correlations $\langle v_n(t) v_{n'}(t') \rangle = \delta_{nn'}(D/\tau)\mathrm{e}^{-|t-t'|/\tau}$ if $t, t' \gg \tau$. In the limit $\tau \to 0$, the active velocities v_n reduce to Gaussian white noises—in this limit the system is in thermal equilibrium, but will be out of equilibrium for a finite τ.

What is the mean first time to rupture for a given link? Our initial intuition was that the rupture rate within a cluster would be the same as that of a two-particle link, as in thermal equilibrium. To explore this, we first compute the variance of the stretched length $\delta\ell = \ell - \ell_0$ (where ℓ is the distance between the particles) in the two-particle case. Extending the approach of Ref. [134], we can map this problem to an inertial Brownian particle in a harmonic potential $U(\delta\ell) = \mu k \delta\ell^2$ experiencing a friction $\eta = 1 + 2\mu k \tau$, and an effective temperature $k_B T_{\text{eff}} = 2D/\eta$ (see Sec. 3.5.1.1). The distribution of $\delta\ell$ in the steady state is Boltzmann-like $\sim \exp(-U/k_B T_{\text{eff}})$, i.e., a Gaussian distribution with zero mean and variance

$$\langle \delta\ell^2 \rangle_2 = \frac{k_B T_{\text{eff}}}{U''} = \frac{D}{\mu k(1 + 2\mu k\tau)}. \tag{3.3}$$

The subscript indicates this is the two-particle result. When $\tau = 0$, $\langle \delta\ell^2 \rangle_2$ approaches the thermal equilibrium solution $\langle \delta\ell^2 \rangle_{\text{th}} = D/\mu k$. If the spring breaks when stretched beyond a critical length $\ell_0 + \delta\ell_b$, the mean escape time can be estimated using standard Kramers' theory. For small effective temperatures, the time for the pair to break is given by

$$\tau_{\text{esc}}^{\text{pair}} = \tau_0 \exp\left(\frac{\Delta U}{k_B T_{\text{eff}}}\right) = \tau_0 \exp\left[\frac{\mu k \delta\ell_b^2 (1 + 2\mu k\tau)}{2D}\right], \tag{3.4}$$

where τ_0 is a subexponential correction [134, 167–170].

For a long chain of $N \gg 1$ cells, we assume that $x_n(t) = n\ell_0 + \delta x_n(t)$, where δx_n represents the deviation of x_n around the lattice point $n\ell_0$. We also assume periodic boundary conditions, $\delta x_{n+N} = \delta x_n$—we find this does not influence our central results in the limit $N \to \infty$ and makes calculation much simpler. We use a standard discrete Fourier transform:

$$\tilde{x}_q = \frac{1}{N} \sum_{n=0}^{N-1} \delta x_n \mathrm{e}^{-iqn}, \tag{3.5}$$

where $q = 2\pi m/N$ and the index $m = 0, 1, \cdots, N-1$. The inverse transform is $\delta x_n = \sum_q \tilde{x}_q \mathrm{e}^{iqn}$. By substituting the inverse transforms into Eq. (3.1) and matching terms for each mode q, we can then obtain (see Sec. 3.5.1.2 for more details):

$$\dot{\tilde{x}}_q = -\mu k_q \tilde{x}_q + \tilde{v}_q, \tag{3.6}$$

where $k_q = 2k(1 - \cos q)$ and $\tilde{v}_q = (1/N) \sum_{n=0}^{N-1} v_n \mathrm{e}^{-iqn}$ is the Fourier amplitude of v_n, with mean $\langle \tilde{v}_q \rangle = 0$, and time correlations:

$$\langle \tilde{v}_q(t) \tilde{v}_{q'}^*(t') \rangle = \delta_{qq'} \frac{D}{N\tau} \mathrm{e}^{-|t-t'|/\tau}. \tag{3.7}$$

By solving Eq. (3.6), the zeroth mode (center of mass) evolves as $\langle |\tilde{x}_0(t)|^2 \rangle = 2Dt/N$, while for the

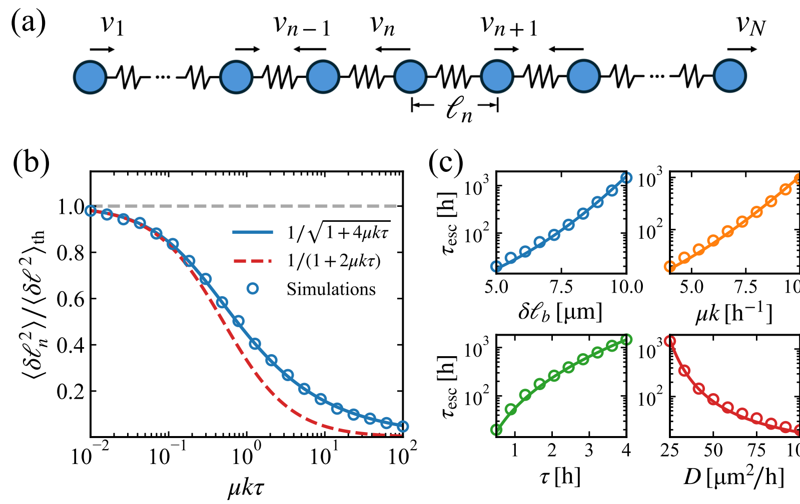

Figure 3.2: (a) Illustration of the active chain model. Cells are connected by springs; each cell has active velocity v_n. (b) Variance of the spring stretch $\langle \delta\ell_n^2\rangle$ as a function of $\mu k\tau$. Gray dashed line is the thermal variance $\langle \delta\ell^2\rangle_{\rm th} = D/\mu k$. We change $\mu k\tau$ by varying τ while fixing μk in the simulations. Red dashed line is the two-particle result $\langle \delta\ell^2\rangle_2$. (c) Mean escape time $\tau_{\rm esc} = 1/k_b$. Empty circles are simulation results with $N = 1000$ cells; solid lines are theory, Eq. (3.10).

nonzero modes:

$$\langle \tilde{x}_q(t)\tilde{x}^*_{q'}(t)\rangle = \delta_{qq'}\frac{D}{N\mu k_q(1+\mu k_q\tau)}. \tag{3.8}$$

We can then calculate the stretched lengths of the springs by $\delta\ell_n = \delta x_{n+1} - \delta x_n$, where the mean is simply $\langle \delta\ell_n\rangle = 0$, and the variance is

$$\langle \delta\ell_n^2\rangle = \sum_{q\neq 0}\sum_{q'\neq 0}\langle \tilde{x}_q\tilde{x}^*_{q'}\rangle \mathrm{e}^{i(q-q')n}(\mathrm{e}^{iq}-1)(\mathrm{e}^{-iq'}-1) = \frac{D}{N\mu k}\sum_{q\neq 0}\frac{1}{1+2\mu k\tau(1-\cos q)}.$$

Replacing the sum by an integral, we obtain

$$\langle \delta\ell_n^2\rangle = \frac{D}{\mu k\sqrt{1+4\mu k\tau}}, \tag{3.9}$$

which is different from the two-particle result $\langle \delta\ell^2\rangle_2$ at large $\mu k\tau$ [Fig. 3.2(b)].

Notably, $\langle \delta\ell_n^2\rangle = \langle \delta\ell^2\rangle_{\rm th}$ when $\tau \to 0$, but $\langle \delta\ell_n^2\rangle$ also converges to the two-particle result $\langle \delta\ell^2\rangle_2$ when $0 < \mu k\tau \ll 1$, as shown in Fig. 3.2(b). This implies that when $\mu k\tau$ is small but finite, the system resides in an *effective equilibrium* regime [64], and such a convergence can be understood through a modified equipartition theorem (see Sec. 3.5.1.3). In the steady state, the distribution of $\delta\ell_n$ follows a Gaussian form, $P_s(\delta\ell_n) \sim \exp(-\delta\ell_n^2/2\langle\delta\ell_n^2\rangle)$. The escape rate will be proportional to the probability density at the breaking length $P_s(\delta\ell_b)$ [134], yielding

$$k_b = \tau_0^{-1} \exp\left(-\frac{\mu k \delta\ell_b^2 \sqrt{1+4\mu k\tau}}{2D}\right), \tag{3.10}$$

where we have found the subexponential correction $\tau_0 = \pi\eta/\mu k$ leads to a good fit to the escape time from our simulation of active cell motion [see Fig. 3.2(c)]. Time to rupture increases with a higher break threshold $\delta\ell_b$, stiffer springs, or greater cell persistence, while it decreases if cells are faster (larger D). Unlike the two-particle problem, where the mean escape time grows exponentially with k in equilibrium and k^2 in the $\mu k\tau \gg 1$ nonequilibrium limit, it grows here exponentially in $k^{3/2}$ for large k.

3.2.2 Growth models

We have described an active chain in which links can break at a rate k_b, while cells can also divide, leading to an increase in the chain length. What controls the cluster size, and how broad is its distribution? As in germline cysts, where all cells can divide [12], we first assume that each cell divides independently with the same division rate k_d. To analyze the distribution of the chain length, we randomly pick one of the daughter chains to track once fragmentation occurs. The selected daughter chain then continues to grow and rupture in the same manner as the original chain, as illustrated in Fig. 3.3(a).

Assuming $p_n(t)$ gives the probability that the length of the tracked chain at time t is n, the master equation governing such a process is, generalizing models of polymerization and fragmentation [171],

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = \sum_{m>n} k_b p_m - p_n \sum_{i+j=n} k_b + k_d(n-1)p_{n-1} - k_d n p_n, \tag{3.11}$$

where the first two terms on the right-hand side describe the fragmentation process while the last two

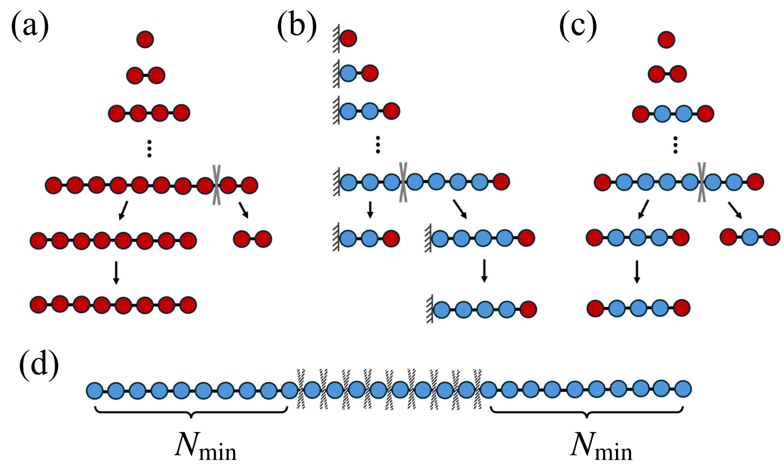

Figure 3.3: Growth and fracture of cell groups under different assumptions. (a) All cells can divide. After a rupture occurs, one daughter chain is selected at random. (b) Only the cell at one end of the chain can divide. (c) Only the cells at the two ends can divide. (d) Potential rupture points when a minimum cluster size N_{min} is enforced. Rupture can only occur when the chain length $n \geqslant 2N_{\text{min}}$. Red cells have a non-zero division rate k_d, while blue cells have a division rate of zero; gray crosses indicate actual or potential rupture points.

terms represent the growth of the chain through cell division. $\sum_{m>n} k_b p_m$ corresponds to the rate of obtaining an n-mer through the fragmentation of a longer chain. Though there are two possible rupture points when an m-mer breaks to form an n-mer, the random selection of one daughter chain causes the prefactors to cancel out. In the second term, $\sum_{i+j=n} k_b = (n-1)k_b$ describes the rupture of the n-mer chain into two daughter fragments, where $(n-1)$ is exactly the number of connections within an n-mer chain. The two terms for the growth process give the rates that an $(n-1)$-mer grows into an n-mer, and an n-mer grows into an $(n+1)$-mer, respectively. Using generating functions, we find the steady-state solution:

$$p_n = \frac{\gamma}{(1+\gamma)^n}, \qquad \text{(all-cell growth)} \tag{3.12}$$

where $\gamma \equiv k_b/k_d$ is the ratio of the break rate to the division rate (see Sec. 3.5.2.1).

So far, we have assumed that all cells in the chain can divide. However, cell division may be regulated by mechanical and spatial constraints. Cells with fewer neighbors or experiencing lower compression

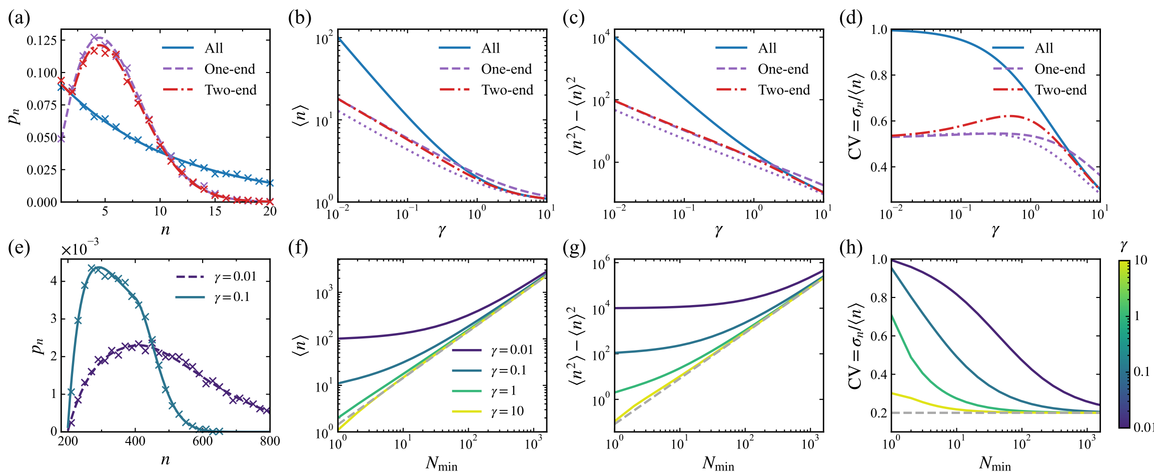

Figure 3.4: (a) Steady-state distribution of chain length p_n for the three growth modes, with $\gamma = 0.1$ for the all-cell growth and two-end growth models, and γ replaced by $\gamma' = \gamma/2 = 0.05$ for the one-end growth model (doubled division rate). Crosses represent simulation results, while lines correspond to theoretical predictions. Panels (b)–(d) display the mean $\langle n \rangle$, variance $\langle n^2 \rangle - \langle n \rangle^2$, and coefficient of variation $\mathrm{CV} = \sigma_n/\langle n \rangle$, where $\sigma_n = \sqrt{\langle n^2 \rangle - \langle n \rangle^2}$. Purple dashed lines are the one-end result after doubling the division rate [$\gamma \to \gamma' = \gamma/2$ in Eq. (3.14)], while the purple dotted lines show Eq. (3.14). (e) Steady-state distribution p_n when minimum cluster size $N_{\min} = 200$. Crosses are simulation results, lines are theoretical predictions by the matrix method. Panels (f)–(h) show how mean, variance, and CV vary with $N_{\min}$ in the all-cell growth model. Gray dashed lines show predictions assuming log-uniform distributions [Eq. (3.18)].

are more likely to divide—"contact inhibition of proliferation" [172]. As an extreme limit of contact inhibition, we develop alternate models where only cells at the chain ends divide. First, consider the case where only one end of the chain grows—e.g., if the other end is constrained by a barrier. In contrast to the all-cell growth model, where the total growth rate scales as $k_d n$ (proportional to chain length), in one-end growth, the total growth rate is fixed at k_d [Fig. 3.3(b)]. Here, the master equation becomes

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = \sum_{m>n} k_b p_m - p_n \sum_{i+j=n} k_b + k_d p_{n-1} - k_d p_n. \tag{3.13}$$

We find that the steady-state solution is

$$p_n = n\gamma^{1-n} \Big/ \left(\frac{1+\gamma}{\gamma}\right)_n, \qquad \text{(one-end growth)} \tag{3.14}$$

where $(x)_n = x(x+1)(x+2)\cdots(x+n-1)$ is the Pochhammer symbol (see Sec. 3.5.2.2). When

$\gamma \ll 1$, p_n can be approximated by a Rayleigh distribution

$$f(n;\sigma) = n\gamma e^{-\gamma n^2/2}, \tag{3.15}$$

with mode $\sigma = 1/\sqrt{\gamma}$.

If the chain can grow from both ends, i.e., only the two cells at the two ends can divide, the total growth rate is $2k_d$, except when $n = 1$ where the total growth rate is k_d [Fig. 3.3(c)]. For this case, the steady-state solution is (see Sec. 3.5.2.3)

$$p_n = \begin{cases} \gamma/(1+\gamma)^n, & n = 1, 2, \\ \dfrac{2+\gamma}{2(1+\gamma)} \dfrac{n(\gamma/2)^{1-n}}{((2+\gamma)/\gamma)_n}, & n > 2. \end{cases} \qquad \text{(two-end)} \tag{3.16}$$

In the limit of rare breaking $\gamma \ll 1$ when clusters almost always have two ends, p_n converges to Eq. (3.14) with a doubled division rate.

Fig. 3.4(a) shows our theoretical predictions p_n for the three growth models, validated by Monte Carlo simulations of chain evolution (see Sec. 3.5.6.2). If all cells can divide, the cluster size distribution is extremely broad with a peak at $n = 1$; if only end cells divide, we have a better-defined peak at $n \approx 1/\sqrt{\gamma}$.

Larger growth rates relative to break rates lead to larger clusters, but does this also result in greater variability of cluster size? With the distributions p_n, we can compute the means and variances of n [Figs. 3.4(b) and 3.4(c)]. In the limit $\gamma \to 0$, fragmentation is rare, leading to both increased growth and increased variability in all models, while in the limit $\gamma \gg 1$, fragmentation dominates, and clusters become single-celled. To quantify the relative variability, we plot the ratio of the standard deviation of cluster size to the mean (coefficient of variation CV) in Fig. 3.4(d). We see a key difference between the end-growth models and the all-growth model when $\gamma \ll 1$ and cluster sizes are large—CV approaches 1 for the all-growth model, while it approaches $\sqrt{(4-\pi)/\pi} \approx 0.523$ for end-growth. The CV for end-growth is roughly half that for all-growth, indicating better size control.

If the interior cells can still divide, but at a lower rate, the CV is intermediate between end- and all-growth (see Sec. 3.5.4).

We have so far assumed all rupture rates to be identical, but secreted factors [173] or stress [174] may vary across the cluster. For instance, it is well established that in large epithelial sheets, stress is peaked at the center of the tissue [174]—suggesting that fractures are more likely to happen at the center of the cluster. Instead of explicitly modeling the stress profile from contact inhibition of locomotion (CIL) [47] and other factors, we include a somewhat *ad hoc* extension of the model, making clusters more likely to break in the center by assuming that only the junctions sufficiently far from the cluster edge break. This is equivalent to introducing a minimum cluster size $N_{\min}$ [Fig. 3.3(d)]. Once we include the constraint that all clusters are larger than $N_{\min}$, we cannot solve the master equation analytically. However, in the steady state, we can write it simply as a matrix equation

$$\boldsymbol{K}\mathbf{p} = \mathbf{0}, \tag{3.17}$$

which can be easily solved numerically to find $\mathbf{p}$ (see Sec. 3.5.3). This method leads to good agreement with stochastic simulations [Fig. 3.4(e)].

In the absence of a minimum cluster size, large cluster sizes always come with an unavoidable variability [Figs. 3.4(b)–(d)], while large rupture rates $\gamma \gg 1$ lead to sizes being precise but $\langle n \rangle \to 1$. Given a minimum cluster size, it is possible to have precise size control and finite clusters [Figs. 3.4(e)–(h)]. In the limit $\gamma \gg 1$, cell clusters grow from size $N_{\min}$ to $2N_{\min}$, then fragment in half. This is very similar to the dynamics of a single cell's growth, akin to a sizer model [175, 176], and leads to distributions p_n [Fig. 3.4(e)] like those of single cell size distributions [175–178]. During each growth cycle $t \in [0, t_c]$, we have $\langle n \rangle = N_{\min} \mathrm{e}^{k_d t}$ if all cells can divide, where $t_c = \ln 2 / k_d$ is the time required to double the chain's length. Over a long trajectory, any moment within the chain's lifecycle is equally likely to be sampled [175]. If t is uniformly distributed in one cycle $[0, t_c]$, then p_n should obey a log-uniform distribution, as shown in Fig. 3.8(e):

$$p_n = \frac{1}{n \ln 2}, \quad n \in [N_{\min}, 2N_{\min}], \tag{3.18}$$

where the mean is $\langle n \rangle = N_{\min}/\ln 2$ and the variance is $\langle n^2 \rangle - \langle n \rangle^2 = (3\ln 2 - 2)\langle n \rangle^2/2$, i.e., the coefficient of variation CV is independent of $N_{\min}$, as shown in Figs. 3.4(f)–(h). Similarly, if only edge cells can divide, the chain length increases linearly with time. p_n is then uniformly distributed in $[N_{\min}, 2N_{\min}]$, leading to a mean of $3N_{\min}/2$ and a variance of $N_{\min}^2/12$. When $\gamma \gg 1$, mean cluster size scales as $N_{\min}$ and the variance scales as $N_{\min}^2$, but the CV goes to a nonzero constant, and is smaller than for clusters with no minimum size. Interestingly, even given this strict cluster size constraint, the coefficient of variation is still not so small, reaching a minimum of $\sqrt{(3\ln 2 - 2)/2} \approx 0.2$ for the all-growth model. This extreme example shows that the decrease in CV from 1 to ≈ 0.523 between all-growth and end-growth is a very relevant change, and even the extremely precise control implied by the minimum cluster size cannot reduce the CV much further.

Our example of the minimum cluster size is more motivated by large epithelial tissues, where the role of CIL and spatially-varying traction forces are better established. This is why we focus on large $N_{\min}$ in Fig. 3.4. However a minimum cluster size does reduce variability even if the minimum is quite small—compare finite $N_{\min}$ in Fig. 3.4(h) with no minimum ($N_{\min} = 1$).

We note, however, that the strict minimum cluster size is a vast oversimplification. If we weaken the strict assumption of a minimum cluster size by assuming a basal break rate k_b^0 for all junctions, even a small k_b^0 creates a relevant population of small clusters (see Sec. 3.5.4), increasing the variability.

3.2.3 Combining mechanics and statistics

According to our theory, the break rate k_b depends on four key parameters related to cell motility and mechanical properties of the cluster—τ, μk, D, and $\delta\ell_b$. While these parameters have not yet been measured simultaneously for a single cell type, we can provide reasonable estimates. Experiments measuring velocity correlations of HaCaT cells [179] give a persistence time τ of roughly 0.5 hour and D of around $\sim 100\,\mu\mathrm{m}^2/\mathrm{h}$, leading to an active velocity v_n on the order of $\sqrt{D/\tau} \approx 15\,\mu\mathrm{m/h}$. The mobility μ and spring constant k together create a relaxation timescale $1/\mu k$, estimated to be approximately 15 min based on measurements for MDCK cells [166] (details in Sec. 3.5.6.1). According to Eq. (3.10), the break rate k_b is highly sensitive to the critical break length $\delta\ell_b$. As a

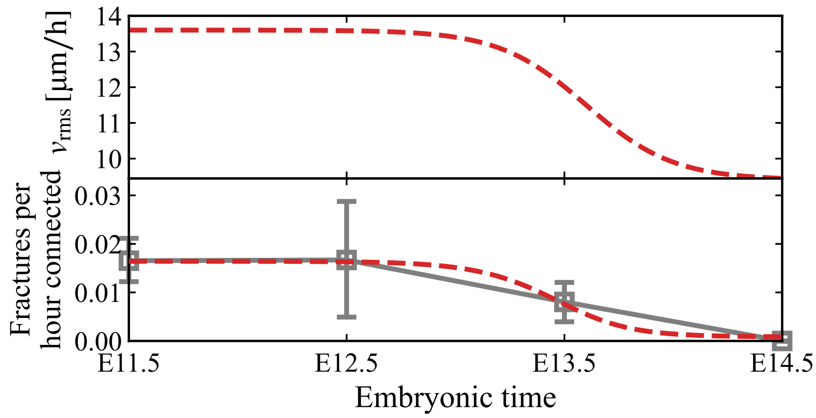

Figure 3.5: A modest change in cell motility can account for the observed decrease in fracture rates over embryonic time. The upper panel shows a logistic decrease in the typical cell speed, $v_{\mathrm{rms}} = \sqrt{D/\tau}$, which we hypothesize underlies the fracture rate reduction. In the lower panel, symbols with error bars are experimental data from Fig. 4C of Ref. [12]; the red dashed line shows the corresponding theoretical predictions for the break rate k_b, calculated from the upper panel's v_{rms} using Eq. (3.10).

rough guess, $\delta\ell_b = 5\,\mu$m yields a mean escape time $\tau_{\mathrm{esc}} \approx 17.6\,$h, or a break rate of approximately $0.057\,\mathrm{h}^{-1}$. This is the right order of magnitude for experiments in germline cyst fracture [12] and cancer cell dissociation [39, 112]. We use these values as our default parameters (Table 3.1).

Our results suggest that clusters of cells may regulate their size not only by changing division rate but also by changing cell motility or persistence or cell-cell adhesion. Recent experiments found that germline cells have motility that decreases as cysts develop, decreasing fracture over time and controlling cyst size [12]. In Ref. [12], Levy *et al.* measured the break rates as the *fractures per hour connected* for germline cysts at different embryonic stages (lower panel, Fig. 3.5). They also reported that germline cysts have a cell cycle length of ~ 16 hours. Thus, the break rate and division rate of germline cysts are approximately $0.02\,\mathrm{h}^{-1}$ and $\ln 2/16 \approx 0.04\,\mathrm{h}^{-1}$, i.e., the ratio $\gamma \approx 0.5$, corresponding to a mean cluster size $\langle n \rangle \approx 3$ in the all-cell growth model. (We note this is not the value found by Ref. [12]—their germline cysts do not reach the steady state. If we simulate for only 5 division times, we get similar results to their experiment and model; see Sec. 3.5.6.2.) For germline cysts, we set $\delta\ell_b = 6.5\,\mu$m to match k_b to experiment.

Cell motility and cell-cell adhesion change k_b and thus change γ, changing cluster size distributions. For instance, if cells double their adhesiveness by increasing k, γ decreases to approximately 0.0067, leading to a larger cluster size $\langle n \rangle \approx 150$ with significantly higher variability $\sigma_n \approx 150$. At long developmental times, the break rate of germline cysts drops to under $1/190\,\mathrm{h}^{-1}$, which Ref. [12] attributes to changes in cell motility. We find this drop does not require a dramatic change in cell speed. To decrease the break rate to this level, D must decrease by more than a factor of 1.5, i.e., the typical speed of cells needs to decrease by more than a factor of 1.2, as shown in Fig. 3.5, where a small decrease in $v_{\mathrm{rms}} = \sqrt{D/\tau}$ captures the observed drop in break rates.

3.2.4 Survival probability has a universal form for all-cell growth

At steady state, how long does a given cluster stay intact before it fractures into two pieces? Our initial intuition for this survival problem was that survival time should depend strongly on the break rate k_b of an individual junction. However, the break rate k_b and division rate k_d affect this time scale in several ways, making their effects not immediately obvious. For instance, a larger k_b makes junctions more likely to break, but it also shifts the steady-state distribution toward smaller clusters, which are less likely to fracture—simply because they contain fewer cell-cell junctions. To tackle these questions, we compute the survival probability, i.e., the probability that the time T of the first fracture event is larger than the current time, $S(t) = \mathbb{P}(T > t)$, for a given cluster which we draw from the steady-state distribution of clusters at $t = 0$. (We used similar approaches to quantify experimental data on cancer cell rupture in Refs. [39, 112] and Chapter 2.) Our Monte Carlo simulation results reveal that, for the all-cell growth model, the survival probability follows an exponential form:

$$S(t) = \mathrm{e}^{-k_d t}, \tag{3.19}$$

which is surprisingly independent of the break rate k_b, and depends solely on the division rate k_d, as shown in Figs. 3.6(a) and 3.6(b).

We can show that this form of the survival probability holds not just for our simplest model, but a broader generalization of the all-cell growth model [Eq. (3.11)]:

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = \sum_{m>n} q_{mn}p_m - \lambda(n)p_n + (n-1)k_d p_{n-1} - nk_d p_n, \tag{3.20}$$

where q_{mn} is the transition rate at which an m-cell cluster breaks into an n-cell cluster. Our earlier all-cell growth model is the simple case where q_{mn} is just set to a constant k_b. The transition rate is symmetric in fragment size: $q_{mn} = q_{m,m-n}$. The total "hazard rate" for an m-cell cluster $\lambda(m)$—i.e., the total rate of fragmentation for a cluster of size m to any size—is then given by $\lambda(m) = \sum_{n\geqslant 1}^{m-1} q_{mn}$. Now, let us consider a more general class than our earlier all-cell growth model, in which the transition rate depends on the pre-fracture cluster size m, but not on the post-fracture fragment size n:

$$q_{mn} = \mathcal{Q}(m). \tag{3.21}$$

This would include fracture rates in models where, for instance, each bond in a larger cluster is less likely to break. Given Eq. (3.21), the total hazard rate for an n-cell cluster becomes $\lambda(n) = (n-1)\mathcal{Q}(n)$. In the steady state, we find a recurrence relation between successive probabilities:

$$\frac{p_n}{p_{n-1}} = \frac{(n-1)k_d}{(n-1)k_d + \lambda(n)} = \frac{k_d}{k_d + \mathcal{Q}(n)}. \tag{3.22}$$

As this is a generalization of the all-cell growth model, the solution in Eq. (3.12) automatically satisfies the recurrence relation.

To compute the total survival probability $S(t)$, we first consider the conditional survival probability $u_n(t) = \mathbb{P}(T > t|N(0) = n)$, defined as the probability that no fracture occurs by time t given an initial chain length $N(0) = n$. For a Markov process with a time-varying hazard rate $\lambda(t)$, the probability that no failure occurs over a time interval can be generalized with an integral $\exp\left(-\int_0^{\Delta t} \lambda(t')\mathrm{d}t'\right)$ [180]. In our case, the total hazard rate $\lambda(t) = \lambda\big(N(t)\big)$ depends on a stochastic process $N(t)$, so we must average over all possible trajectories of $N(t)$:

$$u_n(t) = \mathbb{E}^N\left[\exp\left(-\int_0^t \lambda(t')\mathrm{d}t'\right)\middle| N(0) = n\right], \tag{3.23}$$

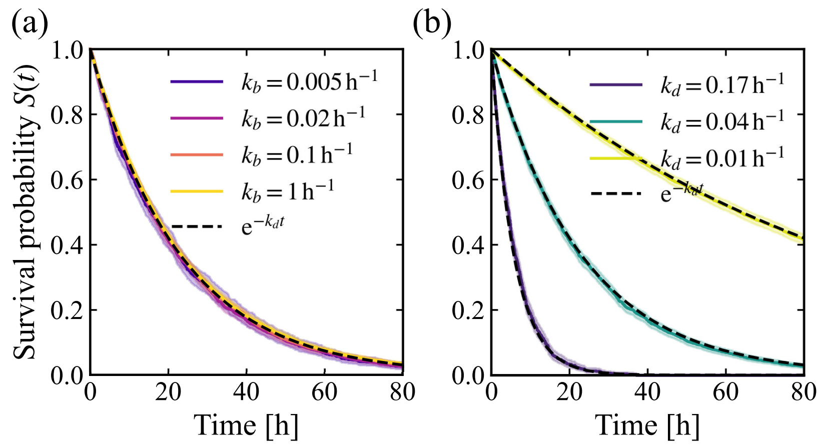

Figure 3.6: Dependence of survival probability $S(t)$ on break and division rates in the one-dimensional all-cell growth model. (a) Survival curves for varying break rates k_b, with fixed division rate k_d, all collapse onto the exponential decay $S(t) = \mathrm{e}^{-k_d t}$ (dashed line). (b) Survival probabilities $S(t)$ for three different division rates k_d, corresponding to cell cycles of 4, 16, and 64 hours, respectively. We apply standard Kaplan-Meier survival analysis [115] to the data from our rejection-free kinetic Monte Carlo simulations; error bars represent 95% confidence intervals.

where $\mathbb{E}^N[\cdot]$ represents the expectation over all realizations of the stochastic process $N(t)$.

We can work out the average in Eq. (3.23) using the Kolmogorov backward equation (KBE). For such a discrete-state, continuous-time Markov jump process, the KBE is [181, 182]:

$$\partial_t P_{ij}(t) = \sum_k w_{ik} P_{kj}(t), \tag{3.24}$$

where $P_{ij}(t) = \mathbb{P}(N(s+t) = j | N(s) = i)$ and w_{ij} is the transition rate from state i to state j. To compute the survival time, we are interested in the dynamics of the cluster to reach an absorbing state of "ruptured", which we denote as the "0" state. Following the KBE, one can show that

$$\partial_t u_n(t) = \sum_{k>0} w_{nk} u_k(t), \tag{3.25}$$

which after separating the rates of rupture and division processes yields

$$\partial_t u_n = -\lambda(n) u_n + n k_d (u_{n+1} - u_n). \tag{3.26}$$

The total survival probability is then obtained by averaging $u_n(t)$ over the distribution of initial chain lengths. If we assume that the system is at steady state, it is then reasonable to use the steady-state p_n as the distribution of initial chain lengths, and we therefore write

$$S(t) = \sum_{n\geqslant 1} p_n \cdot u_n(t). \tag{3.27}$$

By computing the time derivative of $S(t)$ and substituting Eq. (3.25) and the recurrence relation [Eq. (3.22)], we obtain

$$\dot{S}(t) = -k_d \sum_{n\geqslant 1} p_n u_n = -k_d S(t). \tag{3.28}$$

Therefore, we arrive at the striking result that the survival probability decays exponentially as $S(t) = \mathrm{e}^{-k_d t}$, independent of the break rate k_b, and determined solely by the division rate k_d. We note that this derivation is independent of dimensionality and relies only on the assumption in Eq. (3.21) and the all-cell growth mechanism. In our one-dimensional all-cell growth model, where $q_{mn} = k_b$ is constant, the derived $S(t) = \mathrm{e}^{-k_d t}$ is the exact solution, as confirmed by the Monte Carlo simulation results shown in Fig. 3.6.

The intriguing result that the survival probability depends only on k_d depends on the assumption of steady state. However, germline cysts do not reach steady state—so for a comparison to experiment, an ensemble starting from a fixed number of cells corresponding to the initially observed condition would be easier to observe and compare to experiment. The quantity $u_1(t)$—the survival probability conditional on starting from a single cell—might be of particular interest. We present the analytical expression for the conditional survival probability $u_n(t)$ in our one-dimensional all-cell growth model in Sec. 3.5.5.

An essential assumption in this derivation is the all-cell growth mechanism. When alternative growth mechanisms are considered, the universal exponential form no longer holds [Fig. 3.13(a)]. In this sense, the connection between the survival probability $S(t)$ and the single-cell division rate is a signature of simple all-cell growth. Additionally, imposing a minimum cluster size $N_{\min}$—which breaks our assumption of fracture rates q_{mn} being independent of n—also violates the universal $\mathrm{e}^{-k_d t}$

decay [Fig. 3.13(b)].

3.3 Two-dimensional simulations

Most of the preceding results are derived in a one-dimensional geometry, which simplifies analysis but does not capture the spatial complexity of most biological tissues—except for special cases such as germline cysts or tissues under strong confinement. Fracture of a two- or three-dimensional tissue is fundamentally different from our 1D model, because the fracture must cross the entire tissue, rather than break a single link. We extend our model to two dimensions using simulations. Specifically, we implement a two-dimensional model treating cells as active Ornstein-Uhlenbeck particles, governed by the equations:

$$\dot{\mathbf{x}}_n = -\mu \boldsymbol{\nabla}\Phi + \mathbf{v}_n(t), \tag{3.29a}$$

$$\tau \dot{\mathbf{v}}_n = -\mathbf{v}_n + \sqrt{2D}\boldsymbol{\xi}_n(t), \tag{3.29b}$$

where the Gaussian noises satisfy $\langle \xi_{n\alpha}(t)\xi_{n'\beta}(t')\rangle = \delta_{nn'}\delta_{\alpha\beta}\delta(t-t')$, with Greek indices corresponding to spatial components. As in 1D, we have a springlike potential between cells, $\Phi = \frac{1}{2}k\sum_{(i,j)}(|\mathbf{x}_i - \mathbf{x}_j| - \ell_0)^2$, where (i,j) denotes the summation over all connected pairs. In this 2D setting, the root-mean-square speed of an isolated cell is given by

$$v_{\mathrm{rms}} = \sqrt{2D/\tau}. \tag{3.30}$$

Upon division, the original cell is replaced by a mother-daughter pair positioned at the same location [183], with a small displacement $2\ell_0/3$ between them introduced along a random direction. The two daughter cells are initially given velocities with magnitude v_{rms} pointing away from the cell-cell contact. Intercellular connections break when stretched beyond the threshold length $\ell_0 + \delta\ell_b$, and we allow connections to be re-established when the distance between two cells becomes $\leqslant \ell_0 + \epsilon$, where $\epsilon = 10^{-3}\ell_0$. When a cluster has broken into two separate pieces (established by finding the connected components of the cluster), we randomly select one of the two clusters as the daughter cluster we are following. Fig. 3.7(a) shows representative snapshots of a cell division event and a

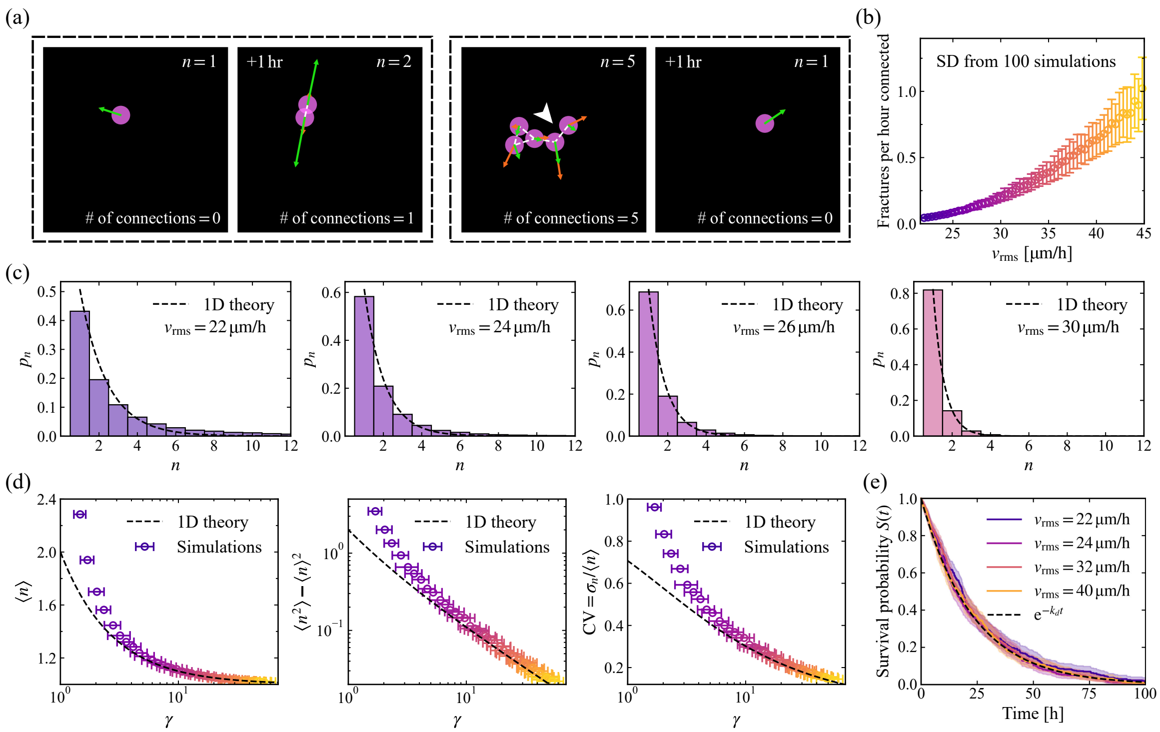

Figure 3.7: Two-dimensional simulation results. (a) Representative snapshots from a simulation with $v_{\rm rms} = 22\,\mu$m/h. The first dashed box shows a cell division event; the second dashed box shows a fracture event (highlighted by the white arrow), where only one of the two resulting sub-clusters is retained. Green arrows indicate instantaneous cell velocities, orange arrows indicate active self-propulsion velocity, and white dashed lines denote intercellular junctions. (b) Fractures per hour connected increase with the root-mean-square cell speed $v_{\rm rms}$. (c) Lineage size distributions p_n for different values of $v_{\rm rms}$, compared to the 1D theoretical prediction $\gamma/(1+\gamma)^n$ (dashed lines), where we have used the corresponding fractures per hour connected in panel (b) as the break rate k_b to compute $\gamma = k_b/k_d$. (d) How the mean, variance, and CV of cluster size vary with γ. (e) Survival probabilities for a wide range of $v_{\rm rms}$ collapse to $S(t) = \mathrm{e}^{-k_d t}$. Kaplan-Meier analysis is used to generate survival curves, and 95% confidence intervals are shown as error bars. Division rate in the simulations is fixed at $k_d = (\ln 2/16)\,\mathrm{h}^{-1}$.

cluster undergoing fracture.

To explore the effect of cell motility, we vary the typical cell speed $v_{\rm rms}$ by tuning D while keeping τ fixed. As shown in Fig. 3.7(b), the mean number of fractures per hour connected increases monotonically with $v_{\rm rms}$. We then use this fracture rate from simulations as the break rate k_b; higher cell speeds result in larger values of k_b, and consequently, a larger break-to-division ratio $\gamma = k_b/k_d$. By measuring the lineage cluster size distribution from simulations, we find that, for the all-cell growth model, the one-dimensional theoretical prediction $p_n = \gamma/(1+\gamma)^n$ [Eq. (3.12)] can still describe the 2D results in a wide range of cell motility, as shown in Fig. 3.7(c). Noticeable deviations occur mainly at small cluster sizes n for low cell speeds. Further comparison of the mean, variance, and coefficient of variation (CV) of cluster sizes as functions of γ [Fig. 3.7(d)] confirms that the 1D theory can describe the 2D simulation outcomes, especially in the regime of large γ (high motility). This agreement can be understood by noting that in the large γ limit—where cell-cell junctions frequently rupture—only small clusters are likely to persist, so the system effectively behaves as quasi-one-dimensional.

Interestingly, the exponential survival probability $S(t) = \mathrm{e}^{-k_d t}$ also shows up in our 2D simulation results [Fig. 3.7(e)]. Is this reasonable? The derivation of this result in Sec. 3.2.4 assumes that the transition rate from clusters of size m to n depends only on m. This is more general than our earlier results assuming a strict 1D model, and in principle, this assumption is independent of dimension. However, this choice of assumption is most natural for a generalization of the 1D model—summarizing the clusters solely by size in 2D is potentially suspect given the potential roles of cluster geometry and topology. In fact, the assumption that the fracture rate does not depend on the final cluster size [Eq. (3.21)] is not strictly satisfied in the 2D simulations. We have quantified q_{mn} by counting the number of occurrences where an m-cell cluster fragments into an n-cell cluster in our simulations. We find that, for a fixed pre-fracture size m, the values of q_{mn} are not uniformly distributed across different n—the cluster is more likely to break asymmetrically, producing either small or large fragments rather than equal-sized ones, as shown in Fig. 3.13(c). Nevertheless, the exponential form remains valid across a broad range of cell motility in 2D simulations, as shown in

Fig. 3.7(e).

A caveat in our analysis of the 2D system is that we are not using the analytical expression for the break rate [Eq. (3.10)] to directly predict the break rate in our 2D simulations. Instead, we extract the break rate k_b using the measured fractures per hour connected from simulations. This is because Eq. (3.10) is derived in one dimension and assumes the limit $N \to \infty$, making its prediction quantitatively incorrect for the 2D system. We also note that the model does not include some features that would be appropriate for a full model of the germline cysts. For instance, our cell-cell interactions are non-specific intercellular connections that can break and reform based on intercellular distances—effectively modeling E-cadherin-based adhesion. This differs from germline cysts, where intercellular bridges (ICBs) form specifically between sister cells during division and do not reconnect once severed [12, 44].

3.4 Discussion

Our work provides a route to understand quantitatively how the size of groups of cells ranging from cancer to organs to organisms can emerge from balancing growth and random cell motility. Our model shows that differences in where cell division and rupture occur can lead to different characteristic distributions of cluster sizes, and that biologically relevant regulation of cluster size can occur from relatively small changes in cell motility. From a physics standpoint, our results show that even in a very simple one-dimensional active material, the collective rupture of a link is nontrivial and can't be understood from the properties of a single pair of cells—unlike in an equilibrium version of our model. This qualitative difference with the simple active trap model [134] suggests that rupture rates may be sensitive to other collective features, e.g., differing between branched and linear chains of cells, or reflecting the degree of cell-cell correlation of velocities.

Though the remarkable exponential decay of the survival probability $S(t) = \mathrm{e}^{-k_d t}$, derived in Sec. 3.2.4, is independent of dimensionality and break rate k_b, and depends solely on the division rate k_d, it relies on two important underlying assumptions. First, the rate of a cluster splitting into two pieces must depend only on the pre-fracture cluster size m, and not on the post-fracture fragment size n, though the functional dependence on m can be arbitrary. This first assumption may be able to be relaxed somewhat—even though in our 2D simulations, clusters of $m = 10$ cells are more likely to have single cells break off than split equally [Fig. 3.13(c)], the total survival probability still follows the predicted exponential form $\mathrm{e}^{-k_d t}$ across a wide range of cell motility. If this first assumption is more dramatically violated, e.g., with a minimum cluster size, $S(t)$ will depend on the break rate k_b [Fig. 3.13(b)]. The second key assumption is the all-cell growth mechanism. When alternative growth mechanisms are considered, the universal exponential form no longer holds [Fig. 3.13(a)]. In this sense, the connection between the survival probability $S(t)$ and the single-cell division rate is a signature of simple all-cell growth.

We have generally chosen analytic tractability over biological detail in our approach to capture the key elements of this problem. There are many possible features that could be included in more general models of size control via fracture. These include the presence of polymer networks within the cell which can alter its ability to load stress onto the cell-cell junction [135], cell-matrix interactions [74, 184], cell shape [112, 136] and its coupling to division [172, 185], or mechanosensitive feedback [151], as well as cell-cell interactions like contact inhibition of locomotion [21, 47, 71, 158] and collective alignment of cell polarity [52, 166]. Determining to what extent these features only change the relevant energy barrier, rescaling k, or qualitatively change our picture is an important open question.

The data and code that support the findings of this chapter are openly available at Zenodo, doi: 10.5281/zenodo.18485014.

3.5 Appendices

3.5.1 Rates for breaking chains using Kramers' theory

3.5.1.1 Results for two particles

The rupture rate of a pair of cells can be found straightforwardly through a very simple generalization of the approach of Ref. [134], who mapped overdamped active Ornstein-Uhlenbeck particles (AOUPs) in a harmonic well to an equivalent thermal system. Here, we show that the same holds for a pair of harmonically bound AOUPs.

For a pair of particles $\{x_1, x_2\}$ with a distance $\ell = x_2 - x_1$ between them, we have

$$\dot{x}_1 = \mu k \delta\ell + v_1,$$
$$\dot{x}_2 = -\mu k \delta\ell + v_2,$$

where $\delta\ell = \ell - \ell_0$ is the stretched length of the spring. Thus we can obtain $\dot{\delta\ell} = -2\mu k \delta\ell + \delta v$, where $\delta v = v_2 - v_1$ is still an Ornstein-Uhlenbeck process but with correlation $\langle \delta v(t) \delta v(t') \rangle = (2D/\tau)\mathrm{e}^{-|t-t'|/\tau}$. By a standard change of variable $p = -2\mu k \delta\ell + \delta v$, we obtain a new set of Langevin equations for a thermal Brownian particle with displacement $\delta\ell$ [134, 186]:

$$\begin{cases} \dot{\delta\ell} = p, \\ \tau \dot{p} = -\eta p - U' + \sqrt{4D}\xi(t), \end{cases} \tag{3.31}$$

where the potential $U(\delta\ell) = \mu k \delta\ell^2$ is harmonic, the persistence time τ is acting as a mass, and the drag $\eta = 1 + 2\mu k \tau$. Thus an effective temperature can be defined by

$$k_B T_{\text{eff}} \coloneqq \frac{2D}{\eta} = \frac{2D}{(1 + 2\mu k \tau)}. \tag{3.32}$$

The Fokker-Planck equation from Eq. (3.31) has a stationary solution given by a Boltzmann-like measure

$$P_s(\delta\ell, p) = \mathcal{Z} \exp\left(-\frac{U}{k_B T_{\text{eff}}} - \frac{\tau p^2}{2 k_B T_{\text{eff}}} \right), \tag{3.33}$$

where $\mathcal{Z}$ is the partition function. The variance of $\delta\ell$ follows directly:

$$\langle \delta\ell^2 \rangle = \frac{k_B T_{\text{eff}}}{U''} = \frac{D}{\mu k (1 + 2\mu k \tau)}. \tag{3.34}$$

From standard Kramers' theory, for a trap in the potential $U(\delta\ell)$ of size $\delta\ell_b$, the mean escape time, when the effective temperature is small compared to the potential barrier height ΔU, is given by [167–169]

$$\tau_{\text{esc}} = \frac{2\pi}{\omega_0} \exp\left(\frac{\Delta U}{k_B T_{\text{eff}}}\right) = \pi\sqrt{\frac{2\tau}{\mu k}} \exp\left[\frac{\mu k \delta \ell_b^2 (1 + 2\mu k \tau)}{2D}\right], \tag{3.35}$$

where $\omega_0 = \sqrt{2\mu k/\tau}$ is the undamped angular frequency of the system. The subexponential correction $\tau_0 = 2\pi/\omega_0$ applies in the limit of moderate friction but may vary under different friction regimes [168, 170].

3.5.1.2 Results for a long chain

We can explicitly write Eq. (3.1) as $\dot{x}_n = \mu k(x_{n+1} - 2x_n + x_{n-1}) + v_n$. Assuming that $x_n(t) = n\ell_0 + \delta x_n(t)$, we find that δx_n obeys the same equation:

$$\dot{\delta x}_n = \mu k(\delta x_{n+1} - 2\delta x_n + \delta x_{n-1}) + v_n. \tag{3.36}$$

For a long chain with $N \gg 1$ particles, we impose periodic boundary conditions, $\delta x_{n+N} = \delta x_n$, and introduce a standard discrete Fourier transform:

$$\tilde{x}_q = \frac{1}{N} \sum_{n=0}^{N-1} \delta x_n \mathrm{e}^{-iqn}, \tag{3.37}$$

where $q = 2\pi m/N$ and the index $m = 0, 1, \cdots, N-1$. By substituting the inverse transforms into Eq. (3.36) and matching terms for each mode q, we obtain

$$\dot{\tilde{x}}_q = -\mu k_q \tilde{x}_q + \tilde{v}_q, \tag{3.38}$$

where $k_q = 2k(1 - \cos q)$ and $\tilde{v}_q = (1/N)\sum_{n=0}^{N-1} v_n \mathrm{e}^{-iqn}$. The mean is simply $\langle \tilde{v}_q \rangle = 0$, and the time correlation is

$$\langle \tilde{v}_q(t) \tilde{v}^*_{q'}(t') \rangle = \frac{1}{N^2} \sum_n \sum_{n'} \langle v_n(t) v_{n'}(t') \rangle \mathrm{e}^{-iqn + iq'n'} = \delta_{qq'} \frac{D}{N\tau} \mathrm{e}^{-|t-t'|/\tau}, \tag{3.39}$$

where we have used the orthogonality relation $\sum_{n=0}^{N-1} \mathrm{e}^{i(q-q')n} = N\delta_{qq'}$. Equation (3.38) can be solved by using the exponential ansatz. When $t \gg 1/\mu k_q$:

$$\langle \tilde{x}_q(t)\tilde{x}^*_{q'}(t)\rangle = \delta_{qq'} \frac{D}{N\mu k_q(1+\mu k_q \tau)}. \tag{3.40}$$

Ref. [65] shows that active Brownian particles (ABPs) exhibit a similar relationship, as their active velocities share the same Gaussian colored noise temporal correlations as those of AOUPs. Then we can calculate the stretched lengths of the springs, where the mean is simply $\langle \delta\ell_n \rangle = 0$, and the variance is

$$\langle \delta\ell_n^2 \rangle = \frac{D}{N\mu k} \sum_{q\neq 0} \frac{1}{1+2\mu k\tau(1-\cos q)} \approx \frac{D}{\mu k\sqrt{1+4\mu k\tau}}, \tag{3.41}$$

where in the limit $N \gg 1$ we have replaced the sum by an integral using $\frac{1}{2\pi}\int_0^{2\pi} \frac{\mathrm{d}q}{1+2\mu k\tau(1-\cos q)} = 1/\sqrt{1+4\mu k\tau}$.

Thus, in the steady state, the marginal distribution of $\delta\ell_n$ follows a Gaussian distribution since each $\delta\ell_n$ is the sum of Gaussian processes. In the small noise limit, the system rapidly relaxes to a quasi-stationary state described by $P_{\mathrm{QS}}(\delta\ell_n, t) \sim P_s(\delta\ell_n)\,\mathrm{e}^{-t/\tau_{\mathrm{esc}}}$, indicating that the escape process is Poissonian with a rate of $1/\tau_{\mathrm{esc}}$ [134]. Therefore, the mean escape time follows an effective "Arrhenius law" where the rate of rupture is proportional to the probability density at the breaking stretch $\delta\ell_b$:

$$\tau_{\mathrm{esc}} = \frac{2\pi\eta}{\omega_0^2\tau} \exp\left(\frac{\mu k\delta\ell_b^2\sqrt{1+4\mu k\tau}}{2D}\right). \tag{3.42}$$

We view this prefactor as a reasonable first approximation and it has a good fit to the simulated rates. The leading behavior for small effective temperatures is dominated by the exponential term [134].

3.5.1.3 When is the two-cell result the N-cell result?—an effective equipartition theorem

For the thermal equilibrium case where $\tau = 0$, the variances of stretched lengths for both a cell pair and a chain are identical, given by $\langle \delta\ell^2 \rangle_{\mathrm{th}} = D/\mu k$. This result can be easily verified using the equipartition theorem.

The results for both the pair and the chain also converge to the same non-equilibrium value when $\mu k\tau \ll 1$ because $\sqrt{1+4\mu k\tau} \to 1 + 2\mu k\tau$ for small $\mu k\tau$. In Ref. [64], Fodor *et al.* found that for a system of interacting AOUPs, at small but finite persistence time τ, there exists an *effective equilibrium* regime—the particle dynamics still respects detailed balance and time-reversal symmetry. Rather than defining an effective temperature, the authors stated that the system could be equivalently described using an effective potential, while defining $k_B T = D/\mu$ [64, 187]:

$$\tilde{\Phi} = \Phi + \tau \left[\frac{\mu}{2}(\nabla_n \Phi)^2 - D\nabla_n^2 \Phi\right] + \mathcal{O}(\tau^2), \tag{3.43}$$

where $\nabla_n \equiv \partial/\partial x_n$, and the Einstein summation convention is applied. Applying equipartition on this effective potential yields the variance derived in Sec. 3.5.1.1. For a long chain, similar analysis shows that $\langle \delta\ell_n^2 \rangle \approx D/\mu k(1 + 2\mu k\tau)$ when $\mu k\tau \ll 1$, matching the two-particle result.

3.5.2 Solving master equations with generating functions

3.5.2.1 All particles can divide

To solve the master equation (3.11), we first define a generating function $G(z,t) \coloneqq \sum_{n\geqslant 1} p_n(t) z^n$. After multiplying the master equation by z^n and summing over all n, we obtain a PDE for the generating function:

$$\frac{\partial G}{\partial t} = (k_d z - k_b - k_d) z \frac{\partial G}{\partial z} - k_b \frac{z}{1-z} G + k_b \frac{z}{1-z}, \tag{3.44}$$

whose steady-state solution is

$$G(z) = \frac{k_b - (1-z)C}{k_b + k_d - k_d z}, \tag{3.45}$$

where C is a constant. By construction, $G(z)$ has no zeroth-order term, so $C = k_b$. Using this value, we find the steady-state generating function:

$$G(z) = \frac{k_b z}{k_b + k_d - k_d z} = \sum_{n\geqslant 1} \frac{k_b}{k_d} \left(\frac{k_d}{k_b + k_d}\right)^n z^n. \tag{3.46}$$

We can read the value of p_n directly from this formal series, finding Eq. (3.12):

$$p_n = \frac{k_b}{k_d}\left(\frac{k_d}{k_b + k_d}\right)^n = \frac{\gamma}{(1+\gamma)^n}, \tag{3.47}$$

where $\gamma \coloneqq k_b/k_d$. The first two moments are

$$\langle n \rangle = \sum_n p_n n = \frac{1+\gamma}{\gamma}, \tag{3.48a}$$

$$\langle n^2 \rangle = \sum_n p_n n^2 = \frac{(1+\gamma)(2+\gamma)}{\gamma^2}, \tag{3.48b}$$

and the variance is $\langle n^2\rangle - \langle n\rangle^2 = (1+\gamma)/\gamma^2 = \langle n\rangle/\gamma$. We can also calculate the coefficient of variation:

$$\mathrm{CV} \coloneqq \frac{\sigma_n}{\mathbb{E}[n]} = \frac{\sqrt{\langle n^2\rangle - \langle n\rangle^2}}{\langle n\rangle} = \frac{1}{\sqrt{1+\gamma}}. \tag{3.49}$$

3.5.2.2 *One-end growth*

To solve Eq. (3.13), we apply the generating function method. The PDE for $G(z,t)$ for the one-end-growth model is:

$$\frac{\partial G}{\partial t} = -k_b z \frac{\partial G}{\partial z} - \left(k_d(1-z) + k_b\frac{z}{1-z}\right)G + k_b\frac{z}{1-z}. \tag{3.50}$$

The steady-state solution is

$$G(z) = \frac{(1-z)}{z^{1/\gamma}} \mathrm{e}^{z/\gamma} \int_0^z \frac{x^{1/\gamma}}{(1-x)^2}\mathrm{e}^{-x/\gamma}\mathrm{d}x. \tag{3.51}$$

By expanding $G(z)$ around $z=0$, we obtain Eq. (3.14):

$$p_n = \frac{n\gamma}{\prod_{m=1}^{n}(1+m\gamma)} = \frac{n\gamma^{1-n}}{((1+\gamma)/\gamma)_n}, \tag{3.52}$$

where $(x)_n = x(x+1)(x+2)\cdots(x+n-1)$ denotes the Pochhammer symbol. The distribution p_n for one-end growth is shown in Fig. 3.8(b), where a distinct peak is observed. For small γ, the peak location $n^* \approx 1/\sqrt{\gamma}$. When $\gamma \ll 1$, the distribution near the peak can be approximated as a Rayleigh

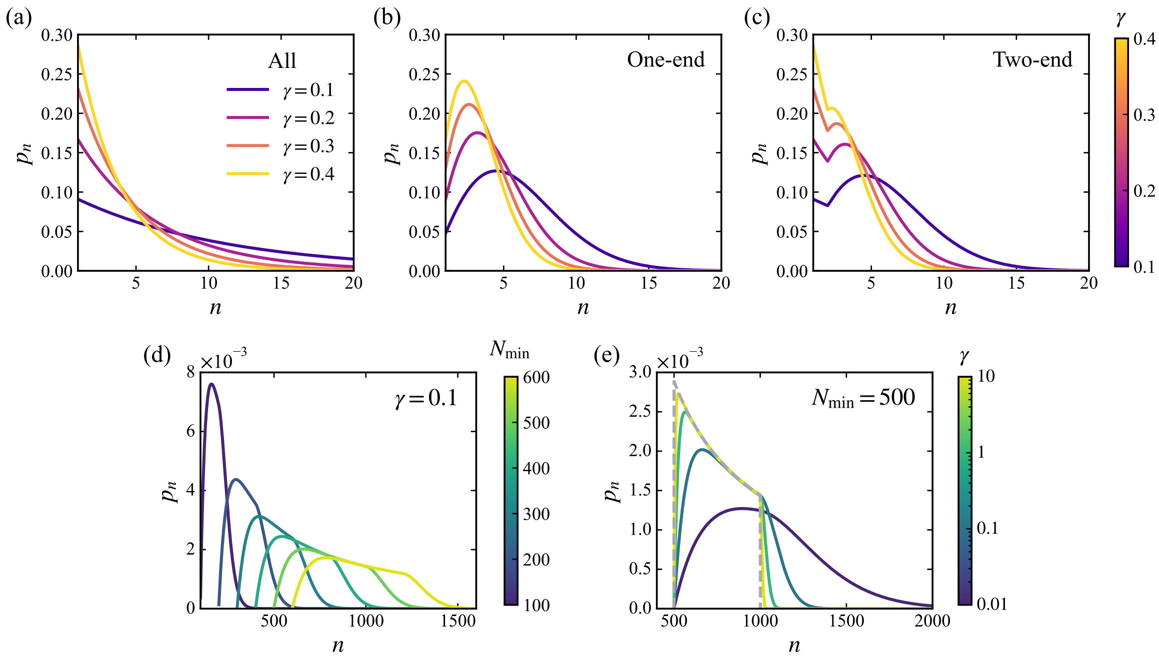

Figure 3.8: How the steady-state distributions p_n vary with γ and $N_{\min}$ across different growth models. (a) Steady-state distribution p_n for various values of γ in the all-cell growth model [Eq. (3.47)]. (b) Steady-state distribution p_n for different γ in the one-end growth model, where we have doubled the division rate [$\gamma \to \gamma' = \gamma/2$ in Eq. (3.52)]. (c) Steady-state distribution p_n for different γ in the two-end growth model from Eq. (3.56). (d) Steady-state distribution p_n obtained from the matrix equation $\boldsymbol{K}\mathbf{p} = \mathbf{0}$ incorporating different values of minimum cluster size $N_{\min}$ with $\gamma = 0.1$. (e) Steady-state distribution p_n obtained from the matrix equation method for different values of γ with $N_{\min} = 500$. The gray dashed line represents the corresponding log-uniform distribution $1/(n \ln 2)$ within the range $[N_{\min}, 2N_{\min}]$.

distribution $p_n \approx n\gamma \mathrm{e}^{-\gamma n^2/2}$ with mode $\sigma = 1/\sqrt{\gamma}$. Using the saddle-point approximation, the mean and variance of n are directly obtained from the Rayleigh distribution:

$$\langle n \rangle = \sigma \sqrt{\frac{\pi}{2}}, \qquad \langle n^2 \rangle - \langle n \rangle^2 = \frac{4-\pi}{2}\sigma^2, \tag{3.53}$$

and the coefficient of variation is $\mathrm{CV} = \sqrt{(4-\pi)/\pi}$.

3.5.2.3 Two-end growth

For the two-end growth scenario, the master equation differs from the all-cell and one-end cases for $n = 1$ and $n = 2$ (where only one cell can divide). Using the generating function method, the steady-state ODE for $G(z)$ is:

$$k_b z \frac{\partial G}{\partial z} = -\left(2k_d(1-z) + k_b \frac{z}{1-z}\right) G + k_b \frac{z}{1-z} + \frac{k_b k_d}{k_b + k_d} z(1-z). \tag{3.54}$$

The solution is given by

$$G(z) = \frac{(1-z)}{z^{2/\gamma}} \mathrm{e}^{2z/\gamma} \int_0^z \frac{x^{2/\gamma}(x^2 - 2x + 2 + \gamma)}{(1+\gamma)(1-x)^2} \mathrm{e}^{-2x/\gamma} \mathrm{d}x. \tag{3.55}$$

By expanding $G(z)$ around $z = 0$, we derive the exact analytical solution for two-end growth:

$$p_n = \begin{cases} \gamma/(1+\gamma), & n = 1, \\ \dfrac{2+\gamma}{2(1+\gamma)} \dfrac{n(\gamma/2)^{1-n}}{((2+\gamma)/\gamma)_n}, & n \geqslant 2. \end{cases} \tag{3.56}$$

Observe that the results are quite similar to the one-end growth case, but with a doubled division rate $2k_d$. When $\gamma \to 0$, meaning only growth without fragmentation, the two scenarios converge. In contrast, when $\gamma \gg 1$, the chain is most likely to be a monomer ($p_1 \approx 1$), and p_n is reduced by half compared to the one-end growth scenario with $2k_d$ for $n \geqslant 2$.

3.5.3 Generalizations with a matrix equation method

A general master equation with a minimum cluster size $N_{\min}$—i.e., the cluster size $n \geqslant N_{\min}$—and different growth mechanisms can be written as

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = k_b \sum_{m \geqslant n + N_{\min}} p_m - k_b \max(n - 2N_{\min} + 1, 0) p_n + g(n-1) p_{n-1} - g(n) p_n, \tag{3.57}$$

where in the first term a cluster of size m can only break into one of size n if $m \geqslant n + N_{\min}$, and in the second term we require that both daughter clusters satisfy $i, j \geqslant N_{\min}$. In the third and fourth

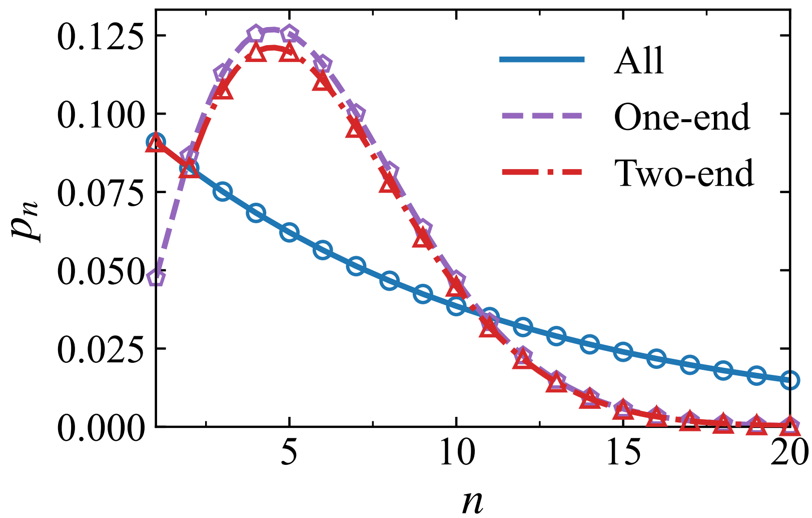

Figure 3.9: Comparison of the analytical solution obtained using the generating function method with numerical results from the matrix equation method. Lines represent the same analytical solutions for $\gamma = 0.1$ as shown in Fig. 3.4(a) ($\gamma' = \gamma/2 = 0.05$ for the one-end growth model). Empty symbols denote exact numerical results from the matrix equation $\boldsymbol{K}\mathbf{p} = \mathbf{0}$.

terms, $g(n)$ describes how the total growth rate scales with the length of the chain:

$$g(n) \coloneqq \begin{cases} k_d n, & \text{all-cell,} \\ k_d, & \text{one-end,} \\ k_d[1 + \Theta(n-2)], & \text{two-end,} \end{cases} \tag{3.58}$$

where $\Theta(x)$ is the Heaviside step function. The master equation can be organized as $\dot{\mathbf{p}} = \boldsymbol{K}\mathbf{p}$, where $\boldsymbol{K}$ is the generator on the state space $\{N_{\min}, N_{\min}+1, \ldots\}$. We decompose $\boldsymbol{K} = \boldsymbol{A} + \boldsymbol{B}$, where $\boldsymbol{A}$ is an upper triangular matrix corresponding to fragmentation events, and $\boldsymbol{B}$ is a lower bidiagonal matrix with main diagonal $B_{nn} = -\lambda(n)$ and subdiagonal $B_{n+1,n} = g(n)$, where $\lambda(n) \coloneqq k_b \max(n - 2N_{\min} + 1, 0) + g(n)$ is the total transition rate from the n-mer state. In the steady state, we obtain the homogeneous equation $\boldsymbol{K}\mathbf{p} = \mathbf{0}$. We find the null space of $\boldsymbol{K}$ and normalize to obtain the steady-state distribution $\mathbf{p}$. We must truncate to a maximum possible cluster size $N_{\max} \gg 2N_{\min}$, where p_n is negligible for $n > N_{\max}$; $N_{\max}$ is set to $N_{\min} + 5000$ by default. When setting $N_{\min} = 1$, the solutions $\mathbf{p}$ match those derived in Sec. 3.5.2 using the generating function method (see Fig. 3.9).

3.5.4 Basal rates

So far we have assumed that both the end-growth and minimum cluster size models are strictly enforced. However, in real cell clusters, it is not clear whether breakage would be completely absent even if its rate is larger near the cluster center. Similarly, cells within a cluster may still divide despite contact inhibition of proliferation. Therefore, it is reasonable to incorporate basal rates into our model.

We first consider a scenario where, in addition to the regulated break rate k_b (restricted by the minimum cluster size), there exists a basal break rate k_b^0 that applies uniformly to all links. In the matrix equation formulation, we solve $\boldsymbol{K}\mathbf{p} = \mathbf{0}$, where now $\mathbf{p} = (p_1, p_2, p_3, \cdots)^\top$. We decompose the transition-rate matrix as $\boldsymbol{K} = \boldsymbol{A} + \boldsymbol{B}$, where $\boldsymbol{A}$ remains an upper triangular matrix but with modified elements $A_{mn} = \Theta(n - m - 1)k_b^0 + \Theta(m - N_{\min})\Theta(n - m - N_{\min})k_b$, and $\boldsymbol{B}$ is updated with the total transition rate $\lambda(n) = k_b^0(n-1) + k_b \max(n - 2N_{\min} + 1, 0) + g(n)$.

Fig. 3.10(a) illustrates how the steady-state distributions change when a small basal break rate $k_b^0 = 10^{-3}k_b$ is applied to all links, while keeping the definition $\gamma \coloneqq k_b/k_d$. For small k_b^0 (low γ), the distribution remains nearly unchanged, but as k_b^0 increases with increasing γ (near-deterministic), it effectively relaxes the minimum cluster size constraint, shifting the distribution toward an exponential decay. Even if the basal break rate is a factor of 100 lower than k_b, it can overwhelm the "controlled" break rate that respects $N_{\min}$, leading to a system dominated by small clusters. This may argue that the limit of fast rupture $\gamma \gg 1$ is not a biologically robust way to control size, since it requires a very small basal break rate.

If a basal division rate k_d^0 is introduced, the results are relatively straightforward—the system simply transitions from the end-growth case to the all-growth case. In the near-deterministic limit $\gamma \gg 1$, the distribution transitions from a uniform distribution in the range $[N_{\min}, 2N_{\min}]$ to the predicted log-uniform distribution in the all-cell growth model. As shown in Fig. 3.10(c)–(d), the key factor in the relevance of the basal growth rate is whether growth is dominated by basal growth or end-growth.

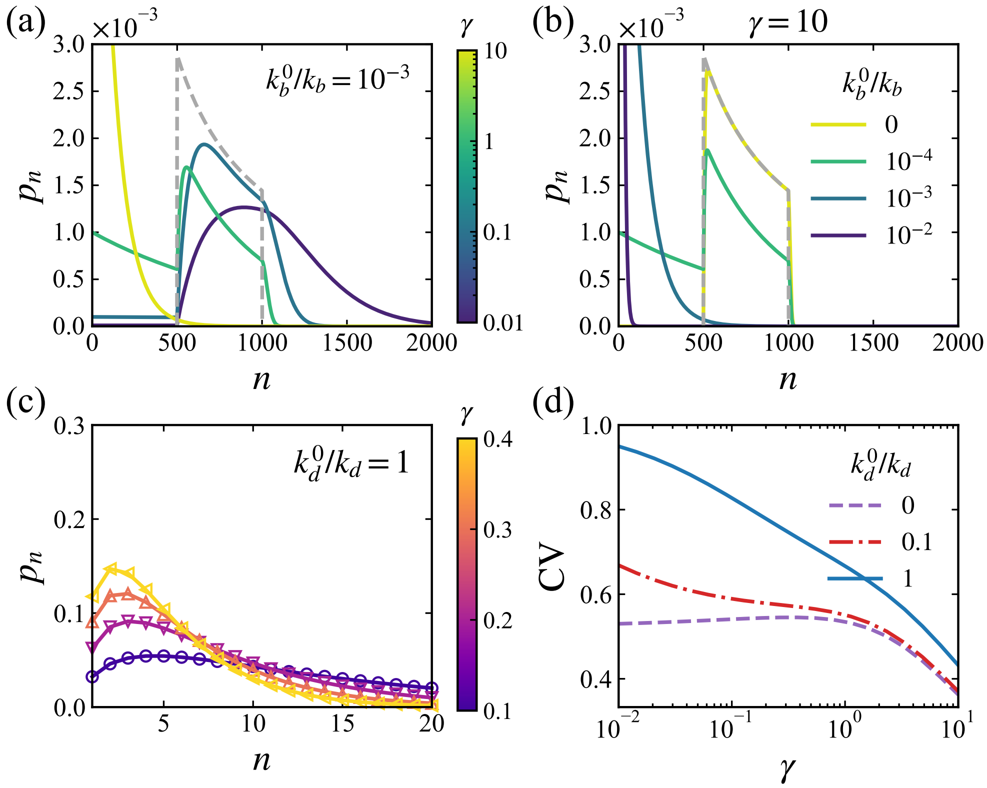

Figure 3.10: Effect of basal break and division rates on the steady-state distributions p_n. (a) Cluster sizes when $N_{\min} = 500$ and γ is varied, but with a basal break rate $k_b^0 = 10^{-3} k_b$; this figure should be compared to Fig. 3.8(e). (b) The near-deterministic case $\gamma = 10$ with varying basal break rates k_b^0. Minimum cluster size is set to $N_{\min} = 500$; gray dashed lines are the corresponding log-uniform distributions. (c) The same distribution as in Fig. 3.8(b), but with a basal division rate $k_d^0 = k_d$. (d) Coefficients of variation (CV) as a function of γ for different basal division rates k_d^0. Note that we have used $\gamma \to \gamma' = \gamma/2$, as in Figs. 3.8(b) and 3.4(d).

3.5.5 Conditional survival probability

For systems that do not reach a steady state, an ensemble initialized from a single cell is more directly comparable to experiments. Accordingly, the first conditional survival probability $u_1(t)$ is of particular interest. Solving Eq. (3.25) analytically for u_1, however, requires a closed-form expression for the total hazard rate $\lambda(n)$. While such an expression is not available in full generality, the problem can be solved explicitly under the simplest assumption $q_{mn} = k_b$, corresponding to our one-dimensional all-cell growth model, using a generating-function approach.

With $q_{mn} = k_b$, the total hazard rate is $\lambda(n) = (n-1)k_b$, and Eq. (3.25) reduces to

$$\partial_t u_n = -(n-1)k_b u_n + nk_d(u_{n+1} - u_n). \tag{3.59}$$

We write $u_n(t) = a_n(t)\mathrm{e}^{k_b t}$, which yields

$$\partial_t a_n = -n(k_b + k_d)a_n + nk_d a_{n+1}. \tag{3.60}$$

We define the generating function

$$G(z,t) \coloneqq \sum_{n \geqslant 1} a_n(t) z^n. \tag{3.61}$$

Multiplying the evolution equation for a_n by z^n and summing over $n \geqslant 1$, we obtain

$$\partial_t G = -(k_b + k_d)z\partial_z G + k_d \partial_z G - (k_d/z)G. \tag{3.62}$$

Solving this first-order PDE with the initial condition $G(z,0) = \sum_{n\geqslant 1} z^n$, which follows from the definition $u_n(0) = 1$ and hence $a_n(0) = 1$, gives

$$G(z,t) = \frac{(k_b + k_d)z}{k_b \mathrm{e}^{(k_b+k_d)t} + k_d - (k_b + k_d)z}. \tag{3.63}$$

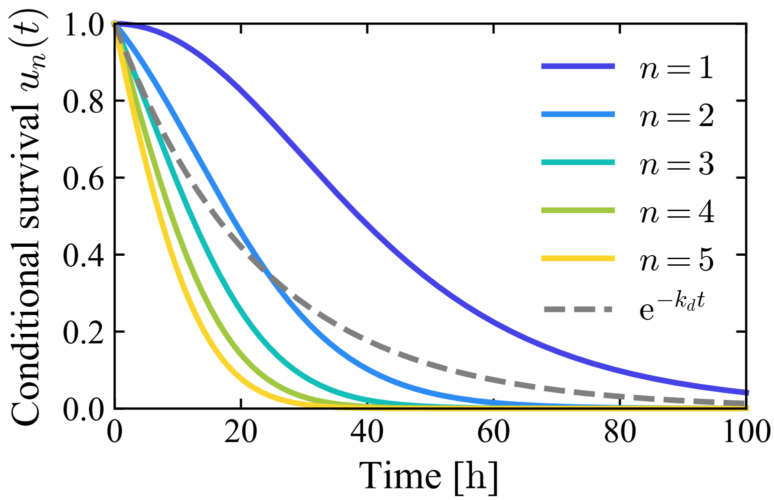

Figure 3.11: Conditional survival probability $u_n(t)$ for different initial cluster sizes $N(0) = n$ in the one-dimensional all-cell growth model, using the estimated break and division rates for germline cysts: $k_b = 0.02\,\mathrm{h}^{-1}$ and $k_d = (\ln 2/16)\,\mathrm{h}^{-1}$. The gray dashed line shows the total survival probability $S(t) = \mathrm{e}^{-k_d t}$, averaged over the corresponding steady-state distribution in Eq. (3.47).

The coefficients $a_n(t)$ are then obtained by expanding $G(z,t)$ as a power series in z:

$$G(z,t) = \sum_{n \geqslant 1} \left(\frac{k_b + k_d}{k_b \mathrm{e}^{(k_b+k_d)t} + k_d} \right)^n z^n. \tag{3.64}$$

Consequently,

$$u_n(t) = a_n(t)\mathrm{e}^{k_b t} = \left(\frac{1+\gamma}{1+\gamma \mathrm{e}^{(1+\gamma)k_d t}} \right)^n \mathrm{e}^{k_b t}, \tag{3.65}$$

which depends explicitly on both k_b and k_d, as shown in Fig. 3.11. We can verify that the total survival probability, averaged over the steady-state distribution [Eq. (3.47)], is

$$S(t) = \sum_{n \geqslant 1} p_n \cdot u_n(t) = \mathrm{e}^{-k_d t}. \tag{3.66}$$

3.5.6 Monte Carlo simulations

3.5.6.1 Active-particle simulations

We simulate the dynamics of N active Ornstein-Uhlenbeck particles (AOUPs) described by Eq. (3.1) using the forward Euler method (Euler-Maruyama method [188]) as in Ref. [64]. We integrate the dynamics with a time step $\Delta t = 10^{-3}\,\mathrm{h}$ over a total simulation time $T = 10^3\,\mathrm{h}$. The variances of the stretched lengths of the springs $\langle \delta\ell_n^2 \rangle$ are calculated from 4800 independent simulation runs. We have verified that $\langle \delta\ell_n^2 \rangle$ does not depend on the position n within the chain when N is large. Therefore, $\langle \delta\ell_{N/2}^2 \rangle$ for the central link is used to represent the variance in Fig. 3.2(b).

To measure the mean escape time τ_{esc}, we first allow the chain to relax to its steady state by running the simulation for $10^3\,\mathrm{h}$. After relaxation, we track the stretched length of the central link $\delta\ell_{N/2}$, starting the timer when $|\delta\ell_{N/2}| < \varepsilon = 0.1\,\mu\mathrm{m}$. The break time τ_{esc} is recorded as the time at which $\delta\ell_{N/2}$ reaches the critical break length $\delta\ell_b$. The final results shown in Fig. 3.2(c) are averaged over 10^3 independent simulations.

Details of parameter estimates The default parameter values used in our model are listed in Table 3.1. We estimated $D = 100\,\mu\mathrm{m}^2/\mathrm{h}$ and $\tau \approx 0.5\,\mathrm{h}$ from data on HaCaT cells (a human keratinocyte cell line) from Ref. [179] (Fig. 4E of that paper). We estimate the spring relaxation time $1/\mu k$ based on the results of Jain *et al.*, who calibrated a self-propelled particle model of collective cell migration using their experimental measurements [166]. Within our model, the force between cells is $-k\delta\ell$—so with a typical spacing of $\ell_0 = 2R$ of twice the radius of the cell, our k value corresponds to $F_C/2R$ in the notation of Ref. [166]. By measuring how much a cell is compressed when another cell is pushing against it, Ref. [166] estimated the ratio of cell stiffness F_C to the persistent motile force F_M to be approximately $F_C/F_M \approx 5.5$. Using the speed of the cell for F_M/ξ and the value from Ref. [166] of $R = 20\,\mu\mathrm{m}$ for MDCK cells, we find $\mu k \approx 4\,\mathrm{h}^{-1}$, or the timescale $1/\mu k \approx 0.25\,\mathrm{h} = 15\,\mathrm{min}$.

2D AOUP simulations For the 2D AOUP simulations in Sec. 3.3, we integrate the dynamics using a time step $\Delta t = 0.01\,\mathrm{h}$ over a total simulation time of $T = 10^3\,\mathrm{h}$. We also perform a time step

Table 3.1: Table of simulation parameters. These serve as the default parameter values for the AOUP simulations; any deviations from them are explicitly specified.

Parameter	Description	Dimension	Value
N	Number of particles	1	1000
Δt	Simulation time step	T	10^{-3} h
$1/\mu k$	Spring relaxation time	T	0.25 h [166]
τ	Persistence time	T	0.5 h [179]
D	Diffusion coefficient	L^2/T	100 μm^2/h [179]
$\delta\ell_b$	Critical break length	L	5 μm

convergence check by reducing the time step to $\Delta t = 0.005$ h, which yields results consistent with Fig. 3.7 within statistical error. In each run, we track the lineage of a single cluster by randomly selecting one of the two daughter clusters once the cluster has broken into two separate pieces, and continuing the simulation with that daughter cluster. We fix the cell division rate $k_d = (\ln 2/16)\,\mathrm{h}^{-1}$, and set cell radius to $R = 20\,\mu$m, so the natural length of the spring $\ell_0 = 2R = 40\,\mu$m. The critical breaking length is set to $\delta\ell_b = 6.5\,\mu$m to match germline cysts. We vary the typical cell speed v_{rms} by changing D while keeping the persistence time τ fixed. The final results shown in Fig. 3.7 are from 100 independent simulations.

3.5.6.2 Fragmentation-growth simulations

We implement a rejection-free kinetic Monte Carlo (rfKMC) algorithm to simulate the fragmentation-growth processes. The initial chain length is set to $n_0 \gg N_{\mathrm{min}}$. For a chain of N particles, there are $N-1$ connections, each with a break rate. If breaking a connection violates the minimum cluster size N_{min}, the break rate is set to zero. Each particle has a division rate, which is set to zero for particles not allowed to divide (e.g., in the end-growth mechanism).

In our simulations, we fix the division rate $k_d = 1$ and vary the ratio γ by adjusting the break rate $k_b = \gamma$, i.e., the time unit is $1/k_d$. In the rfKMC algorithm, the time step for each MC step is calculated as $\Delta t = Q_k^{-1}\ln(1/u)$, where the total rate Q_k is the sum of all break and division rates,

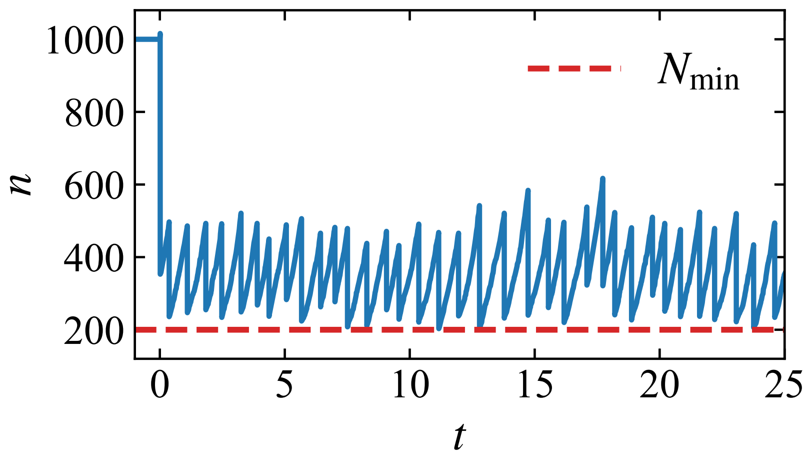

Figure 3.12: A representative time trajectory of chain length $n(t)$ from rfKMC simulation with $N_{\min} = 200$ and $\gamma = 0.1$. The initial chain length is set to $n_0 = 1000$, and the simulation begins at $t = 0$. The time unit is $1/k_d$.

and u is a uniform random number in $(0, 1]$. Each step permits either a fragmentation or a division event. After fragmentation, one of the daughter chains is randomly selected as the new tracked chain. To obtain the steady-state distribution p_n, we must wait for a sufficiently long time T for the chain length $n(t)$ to reach a stable increase-then-drop pattern (as shown in Fig. 3.12). The distribution is then generated from the results of 10^4 independent simulation runs.

We note that the cluster sizes of germline cysts shown in Fig. 5 of Ref. [12] are smaller than our steady-state predictions using their measured break and division rates, as their system has not yet reached steady state. For instance, the break rate is approximately $0.015\,\mathrm{h}^{-1}$ at E11.5 and E12.5 ovaries (Fig. 4C of Ref. [12]), while the division rate for germline cysts is about $\ln 2/16\,\mathrm{h}^{-1}$. Our steady-state solution predicts a mean cluster size $(1+\gamma)/\gamma \approx 4$, whereas their simulated average cyst sizes are 1.73 and 3.26 at E11.5 and E12.5, respectively. If we similarly evolve a single cell only from E10.5 to E11.5 and E12.5, as in Ref. [12], we obtain mean cluster sizes of approximately 2 and 3, which closely match their results.

3.5.7 Additional figures

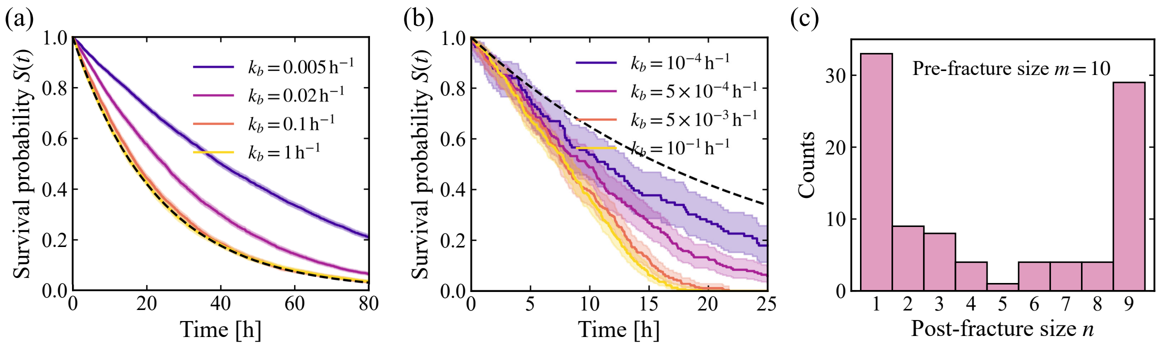

Figure 3.13: (a) Survival probability $S(t)$ from rfKMC simulations of the one-end growth model at fixed division rate k_d with varying break rate k_b. (b) Survival probability $S(t)$ from rfKMC simulations of the all-cell growth model with minimum cluster size $N_{\min} = 200$, again varying k_b at fixed k_d. Dashed lines are the exponential decay $e^{-k_d t}$. Error bars on $S(t)$ represent 95% confidence intervals. (c) Distribution of post-fracture size n for pre-fracture size $m = 10$ in 2D AOUP simulations with cell motility $v_{\rm rms} = 22\,\mu$m/h. Division rate $k_d = (\ln 2/16)\,h^{-1}$.

Chapter 4
Divergence of detachment forces in the finite Voronoi model

This chapter is adapted from the following preprint:

- Wang, W. & Camley, B. A. Divergence of detachment forces in the finite Voronoi model. *arXiv preprint arXiv:2604.15481.* DOI: 10.48550/arXiv.2604.15481 (2026).

4.1 Introduction

Both confluent tissues, in which cells densely pack and tile space, and nonconfluent cell populations, in which cells or clusters of cells are separated by gaps, are common in biology. Many systems transition between these two states. A prominent example is epithelial-mesenchymal transition (EMT), in which epithelial tissues lose cohesion and give rise to dispersed, migratory cells or clusters [28, 29]. Similarly, during cancer dissemination, tumor cell collectives can transition from a compact, confluent mass to a nonconfluent, invasive state [6, 190]. Understanding such transitions is central to describing collective cell behavior, and these approaches require simulation methods that can correctly resolve the mechanics of both confluent and nonconfluent tissues. One popular and powerful approach for simulating confluent tissues is the active Voronoi model [25], in which cells have a polygonal shape given by the Voronoi tessellation of cell centers, and cells evolve under a mechanical energy that depends on this geometry. These models have been particularly successful in explaining the jamming transition of confluent monolayers, where cells tile space without gaps and neighbor exchanges occur through topological rearrangements [25, 67, 136]—but the commonly simulated active Voronoi model [25] does not allow for simulating nonconfluent tissues. However, recent work has developed so-called "finite Voronoi" models, originating from the concept of free Dirichlet domains [79], to extend the conventional Voronoi model to describe nonconfluent monolayers [80, 81, 191, 192]. Finite Voronoi models retain the advantages of Voronoi-based descriptions in characterizing cell shape, while remaining significantly less computationally demanding than approaches like phase-field [72, 76, 112, 116], subcellular element [84, 193, 194], or deformable polygon [82, 83, 150] models, making them promising candidates for simulations of cohesive clusters, active fragmentation, and tissue rupture [80, 81].

Here, we study the finite Voronoi model of Refs. [80, 81], and demonstrate that in some circumstances the force required to detach two cells can diverge unphysically due to the imposed geometry of cell-cell contacts, preventing cells from rearranging. This is akin to—though more dramatic than—earlier results showing that the assumption of a strict Voronoi shape can eliminate the unjamming transition observed in vertex models [70] and change the dynamics of heterotypic interfaces [195, 196].

We introduce the active finite Voronoi model in Sec. 4.2. We then demonstrate that when simulating tissue fracture, the results systematically and unavoidably depend on the time step used in numerical simulation, with fracture vanishing at small time steps (Sec. 4.3.1). We then show that this pathology is not a numerical accident but a property of the finite Voronoi geometry. In Sec. 4.3.2, we analyze the simplest case of cell doublet detachment and show that the detachment force will diverge if there is an interfacial tension that differs between cell-cell and cell-medium interfaces. Based on this analysis, we propose a regularization to address this divergence in Sec. 4.3.3. In Sec. 4.4, we compare the finite Voronoi model with a deformable polygon model that has the same energy function, but does not assume the strict Voronoi shape, and analyze two different calibration strategies in Secs. 4.4.3 and 4.4.4.

4.2 Model

We consider N cells in two dimensions, labeled by $i = 1, \ldots, N$, with cell centers at positions $\{\mathbf{r}_i\}$. In a conventional Voronoi model for confluent monolayers [25], Voronoi tessellations tile all of space, and the Voronoi region of cell i is the set of points closer to $\mathbf{r}_i$ than to any other center. In the finite Voronoi (FV) model, we assume that the cell cannot extend further than a distance ℓ away from its center. In this way, the shape of a cell is obtained by truncating the Voronoi region at a maximum radius ℓ from the cell center [80, 81]. More formally, the domain of cell i is the intersection of its Voronoi region with the disk $\{\mathbf{x} : |\mathbf{x} - \mathbf{r}_i| \leqslant \ell\}$. As a result, FV boundaries are composed of (i) straight Voronoi segments where two cells are in contact and (ii) circular arcs of radius ℓ where the cell is exposed to the surrounding medium (Fig. 4.1). This construction allows two cells to separate once

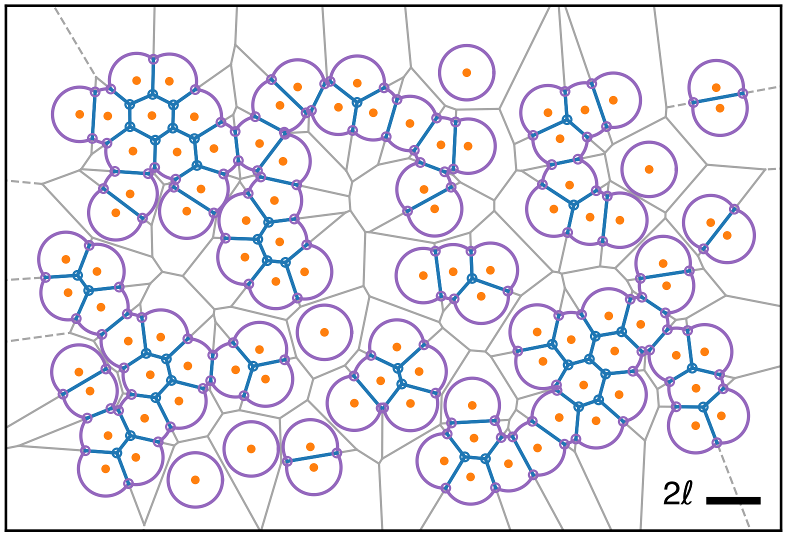

Figure 4.1: Illustration of the finite Voronoi (FV) model. Cell center positions $\{\mathbf{r}_i\}$ are the orange points, and gray lines are the corresponding Voronoi diagram generated by $\{\mathbf{r}_i\}$ while the dashed lines represent rays of Voronoi edges that extend to infinity. Empty circles represent vertices $\{\mathbf{h}_m\}$ in the FV model: Blue circles are the triple junction vertices connecting three cells, while purple circles are the outer vertices $\mathbf{h}^{\text{out}}$ connecting two cells. Scale bar: maximum diameter 2ℓ.

their centers move sufficiently far apart, creating gaps between cells without introducing additional degrees of freedom.

The energy of the cells is defined similarly to that in the conventional Voronoi/vertex-based model: a quadratic elastic energy penalizing area deviation away from a preferred area, a quadratic energy penalizing cell perimeter, and line tensions that are potentially different between cell-cell and cell-medium interfaces [80, 81, 136]:

$$E = \sum_i K_A(A_i - A_0)^2 + K_P P_i^2 + \lambda^{(c)} P_i^{(c)} + \lambda^{(n)} P_i^{(n)}, \tag{4.1}$$

where K_A and K_P are elastic moduli for cell area A_i and perimeter P_i, respectively, and A_0 is the preferred cell area. $\lambda^{(c)}$ and $\lambda^{(n)}$ are the cortical tensions for contacting edges (cell-cell) and non-contacting edges (cell-medium interface), and the total circumference $P_i = P_i^{(c)} + P_i^{(n)}$ is the sum of contacting length and non-contacting length. Up to an additive constant, this can be rewritten in a “preferred perimeter” form:

$$E = \sum_i K_A(A_i - A_0)^2 + K_P(P_i - P_0)^2 + \Lambda P_i^{(n)}, \tag{4.2}$$

where $P_0 = -\lambda^{(c)}/2K_P$ is the preferred perimeter, and $\Lambda \equiv \lambda^{(n)} - \lambda^{(c)}$ measures the tension difference between contacting and non-contacting edges.

We assume that the cell centers are overdamped, so cell i has one term in its velocity proportional to the force $-\nabla_i E$ on cell i and one arising from self-propulsion [81],

$$\dot{\mathbf{r}}_i = -\mu \boldsymbol{\nabla}_i E + v_0 \mathbf{n}_i, \tag{4.3}$$

where μ is the cell mobility, v_0 sets the self-propulsion speed, and $\mathbf{n}_i = (\cos\theta_i, \sin\theta_i)$ is the polarity direction—the direction the cell would travel in the absence of cell-cell interactions. The polarity undergoes rotational diffusion,

$$\dot{\theta}_i = \sqrt{2D_r}\eta_i(t), \tag{4.4}$$

with zero-mean, unit-variance Gaussian white noise satisfying $\langle \eta_i(t) \rangle = 0$ and $\langle \eta_i(t)\eta_j(t')\rangle = \delta_{ij}\delta(t - t')$.

All equations are nondimensionalized using K_A and a characteristic length scale L, e.g., K_P is scaled by $K_A L^2$ and Λ is scaled by $K_A L^3$ [1]. Following Ref. [81], we choose the length scale $L = \sqrt{A_0/\pi}$ so that the nondimensional preferred area is $A_0 = \pi$. Time is scaled by the mechanical relaxation time $1/\mu K_A L^2$. Unless otherwise stated, we set the scaled maximum radius $\ell = 1$ and the scaled $K_P = 1$ (see Table 4.1), and we vary the control parameters P_0 and Λ to explore the detachment forces. Though we have implemented the model independently, our model assumptions are identical to those of Ref. [81], except that we simulate with open boundaries rather than periodic boundaries.

[1]This corrects a typo in Eq. (E4) of Ref. [80], where Λ should be scaled by $K_A L^3$ rather than $K_A L^4$.

4.3 Divergence of the detachment forces

4.3.1 Tissue fracture timescale depends systematically on time step

When performing a simple time-step convergence check of the AFV model, we find a strange phenomenon: when the tension difference $\Lambda > 0$ and a smaller time step Δt is used to evolve the dynamics, cells appear to be more adherent to each other, and fewer gaps form between cells. As shown in Fig. 4.2a, we evolve a cluster of $N = 100$ cells using different time steps, $\Delta t = 0.01$ and $\Delta t = 0.005$. After a simulation of 100 time units, the cluster evolved with the smaller Δt is more compact than the one evolved with the larger Δt, and unlike the large-Δt cluster, it has not fractured or formed holes. We have also confirmed that this behavior occurs in the code released by Ref. [81]. One way to quantify cluster cohesion is to consider the time scale of rupture of the cluster—i.e., for an initially N-cell cluster, due to the random motility of individual cells, when does it rupture into two disconnected clusters? This can be measured using the survival probability $S(t) = \mathbb{P}(T > t)$, where we monitor the time T of the *first* rupture event [66, 112]. Fig. 4.2b shows $S(t)$ for a range of Δt: as Δt decreases, rupture is delayed, and the survival curves shift to longer times. A convenient summary statistic is the median survival time $t_{1/2} = S^{-1}(1/2)$—the time at which the survival probability drops to one half—which grows rapidly as Δt is reduced (blue curve in Fig. 4.2c).

Fig. 4.2b,c shows that our qualitative visual analysis is correct: the cells become more adherent as the time step Δt decreases. Why does this happen? Does this reflect an issue with numerical algorithms, such as an incorrect choice of time integration schemes, or is it something more fundamental? In the next section, we show that this is an unavoidable consequence of the underlying geometric assumptions of the model.

4.3.2 Forces between a cell doublet

To identify the origin of the divergence, we begin with the simplest detachment event: two identical cells that share a single contact. As shown in Fig. 4.3a, we place the two cell centers symmetrically at $\mathbf{r}_{\pm} = \pm(\ell - \epsilon)\hat{x}$, so that the center-to-center distance is $d = 2(\ell - \epsilon)$ [80]. The parameter ϵ therefore measures how close the pair is to detachment: $\epsilon > 0$ corresponds to a finite contact ($d < 2\ell$), and

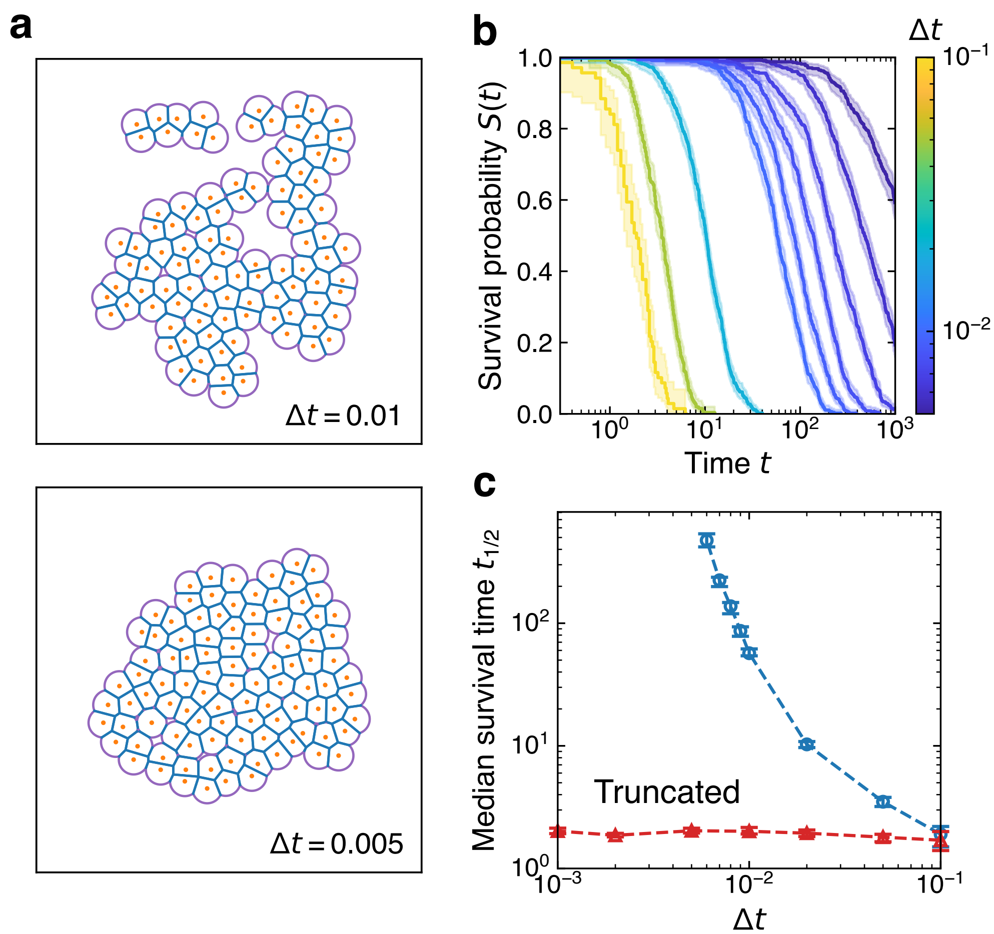

Figure 4.2: Tissue cohesion is strongly dependent on simulation timestep. **a**, Simulation snapshots of an $N = 100$ cell cluster at $t = 100$ for time steps $\Delta t = 0.01$ (upper) and $\Delta t = 0.005$ (lower). Each simulation was first evolved for 20 time units with zero motility to reach a steady state, followed by 100 time units of active dynamics. **b**, Survival probability $S(t)$ of an initially intact monolayer with time steps $\Delta t = 0.1$, 0.05, 0.02, 0.01, 0.009, 0.008, 0.007, 0.006, and 0.005. **c**, Median survival time $t_{1/2} = S^{-1}(1/2)$ as a function of Δt; the cutoff is set to $\delta = 0.45$ for the truncated curve. For each value of Δt, we apply standard Kaplan-Meier survival analysis [115] to the results of 480 independent simulations; error bars represent 95% confidence intervals. Parameters: number of cells $N = 100$, initial packing fraction $\phi = 0.5$, $\ell = 1$, $K_P = 1$, $A_0 = \pi$, $P_0 = 4.8$, $\Lambda = 0.1$, angular diffusion coefficient $D_r = 1.33$, and active velocity $v_0 = 1.5$.

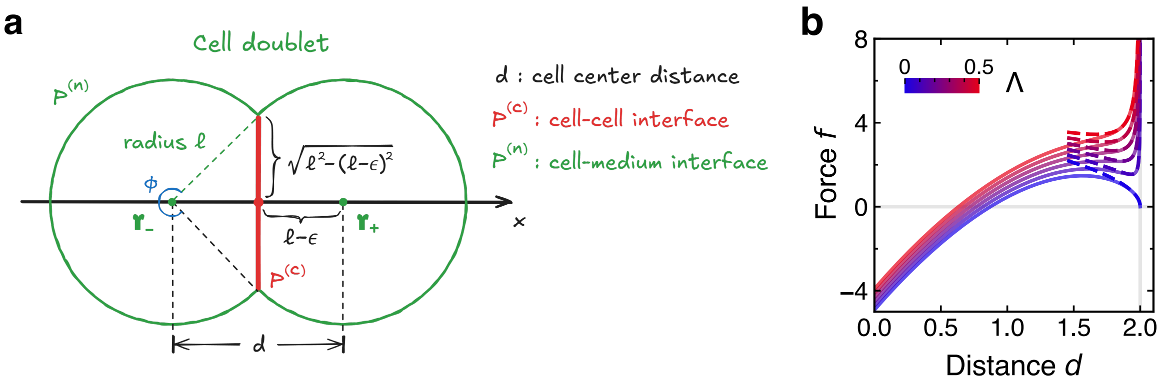

Figure 4.3: Cell doublet. **a**, Schematic of a cell doublet with centers positioned symmetrically at $\mathbf{r}_\pm = \pm(\ell - \epsilon)\hat{x}$. The arc radius of each cell is ℓ, and the distance between the two cell centers is given by $d = 2(\ell - \epsilon)$, thus the contact length (red) is $2\sqrt{\ell^2 - (\ell - \epsilon)^2}$. **b**, Force between a cell doublet as a function of their separation distance for $\Lambda = 0$, 0.1, 0.2, 0.3, 0.4, and 0.5 (blue to red). Positive values $f > 0$ correspond to attractive forces. Solid lines show Eq. (4.6), and dashed lines are the asymptotic form for $d \to 2\ell$ from Eq. (4.30). Default parameters: $\ell = 1$, $K_P = 1$, $A_0 = \pi$, and $P_0 = 4.8$.

$\epsilon \to 0^+$ is the limit in which cells detach. In this geometry, each cell boundary consists of a circular arc of length $P^{(n)} = \ell\phi$ together with a straight contact segment of length $P^{(c)} = 2\sqrt{\ell^2 - (\ell - \epsilon)^2}$, where $\phi = 2\pi - 2\,\mathrm{atan2}\big(\sqrt{\ell^2 - (\ell - \epsilon)^2}, \ell - \epsilon\big)$ is the angle spanning the region not in contact.

Because the doublet is symmetric, it is convenient to use the energy of a single cell:

$$E_\mathrm{s} = K_A(A - A_0)^2 + K_P(P - P_0)^2 + \Lambda P^{(n)}, \tag{4.5}$$

where the area and perimeter of each cell are

$$A = (\ell - \epsilon)\sqrt{\ell^2 - (\ell - \epsilon)^2} + \frac{\ell^2\phi}{2},$$
$$P = 2\sqrt{\ell^2 - (\ell - \epsilon)^2} + \ell\phi.$$

The (passive) interaction force between the two cells follows from the derivative of E_s with respect to the separation parameter, $f = -\partial E_\mathrm{s}/\partial\epsilon$. Evaluating this derivative gives (see Sec. 4.6 for details and some asymptotic limits)

$$f = 4\sqrt{(2\ell-\epsilon)\epsilon}\left[K_A(A-A_0) + K_P\frac{(P-P_0)}{2\ell-\epsilon}\right] + \Lambda\frac{2\ell}{\sqrt{(2\ell-\epsilon)\epsilon}}, \quad (\ell \geqslant \epsilon > 0) \tag{4.6}$$

where in our sign convention $f > 0$ corresponds to attractive forces (i.e., forces that tend to increase ϵ). We see that the last term in Eq. (4.6) diverges as the two cells separate, $\epsilon \to 0^+$. This term comes from the diverging derivative of the non-contacting perimeter $P^{(n)}$ with ϵ—in the finite Voronoi representation, small changes of the cell centers can lead to very large changes of perimeters.

The divergence of the detachment forces explains the time-step dependence of the rupture time scale: when $\Lambda > 0$, cells *should not be able to detach from one another* in the FV model of Refs. [80, 81]. The presence of detachment occurs only because we are using a finite time step Δt, where the numerical evolution can allow the cell to "skip" the divergence to reach full cell-cell separation, where the force is zero. It is possible for numerical errors to skip the divergence in part because the rapid increase in force is only apparent at distances very close to full separation (Fig. 4.3b)—if the cells move from distance $d \approx 1.95$ to over 2 in a single time step, the divergence would not be felt. Given the value $v_0 = 1.5$ and a time step $\Delta t = 0.05$, this could be quite common. By decreasing the time step Δt, we are approaching the "correct" no-detachment limit where $t_{1/2} \to \infty$ (Fig. 4.2c). Clearly, this behavior is not desired in a model of cells that can detach from one another, and we attempt to regularize these divergences in the next section.

4.3.3 Regularization by introducing a cutoff

To regularize the divergent detachment force, it is useful to identify precisely where the divergence enters the force calculation in the active finite Voronoi model. Given the cell-center positions of N cells $\{\mathbf{r}_i\}$, the finite Voronoi structure produces a set of vertices $\{\mathbf{h}_m\}$ (including inner triple-junction vertices and outer vertices where straight edges meet circular arcs; see Fig. 4.1). Forces on cell centers follow from the chain rule:

$$f_{i,x} = -\frac{\partial E}{\partial x_i} = -\sum_m \frac{\partial E}{\partial \mathbf{h}_m} \cdot \frac{\partial \mathbf{h}_m}{\partial x_i}, \tag{4.7}$$

where x_i is the x component of $\mathbf{r}_i$, with an analogous expression for the y component.

The divergence does *not* originate from the inner vertices (circumcenters of Delaunay triangles), whose derivatives remain finite for generic configurations. Instead, it comes from *outer* vertices $\mathbf{h}^{\text{out}}$ that connect a straight Voronoi segment to a circular arc. For two cells i and j sharing an outer vertex, there are two points $\mathbf{h}^{\text{out}}_{\pm}$ that lie at a distance ℓ from both centers, which are [80]

$$\mathbf{h}^{\text{out}}_{\pm} = \frac{\mathbf{r}_i + \mathbf{r}_j}{2} \pm \frac{\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2}}{2|\mathbf{r}_i - \mathbf{r}_j|}(\mathbf{r}_i - \mathbf{r}_j) \times \hat{z}. \tag{4.8}$$

The $\pm$ reflects the two symmetric intersection points of the circles of radius ℓ centered at $\mathbf{r}_i$ and $\mathbf{r}_j$, lying on opposite sides of the line ij. Their derivatives with respect to x_i are given by

$$\frac{\partial \mathbf{h}^{\text{out}}_{\pm}}{\partial x_i} = \frac{\hat{x}}{2} \mp \frac{2\ell^2(x_i - x_j)}{\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2}} \frac{(\mathbf{r}_i - \mathbf{r}_j)}{|\mathbf{r}_i - \mathbf{r}_j|^3} \times \hat{z} \mp \frac{\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2}}{2|\mathbf{r}_i - \mathbf{r}_j|}\hat{y}. \tag{4.9}$$

We can see that $\partial \mathbf{h}^{\text{out}}_{\pm}/\partial x_i$ contains a term with $\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2}$ in the denominator, which diverges as $|\mathbf{r}_i - \mathbf{r}_j| \to 2\ell$. This geometric divergence is the origin of the divergent detachment force in Eq. (4.6).

Physically, $\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2} = 2\sqrt{\ell^2 - (|\mathbf{r}_i - \mathbf{r}_j|/2)^2}$ is exactly the distance between the two outer vertices $\mathbf{h}^{\text{out}}_{\pm}$ generated by the pair $\{\mathbf{r}_i, \mathbf{r}_j\}$, as defined in Eq. (4.8). For the cell doublet shown in Fig. 4.3a, this quantity is simply the contact length $P^{(c)}$ (red). Thus, a minimal practical regularization is to bound the vanishing denominator in Eq. (4.9) by a small cutoff δ:

$$\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2} \to \max\left(\sqrt{4\ell^2 - |\mathbf{r}_i - \mathbf{r}_j|^2}, \delta\right). \tag{4.10}$$

This prescription renders forces finite at detachment while leaving them unchanged for $|\mathbf{r}_i - \mathbf{r}_j| < \sqrt{4\ell^2 - \delta^2} = 2\ell - (\delta^2/4\ell) + \mathcal{O}(\delta^4)$, at the cost of introducing a new microscopic length scale δ. This cutoff can be interpreted as a threshold contact length below which the Voronoi shape assumption is faulty, analogous to the finite edge-length criterion used to trigger T1 transitions in vertex models [67, 68, 136, 197].

From the shapes of the curves in Fig. 4.3b, for $\Lambda > 0$ the force-distance relation exhibits a local

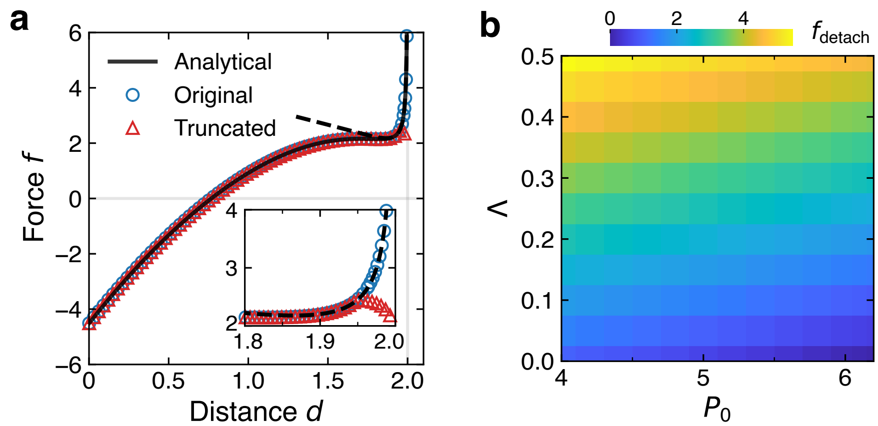

Figure 4.4: Regularization by introducing a cutoff. **a**, Force between a cell doublet as a function of cell-center distance for $\Lambda = 0.2$ and $P_0 = 4.8$. Empty circles and triangles represent forces obtained from simulations before and after implementing the truncation; analytical results are the corresponding curves in Fig. 4.3b. Inset: enlarged view around the cutoff $\delta = 0.45$, which corresponds to $d_c = \sqrt{4\ell^2 - \delta^2} \approx 1.95$. **b**, Phase diagram for the final detachment force f_{detach} when $d = 2\ell$ in the P_0–Λ plane, with cutoff $\delta = 0.45$. Default parameters: $\ell = 1$, $K_P = 1$, and $A_0 = \pi$.

minimum near $d \approx 2\ell$, after which the force increases rapidly toward divergence. For typical parameters, this rapid growth sets in around $d = 1.95\ell$, corresponding to a threshold value $\delta = \sqrt{4\ell^2 - d^2} \approx 0.45\ell$. Applying the cutoff of Eq. (4.10) with $\delta = 0.45$ (as $\ell = 1$), we successfully regularize the detachment forces around $d = 2\ell$ (see Fig. 4.4a), and the truncation effect kicks in at $d_c \approx 1.95$ as expected (inset of Fig. 4.4a). With this truncation, we eliminate the time-step dependence of the tissue fracture timescale (Fig. 4.2c, red curve).

In the finite Voronoi model, cell doublets are forced to detach when the cell-center distance d exceeds 2ℓ. We extract the final detachment forces f_{detach} at $d = 2\ell$ numerically, and we find that, after introducing the regularization, this force is no longer divergent even when sweeping through the parameter space of P_0 and Λ (Fig. 4.4b). The detachment forces in Fig. 4.4b increase strongly with Λ, as we would expect—increasing Λ corresponds to a higher energy cost for the cell-medium interface, increasing the cost for separation. We also see that increasing P_0 while holding other parameters constant weakly decreases the detachment force. This is consistent with, e.g., looking at the analytical detachment force in Eq. (4.6) at the distance at which regularization kicks in, $\epsilon_c = \ell - d_c/2$.

4.4 Calibration with a deformable polygon model

The origin of the divergence we found in the finite Voronoi model is the assumption that, even as the cells are almost completely separated, the cells remain circular with a flat contact. During detachment, however, real cells exhibit appreciable shape deformation [198–203]. While we are able to regularize the divergence by introducing a new scale δ, we want to better understand how to set this value. To calibrate our regularized finite Voronoi model, we compare it against a deformable polygon (DP) model that has the same energy function but in which cell boundaries are explicitly represented by polygons with large numbers of vertices that can deform continuously [82, 83, 150]—i.e., the same energetics as our FV model but without the Voronoi geometry assumption. During calibration, we focus on passive mechanical forces and set $v_0 = 0$.

4.4.1 Matching passive steady states between the deformable polygon model and finite Voronoi model

In the DP model, each cell is represented by a closed polygon with an ordered set of vertices $\{\mathbf{h}_m\}_{m=1}^{M}$. We use the same energy as in the FV model: quadratic area and perimeter elasticity together with distinct line tensions on cell-cell contacts and cell-medium boundary. However, the degrees of freedom are not the cell centers, but the vertices on the cell boundaries. Vertex dynamics are evolved by overdamped relaxation:

$$\frac{\mathrm{d}\mathbf{h}_m}{\mathrm{d}t} = -\mu \frac{\partial E}{\partial \mathbf{h}_m}. \tag{4.11}$$

To place the two models on a comparable footing, we first choose parameters so that the steady states of passive cell doublets match between the models. Minimizing the energy in the DP model using Eq. (4.11), we see that the minimal-energy state is, as the finite Voronoi model assumes, generally a pair of circular arcs with a straight boundary between them. However, the resulting steady-state radius of the arcs is the value that minimizes the energy, which will depend on the mechanical parameters of the model—and not the default $\ell = 1$ used in the FV model. To match the FV and DP models, we must vary ℓ when we vary the FV model parameters, choosing the value that minimizes the energy. In this case, as long as the steady state of the DP model fits the finite Voronoi assumption,

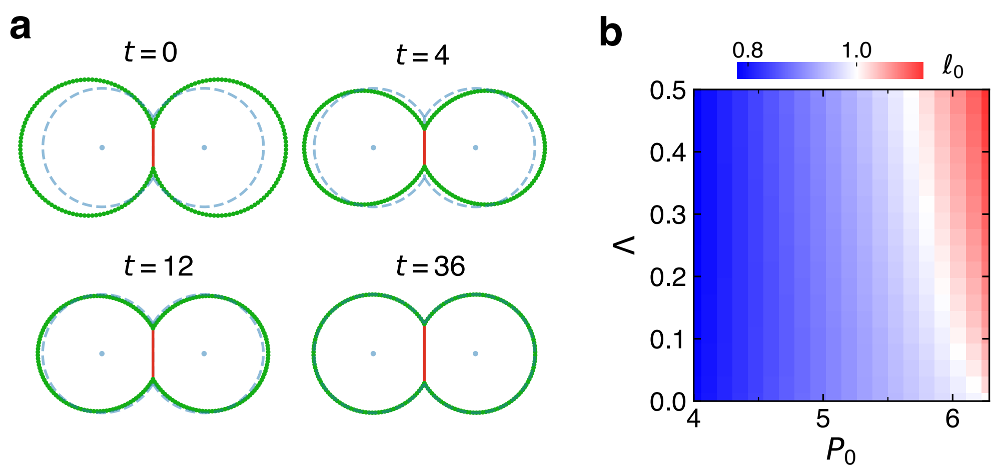

Figure 4.5: Steady states of deformable polygon model. **a**, Simulation snapshots of how the cell doublet relaxes from its initial shape to the steady state in DP model. Blue shapes indicate the optimal $(\ell_0, \epsilon_0) \approx (0.87, 0.12)$ computed from Eq. (4.12) for $P_0 = 4.8$ and $\Lambda = 0.2$. **b**, Phase diagram for optimal ℓ_0 in the P_0–Λ plane. Default parameters: $K_P = 1$, $A_0 = \pi$.

the FV model should recover the same steady state. Therefore, in our calibration process, we find the minimal-energy separation between cell centers and the minimal-energy value of ℓ,

$$(\ell_0, \epsilon_0) = \underset{\ell \geqslant \epsilon > 0}{\arg\min}\, E_{\mathrm{s}}(\ell, \epsilon), \tag{4.12}$$

and then set the FV radius to $\ell = \ell_0$. We see that in the absence of external forcing, FV and DP predict the same relaxed doublet shape (Fig. 4.5a). The values of the steady-state radius ℓ_0 for different sets of (P_0, Λ) are shown in Fig. 4.5b. We see that ℓ_0 increases with the preferred perimeter P_0, while Λ has a relatively weaker effect on ℓ_0.

4.4.2 Force-distance curves in the deformable polygon model

We want to find the relationship of force between cell pairs and distance in the DP model, akin to the values predicted for the finite Voronoi model in Fig. 4.3b. To do this, we apply equal and opposite forces to each cell in an initially relaxed doublet in the DP model, and measure the resulting displacement of the cell centroids d. (Note that to be consistent in our sense of cell-cell separation between the DP and FV model, here and throughout this section, we use d to denote the centroid-centroid separation of both the DP and FV cells—cells in the DP model are not perfectly circular and do not have a "center" defined as in the FV model.) We distribute the force on one cell uniformly over all vertices of each cell to avoid spurious torques [198], as shown in Fig. 4.6a. We then record the centroid separation and contact length $P^{(c)}$ (red segments in Fig. 4.3a) as functions of the external force f. We define the detachment force $f^{\mathrm{DP}}_{\mathrm{detach}}$ as the force at which the contact length falls below the typical distance $\ell_c = 2\pi\ell_0/M$ between neighboring vertices (see Sec. 4.6). The detachment force increases strongly with Λ as expected, and decreases weakly with P_0 (Fig. 4.6b). If we choose parameters of the FV model such that the steady states of the FV and DP models match (i.e., ℓ is the energy-minimizing ℓ_0), we find that the FV detachment force exhibits the opposite trend with P_0: for fixed Λ, increasing P_0 leads to larger $f^{\mathrm{FV}}_{\mathrm{detach}}$ (Fig. 4.6c), in contrast to the behavior in the DP model (Fig. 4.6b) and in the FV model with a fixed maximum radius ($\ell = 1$, Fig. 4.4b).

What behavior should we expect for detachment forces' dependence on P_0? In our rescaled units, $P_0 = -\lambda^{(c)}/2K_P$, so increasing P_0 typically corresponds to stronger cell-cell adhesion (i.e., making cell-cell line tension $\lambda^{(c)}$ more negative). Stronger cell-cell adhesion must increase the detachment force, so at first glance, it initially appears that only Fig. 4.6c exhibits the physically correct trend. However, P_0 and $\Lambda = \lambda^{(n)} - \lambda^{(c)}$ both depend on the more fundamental parameter $\lambda^{(c)}$. If $\lambda^{(n)}$ is held fixed and only $\lambda^{(c)}$ is varied, increasing adhesion corresponds to moving along the line $\Lambda = \lambda^{(n)} + 2K_P P_0$ in the Λ–P_0 plane. Along this path, the detachment forces in both the finite Voronoi model and the deformable polygon model increase as cell-cell adhesion is made stronger (Fig. 4.6d).

We compute the force-distance curve in the DP model and track the contact length as the cells are pulled apart from one another. We show these curves in Fig. 4.7a,b for two different values of P_0. Here $f > 0$ corresponds to a pull-off force dipole while $f < 0$ corresponds to a pushing force dipole. The contact length $P^{(c)}$ of the cell doublet decreases roughly linearly with the cell-centroid distance, and in the pulling limit ($f > 0$) the force for the DP model increases with centroid distance almost linearly, consistent with Ref. [198], where cells are modeled as three-dimensional elastic triangulated shells.

Comparing the analytical curves of the FV model and the DP simulation results in Fig. 4.7a,b, we find two key results: 1) for small perturbations away from the relaxed state (zero force), the DP and FV models agree well, and 2) the DP model does not have a divergent detachment force, staying attached over a much larger cell-cell distance than the FV model. We also see by looking at the cell doublet shapes in Fig. 4.6a that at larger forces, the finite Voronoi shape assumption starts to fail—cells can have centroids separated by more than $2\ell_0$ without losing contact, and at higher forces cells are no longer well described by circular arcs (rightmost panel of Fig. 4.6a).

We have found that, simply by choosing the energy-minimizing ℓ_0, the small forces near the equilibrium are in agreement between FV and DP models. However, the detachment forces are clearly not in good agreement. Fig. 4.7 shows two potential approaches to address this.

4.4.3 Calibration strategy 1: setting a critical length

As a first calibration strategy for near-detachment behavior, we choose the energy-minimizing value $\ell = \ell_0$ and adopt a fixed value of δ, using the default value $\delta = 0.45$ as in Fig. 4.4. Choosing a fixed value of δ corresponds to the idea that there is a minimal contact length beyond which cell-cell forces cannot increase, or equivalently a maximal cell-cell separation. We choose the value $\delta = 0.45$ to capture a reasonable scale at which the force-distance curve flattens as in Fig. 4.4a—in strategy 1 we do not use any information from the DP model to set this value. After setting $\ell = \ell_0$ and applying this fixed cutoff to truncate the rapidly growing force near detachment $d \approx 2\ell_0$ in the FV model, the simulation results (empty squares in Fig. 4.7a,b) follow the analytical curves but lack the divergence near $2\ell_0$, as expected. Therefore, consistent with the comparison in the previous section between the analytical FV results and the DP model, we observe reasonable agreement between the FV and DP force-distance relations for $d \leqslant 2\ell_0$, with only a few deviations at the smallest cell-cell distances where cells are highly compressed. However, the agreement in the force-distance curves necessarily means that the detachment forces differ between DP and FV in strategy 1—since cells remain connected even when $d > 2\ell_0$ in the DP model, the DP model exhibits a much larger detachment force.

Given this strategy, how does cluster rupture, as we explored in Fig. 4.2b,c, change with mechanical parameters like P_0, v_0, and Λ? We study the full active finite Voronoi model with this truncation

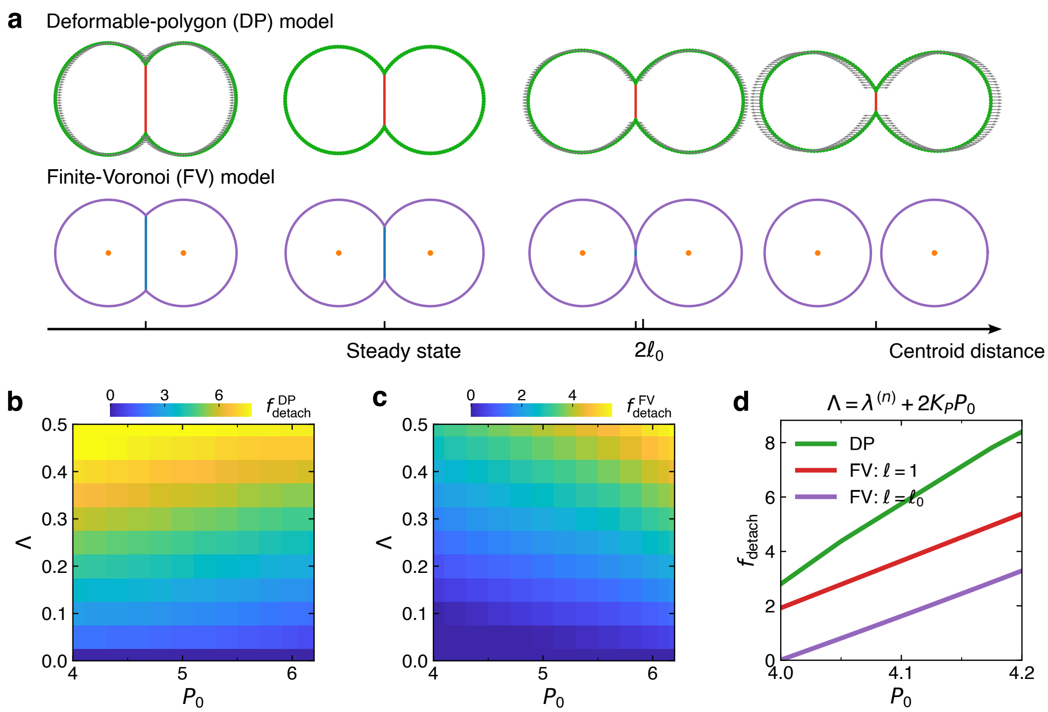

Figure 4.6: Detachment of cell doublets in the finite Voronoi model (FV) and deformable polygon (DP) models. **a**, Snapshots of a cell doublet for both deformable polygon model and finite Voronoi model at different centroid distances. Preferred perimeter is set to $P_0 = 4.8$ and $\Lambda = 0.2$, yielding the optimal $(\ell_0, \epsilon_0) \approx (0.87, 0.12)$ which has been used to set the maximum radius $\ell = \ell_0$ in FV model. The four centroid distances from left to right are 1.374 (compressed), 1.550 (steady state), 1.735 (approaching detachment threshold $2\ell_0$ in FV), and 1.912 ($> 2\ell_0$). **b**, The detachment forces $f^{\mathrm{DP}}_{\mathrm{detach}}$ for the DP model where contact length $P^{(c)}$ of the cell doublet approaches zero. **c**, The detachment forces $f^{\mathrm{FV}}_{\mathrm{detach}}$ for the FV model when $d = 2\ell$, where we have set the maximum radius to the steady-state radius from the DP model, i.e., $\ell = \ell_0$, and the threshold value $\delta = 0.45$. **d**, Detachment forces along the path $\Lambda = \lambda^{(n)} + 2K_P P_0$ in the previous phase diagrams, with the line tension on cell-medium interfaces fixed at $\lambda^{(n)} = -7.9$. Default parameters: $K_P = 1$, $A_0 = \pi$.

and calibration procedure. As shown in Fig. 4.7c, for fixed v_0 and P_0, increasing Λ consistently makes clusters harder to fracture. This is because increasing Λ effectively increases $\lambda^{(n)}$, thereby penalizing the creation of non-contacting free boundaries. In standard active Voronoi and vertex models, it is well established that larger preferred perimeter or higher motility v_0 leads to more fluid-like behavior. Our result shows that holding all other parameters constant, a larger preferred perimeter P_0—corresponding to a more fluid-like state with a lower line tension $\lambda^{(c)}$ for contacting edges—makes clusters harder to rupture (Fig. 4.7d). This differs from Ref. [81], where clusters with larger P_0 were more likely to be in dispersed states. We attribute this discrepancy not to the choice of cutoff δ, but to calibrating the maximum cell radius ℓ to its energy-minimizing value ℓ_0, which leads to detachment forces that increase with P_0 (Fig. 4.6c).

4.4.4 Calibration strategy 2: matching detachment forces

As discussed above, the previous strategy aligns the FV and DP models only for $d \leqslant 2\ell_0$, and this results in the FV model requiring a much smaller pull-off force to detach two cells. Thus, we also consider an alternative calibration based on the detachment force f_{detach}. In this approach, we set ℓ to be the energy-minimizing ℓ_0 and adjust δ such that the detachment force in the FV model matches that of the DP model (Fig. 4.6b), i.e., $f^{\text{FV}}_{\text{detach}} = f^{\text{DP}}_{\text{detach}}$. We numerically compute the value of d_c required for Eq. (4.6) to equal $f^{\text{DP}}_{\text{detach}}$, and obtain the corresponding δ_c. The results are shown in Fig. 4.7e,f, where the horizontal dashed lines indicate the matched detachment forces.

As shown in Fig. 4.7e,f, this calibration strategy introduces sharply increasing forces near detachment which will naturally require finer timestepping; to obtain reliable fracture statistics, we therefore gradually reduce the time step Δt from the default value of 0.01 to 0.002 until the results converge. In the phase diagram of the median survival time $t_{1/2}$ (Fig. 4.7g,h), this calibration generally makes clusters harder to detach than under the previous strategy (more points appear as stars), since the detachment forces $f^{\text{FV}}_{\text{detach}}$ are much larger now. Moreover, because the detachment forces are matched to those of the DP model, the transition lines between no-rupture and rupture states differ from those in Fig. 4.7c,d: the fracture timescale depends only weakly on the preferred perimeter P_0 now, consistent with the detachment-force phase diagram in Fig. 4.6b, whereas the previous

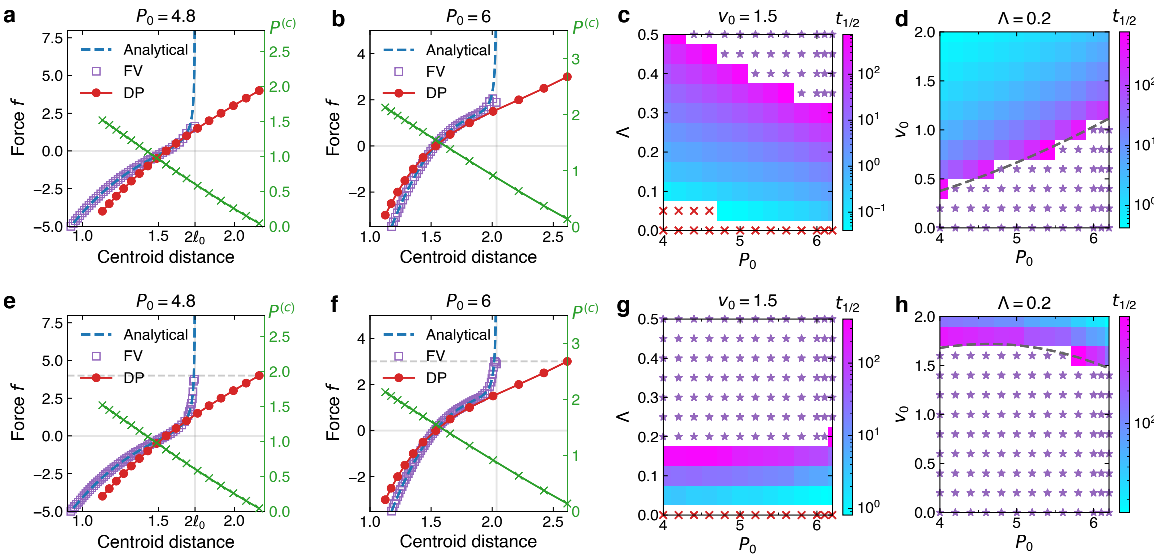

Figure 4.7: Calibrating the finite Voronoi model (FV) to a deformable polygon (DP) model. **a**,**b** show how the forces in both models and also the contact length $P^{(c)}$ in DP model (green) vary with increasing centroid distance at two different P_0 values ($\Lambda = 0.2$ and $\delta = 0.45$). **c**,**d** are the phase diagrams of median survival time $t_{1/2}$ on P_0–Λ and P_0–v_0 planes, where red crosses ($t_{1/2} = 0$) indicate tissues are already ruptured at the start (after initial relaxation time), and purple stars indicate that $t_{1/2}$ exceeds the total simulation time. Threshold value is set to $\delta = 0.45$. **e**–**h**, Calibrate FV model by matching the detachment forces f_{detach} to DP model with varying cutoff δ. Gray dashed lines in **e**,**f** indicate the match of detachment forces in both models. Each data point of $t_{1/2}$ in the phase diagrams is obtained by computing the survival probability from the results of 480 independent simulations. Default parameters: $K_P = 1$, $A_0 = \pi$, $D_r = 1.33$.

strategy (Fig. 4.7c,d) corresponds to Fig. 4.6c. The other trends remain similar to the earlier results: decreasing Λ or increasing v_0 both promote fracture.

4.5 Discussion

Our results show that the finite Voronoi model contains a pathological behavior when cell-cell and cell-medium tensions differ: the force required to separate two cells diverges, so fracture in active clusters becomes spuriously controlled by the numerical time step rather than by physical parameters. We trace the origin of this divergence analytically by analyzing a two-cell system and introduce a cutoff threshold that removes the divergence and restores a well-defined fracture timescale. We then compare the finite Voronoi model with a deformable polygon model and propose two calibration strategies, providing a practical framework for regularizing and calibrating the model in studies of tissue fracture and cell separation.

The cutoff regularization introduced in Eq. (4.10) removes the time step dependence of the AFV model, but it modifies near-detachment mechanics by introducing a new length scale δ. We think of δ as a truncation threshold for the contact length $P^{(c)}$: once the contact shrinks below δ, the intercellular force no longer increases. However, the precise value of this threshold is both model- and parameter-dependent, and cannot be uniquely fixed within the FV framework. Recent work using finite-element descriptions of cell mechanics has emphasized that tissue-level phenomena such as jamming can depend sensitively on the force required to separate neighboring cells [199], suggesting that the detachment force should be viewed as an important physical control parameter rather than a purely numerical detail. From this perspective, the regularization parameter δ provides a practical way to tune the FV model so that its near-detachment mechanics match those of more detailed descriptions. For the two calibration strategies introduced in Sec. 4.4, the appropriate choice depends on the physical regime of interest. If detachment events are rare—for example, when the tissue remains largely in a solid-like state—strategy 1 is typically sufficient, since it already reproduces the force-distance relationship well near the steady-state configuration. By contrast, when frequent cell rearrangements or fracture events occur, strategy 2 is more appropriate because it matches the detachment forces between the FV and DP models. More generally, the cutoff δ can also be calibrated to match detachment forces obtained from more detailed models, such as those in Refs. [82, 83], or even from experimental measurements of cell-cell pull-off forces.

The divergence of detachment forces in the finite Voronoi model may affect the results of Ref. [81] for simulations done with $\Lambda > 0$. First, because no force cutoff was implemented, clusters can fracture only due to the finite simulation time step Δt. Although the numerical dynamics with finite Δt will act as an effective regulator similar to the threshold δ introduced here, we think the use of the regularizing δ avoids possible complications in which varying numerical details like Δt can have spurious physical effects. We also expect some quantitative differences between regularization with a finite Δt and our approach. Secondly, choices about the calibration of ℓ and regularization can qualitatively change results. Using a fixed ℓ rather than the steady-state radius ℓ_0 from the DP model reverses the trend of fracture behavior when varying P_0 (Figs. 4.4b and 4.6c). Interestingly, in our model using a varying ℓ_0 and calibrating to the detachment forces (Fig. 4.7h), which we think is the most reasonable approach, we do see the same qualitative trends in the effect of P_0 that are observed by Ref. [81] in their phase diagrams (e.g., their Fig. 2b). However, we believe the origin of these behaviors may be subtler than expected, and they depend on our calibration choices.

The associated purpose-built Python package `PyAFV` is available on GitHub at wwang721/pyafv. A snapshot of the package and other data and code required to reproduce this chapter have been archived on Zenodo, doi: 10.5281/zenodo.18091659.

4.6 Appendices

4.6.1 Simulation details

We simulate the dynamics of both the FV and DP models using the forward Euler (Euler-Maruyama) method [188]. Default parameters are given in Table 4.1. All parameters are nondimensionalized using the length scale $L = \sqrt{A_0/\pi}$, energy scale $K_A L^4$, and time scale $1/\mu K_A L^2$. Note that we are not using the maximum radius ℓ as the scaling length unit so that ℓ can be set to other values than 1, such as the optimal ℓ_0 from minimizing E_s from Eq. (4.5).

When simulating the dynamics of the finite Voronoi model, before activating the active motility v_0, the initial cell center positions are randomly distributed within a square domain at a packing fraction of about 0.5 [81]. The system is then evolved for $T_{\text{relax}} = 20$ time units to reach a steady state. Subsequently, active motility is turned on, and the simulation is run for a duration T. The time since relaxation is used in the survival probability and mean survival time calculations.

Table 4.1: Table of simulation parameters. These parameters are used throughout this chapter; any deviations from them are explicitly specified.

Parameter	Description	Scale	Value
A_0	Preferred area	L^2	π
P_0	Preferred perimeter	L	4.8
ℓ	Maximum radius	L	1
K_P	Perimeter elastic modulus	$K_A L^2$	1
Λ	Tension difference $\lambda^{(n)} - \lambda^{(c)}$	$K_A L^3$	0.2
Δt	Time step	$1/\mu K_A L^2$	0.01
T_{relax}	Initial relaxation time	$1/\mu K_A L^2$	20
T	Total simulation time	$1/\mu K_A L^2$	1000
D_r	Rotational noise	$\mu K_A L^2$	1.33
N	Number of cells	1	100

4.6.1.1 Details of finite Voronoi model force calculations

In the standard/finite Voronoi model, as given in Refs. [25, 80], an inner vertex connecting three cells i, j, and k is given by

$$\mathbf{h}^{\text{in}} = \alpha_i \mathbf{r}_i + \alpha_j \mathbf{r}_j + \alpha_k \mathbf{r}_k. \tag{4.13}$$

The three barycentric coordinates of the circumcenter are

$$\alpha_i = |\mathbf{r}_j - \mathbf{r}_k|^2 (\mathbf{r}_i - \mathbf{r}_j) \cdot (\mathbf{r}_i - \mathbf{r}_k)/D, \tag{4.14a}$$

$$\alpha_j = |\mathbf{r}_i - \mathbf{r}_k|^2 (\mathbf{r}_j - \mathbf{r}_i) \cdot (\mathbf{r}_j - \mathbf{r}_k)/D, \tag{4.14b}$$

$$\alpha_k = |\mathbf{r}_i - \mathbf{r}_j|^2 (\mathbf{r}_k - \mathbf{r}_i) \cdot (\mathbf{r}_k - \mathbf{r}_j)/D, \tag{4.14c}$$

where $D = 2|(\mathbf{r}_i - \mathbf{r}_j) \times (\mathbf{r}_j - \mathbf{r}_k)|^2$. Thus, its derivative with respect to x_i is given by

$$\frac{\partial \mathbf{h}^{\text{in}}}{\partial x_i} = \alpha_i \hat{x} + \frac{\partial \alpha_i}{\partial x_i}\mathbf{r}_i + \frac{\partial \alpha_j}{\partial x_i}\mathbf{r}_j + \frac{\partial \alpha_k}{\partial x_i}\mathbf{r}_k = \alpha_i \hat{x} + \frac{\partial \alpha_j}{\partial x_i}(\mathbf{r}_j - \mathbf{r}_i) + \frac{\partial \alpha_k}{\partial x_i}(\mathbf{r}_k - \mathbf{r}_i), \tag{4.15}$$

where in the second equation we have used the relation $\alpha_i + \alpha_j + \alpha_k = 1$. We note this corrects a typo in Ref. [80]. A similar formula naturally holds for the derivative with respect to y_i. Then we need to compute

$$\frac{\partial \alpha_j}{\partial \mathbf{r}_i} = \alpha_j \left[\frac{\mathbf{r}_k - \mathbf{r}_j}{(\mathbf{r}_j - \mathbf{r}_i) \cdot (\mathbf{r}_j - \mathbf{r}_k)} + 2\frac{\mathbf{r}_i - \mathbf{r}_k}{|\mathbf{r}_i - \mathbf{r}_k|^2} - 2\frac{(\mathbf{r}_j - \mathbf{r}_k) \times \hat{z}}{[(\mathbf{r}_i - \mathbf{r}_j) \times (\mathbf{r}_j - \mathbf{r}_k)]_z} \right]. \tag{4.16}$$

The expression for $\partial \alpha_k / \partial \mathbf{r}_i$ follows by exchanging the indices j and k. We have also explicitly shown the expressions for outer vertices in Eqs. (4.8) and (4.9).

The mechanical force experienced by cell i is given by the gradient of the energy:

$$f_{i,x} = -\frac{\partial E}{\partial x_i} = -\sum_j \left[2K_A(A_j - A_0)\frac{\partial A_j}{\partial x_i} + 2K_P(P_j - P_0)\frac{\partial P_j}{\partial x_i} + \Lambda \frac{\partial P_j^{(n)}}{\partial x_i} \right]. \tag{4.17}$$

We note that each cell can be decomposed into a polygon (connecting all vertices belonging to that cell), together with additional circular segments (a segment is defined as the circular sector minus the isosceles triangle formed by the two radii and the chord), i.e., $A_j = A_j^{\text{poly}} + A_j^{\text{seg}}$ and $P_j = P_j^{\text{poly}} + P_j^{(n)}$. Thus, the derivatives of the area and perimeter can be decomposed into two corresponding parts:

$$\frac{\partial A_j}{\partial x_i} = \frac{\partial A_j^{\text{poly}}}{\partial x_i} + \frac{\partial A_j^{\text{seg}}}{\partial x_i}, \qquad \frac{\partial P_j}{\partial x_i} = \frac{\partial P_j^{\text{poly}}}{\partial x_i} + \frac{\partial P_j^{(n)}}{\partial x_i}. \tag{4.18}$$

For the polygon part,

$$\frac{\partial A_j^{\text{poly}}}{\partial x_i} = \sum_m \frac{\partial A_j^{\text{poly}}}{\partial \mathbf{h}_m} \cdot \frac{\partial \mathbf{h}_m}{\partial x_i}, \qquad \frac{\partial P_j^{\text{poly}}}{\partial x_i} = \sum_m \frac{\partial P_j^{\text{poly}}}{\partial \mathbf{h}_m} \cdot \frac{\partial \mathbf{h}_m}{\partial x_i}, \tag{4.19}$$

where the sum runs over all vertices (both inner and outer), as in Eq. (4.7). The derivatives $\partial \mathbf{h}_m / \partial x_i$ are already given above, while the derivatives $\partial A_j^{\text{poly}} / \partial \mathbf{h}_m$ and $\partial P_j^{\text{poly}} / \partial \mathbf{h}_m$ are provided in Eq. (B15) of Ref. [80].

We now compute the contribution from the segment part. For a cell j with a total segment area A_j^{seg} and total arc length $P_j^{(n)}$, consider a segment between two neighboring outer vertices $\mathbf{h}_m^{\text{out}}$ and $\mathbf{h}_{m+1}^{\text{out}}$

(ordered clockwise). Let ϕ_m^j denote the angle between the vector $\mathbf{u} = \mathbf{h}_m^{\mathrm{out}} - \mathbf{r}_j$ and the x-axis, i.e., $\phi_m^j = \mathrm{atan2}(u_y, u_x) \in (-\pi, \pi]$. The arc length between the two vertices is then $\Delta P = \ell \Delta\phi$, and the area of the corresponding segment is

$$\Delta A = \frac{\ell^2}{2}(\Delta\phi - \sin\Delta\phi), \tag{4.20}$$

where $\Delta\phi \equiv \phi_m^j - \phi_{m+1}^j \pmod{2\pi}$. Thus we have

$$\frac{\partial P_j^{(n)}}{\partial \phi_m^j} = \ell, \qquad \frac{\partial A_j^{\mathrm{seg}}}{\partial \phi_m^j} = \frac{\ell^2}{2}(1 - \cos\Delta\phi), \tag{4.21}$$

and $\partial P_j^{(n)}/\partial\phi_{m+1}^j = -\partial P_j^{(n)}/\partial\phi_m^j$, $\partial A_j^{\mathrm{seg}}/\partial\phi_{m+1}^j = -\partial A_j^{\mathrm{seg}}/\partial\phi_m^j$. Because an outer vertex such as $\mathbf{h}_m^{\mathrm{out}}$ is shared by cell j and a neighboring cell i [see Eq. (4.8)], the angle ϕ_m^j depends on both $\mathbf{r}_i$ and $\mathbf{r}_j$. With the definition $\mathbf{u} = \mathbf{h}_m^{\mathrm{out}} - \mathbf{r}_j = |\mathbf{u}|(\cos\phi_m^j, \sin\phi_m^j)$ and $|\mathbf{u}| = \ell$, we have

$$\frac{\partial \mathbf{u}}{\partial x_i} = \frac{\partial \mathbf{h}_m^{\mathrm{out}}}{\partial x_i} = \ell(-\sin\phi_m^j, \cos\phi_m^j)\frac{\partial \phi_m^j}{\partial x_i}, \tag{4.22}$$

where $(-\sin\phi_m^j, \cos\phi_m^j) = -(\mathbf{u} \times \hat{z})/|\mathbf{u}|$ is a unit vector perpendicular to $\mathbf{u}$. Thus, taking the dot product with $(-\sin\phi_m^j, \cos\phi_m^j)/\ell$, we obtain the derivative of ϕ_m^j with respect to x_i:

$$\frac{\partial \phi_m^j}{\partial x_i} = -\frac{1}{\ell^2}[(\mathbf{h}_m^{\mathrm{out}} - \mathbf{r}_j) \times \hat{z}] \cdot \frac{\partial \mathbf{h}_m^{\mathrm{out}}}{\partial x_i}, \tag{4.23}$$

and the y-component has an analogous form. Since $\mathbf{u}$ (and hence ϕ_m^j) depends only on the relative position $\mathbf{r}_i - \mathbf{r}_j$, we also have $\partial\phi_m^j/\partial\mathbf{r}_j = -\partial\phi_m^j/\partial\mathbf{r}_i$. With all these derivatives in hand, we can then compute the segment contributions to the derivative of the area and perimeter:

$$\frac{\partial A_j^{\mathrm{seg}}}{\partial x_i} = \sum_{\mathbf{h}_m^{\mathrm{out}}} \frac{\partial A_j^{\mathrm{seg}}}{\partial \phi_m^j}\frac{\partial \phi_m^j}{\partial x_i}, \qquad \frac{\partial P_j^{(n)}}{\partial x_i} = \sum_{\mathbf{h}_m^{\mathrm{out}}} \frac{\partial P_j^{(n)}}{\partial \phi_m^j}\frac{\partial \phi_m^j}{\partial x_i}. \tag{4.24}$$

4.6.1.2 Details of deformable polygon model

In our simulations of the deformable polygon model, each cell is represented by $M = 100$ vertices. Enumerating the vertices $\{\mathbf{h}_m\}$ of cell i in clockwise order, the force exerted on vertex m is given by

$$\mathbf{f}_m = -\frac{\partial E}{\partial \mathbf{h}_m}. \tag{4.25}$$

We need the derivative of area A_i with respect to $\mathbf{h}_m$ [80]:

$$\frac{\partial A_i}{\partial \mathbf{h}_m} = \frac{(\mathbf{h}_{m-1} - \mathbf{h}_{m+1}) \times \hat{z}}{2}, \tag{4.26}$$

and derivatives of P_m and P_{m-1} with respect to $\mathbf{h}_m$ [80]:

$$\frac{\partial P_m}{\partial \mathbf{h}_m} = \frac{\mathbf{h}_m - \mathbf{h}_{m+1}}{|\mathbf{h}_m - \mathbf{h}_{m+1}|}, \qquad \frac{\partial P_{m-1}}{\partial \mathbf{h}_m} = \frac{\mathbf{h}_m - \mathbf{h}_{m-1}}{|\mathbf{h}_m - \mathbf{h}_{m-1}|}, \tag{4.27}$$

where $P_m = |\mathbf{h}_m - \mathbf{h}_{m+1}|$, and $\mathbf{h}_{m-1}$ and $\mathbf{h}_{m+1}$ are two adjacent vertices of $\mathbf{h}_m$ in cell i. External forces are applied quasi-statically, starting from zero, and the system is evolved for 5×10^4 steps (step size $\Delta t = 0.001$) to reach a sequence of steady states. Vertices are resampled every 1000 steps by redistributing them along the free-boundary polyline to achieve approximately uniform spacing between adjacent vertices [150]. We record the force at which the contact length $P^{(c)}$ falls below the typical segment length between vertices, $\ell_c = 2\pi\ell_0/M$, as the detachment force $f^{\mathrm{DP}}_{\mathrm{detach}}$.

4.6.2 Analytical details

To derive the interaction force $f = -\partial E_s/\partial \epsilon$ of a cell doublet [Eq. (4.6)], we first compute the necessary derivatives. For the angle spanning the non-contacting region $\phi = 2\pi - 2\,\mathrm{atan2}\big(\sqrt{\ell^2 - (\ell-\epsilon)^2}, \ell - \epsilon\big)$, we obtain

$$\frac{\partial \phi}{\partial \epsilon} = -\frac{2}{\sqrt{(2\ell - \epsilon)\epsilon}}. \tag{4.28}$$

This gives

$$\frac{\partial A}{\partial \epsilon} = -2\sqrt{(2\ell - \epsilon)\epsilon}, \qquad \frac{\partial P}{\partial \epsilon} = -\frac{2\epsilon}{\sqrt{(2\ell - \epsilon)\epsilon}}. \tag{4.29}$$

We therefore see that the divergent contribution arises from the non-contacting perimeter $P^{(n)} = \ell\phi$ rather than from the area A or the total perimeter P.

We can further take the limit $\epsilon \to 0^+$ in Eq. (4.6), yielding the asymptotic form

$$2\sqrt{2\ell\epsilon}\left[2K_A(\pi\ell^2 - A_0) + K_P\frac{(2\pi\ell - P_0)}{\ell}\right] + \Lambda\sqrt{\frac{2\ell}{\epsilon}}, \tag{4.30}$$

which diverges for $\Lambda \neq 0$, as shown in Fig. 4.3b. This implies that, with $\Lambda \neq 0$, contacts become increasingly hard to break as their length shrinks to zero in the AFV model. Note that by using K_A and $L = \ell$ to nondimensionalize terms in the square bracket, we recover the dimensionless criterion for attractive forces given in Ref. [80] for the $\Lambda = 0$ case: $2(\pi - \tilde{A}_0) + \tilde{K}_P(2\pi - \tilde{P}_0) > 0$.

Chapter 5
Limits on the accuracy of contact inhibition of locomotion

This chapter is adapted from the following article with permission from the American Physical Society:

- Wang, W. & Camley, B. A. Limits on the accuracy of contact inhibition of locomotion. *Phys. Rev. E* **109,** 054408. DOI: 10.1103/PhysRevE.109.054408 (2024).

5.1 Introduction

More than half a century ago, researchers first observed contact inhibition of locomotion (CIL), when two colliding fibroblasts changed direction rapidly and migrated away from the collision [20–22, 46, 122]. During the following years, different variants of CIL were found in a large number of healthy and cancerous cell types [21]. The outcomes of cell-cell collisions that can be called CIL may include both cells reversing or a single cell reversing, leading to cell train formation, depending on the cell environment and cell type [22, 123, 165, 204–206]. The outcomes of cell-cell collisions are also known to be stochastic in nature [22, 165, 204]. We will think of CIL as the phenomenon of a cell actively changing its polarization upon encountering—and specifically recognizing—another cell. CIL can be classified into homotypic (occurring between cells of the same type) and heterotypic CIL (occurring between cells of different types), which is usually lost in cancer cells when confronted with normal cells [21]. CIL is now a well-known characteristic of normal cells and plays an important role in regulating cell motility [22, 122], tissue growth [47], and development [48, 50, 51]. For example, experiments revealed that *Drosophila* macrophages (haemocytes) require CIL for their uniform embryonic dispersal [48, 49]. CIL can happen—though rarely—even when cells are crawling on isolated nanofibers, leading to very small ($\sim \mu$m) contact regions [22]. How can cells reliably observe the presence of another cell just by such a small contact?

CIL is regulated by the Eph-ephrin signaling pathway [48, 50, 207, 208]. Contact detected by Eph-ephrin or cadherin binding [86, 209] reorganizes the actin cytoskeleton [87] by modulating the main elements of the Rho GTPase family [106, 108], resulting in the increase of actomyosin contractility and the inhibition of lamellipodia formation at the cell-cell contact [50, 107]. Eph receptors are a family of transmembrane receptor tyrosine kinases (RTK) [50, 207] that regulate tissue boundary formation in development [210] and tissue organization maintenance in adult organisms [50]. Their ligands, known as ephrins, are normally anchored to the cell surface [50, 51, 207]—Eph-ephrin signaling occurs when cells come into contact with one another. Ephrin cues can be used to reorganize and repolarize cells. Ephrin-coated beads lead to CIL when cells collide with them [51], and micropatterned ephrin spatial cues can be used to align intestinal crypt cells [50].

Cells can use receptors on their membrane to estimate external ligand concentrations [19, 211–213] and also to sense spatial [15, 18, 90, 96, 97, 214] or temporal [98] gradients. The accuracy limits of these estimations have been extensively studied [15, 96, 97, 212–214], including related measurements of spatially localized extracellular signals [97]. For a cell to undergo CIL, it must detect the presence and location of the cell-cell contact—akin to measuring a spatially localized chemical. The cell's ability to make this detection will be challenged if ligands other than ephrin, i.e., spurious ligands, bind non-specifically to the Eph receptors [92, 104, 105, 215]. Potential candidates for the spurious ligands are soluble ephrins [109], ephrins on extracellular vesicles [110], or ligands for other RTKs, e.g., growth factors [111]. In scenarios where there is only a small patch of contact [22], such interference from other ligands poses a significant threat to the precision of cell sensing.

In this chapter, we develop a theory for sensing cell-cell contact via Eph receptor-ephrin contact in the presence of spurious ligands using a multi-ligand model [104, 105, 212, 216]. To analyze the optimal performance of such a sensing mechanism, we adopt the method of maximum likelihood estimation (MLE) and derive the fundamental sensing limits by using the Cramér-Rao bound. We find that the accuracy of detecting the location of the other cell is degraded when the binding affinity of the spurious ligands is close to the true ephrin's affinity, or when the fraction of correct ligands is small. Surprisingly, accuracy does not depend strongly on the contact width. These results are also supported by Monte Carlo simulations. We use statistical decision theory to study the problem of whether the cell can effectively detect the presence of another cell via a small cell-cell contact, and show how cells can trade off false negatives, where they fail to respond to another cell, and false

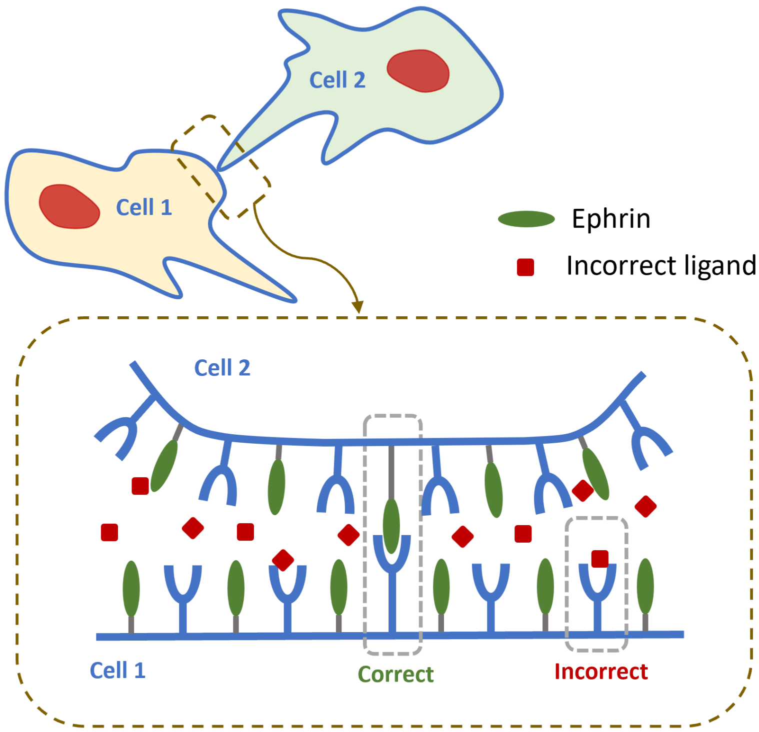

Figure 5.1: Illustration of two cells in contact where they can sense each other through Eph-ephrin signaling. Both Eph receptors (blue receptors) and their ligands—ephrins (green ellipses), are distributed on the cell membranes. Besides the cognate ligands, other spurious ligands (red squares) can also bind to the Eph receptors. These incorrect binding events mixed with the correct Eph-ephrin binding will affect the sensing accuracy of contact inhibition of locomotion.

positives, where they react even in the absence of a contacting cell. We find that cell-cell contact can be highly reliably detected even at large levels of spurious ligand concentrations.

5.2 Model

Consider two cells coming into contact as shown in Fig. 5.1, where both cells can sense each other through Eph-ephrin binding [50, 51]. We study the sensing accuracy for cell 1 detecting cell 2, as the mirror problem is equivalent.

As an initial model for the fundamental sensing limits of contact inhibition of locomotion, we work in an effectively one-dimensional model, with Eph receptors characterized by their location x around the perimeter of cell 1. We assume ephrins from cell 2 can bind to the Eph receptors on cell 1 with a rate $kc(x)$, i.e., a base rate k times an effective ephrin concentration $c(x)$. The rate $kc(x)$ combines both the area density of the ephrin on the membrane with the reduced likelihood of binding for ephrins which are further away from the membrane of cell 1—so if we treat k as constant we expect $c(x)$ to vary from its maximum value at the cell-cell contact to essentially zero far away from the contact. The details of the shape of $c(x)$ will depend on the details of the cell-cell junction and the cell-cell contact, but we make an initial hypothesis that $c(x)$ is effectively constant over the cell-cell contact width and then smoothly decreases to zero (Fig. 5.2, top). We use a smooth rectangular function for the effective ephrin concentration that cell 1 senses on cell 2's surface:

$$c(x) = c_0 S\big(x - (x_0 - \sigma)\big) S\big((x_0 + \sigma) - x\big), \tag{5.1}$$

where $S(x) = [1 + \tanh(x/\omega)]/2$, and ω controls the steepness at the transition between 0 and c_0. x_0 is the center of the region of cell-cell contact and 2σ is the contact width. Receptors bound with the correct ligand (ephrin) will unbind with rate r. Receptors can also bind the incorrect/spurious ligand with rate kc' and unbind with rate r' (Fig. 5.2, bottom), where we assume the effective concentration of incorrect ligands c' is constant.

In our model, we adopt the assumption of Ref. [104], where the only distinction between the two ligands is their binding kinetics, neglecting features like potential ligand bias [217, 218]. This means that the cell only knows whether a receptor is bound, not what ligand is bound to it. How, then, can the cell detect the presence of another cell and repolarize away from it? First, even if ligands had identical unbinding rates, there would be more bound receptors at the cell-cell contact. Secondly, if the correct and incorrect ligands have different affinities ($r \neq r'$), the cell can discriminate between them statistically. Ligands with a larger unbinding rate remain bound to receptors for a shorter time compared to ligands with a smaller unbinding rate. Thus, the cell can discern the mixture of two different ligands if the detailed occupancy history of each receptor is available [104]. This record can be summarized by the times the receptor spends unbound $\tau_{u,i}$ and the times it spends bound $\tau_{b,i}$ [98, 101, 104], e.g., $\{\tau_{u,1}, \tau_{b,1}, \tau_{u,2}, \tau_{b,2}, \cdots\}$ (as shown in Fig. 5.2, bottom). Given the receptor record,

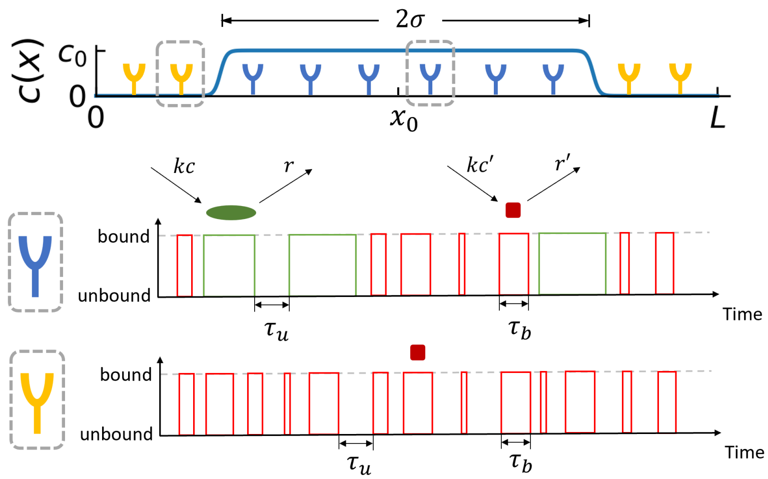

Figure 5.2: Extracting information from bound-unbound time series of receptors. Top: The ephrin concentration is a smooth rectangular function $c(x)$ that is high at the region of contact ($\sim \pm\sigma$ of the contact center at x_0). Bottom: The temporal record of the binding states of example receptors inside and outside the contact region. Receptors within the contact region (blue) can bind to both ephrins (green) and incorrect ligands (red) while receptors outside (yellow) can only bind to the spurious ligands.

$c(x)$ can be estimated, which allows the cell to identify the location of the cell-cell contact x_0.

The probability densities to have a bound interval of length τ_b and an unbound interval of length τ_u for a receptor seeing concentration c of correct ligand are [104, 214]

$$P(\tau_u) = k(c + c')\mathrm{e}^{-k(c+c')\tau_u}, \tag{5.2}$$

$$P(\tau_b) = \frac{c}{c + c'} r\mathrm{e}^{-r\tau_b} + \frac{c'}{c + c'} r'\mathrm{e}^{-r'\tau_b}, \tag{5.3}$$

where the factor $(c+c')$ in Eq. (5.2) is the total concentration, and the factors $c/(c+c')$ and $c'/(c+c')$ in Eq. (5.3) are the probabilities of the binding event being the correct ligand and incorrect ligand, respectively. We assume that there are N receptors evenly spaced over the cell perimeter L, each with position x_n so the n-th receptor has ephrin concentration $c(x_n)$.

We assume that the cells observe the binding and unbinding events over a "measuring time" T and use this to make a CIL decision—estimating the position of the cell-cell contact. The probability of a particular time record $\{\tau_{u,1}, \tau_{b,1}, \tau_{u,2}, \tau_{b,2}, \cdots\}$ for the n-th receptor is $P_n = \prod_{i=1}^{M_n} P(\tau_{u,i}^{(n)}) \cdot P(\tau_{b,i}^{(n)})$ where M_n is the number of pairs of binding-unbinding events during the time T. Therefore, given the three parameters $\boldsymbol{\theta} \equiv (c_0, c', x_0)$ in our model, the total probability of a trajectory of bound and unbound times arising from N independent receptors is $P(\{\tau\}|\boldsymbol{\theta}) = \prod_{n=1}^{N} P_n$.

To derive the fundamental sensing limits, we use the method of maximum likelihood estimation (MLE) [96, 101, 219]. The log-likelihood function is $\ln \mathcal{L}(\boldsymbol{\theta}|\{\tau\}) = \ln P(\{\tau\}|\boldsymbol{\theta})$. The MLE estimate of the parameters $\hat{\boldsymbol{\theta}}$ is obtained by maximizing the log-likelihood, e.g., by solving $\partial \ln \mathcal{L}/\partial\theta_\mu|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} = 0$, where the index $\mu = 1, 2, 3$, or by numerical maximization of $\ln \mathcal{L}$. The error between any unbiased estimate of the parameters that the cell can make, $\hat{\boldsymbol{\theta}}$, and the true parameters is limited by the Fisher information matrix and the Cramér-Rao bound [Eq. (5.4)]. We can compute the Fisher information matrix $I(\boldsymbol{\theta})_{\mu\nu} = -\langle \partial^2 \ln \mathcal{L}/\partial\theta_\mu \partial\theta_\nu \rangle$ for our model in a partially-analytic way. After taking the derivatives of the log-likelihood necessary to compute the Fisher information matrix, we get an equation that is a complicated function of the bound/unbound times $\{\tau_b, \tau_u\}$. This reduces the problem to quadrature—we can calculate the average $\langle \cdots \rangle$ by numerical integration using the probability densities in Eq. (5.2) and (5.3) (see Sec. 5.5.2 for details). The Cramér-Rao bound (CRB) then gives the lower bound for the variance of an unbiased estimator $\hat{\boldsymbol{\theta}}$ of parameters [15]:

$$\mathrm{Var}(\hat{\theta}_\mu) \geqslant [I(\boldsymbol{\theta})^{-1}]_{\mu\mu}, \tag{5.4}$$

where $I(\boldsymbol{\theta})^{-1}$ is the matrix inverse. Eq. (5.4) sets the best precision with which a cell could measure, e.g., the location of the cell-cell contact, given the unavoidable stochasticity arising from the ligand-receptor interactions.

5.3 Results

How precisely can a cell detect the position of cell-cell contact, or the concentrations of ligands? This is set by Eq. (5.4)—the best possible unbiased estimates we can make of our parameters. We will display our results in terms of the variables $\boldsymbol{\theta}' \equiv (c_t, \chi, x_0)$, where $c_t \equiv c_0 + c'$ is the total concentration, i.e., the concentration that sets the total binding rate at the cell-cell contact and

$\chi = c_0/c_t$ is the fraction of correct ligands. When $\chi \ll 1$, most bindings are the incorrect ligand, and we expect the cell to struggle to perform CIL. We also define the unbinding rate ratio of the two ligands as $\alpha \equiv r/r'$. The correct ligands are expected to have a higher affinity to the receptors and thus take longer to unbind, i.e., $r \leqslant r'$, so we assume $\alpha \leqslant 1$.

In Fig. 5.3, we plot both our calculations of the minimal standard deviations required from the Cramér-Rao bound in Eq. (5.4) as well as simulation results from Monte Carlo (MC) simulations where we generate stochastic receptor trajectories according to Eq. (5.2) and Eq. (5.3), and then numerically maximize the likelihood to determine the maximum-likelihood estimators $\hat{c}_t$, $\hat{\chi}$, and $\hat{x}_0$ (Sec. 5.5.3). Broadly, we see good agreement between simulation and theory, except for a few data points at small patch sizes when $\sigma \sim 1$–$2\,\mu$m, which we discuss further below.

5.3.1 How do the sensing limits change when it becomes difficult to distinguish the two ligands?

The unbinding rate ratio $\alpha = r/r'$ quantifies the difference in receptor-ligand affinity between the two ligands. In other words, it characterizes how hard it is for the cell to discriminate the two ligands. When α is close to 0, i.e., in the easy discrimination regime, the standard deviations of all three estimated parameters $(\hat{c}_t, \hat{\chi}, \hat{x}_0)$ are small. When α gets closer to 1, it becomes harder to distinguish the two ligands. If $\alpha = 1$, both ligands have the same unbinding rate and are thus indistinguishable for the cell within our assumptions. The difficulty to tell the two ligands apart leads to a rise in sensing errors for all parameters as α nears 1, i.e., the smaller the discrepancy between ligands, the less accurately the cell can sense the contact [Figs. 5.3(a)–(f)]. The standard deviations of all three parameters are small compared to their relevant scales, which are for Fig. 5.3 total concentration $c_t = 1000\,\mu\text{m}^{-2}$, correct ligand fraction $\chi = 0.1$, and the cell perimeter $L = 100\,\mu$m. We also see that the errors relative to the mean are small [Fig. 5.9]. This means that the limits imposed by stochastic noise of ligand-receptor binding may not be that stringent in practice. For example, if the cell can locate the source of cell-cell contact to $\pm\text{SD}(\hat{x}_0)$ of ~ 0.4 microns [the worst case of Fig. 5.3(f)], this means the cell has an angular uncertainty of the direction to go which is $2\pi \cdot \text{SD}(\hat{x}_0)/L$ of $\sim 1.5°$. This angular uncertainty is much smaller than the typical angular range of acceleration

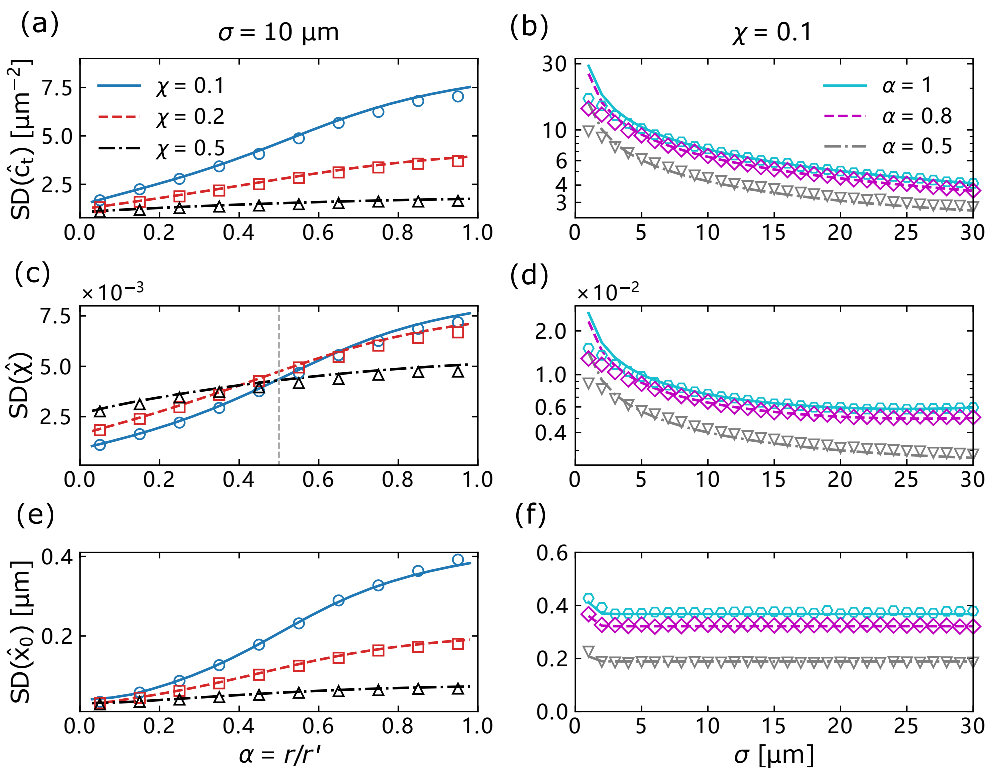

Figure 5.3: Errors of maximum likelihood estimate (MLE) influenced by receptor-ligand affinity, fraction of correct ligands, and contact width. Panels (a), (b) show the standard deviation (SD) of the total concentration $\hat{c}_t$, panels (c), (d) show the SD of correct ligand fraction $\hat{\chi}$, and panels (e), (f) display the SD of contact center position $\hat{x}_0$. In all figures, the lines (solid, dashed, and dash-dot) represent theoretical results obtained from Eq. (5.4), while the points are obtained from Monte Carlo simulations (the standard deviations are computed from 10^4 simulations for each point). Measuring time is $T = 10\,\mathrm{s}$. Corresponding relative errors (SD/average) are shown in Fig. 5.9.

vectors observed in CIL on two-dimensional substrates [48], which indicates that cells likely have more positional information than they really use in CIL—unless the concentration of spurious ligands is extraordinarily high ($\chi \ll 1$) or our model is incomplete or there is additional noise due to one of the other factors we raise later in Sec. 5.4.

In Figs. 5.3(a), 5.3(c), and 5.3(e), we also show the standard deviations for different correct ligand fractions $\chi = c_0/c_t$. We expect that, as the fraction of correct ligands decreases, the error in estimating each parameter should increase. We do see that errors associated with estimating the total concentration c_t and the contact center position x_0 decrease as χ increases [Figs. 5.3(a) and 5.3(e)].

However, estimating the correct ligand fraction χ itself has a more complex behavior [Fig. 5.3(c)]. Estimating χ in our model is akin to a spatially-dependent multi-receptor model of the question asked by Ref. [104]—how precisely can a single receptor detect a rare ligand binding? Our model, though, shows qualitative differences compared to the single receptor model of Ref. [104]. For a single receptor in the easy discrimination regime when the cell is confident about what type of ligand is bound ($\alpha < 1/2$), the uncertainty of sensing χ increases with χ as $\mathrm{SD}(\hat{\chi}) \sim \chi^{\beta/2}$ with $\beta = 1 - \alpha/(1-\alpha)$, though the relative error $\mathrm{SD}(\hat{\chi})/\chi$ always decreases with χ [104], while in the hard discrimination regime ($\alpha > 1/2$), $\mathrm{SD}(\hat{\chi})$ is roughly independent of χ, and diverges as $\alpha \to 1$, where a single receptor cannot discriminate between ligands. In our spatial, multi-receptor model, we find that when $\alpha < 1/2$ the error in χ increases with χ, as seen in Ref. [104]. However, in the hard discrimination regime ($\alpha > 1/2$), we find that $\mathrm{SD}(\hat{\chi})$ has the opposite behavior, increasing when χ decreases. We argue that this occurs because there are two ways for the cell to estimate χ: first, by discriminating between the two ligands by their binding times, and secondly, by comparing the total amount of binding observed at the cell-cell contact (depending on $c_0 + c'$) to the binding outside the contact (depending only on c'). The second mechanism does not require that the cell can discriminate between the two ligands, and will still function as $\alpha \to 1$. Hence, we see that as $\alpha \to 1$, the error of sensing χ decreases as χ increases just like the error of the total concentration c_t. This source of error is essentially arising because at small χ, the difference between $c_0 + c'$ and c_0 will shrink. By contrast, for $\alpha < 1/2$, the estimation of χ is similar to the single-molecule discrimination of Ref. [104] and the error in χ increases with χ.

5.3.2 How does contact width limit CIL?

As the contact width σ expands, more receptors come inside the contact region and the cell can obtain more information about the concentration of the true ligands—we expect that the accuracy of estimating our parameters should generally increase as the contact width increases. In Figs. 5.3(b) and 5.3(d), we observe that both the standard deviation of total concentration $\mathrm{SD}(\hat{c}_t)$ and the standard deviation of the fraction of correct ligands $\mathrm{SD}(\hat{\chi})$ decrease with increasing contact width σ. Surprisingly, we find the error in estimating the contact position $\mathrm{SD}(\hat{x}_0)$ remains almost constant as σ ranges from 3–30 μm [Fig. 5.3(f)]. Why? We argue that the cell's information about the contact location is largely coming from where the edges of the contact are, i.e., the transitions in $c(x)$. To better illustrate this, consider a simpler model that only involves one snapshot in time and one species of ligands, which we denote with a capital $C(x)$. In this toy model, given concentration $C(x)$, the occupancy for each receptor is given by $p_n = C(x_n)/[C(x_n)+K_D]$, with the dissociation constant $K_D = r/k$ [90]. The joint probability for the state of all receptors is $P(\{Z_n\}) = \mathcal{L}^{\text{toy}}$ with

$$\mathcal{L}^{\text{toy}} = \prod_n p_n^{Z_n}(1-p_n)^{1-Z_n}, \tag{5.5}$$

where $Z_n = 0$ indicates the receptor is unbound and $Z_n = 1$ indicates it is bound. In this toy model, we can compute the Fisher information:

$$I_{\mu\nu}^{\text{toy}} = -\left\langle \frac{\partial^2 \ln \mathcal{L}^{\text{toy}}}{\partial\theta_\mu \partial\theta_\nu} \right\rangle = \sum_n \frac{K_D}{[C(x_n)+K_D]^2 C} \frac{\partial C(x_n)}{\partial\theta_\mu} \frac{\partial C(x_n)}{\partial\theta_\nu}. \tag{5.6}$$

If the concentration $C(x)$ takes the form of a smooth rectangular function (Eq. (5.1)), and we compute the x_0-x_0 component of the Fisher information matrix, the only terms that are nonzero in the sum in Eq. (5.6) are those where the derivative $\partial C(x)/\partial x_0$ is nonzero—which are the edges of the contact zone. By contrast, if we were going to estimate the concentration level of the contact, then we would get a non-zero term at every receptor within the contact, and we would expect the error to decrease with contact size.

In Fig. 5.3(f) the uncertainty in measuring the contact center x_0 does increase for small enough σ, in the range of $\sigma = 1$–$2\,\mu$m. This happens when the contact width σ is similar to the scale $\omega = 1\,\mu$m over which $c(x)$ is varying—so then $\partial c/\partial x_0$ will be nonzero over the whole contact patch. If we increase ω, the general behaviors of the errors will not change, but the increase in $\mathrm{SD}(\hat{x}_0)$ at small σ will occur at a larger σ, as the concentration profile becomes strongly dependent on x when $\sigma \sim \omega$ [see Fig. 5.10(f)].

However, the derivative $\partial c/\partial x_0$ depends significantly on the chosen concentration profile—it's zero except at the boundaries because we are using a smooth rectangular function (Eq. (5.1)) here. If we choose a concentration profile where $\partial c/\partial x_0$ is non-zero throughout, such as a Gaussian function, all receptors within the contact region will contribute to the accuracy of position sensing, and the dependence of the estimation errors of $\hat{x}_0$ shown in Fig. 5.3(f) will be different. See Sec. 5.4 and Fig. 5.6 for more details.

For most of Figs. 5.3(a)–(f), we see good agreement between our bound from the Fisher information in Eq. (5.4) and the variance of the maximum likelihood estimator computed from a stochastic simulation. However, at the smallest σ, we do see a larger deviation between Monte Carlo simulations and Fisher information results in Figs. 5.3(b), 5.3(d), and 5.3(f). This discrepancy likely arises from the difficulty of the maximization process when σ is small—when we do numerical optimization to find the maximum-likelihood estimator $\hat{\boldsymbol{\theta}}$, there might be some deviations from the true global maximum point—if the optimizer becomes trapped in a local maximum, or if the number of iterations provided is too few, etc. These numerical problems are more likely to happen for small contact size σ and $\alpha \to 1$, because in such cases, there is less difference between a cell with a very small contact patch and one with essentially no contact ($\chi = 0$). In fact, the reasons that make it difficult for us to do this numerical optimization also make it difficult for the cell to determine if another cell is present, which we address in the next section. In other words, we suspect that any effective biochemical optimization process would also be difficult in cases where numerical optimization fails.

5.3.3 How reliably can the cell detect contact at all?

In many situations, determining the presence of another cell is more useful for the cell than precisely locating the contact point. For instance, a recent experiment [22] studied CIL for cells on suspended nanofibers, which are used to mimic the extracellular matrix. In this biologically relevant case—and in other experiments of CIL on micropatterns [51, 165]—cells only have two possible directions. Thus the precision in estimating the location of contact x_0 is not so important, but deciding whether there is contact or not is very important.

We study the problem of how precisely a cell can detect the presence of a neighbor from a localized signal by using tools from statistical decision theory. First, if we regard the cell as a binary classifier, we can pose a question to the cell: "Is there another cell in proximity?" This question can be formally translated into the task of distinguishing between two hypotheses: the presence vs. absence of another cell. In our model, if we denote the three-dimensional parameter space where $\boldsymbol{\theta}$ lives as Θ, the two hypotheses—presence or absence [104, 216, 220], can be written as:

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \equiv \{(c_0, c', x_0) | c_0 = 0\}, \tag{5.7}$$

$$H_1 : \boldsymbol{\theta} \in \Theta_0^C \equiv \Theta \backslash \Theta_0, \tag{5.8}$$

where H_0 (called the null hypothesis) states $\boldsymbol{\theta}$ lives within a subspace Θ_0, which corresponds to the plane of $c_0 = 0$, i.e., the other cell is not there. And H_1 (called the alternative hypothesis) asserts $\boldsymbol{\theta}$ lives in Θ_0^C, the complement space of Θ_0, i.e., $c_0 \neq 0$ and there is another cell nearby.

This question can be addressed by employing a statistical method known as the likelihood-ratio test (Wilks test) [220] to help the cell make decisions. The basic concept of this test is to calculate the ratio of the maximum likelihood under the two competing hypotheses:

$$\lambda = \ln \frac{\sup_{\boldsymbol{\theta} \in \Theta} \mathcal{L}}{\sup_{\boldsymbol{\theta} \in \Theta_0} \mathcal{L}} = \sup_{\boldsymbol{\theta} \in \Theta} (\ln \mathcal{L}) - \sup_{\boldsymbol{\theta} \in \Theta_0} (\ln \mathcal{L}) \geqslant 0, \tag{5.9}$$

where the sup notation refers to the supremum. Therefore, if λ is large, it indicates the alternative hypothesis H_1 (presence) is more competitive. In contrast, the null hypothesis H_0 (absence) becomes a more compelling choice if λ is small. In general, the ratio λ will tune how much evidence is needed to prefer hypothesis H_1 over hypothesis H_0. Then we can set an adjustable threshold value λ_c for the cell. If $\lambda > \lambda_c$, the cell believes it is contacting another cell. Conversely, if $\lambda < \lambda_c$, the cell concludes that there is no other cell nearby.

To gain deeper insight into the problem, we want to know: How likely is it for the cell to fail in detecting the presence of another cell when it is indeed there? Or how likely is it for the cell to claim the detection of another cell but it is actually absent? In statistics, the former probability is defined as the false negative rate (FNR), and the latter is referred to as the false positive rate (FPR). In our

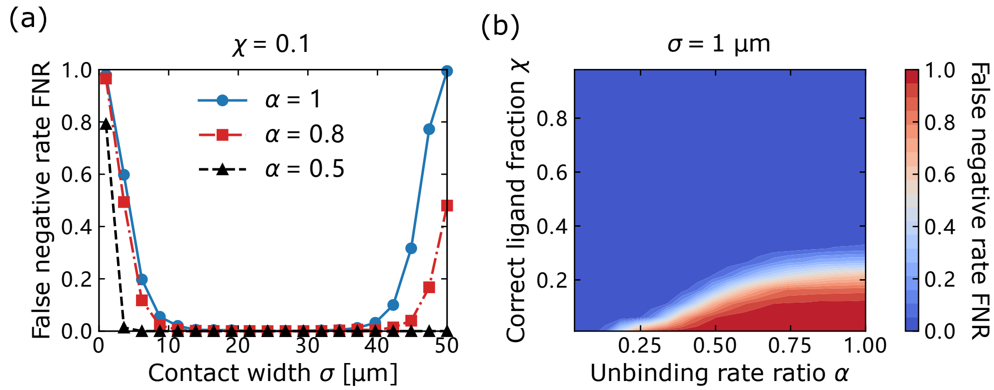

Figure 5.4: False negative rates showing how the probability of a cell missing the presence of a contacting cell is affected by contact width σ, unbinding rate ratio α, and the fraction of correct ligands χ. Panel (a) illustrates the change in FNR as contact width varies, and the phase diagram in panel (b) shows a sharp transition from low α, high χ values to high α, low χ values. We conduct $\sim 10^5$ simulations for each data point. The threshold parameter $\lambda_c = 7$ is chosen somewhat arbitrarily to show when the maximum value of FNR can be close to 1 in these figures.

model, the false negative rate can be computed as [104]

$$\text{FNR} = \langle H(\lambda_c - \lambda) \rangle_+, \tag{5.10}$$

where $H(x)$ is the Heaviside step function. The symbol $\langle \cdots \rangle_+$ denotes the average over all allowed configurations of the time series, and the positive subscript $+$ indicates the average is taken under the condition of another cell being there. Only when the cell misses the presence of the other cell, i.e., $\lambda < \lambda_c$, the step function is non-zero. Thus, Eq. (5.10) gives exactly the probability of the cell making a false negative decision, where it erroneously asserts the absence of the other cell. To compute the FNR, we choose a value λ_c, use our Monte Carlo simulation to generate a set of receptor occupancies over time given the presence of a second cell (Sec. 5.5.3), then numerically maximize the likelihood under the two hypotheses (presence and absence of the contacting cell) to compute λ from Eq. (5.9). We then repeat this many times to compute the FNR from Eq. (5.10).

Figure 5.4(a) shows the Monte Carlo simulation results of FNR for varying contact widths. At small σ, when there are relatively few receptors where the true concentration $c(x)$ is nonzero, the false negative rate is large. Essentially, in this case, the probability of small numbers of correct ligands binding from cell-cell contact being mistaken for a coincidence is large. The FNR decreases as the contact width is increased, leading more receptors to come into the contact region. However, when the contact width 2σ becomes very large and approaches the cell perimeter L, the false negative rate starts to rise again, particularly when $\alpha = 1$. This $2\sigma \to L$ case could happen if the cell is nearly engulfed by another cell, or is in contact with many cells. FNR will increase as the contact region approaches L only in the hard discrimination regime (α approaches 1), when the two ligands are indistinguishable. When 2σ approaches the cell perimeter, $c(x)$ becomes nearly constant $c(x) \approx c_0$—equivalent to a simple change in the background concentration c', if the cell cannot discriminate between the two ligands. The rise in FNR at large σ disappears as the cell becomes increasingly capable of distinguishing between the two ligands, i.e., when $\alpha < 1$.

We then perform a sweep across the unbinding rate ratio α and the correct ligand fraction χ, both of which significantly impact the false negative rate. We show a phase diagram for when cells might fail to recognize a contact in Fig. 5.4(b), which shows a sharp transition in FNR as we go from low α, high χ values to high α, low χ values. Notably, the FNR exhibits non-zero values only when α approaches 1 and $\chi \ll 1$, i.e., the cell is more prone to making mistakes in the hard discrimination regime, where the ligands are nearly indistinguishable, and when there is a higher presence of interfering background ligands. These findings align with the previous results in Figs. 5.3(a), 5.3(c), and 5.3(e).

The false negative rates are influenced by the adjustable threshold parameter λ_c we chose. We can always increase the false positive rate (FPR) and decrease the false negative rate by adjusting the threshold parameter λ_c to a small value, and vice versa. To compare the ability to discriminate independent of the arbitrary λ_c, we use the detection error tradeoff (DET) graph [104, 221, 222]. First, we can compute the false positive rate in a similar manner to FNR:

$$\mathrm{FPR} = \langle H(\lambda - \lambda_c) \rangle_-, \tag{5.11}$$

where $\langle \cdots \rangle_-$ indicates taking the average with no second cell present, i.e., setting the ephrin concentration $c_0 = 0$. We visualize the trade-off between false positives and false negatives by varying the threshold parameter λ_c over the whole range of λ observed in our simulations, and then plotting the FPR found for each λ_c against the FNR, as shown in Figs. 5.5(a)–(c). The figures in Figs. 5.5(a)–(c) show how FPR varies as a function of FNR. For instance, if we look at Fig. 5.5(a), the contours indicate the best false positive rate the cell can achieve given a fixed false negative rate, or vice versa. If the cell can tolerate a false positive rate of 10^{-3}, and the contact width is $\sigma = 20\,\mu$m, the cell can achieve a low false negative rate (down to $\sim 10^{-3}$, red line). However, either decreasing the contact to $\sigma = 10\,\mu$m, or expanding the contact to $\sigma = 40\,\mu$m makes the FNR larger for this FPR—the cell is more likely to overlook the contact.

In Fig. 5.5(b), we see that the ability of the cell to reliably distinguish the presence of contact gets progressively worse as $\alpha \to 1$—the FPR-FNR tradeoff curve gets further from the origin. Increasing the cell's measurement time T improves the accuracy with which the cell can sense cell-cell contact [Fig. 5.5(c)].

We can summarize the efficacy of the cell's ability to distinguish the presence of a contacting cell by computing the area under the curve (AUC) of the tradeoff graphs in Figs. 5.5(a)–(c). A smaller AUC indicates that the cell exhibits higher accuracy in distinguishing between the two hypotheses H_0 and H_1—roughly, that the cell achieves a lower false negative rate for a given false positive rate. We note that tradeoffs of this sort are also often plotted on a receiver operating characteristic (ROC) curve, which plots true positive rate against false positive rate. In our convention, a smaller area under the curve is more predictive, instead of larger; our AUC is $1 - \text{AUC}_{\text{ROC}}$ [221, 222].

Figure 5.5(d) shows the AUC value as a function of contact width, which initially decreases and then increases as 2σ gets closer to L. This confirms the results of Fig. 5.4(a): The cell can better sense the existence or absence of the other cell when the contact width is at some intermediate value. Figs. 5.5(b) and 5.5(e) show how the DET curve and the associated AUC change with the unbinding rate ratio α. The area under the curve increases as $\alpha \to 1$, which means the cell is more likely to make incorrect decisions in the hard discrimination regime. We also observe that the AUC drops to approximately 0 in the easy discrimination regime when $\alpha \approx 0.5$, which is consistent with where the transition happens in the phase diagram of FNR [Fig. 5.4(b)]. As we saw above, accuracy in sensing also depends on the fraction of ligands that are spurious. If we decrease the correct ligand

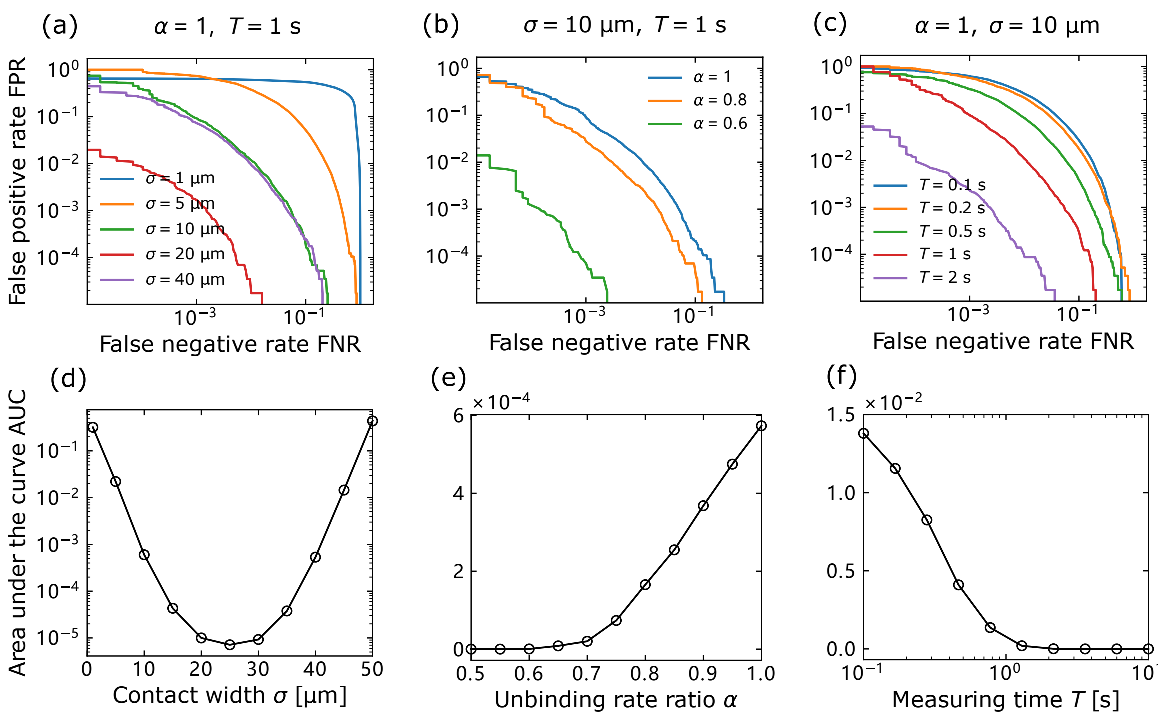

Figure 5.5: Detection error tradeoff (DET) graphs and the associated area under the curve (AUC) showing how the cell's ability of making accurate decisions varies with the contact width σ, unbinding rate ratio α, and measuring time T. Panels (a) and (d) show that the optimal contact width is at some intermediate value. Panels (b) and (e) show that the cell is more likely to make incorrect decisions when it is difficult to distinguish the two ligands. Panels (c) and (f) show that the cell can more reliably detect the presence or absence of another cell with a longer measuring time T. Correct ligand fraction is set to $\chi = 0.1$ in all figures.

fraction to $\chi = 0.01$ [Fig. 5.8], the overall AUC can become much larger, reaching AUC ~ 0.4, i.e., it becomes much harder to make correct decisions. However, even with this extremely small correct ligand fraction χ, the cell can detect contact very accurately once the unbinding rate ratio reaches $\alpha \sim 0.25$ [Fig. 5.8]. So far, we have kept the amount of time that the cell uses to make a decision constant. Increasing the measurement time T monotonically brings the DET curve toward the origin, decreasing the AUC [Figs. 5.5(c) and 5.5(f)]. Cells can more reliably detect the presence or absence of another cell by extending the measuring time T, as we would expect. AUC are mostly large when the measuring time is on the scale of seconds or less. This is a relatively short time compared to the time over which CIL takes place, which can be tens of minutes [165].

5.4 Discussion

Our multi-ligand model highlights the significant impact of various factors on the accuracy of sensing cell-cell contact, including the receptor-ligand affinity, the fraction of correct ligands, and the contact width. It becomes more challenging to sense the cell-cell contact when it's difficult for the cell to distinguish ephrins from spurious ligands, or when there are more interfering spurious ligands present. Moreover, our model reveals that as the contact width σ expands, it becomes easier for the cell to estimate the concentration and the fraction of correct ligands. However, the error in contact position x_0 estimation remains constant while the contact width increases. We argue that this phenomenon occurs because the cell can only obtain spatial information from the two boundaries where the concentration changes. Furthermore, we investigate whether the cell can successfully detect the presence of another cell through a small cell-cell contact. Our finding indicates that the cell is more likely to make incorrect decisions when the contact width is either too small or too large. In addition, our results also suggest that the cell is more probable to make mistakes if it's difficult for the cell to distinguish ephrins from spurious ligands ($\alpha \to 1$). The cell can extend the time it takes to make a decision to improve decision-making robustness.

A core element of our model is the profile $c(x)$ in Eq. (5.1), which represents a cell in contact with another cell with a clear contact region of fixed size and a smoothed rectangular form. This neglects many potential complications—e.g., that the size of cell-cell contacts may evolve over time during CIL (see, e.g., the movies of Ref. [22]), how ligand-receptor binding could drive changes in the size and spacing of the cell-cell contact [223, 224], and how cadherin proteins could facilitate the formation of contacting patches by zipping (or unzipping) regions of the contacting membrane, thereby leading the coalescence of a patch of defined size [225]. To check the robustness of some of our assumptions, we also explore the case where the ephrin concentration takes the form of a Gaussian function $c(x) = c_0 \exp(-(x - x_0)^2/2\sigma^2)$. This would arise if we assume that our effective binding rate $kc(x)$ is suppressed when there is a gap large enough to require ephrin to undergo a strain in order to bind. If there is a gap $h(x)$ between the two cells, we would expect $kc \sim \mathrm{e}^{-\frac{1}{2}\kappa h^2/k_B T}$ with κ an effective spring stiffness [226]. If $h(x)$ increases linearly away from the point of closest approach x_0, we would get a Gaussian profile. Many of our core results are quite similar between the Gaussian and the smooth step profile for $c(x)$, see Figs. 5.6(a)–(e). However, in the scenario of a Gaussian concentration, we find that the error in contact position estimation $\mathrm{SD}(\hat{x}_0)$ increases with increasing contact width [Fig. 5.6(f)]. Unlike the case of rectangular profiles, the cell now can obtain spatial information from receptors away from the contact edges, as $\partial c/\partial x_0 = c(x)(x - x_0)/\sigma^2$ is nonzero throughout the contact and depends systematically on σ—there is more spatial information at any given point as the width σ becomes smaller. Therefore, the error of estimating the contact position x_0 does not remain constant as σ expands; instead, it increases with increasing σ.

Moreover, the concentration profile $c(x)$ may take on a more intricate form, such as comprising several small patches. Our model currently implicitly assumes a single patch. If a cell expecting a single patch had multiple patches in contact with it, it would do its best to interpret this—either lumping together multiple patches or treating one patch as the "real" one and assuming the other was a coincidence. In either case, we expect similar factors would control the cell's ability to detect the presence of a second cell, but the localization error we study here might not be as directly applicable.

Can a cell reasonably reach the limits we suggest here? Ref. [104] proposed a network for sensing the concentration of correct ligands so that biological systems can approach the physical limits given by the Cramér-Rao bound. Other papers have also suggested signal transduction networks to interpret sensing from ligands [105, 227]. However, the current state of the art in this field only treats concentration sensing, and its extension to our spatially-dependent model for position sensing, let alone decision-making, is not straightforward. We know that experimentally, CIL occurs through the regulation of Rho GTPase polarity [51]. We speculate that modeling patterning occurring via Rho GTPases, regulated by the measured local concentrations as proposed by Ref. [104], may serve as a sensible first model for spatial decision-making. However, the details of this model would be much more complex than the approach we take here.

The estimation errors shown in our model (Fig. 5.3) are generally small compared to their relevant scales, suggesting cells may have fairly precise knowledge of the cell-cell contact location and ephrin levels. This may reflect the large number of binding-unbinding events. With the total number of receptors $N = 10^4$ and typical receptor correlation times of $\sim$ s (Sec. 5.5.1), and the measuring time $T = 10\,\mathrm{s}$, there are $\sim 10^5$ binding-unbinding events during the time the cell makes its CIL measurement. Given our results, it is likely that, unless one of our assumptions is incorrect, or that the fraction of spurious ligands is extremely high (e.g., $\chi < 0.1$), stochastic ligand-receptor noise is not a crucial factor in CIL decisions. Where could our assumptions be incomplete? One possibility is that we have neglected details of the Eph receptor interactions. Experiments have found that receptor clustering and other biophysical complexities [207, 228–230] can influence the functioning of receptors in binding kinetics in Eph receptors and other RTKs. For instance, EGF receptors dimerize while binding to their ligands EGF [228]. Moreover, for A549 lung cancer cells, it has been observed that the vast majority of EphA2 molecules exist in clusters [229]. As a result, receptor clustering could significantly reduce the number of distinct receptor locations in our model, thereby substantially increasing the errors when estimating the cell-cell contact. Additionally, the precise values of the binding constant k and unbinding rate r for ligands and receptors tethered to cell membranes, such as Eph-ephrin, are not entirely clear [226, 231–233]. These parameters can also significantly influence the number of binding-unbinding events M that occur during the measuring time T. Sources of noise downstream of the initial ephrin binding, such as stochasticity in the polarization of the cell's Rho GTPases [123, 234–236], might lead to significant additional noise in responses to CIL. Incorporating these factors into our model could provide a more comprehensive understanding of cell-cell interaction and response mechanisms.

Code to reproduce this chapter has been deposited at Zenodo, doi: 10.5281/zenodo.11094134.

5.5 Appendices

5.5.1 Correlation time of binding kinetics

In our analytical model, we assume that the number of binding-unbinding events for each receptor is large. This requires that the time of the observation T is large in comparison to the timescales relevant for binding and unbinding. In this appendix, we determine these timescales. We show that the probability of occupation of a single receptor relaxes to its steady state with two separate timescales $\tau^{\pm}_{\mathrm{corr}}$, where the correlation time depends on the kinetics of binding (rates k, r, and r')

and the effective concentrations c and c'. With the presence of the incorrect ligands, a receptor can be in an unbound state, a correct bound state, or an incorrect bound state. Thus we can write the corresponding three-state master equations:

$$\frac{\mathrm{d}p_{\text{correct}}}{\mathrm{d}t} = (1 - p_{\text{correct}} - p_{\text{incorrect}})kc - p_{\text{correct}}r,$$
$$\frac{\mathrm{d}p_{\text{incorrect}}}{\mathrm{d}t} = (1 - p_{\text{correct}} - p_{\text{incorrect}})kc' - p_{\text{incorrect}}r',$$

where $1 - p_{\text{correct}}(t) - p_{\text{incorrect}}(t)$ is the probability that the receptor is unbound. The steady-state solutions can be easily obtained by setting the time derivatives to zero:

$$p^{\text{SS}}_{\text{correct}} = \frac{kcr^{-1}}{1 + kcr^{-1} + kc'r'^{-1}}, \tag{5.12}$$
$$p^{\text{SS}}_{\text{incorrect}} = \frac{kc'r'^{-1}}{1 + kcr^{-1} + kc'r'^{-1}}. \tag{5.13}$$

If there is only one species of ligands, the occupancy (aka fractional saturation) is given by $p = c/(c + K_D) = kcr^{-1}/(1 + kcr^{-1})$, where $K_D = r/k$ is the dissociation constant [90]. With the presence of the second species of ligands, the probability of being bound at all is [104]:

$$p = p^{\text{SS}}_{\text{correct}} + p^{\text{SS}}_{\text{incorrect}} = \frac{kcr^{-1} + kc'r'^{-1}}{1 + kcr^{-1} + kc'r'^{-1}}. \tag{5.14}$$

However, in order to calculate the receptor correlation timescale, we have to solve the master equations over time. Because the final solutions should converge to the steady state, we assume the complete solutions are in the form of

$$p_{\text{correct}}(t) = A\mathrm{e}^{-t/\tau_1} + B\mathrm{e}^{-t/\tau_2} + p^{\text{SS}}_{\text{correct}},$$
$$p_{\text{incorrect}}(t) = C\mathrm{e}^{-t/\tau_1} + D\mathrm{e}^{-t/\tau_2} + p^{\text{SS}}_{\text{incorrect}},$$

where $p_{\text{correct}} \to p^{\text{SS}}_{\text{correct}}$ and $p_{\text{incorrect}} \to p^{\text{SS}}_{\text{incorrect}}$ when $t \to \infty$. Here we are primarily interested in the two correlation times τ_1 and τ_2—which should not depend on the particular initial conditions. We then assume the receptor is bound to a correct ligand at $t = 0$, so the initial conditions are

$p_{\mathrm{correct}}(0) = 1$ and $p_{\mathrm{incorrect}}(0) = 0$, i.e., $A + B = 1 - p^{\mathrm{SS}}_{\mathrm{correct}}$ and $C + D = -p^{\mathrm{SS}}_{\mathrm{incorrect}}$. Substituting the solutions back to the master equations, we can find the four coefficients:

$$
\begin{aligned}
A &= -\frac{kc}{2\sqrt{\Delta}} p^{\mathrm{SS}}_{\mathrm{incorrect}} + \frac{1}{2}(1 - p^{\mathrm{SS}}_{\mathrm{correct}}) + \frac{(kc + r) - (kc' + r')}{2\sqrt{\Delta}}(1 - p^{\mathrm{SS}}_{\mathrm{correct}}), \\
B &= \frac{kc}{2\sqrt{\Delta}} p^{\mathrm{SS}}_{\mathrm{incorrect}} + \frac{1}{2}(1 - p^{\mathrm{SS}}_{\mathrm{correct}}) - \frac{(kc + r) - (kc' + r')}{2\sqrt{\Delta}}(1 - p^{\mathrm{SS}}_{\mathrm{correct}}), \\
C &= \frac{1}{\tau_2} \frac{p^{\mathrm{SS}}_{\mathrm{incorrect}}}{\sqrt{\Delta}}, \\
D &= -\frac{1}{\tau_1} \frac{p^{\mathrm{SS}}_{\mathrm{incorrect}}}{\sqrt{\Delta}},
\end{aligned}
$$

and the two characteristic times:

$$\tau_1 = \frac{2}{kc + r + kc' + r' + \sqrt{\Delta}}, \tag{5.15}$$

$$\tau_2 = \frac{2}{kc + r + kc' + r' - \sqrt{\Delta}}, \tag{5.16}$$

where the discriminant-like term $\Delta = [k(c + c') + r - r']^2 - 4kc'(r - r')$ is larger than 0 under our assumption $r' \geqslant r$. We can also rewrite Δ in a more symmetric form:

$$\Delta = (kc + r + kc' + r')^2 - 4rr'(1 + kcr^{-1} + kc'r'^{-1}),$$

which ensures $\tau_1, \tau_2 > 0$. Moreover, when there is only one species of ligands, i.e., the correct ligand concentration $c = 0$, the two timescales degenerate to $\tau_2 = 1/r$ and $\tau_1 = 1/(kc' + r')$, which is exactly the correlation timescale for one species of ligands [237]. In brief, we can write the characteristic correlation times τ_1 and τ_2 as

$$\tau^{\pm}_{\mathrm{corr}} = \frac{2}{kc + r + kc' + r' \pm \sqrt{\Delta}}, \tag{5.17}$$

and expect that for our semi-analytical calculation of the Fisher information to be valid, we must be in the limit $T \gg \tau^{\pm}_{\mathrm{corr}}$ for all receptors.

5.5.2 Semi-analytical calculation of Fisher information

If the correct ligand concentration takes the form of $c(x) = c_0 g(x)$, and there are N independent receptors evenly distributed in the cell perimeter L, the n-th receptor will have a different value of the ephrin concentration $c(x_n)$ and a different number of binding-unbinding events M_n during a measuring time T. Then for the n-th receptor, the probability to have a time record $\{\tau_b^{(n)}, \tau_u^{(n)}\}$ is

$$P_n = \prod_{i=1}^{M_n} P(\tau_{u,i}^{(n)}) \cdot P(\tau_{b,i}^{(n)}), \tag{5.18}$$

where the unbinding and binding times are given by

$$P(\tau_u) = k(c + c')\mathrm{e}^{-k(c+c')\tau_u}, \tag{5.19}$$

$$P(\tau_b) = \frac{c}{c+c'} r\mathrm{e}^{-r\tau_b} + \frac{c'}{c+c'} r'\mathrm{e}^{-r'\tau_b}. \tag{5.20}$$

Therefore, given the three parameters $\boldsymbol{\theta} = (c_0, c', x_0)$ in our model, the total probability of all N receptors is $P(\{\tau\}|\boldsymbol{\theta}) = \prod_{n=1}^{N} P_n$, and the likelihood $\mathcal{L}(\boldsymbol{\theta}|\{\tau\}) = P(\{\tau\}|\boldsymbol{\theta})$ is given by

$$\begin{aligned}
\mathcal{L} &= \prod_{n=1}^{N} \prod_{i=1}^{M_n} \left[k(c+c')\mathrm{e}^{-k(c+c')\tau_{u,i}^{(n)}} \times \left(\frac{c}{c+c'} r\mathrm{e}^{-r\tau_{b,i}^{(n)}} + \frac{c'}{c+c'} r'\mathrm{e}^{-r'\tau_{b,i}^{(n)}} \right) \right] \\
&= \prod_{n=1}^{N} \mathrm{e}^{-k(c+c')T_u^{(n)}} \prod_{i=1}^{M_n} k \left(rc\mathrm{e}^{-r\tau_{b,i}^{(n)}} + r'c'\mathrm{e}^{-r'\tau_{b,i}^{(n)}} \right) \\
&= \prod_{n=1}^{N} \mathrm{e}^{-kc_t[\chi g(x_n)+1-\chi]T_u^{(n)}} \prod_{i=1}^{M_n} kr'c_t \mathrm{e}^{-r'\tau_{b,i}^{(n)}} \left[1 - \chi + \alpha\chi g(x_n) \mathrm{e}^{(1-\alpha)r'\tau_{b,i}^{(n)}} \right] \\
&= \prod_{n=1}^{N} \mathrm{e}^{-kc_t[\chi g(x_n)+1-\chi]T_u^{(n)} - r'T_b^{(n)}} \prod_{i=1}^{M_n} kr'c_t \left[1 - \chi + \alpha\chi g(x_n) \mathrm{e}^{(1-\alpha)r'\tau_{b,i}^{(n)}} \right],
\end{aligned}$$

where in the second and last steps we have defined the total amount of unbound and bound time for the n-th receptor as $T_u^{(n)} = \sum_{i=1}^{M_n} \tau_{u,i}^{(n)}$ and $T_b^{(n)} = \sum_{i=1}^{M_n} \tau_{b,i}^{(n)}$, respectively. In the penultimate step, we introduce a transformation from (c_0, c') to the total concentration $c_t = c_0 + c'$ and correct ligand fraction $\chi = c_0/c_t$, i.e., we substitute $c' = c_t(1-\chi)$, $r = \alpha r'$, and $c(x) = c_t \chi g(x)$. After taking a logarithm, $\ln \mathcal{L}$ can be written as a sum of five independent contributions:

$$
\begin{aligned}
\ln \mathcal{L}_0 &= \sum_{n=1}^{N} \sum_{i=1}^{M_n} \ln kr' = \sum_{n=1}^{N} M_n \ln kr', \\
\ln \mathcal{L}_1 &= -r' \sum_{n=1}^{N} \sum_{i=1}^{M_n} \tau_{b,i}^{(n)} = -r' \sum_{n=1}^{N} T_b^{(n)}, \\
\ln \mathcal{L}_2 &= \sum_{n=1}^{N} \sum_{i=1}^{M_n} \ln c_t = \sum_{n=1}^{N} M_n \ln c_t, \\
\ln \mathcal{L}_3 &= \sum_{n=1}^{N} \sum_{i=1}^{M_n} \ln \left[1 - \chi + \alpha \chi g(na) \mathrm{e}^{(1-\alpha) r' \tau_{b,i}^{(n)}} \right], \\
\ln \mathcal{L}_4 &= -\sum_{n=1}^{N} k c_t \left[1 - \chi + \chi g(na) \right] T_u^{(n)},
\end{aligned}
\tag{5.21}
$$

where a is the receptor spacing and $x_n = na$ gives the position of the n-th receptor. Note that $\mathcal{L}_0$ and $\mathcal{L}_1$ are trivial terms since they don't depend on any parameter we are concerned with, and thus we only need to include the other three terms in the following computation. Calculating the derivatives $\partial^2 \ln \mathcal{L} / \partial\theta_\alpha \partial\theta_\beta$ is straightforward but not particularly informative. These derivatives depend on the bound/unbound times through terms like $T_u^{(n)}, M_n$, but also in more complex forms $f(\tau_{b,i}^{(n)})$ like $\mathrm{e}^{(1-\alpha) r' \tau_{b,i}^{(n)}}$.

The complex part during the derivation of the Fisher information matrix $I(\boldsymbol{\theta})_{\mu\nu} = -\langle \partial^2 \ln \mathcal{L} / \partial\theta_\mu \partial\theta_\nu \rangle$ arises from computing the average $\langle \cdots \rangle$. To begin, we can calculate the average of the bound and unbound times for the n-th receptor:

$$
\begin{aligned}
\langle \tau_u^{(n)} \rangle &= \int_0^\infty \tau P(\tau_u^{(n)} = \tau) \mathrm{d}\tau = \frac{1}{k(c_n + c')}, \\
\langle \tau_b^{(n)} \rangle &= \int_0^\infty \tau P(\tau_b^{(n)} = \tau) \mathrm{d}\tau = \frac{c_n}{r(c_n + c')} + \frac{c'}{r'(c_n + c')}.
\end{aligned}
$$

We then assume the average of the total binding-unbinding events M_n during the measuring time T can be approximated as

$$
\langle M_n \rangle = \frac{T}{\langle \tau_u^{(n)} \rangle + \langle \tau_b^{(n)} \rangle}, \tag{5.22}
$$

which should only be valid when $T \gg \tau_{\mathrm{corr}}^{\pm}$ (Sec. 5.5.1). And the average of the total unbound time for the n-th receptor is

$$\langle T_u^{(n)} \rangle = (1 - p_n)T, \tag{5.23}$$

where p_n is the occupancy given by Eq. (5.14). We can readily verify $\langle M_n \rangle = \langle T_u^{(n)} \rangle / \langle \tau_u^{(n)} \rangle$, which is expected to be true if $T \gg \tau_{\mathrm{corr}}^{\pm}$. When computing a sum over all binding-unbinding events, e.g., $\langle \sum_{i=1}^{M_n} f(\tau_{b,i}^{(n)}) \rangle$, we assume that the average of each term is equal, and as a result, it's equivalent to multiply the average of a single term by $\langle M_n \rangle$. This essentially treats the value of the number of binding-unbinding events M_n as fixed, and requires again that we have $T \gg \tau_{\mathrm{corr}}^{\pm}$. To sum up, to go from the analytical derivatives of Eq. (5.21) to computing the averages required for the Fisher information matrix, we apply the rules:

$$\begin{aligned}
\langle \sum_{n=1}^{N} M_n \rangle &\rightarrow \sum_{n=1}^{N} \langle M_n \rangle, \\
\langle \sum_{n=1}^{N} T_u^{(n)} \rangle &\rightarrow \sum_{n=1}^{N} \langle T_u^{(n)} \rangle, \\
\langle \sum_{n=1}^{N} \sum_{i=1}^{M_n} f(\tau_{b,i}^{(n)}) \rangle &\rightarrow \sum_{n=1}^{N} \langle M_n \rangle \int_0^{\infty} \mathrm{d}\tau \ P(\tau_b^{(n)} = \tau) f(\tau),
\end{aligned} \tag{5.24}$$

where the last term is computed by numerical integration using Gaussian quadrature (`integrate.quad` method provided by SciPy [238]).

We can see from Eq. (5.21) that all terms in the Fisher information matrix should contain $\langle M_n \rangle$ or $\langle T_u \rangle$—so we expect that the Fisher information time is thus proportional to the total measuring time T, at least in the limit of $T \gg \tau_{\mathrm{corr}}$ where our analytic theory is appropriate. After obtaining all the terms in the matrix numerically, we can compute its inverse and take the diagonal elements as the lower bound of the variances, namely the Cramér-Rao bound in Eq. (5.4).

5.5.3 Details of stochastic simulations

5.5.3.1 Simulation methods

In the Monte Carlo simulations, we generate a series of random bound/unbound times for each receptor from the probability distribution in Eqs. (5.2) and (5.3). To do this, we note that Eq. (5.3) is essentially

Table 5.1: Table of simulation parameters. These parameters are used throughout this chapter; any deviation from them is explicitly noted.

Parameter	Description	Dimension	Value
c_0	Ephrin concentration	L^{-2}	$10^2\,\mu\text{m}^{-2}$ [239]
χ	Fraction of correct ligands c_0/c_t	1	Varying
k	Eph-ephrin binding constant	L^2/T	$10^{-2}\,\mu\text{m}^2/\text{s}$
r	Eph-ephrin unbinding rate	T^{-1}	$1\,\text{s}^{-1}$
α	Ratio of unbinding rates r/r'	1	Varying
x_0	Contact center position	L	$50\,\mu\text{m}$
σ	Contact width	L	Varying
ω	Boundary steepness	L	$1\,\mu\text{m}$
T	Measuring time	T	$1\,\text{s}$
L	Cell perimeter	L	$100\,\mu\text{m}$
a	Eph receptor spacing	L	$0.01\,\mu\text{m}$
N	Number of receptors	1	10000

a composite of two exponential distributions: We draw samples from $r\text{e}^{-r\tau_b}$ with a probability of $c/(c+c')$ and from $r'\text{e}^{-r'\tau_b}$ with a probability of $c'/(c+c')$, which is equivalent to running a kinetic Monte Carlo simulation to generate these time series. Subsequently, we compute the log-likelihood function $\ln\mathcal{L}(\boldsymbol{\theta}'|\tau)$ using Eq. (5.21) and numerically maximize it to find the optimal parameters $(\hat{c}_t, \hat{\chi}, \hat{x}_0)$. When doing the optimization, we use the modified Powell algorithm [240] (provided by SciPy [238]) to minimize $-\ln\mathcal{L}$ within the bounds $c_t \in [0, 10^8]\,\mu\text{m}^{-2}$, $\chi \in [0,1]$, $x_0 \in [0,100]\,\mu\text{m}$, and set the relative error in solutions acceptable for convergence to 10^{-7}. This gives us the maximum-likelihood estimators $(\hat{c}_t, \hat{\chi}, \hat{x}_0)$ for this individual trajectory. We then repeat this process many times for new trajectories ($\sim 10^4$), thus allowing us to compute distributions of the estimators. We then plot the standard deviations of these estimator distributions in, e.g., Fig. 5.3.

We have not been able to compute the false negative rates in Eq. (5.10) and false positive rates in Eq. (5.11) analytically. However, we can still calculate the FNR and FPR through Monte Carlo simulations. To compute FNR, we assume the other cell is present, i.e., $c_0 \neq 0$, and then generate a time series to compute the likelihood ratio by computing the maximum log-likelihood [Eq. (5.21)] numerically under both the assumption of the other cell being present and it being absent. We then repeat this process many times to compute the FNR for a fixed value of λ. For the FPR, we do simulations by first assuming the other cell is absent (setting $c_0 = 0$), but keep all other parameters the same. We perform approximately $\sim 10^5$ simulations to do averages for each data point shown in Fig. 5.4, which allows us to obtain robust and reliable results for the analysis.

5.5.3.2 Parameters

The default parameter values in our model are shown in Table 5.1. The typical radius of the *Drosophila* haemocytes cell is in the order of $10\,\mu$m [48], which makes the cell perimeter roughly $10^2\,\mu$m. The concentration of EphA2 receptors on the cell surface is reported to be around 100–1000/μm^2 for CHO cells [241] and about 600/μm^2 for A549 lung cancer cells [229]. As we're modeling the perimeter of the cell as a line, we set the contact center $x_0 = L/2$ to avoid boundary effects. Additionally, we choose a receptor spacing of $a = 0.01\,\mu$m to ensure that there are approximately 100 receptors per square micron of the surface area on the cell, assuming a height in the z-direction around the order of $1\,\mu$m. This neglects many potential complications like the possible change in cell shape with time. In our model, we only considered CIL happening between two cells with a fixed height. Increasing the height of the contacting region is equivalent to increasing the number of receptors in the contact region, which would increase the sensing accuracy in general.

5.5.3.3 Consistency between stochastic simulation and semi-analytic theory

The fundamental assumption we utilize when deriving the Fisher information matrix in Sec. 5.5.2 is that $T \gg \tau^{\pm}_{\mathrm{corr}}$ for all receptors on the cell. In this limit, it is reasonable to compare our theory, which has a fixed number of binding/unbinding events for each receptor, to the stochastic simulations, where we fix the measuring time T. This assumption leads to the theoretical Cramér-Rao bound scaling as $\sim 1/T$ (Sec. 5.5.2). When we compare simulation and theory, we should check this dependence, to ensure we are exploring times $T \gg \tau^{\pm}_{\mathrm{corr}}$. We find that the variances obtained from simulations don't depend strongly on T when $T \ll 1$ second and the discrepancy between simulation and theory grows with decreasing measuring time (Fig. 5.7). This makes sense: In the limit $T \ll \tau_{\mathrm{corr}}$, the cell is effectively making a "snapshot" measurement, using the receptor state at one instant to estimate parameters, and the error in measurement will not depend strongly on the measurement time. The correlation timescale of about a second is consistent with our analysis in Sec. 5.5.1. With the chosen binding rate kc and unbinding rate r in the order of $1\,\mathrm{s}^{-1}$ (Table 5.1), the correlation times τ_{corr} given by Eq. (5.17) are $\tau_1 \sim 10^{-1}\,\mathrm{s}$ and $\tau_2 \sim 1\,\mathrm{s}$. Therefore, to effectively compare the simulation results with the theoretical Cramér-Rao bound in Eq. (5.4), we need to increase the measuring time to $T \sim 10\,\mathrm{s}$ in Fig. 5.3 and Fig. 5.6.

5.5.4 Additional figures

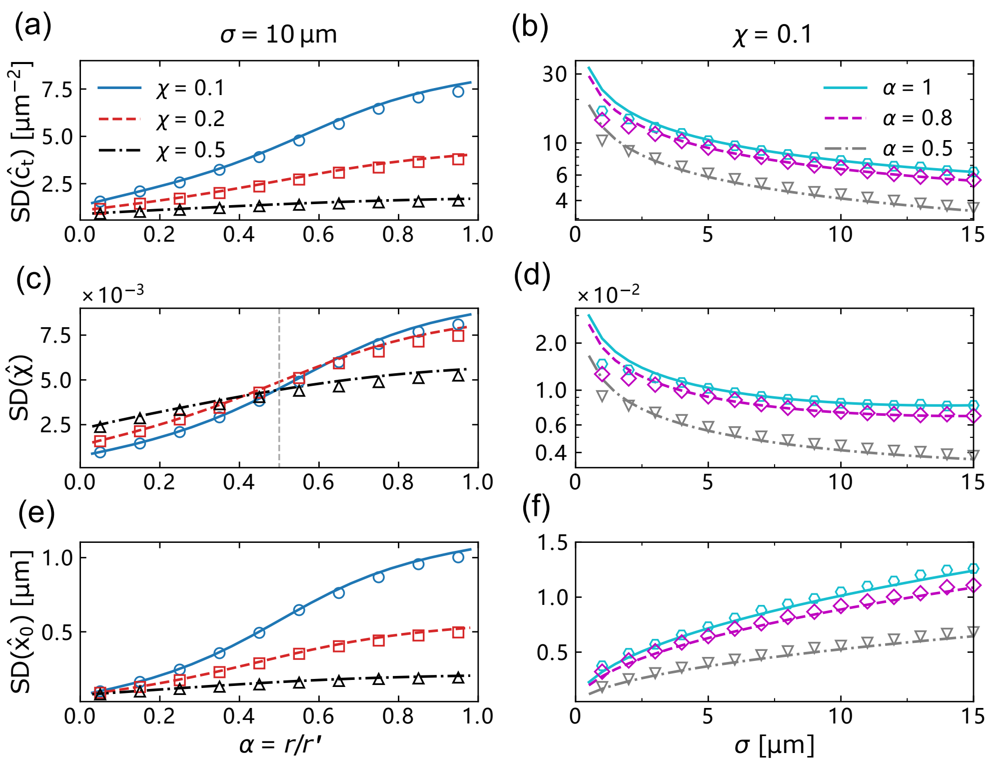

Figure 5.6: Estimating errors of MLE when the correct ligand concentration $c(x)$ takes the form of a Gaussian function $c(x) = c_0 \exp(-(x-x_0)^2/2\sigma^2)$ rather than the rectangular function given in Eq. (5.1). Panels (a), (b) show the standard deviation (SD) of the total concentration $\hat{c}_t$, panels (c), (d) show the SD of correct ligand fraction $\hat{\chi}$, and panels (e), (f) display the SD of contact center position $\hat{x}_0$, where the estimating error of x_0 increases as the contact width σ expands. In all figures, the lines (solid, dashed, and dash-dot) represent theoretical results obtained from Eq. (5.4), while the points are obtained from Monte Carlo simulations (the standard deviations are computed from 10^4 simulations for each point). Measuring time is $T = 10\,\mathrm{s}$.

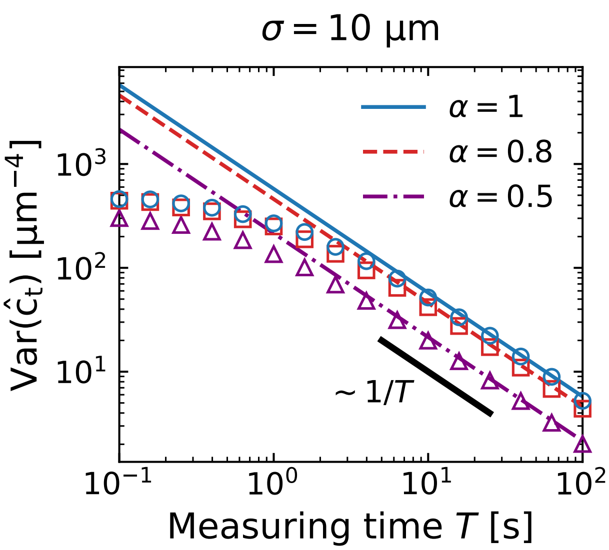

Figure 5.7: Variances of MLE decrease with increasing measuring time T. The variances from the Cramér-Rao bound scale as $\sim 1/T$ in our derivation (Sec. 5.5.2). Though the simulation results show a slower decrease compared to the theoretical predictions at small T, they still converge as the measuring time $T \gg \tau_{\text{corr}}$. Lines represent theoretical results obtained from Eq. (5.4), while the points are obtained from Monte Carlo simulations. Correct ligand fraction is $\chi = 0.1$.

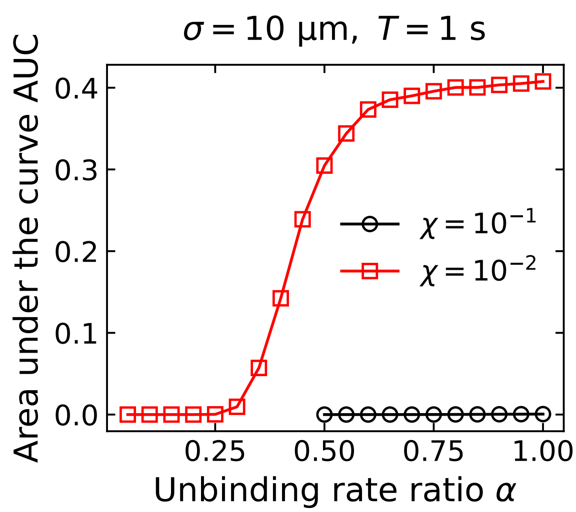

Figure 5.8: Area under the DET curve changing with unbinding rate ratio α when the correct ligand fraction $\chi = 10^{-2}$. Black points are exactly the results in Fig. 5.5(e) when $\chi = 10^{-1}$.

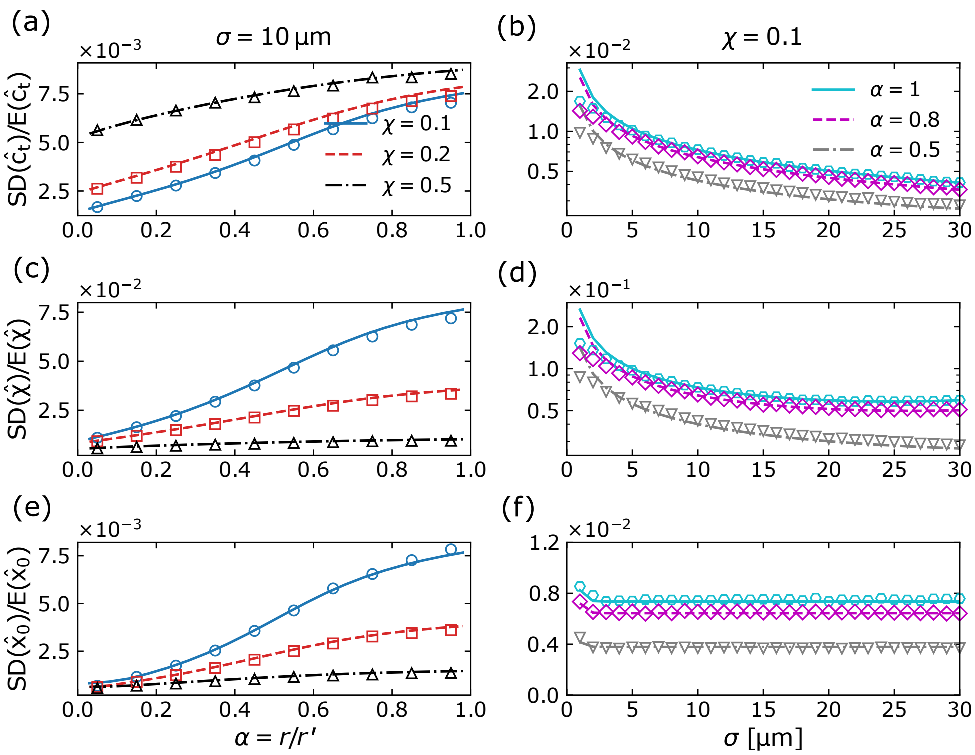

Figure 5.9: Relative errors (SD/average) for Fig. 5.3 in the main text, which are generally small everywhere. In all figures, the lines (solid, dashed, and dash-dot) represent theoretical results obtained from Eq. (5.4), while the points are obtained from Monte Carlo simulations (the standard deviations are computed from 10^4 simulations for each point). Measuring time is $T = 10\,\mathrm{s}$.

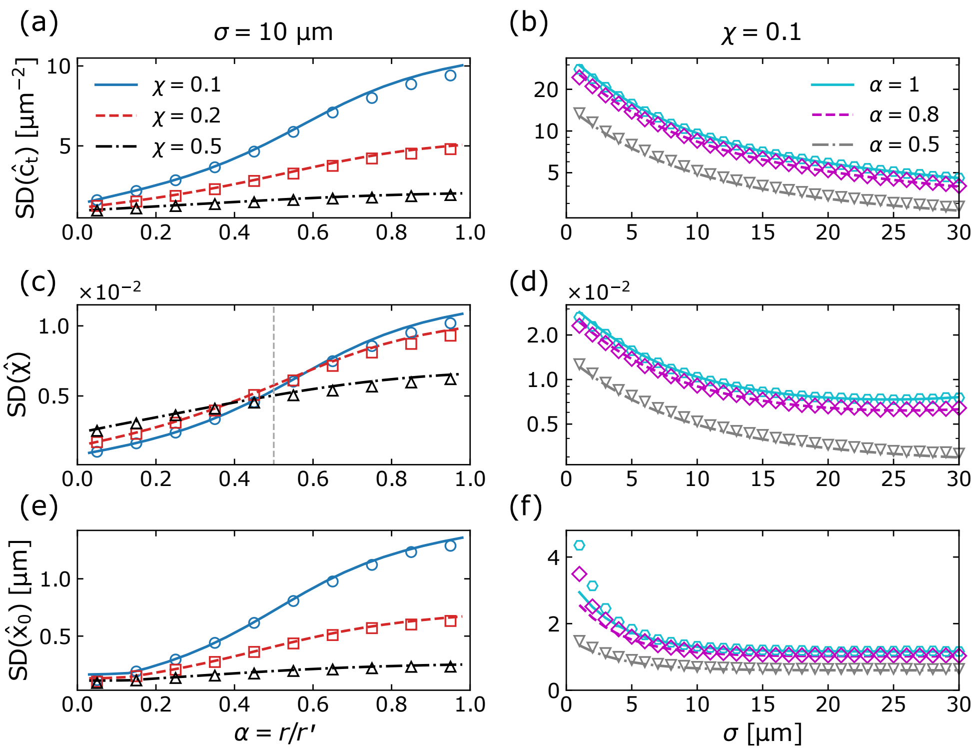

Figure 5.10: Estimating errors of MLE when $\omega = 10\,\mu$m. All SDs have a similar pattern to Fig. 5.3 where $\omega = 1\,\mu$m. Besides, we note that the SD($\hat{x}_0$) also starts to rise as σ decreases to $\sim \omega$.

Chapter 6
Conclusion and future directions

6.1 Summary

The dissociation of cells from a collectively invading strand is an early step in cancer metastasis, but it has been hard to say which mechanical features of the strand and its confinement control how often cells leave, and whether they leave singly or in clusters. The microchannel experiments of Law *et al.* gave several clean readouts: most dissociations are single cells, larger clusters become more common in wider channels, and the time to the first rupture is independent of channel width. In Chapter 2 I reproduced these observations with a phase-field model of cells crawling through the channel. The single-cell predominance arises from rare, highly motile leader cells at the invading front that pull ahead and detach on their own, while the width-independence of the rupture time can be explained by a modest weakening of cell-cell adhesion in the narrower channels. The model also shows that stronger cell-cell adhesion makes ruptures rarer but larger, which may help explain the paradoxical role of E-cadherin in metastasis, where loss of adhesion increases invasion yet larger clusters seed metastases more efficiently. A complementary result concerns the role of tissue fluidization: unjamming is necessary but not sufficient for rupture, because the cells must also be motile enough to invade the channel and pull their junctions apart.

The size of an organ or an organism is usually thought to be set by feedback on cell division, but several systems instead appear to regulate size through a competition between growth and motility-driven fracture, including germline cysts in the mouse, snowflake yeast clusters, and the metazoan *Trichoplax adhaerens*. Can such a competition set a well-defined size, or does it only produce a broad spread of sizes? In Chapter 3 I modeled the cluster as a one-dimensional chain of active particles joined by breakable junctions and solved for the exact steady-state size distribution. The distribution is controlled by a single ratio, the junction break rate over the cell division rate. When every cell can divide, the distribution always peaks at a single cell, and although the mean size grows as fracture becomes rare, the spread grows just as fast, so the size is never sharply defined.

Restricting division to the edge of the cluster improves size control; in the limit of rare fracture, where clusters are large, it roughly halves the coefficient of variation of the cluster size, from about 1 to about 0.5. Allowing fracture only near the cluster middle reduces the variability further. The model also shows that the experimentally observed drop in fracture rate over embryonic development in germline cysts requires only a small reduction in cell motility.

A natural next step is to push this program to the scale of a real organism such as *Trichoplax* (up to $\sim 10^5$ cells), where the finite Voronoi model is an appealing candidate because it captures cell shape, allows both confluent and nonconfluent tissue, and is cheaper than phase-field or deformable polygon models. In Chapter 4 I found that this model has a numerical pathology: when the cell-cell and cell-medium tensions differ, the force needed to separate two cells diverges as they detach, so the simulated fracture time is set by the numerical time step rather than by the physics, which may have affected earlier results. I traced the divergence to the Voronoi geometry analytically, proposed a cutoff that removes it and restores a well-defined fracture timescale, and calibrated the cutoff against a deformable polygon model with the same energy but without the Voronoi shape constraint. It also shows that a simulation method built for confluent tissues can fail in subtle ways when it is pushed to describe fracture and detachment.

Contact inhibition of locomotion (CIL) can fire even when two cells touch over only a micron-scale patch, which raises the question of how a cell reliably registers a neighbor through so small a contact. Earlier work has focused on the downstream signaling, but had not asked whether the contact-detection step itself sets a physical limit. In Chapter 5 I treated CIL as a sensing problem and used maximum-likelihood estimation and the Cramér-Rao bound to find the best possible sensing accuracy. A cell can decide whether a neighbor is present with high reliability even when interfering ligands are nearly ten times as abundant as the correct ephrin ligand, and the accuracy of locating the contact depends only weakly on its width. Both results are consistent with the robustness of CIL across very different contact geometries, and they suggest that stochastic ligand-receptor noise is not the limiting factor for CIL unless the fraction of correct ligands is very small.

6.2 Future directions

A clear extension of Chapters 3 and 4 is to ask how an organism as simple as the metazoan *Trichoplax adhaerens* regulates its size without any centralized genetic control. Several features seen in experiments lie beyond the simplest model of Chapter 3. The active Ornstein-Uhlenbeck particle model almost always sheds small cell clusters, yet *Trichoplax* often splits into two comparable halves, which hints at some alignment of cell polarity across the tissue that the model does not include [242], for example a velocity-alignment interaction that would make neighboring cells tend to crawl in the same direction [140]. Experiments also report occasional rotation of the whole animal, and a wide separation of timescales: individual cells reorient within seconds, but fission events are separated by days [13, 243]. These observations point to a picture in which correlated, collective motion, rather than independent single-junction fluctuations, controls when and where the tissue breaks. The regularized finite Voronoi model of Chapter 4 can simulate large systems, and I am accelerating the code through domain-decomposition-based parallelization to reach the scale of *Trichoplax*, which would make it a natural tool for exploring these questions.

A second direction uses the germline cyst data of Levy *et al.* from Chapter 3 to ask whether motility-induced fracture alone can explain the observed drop in the intercellular-bridge fracture rate over embryonic development, or whether additional mechanisms such as contact inhibition of locomotion (Chapter 5) are needed. Our model uses several parameters that are hard to measure directly in the cyst: the junction relaxation time, the cell motility persistence time, and the velocity-alignment and CIL strengths that we would like to add. These might be inferred from the embryo data using simulation-based inference (SBI), a modern neural-network-based inference approach. Such an analysis would tell us whether motility alone is sufficient, and which measurements would best constrain the parameters.

References

1. Scarpa, E. & Mayor, R. Collective cell migration in development. *Journal of Cell Biology* **212,** 143–155 (2016).

2. Weijer, C. J. Collective cell migration in development. *Journal of cell science* **122,** 3215–3223 (2009).

3. Poujade, M., Grasland-Mongrain, E., Hertzog, A., Jouanneau, J., Chavrier, P., Ladoux, B., Buguin, A. & Silberzan, P. Collective migration of an epithelial monolayer in response to a model wound. *Proceedings of the National Academy of Sciences* **104,** 15988–15993 (2007).

4. Trepat, X., Chen, Z. & Jacobson, K. Cell migration. *Comprehensive Physiology* **2,** 2369–2392 (2012).

5. Haeger, A., Krause, M., Wolf, K. & Friedl, P. Cell jamming: Collective invasion of mesenchymal tumor cells imposed by tissue confinement. *Biochimica et Biophysica Acta (BBA) - General Subjects* **1840,** 2386–2395 (2014).

6. Cheung, K. J. & Ewald, A. J. A collective route to metastasis: seeding by tumor cell clusters. *Science* **352,** 167–169 (2016).

7. Cheung, K. J. & Horne-Badovinac, S. Collective cell migration modes in development, tissue repair and cancer. *Nature Reviews Molecular Cell Biology* **26,** 741–758 (2025).

8. Friedl, P. & Gilmour, D. Collective cell migration in morphogenesis, regeneration and cancer. *Nature reviews Molecular cell biology* **10,** 445–457 (2009).

9. Rørth, P. Collective cell migration. *Annual review of cell and developmental biology* **25,** 407–429 (2009).

10. Alert, R. & Trepat, X. Physical models of collective cell migration. *Annual Review of Condensed Matter Physics* **11,** 77–101 (2020).

11. Hakim, V. & Silberzan, P. Collective cell migration: a physics perspective. *Reports on Progress in Physics* **80,** 076601 (2017).

12. Levy, E. W., Leite, I., Joyce, B. W., Shvartsman, S. Y. & Posfai, E. A tug-of-war between germ cell motility and intercellular bridges controls germline cyst formation in mice. *Current Biology* **34,** 5728–5738.e4 (2024).

13. Prakash, V. N., Bull, M. S. & Prakash, M. Motility-induced fracture reveals a ductile-to-brittle crossover in a simple animal's epithelia. *Nature Physics* **17,** 504–511 (2021).

14. Camley, B. A., Zimmermann, J., Levine, H. & Rappel, W.-J. Emergent collective chemotaxis without single-cell gradient sensing. *Phys. Rev. Lett.* **116,** 098101 (2016).

15. Ipiña, E. P. & Camley, B. A. Collective gradient sensing with limited positional information. *Phys. Rev. E* **105,** 044410 (2022).

16. Pallarès, M. E., Pi-Jaumà, I., Fortunato, I. C., Grazu, V., Gómez-González, M., Roca-Cusachs, P., *et al.* Stiffness-dependent active wetting enables optimal collective cell durotaxis. *Nature Physics* **19,** 279–289 (2023).

17. Sun, Y., Reid, B., Zhang, Y., Zhu, K., Ferreira, F., Estrada, A., Sun, Y., Draper, B. W., Yue, H., Copos, C., *et al.* Electric field–guided collective motility initiation of large epidermal cell groups. *Molecular biology of the cell* **34,** ar48 (2023).

18. Nwogbaga, I., Kim, A. H. & Camley, B. A. Physical limits on galvanotaxis. *Phys. Rev. E* **108,** 064411 (2023).

19. Berg, H. C. & Purcell, E. M. Physics of chemoreception. *Biophysical Journal* **20,** 193–219 (1977).

20. Abercrombie, M. Contact inhibition in tissue culture. *In Vitro* **6,** 128–142 (1970).

21. Mayor, R. & Carmona-Fontaine, C. Keeping in touch with contact inhibition of locomotion. *Trends in cell biology* **20,** 319–328 (2010).

22. Singh, J., Pagulayan, A., Camley, B. A. & Nain, A. S. Rules of contact inhibition of locomotion for cells on suspended nanofibers. *Proceedings of the National Academy of Sciences* **118,** e2011815118 (2021).

23. Montell, D. J., Yoon, W. H. & Starz-Gaiano, M. Group choreography: mechanisms orchestrating the collective movement of border cells. *Nature reviews Molecular cell biology* **13,** 631–645 (2012).

24. Park, J.-A., Kim, J. H., Bi, D., Mitchel, J. A., Qazvini, N. T., Tantisira, K., Park, C. Y., McGill, M., Kim, S.-H., Gweon, B., *et al.* Unjamming and cell shape in the asthmatic airway epithelium. *Nature materials* **14,** 1040–1048 (2015).

25. Bi, D., Yang, X., Marchetti, M. C. & Manning, M. L. Motility-driven glass and jamming transitions in biological tissues. *Phys. Rev. X* **6,** 021011 (2016).

26. Rustarazo-Calvo, L., Pallares-Cartes, C., Aguirre-Tamaral, A., Floris, E., Hingerl, M., Autorino, C., Khan, A. U. M., Corominas-Murtra, B. & Petridou, N. I. Adhesion-driven rigidity transition decoupled from density-driven jamming triggers epithelial organization in embryonic tissues. *Nature Physics,* 1–13 (2026).

27. Cai, G., Rodgers, N. C. & Liu, A. P. Unjamming transition as a paradigm for biomechanical control of cancer metastasis. *Cytoskeleton* **82,** 388–403 (2025).

28. Youssef, K. K. & Nieto, M. A. Epithelial–mesenchymal transition in tissue repair and degeneration. *Nature Reviews Molecular Cell Biology* **25,** 720–739 (2024).

29. Tripathi, S., Levine, H. & Jolly, M. K. The physics of cellular decision making during epithelial–mesenchymal transition. *Annual Review of Biophysics* **49,** 1–18 (2020).

30. Tarle, V., Gauquelin, E., Vedula, S. R. K., D'Alessandro, J., Lim, C. T., Ladoux, B. & Gov, N. S. Modeling collective cell migration in geometric confinement. *Phys. Biol.* **14,** 035001 (2017).

31. Balzer, E. M., Tong, Z., Paul, C. D., Hung, W. C., Stroka, K. M., Boggs, A. E., Martin, S. S. & Konstantopoulos, K. Physical confinement alters tumor cell adhesion and migration phenotypes. *The FASEB Journal* **26,** 4045–4056 (2012).

32. Lin, W.-J., Yu, H. & Pathak, A. Gradients in cell density and shape transitions drive collective cell migration into confining environments. *Soft Matter* **21,** 719–728 (2025).

33. Paul, C. D., Mistriotis, P. & Konstantopoulos, K. Cancer cell motility: lessons from migration in confined spaces. *Nature reviews cancer* **17,** 131–140 (2017).

34. Weigelin, B., Bakker, G.-J. & Friedl, P. Intravital third harmonic generation microscopy of collective melanoma cell invasion: principles of interface guidance and microvesicle dynamics. *IntraVital* **1,** 32–43 (2012).

35. Wolf, K., Alexander, S., Schacht, V., Coussens, L. M., von Andrian, U. H., van Rheenen, J., Deryugina, E. & Friedl, P. *Collagen-based cell migration models in vitro and in vivo* in *Seminars in cell & developmental biology* **20** (2009), 931–941.

36. Green, B. J., Marazzini, M., Hershey, B., Fardin, A., Li, Q., Wang, Z., Giangreco, G., Pisati, F., *et al.* PillarX: A Microfluidic Device to Profile Circulating Tumor Cell Clusters Based on Geometry, Deformability, and Epithelial State. *Small* **18,** 2106097 (2022).

37. Di Russo, S., Liberati, F. R., Riva, A., Di Fonzo, F., Macone, A., Giardina, G., Arese, M., Rinaldo, S., Cutruzzolà, F. & Paone, A. Beyond the barrier: the immune-inspired pathways of tumor extravasation. *Cell Communication and Signaling* **22,** 104 (2024).

38. Saini, M., Szczerba, B. M. & Aceto, N. Circulating tumor cell-neutrophil tango along the metastatic process. *Cancer Research* **79,** 6067–6073 (2019).

39. Law, R. A., Kiepas, A., Desta, H. E., Perez Ipiña, E., Parlani, M., Lee, S. J., Yankaskas, C. L., Zhao, R., Mistriotis, P., Wang, N., Gu, Z., Kalab, P., Friedl, P., Camley, B. A. & Konstantopoulos, K. Cytokinesis machinery promotes cell dissociation from collectively migrating strands in confinement. *Science Advances* **9,** eabq6480 (2023).

40. Ueda, T., Koya, S. & Maruyama, Y. K. Dynamic patterns in the locomotion and feeding behaviors by the placozoan Trichoplax adhaerence. *Biosystems* **54,** 65–70 (1999).

41. Smith, C. L., Reese, T. S., Govezensky, T. & Barrio, R. A. Coherent directed movement toward food modeled in Trichoplax, a ciliated animal lacking a nervous system. *Proceedings of the National Academy of Sciences* **116,** 8901–8908 (2019).

42. Jacobeen, S., Pentz, J. T., Graba, E. C., Brandys, C. G., Ratcliff, W. C. & Yunker, P. J. Cellular packing, mechanical stress and the evolution of multicellularity. *Nature physics* **14,** 286–290 (2018).

43. Lecuit, T. & Le Goff, L. Orchestrating size and shape during morphogenesis. *Nature* **450,** 189–192 (2007).

44. Ikami, K., Shoffner-Beck, S., Tyczynska Weh, M., Schnell, S., Yoshida, S., Diaz Miranda, E. A., Ko, S. & Lei, L. Branched germline cysts and female-specific cyst fragmentation facilitate oocyte determination in mice. *Proceedings of the National Academy of Sciences* **120,** e2219683120 (2023).

45. Lei, L. & Spradling, A. C. Mouse primordial germ cells produce cysts that partially fragment prior to meiosis. *Development* **140,** 2075–2081 (2013).

46. Abercrombie, M. Contact inhibition and malignancy. *Nature* **281,** 259–262 (1979).

47. Zimmermann, J., Camley, B. A., Rappel, W.-J. & Levine, H. Contact inhibition of locomotion determines cell–cell and cell–substrate forces in tissues. *Proceedings of the National Academy of Sciences* **113,** 2660–2665 (2016).

48. Davis, J. R., Huang, C.-Y., Zanet, J., Harrison, S., Rosten, E., Cox, S., Soong, D. Y., Dunn, G. A. & Stramer, B. M. Emergence of embryonic pattern through contact inhibition of locomotion. *Development* **139,** 4555–4560 (2012).

49. Stramer, B., Moreira, S., Millard, T., Evans, I., Huang, C.-Y., Sabet, O., Milner, M., Dunn, G., Martin, P. & Wood, W. Clasp-mediated microtubule bundling regulates persistent motility and contact repulsion in drosophila macrophages in vivo. *Journal of Cell Biology* **189,** 681–689 (2010).

50. Larrañaga, E., Fernández-Majada, V., Ojosnegros, S., Comelles, J. & Martinez, E. Ephrin micropatterns exogenously modulate cell organization in organoid-derived intestinal epithelial monolayers. *Advanced Materials Interfaces,* 2201301 (2022).

51. Lin, B., Yin, T., Wu, Y. I., Inoue, T. & Levchenko, A. Interplay between chemotaxis and contact inhibition of locomotion determines exploratory cell migration. *Nat Commun* **6,** 6619 (2015).

52. Camley, B. A. & Rappel, W.-J. Physical models of collective cell motility: from cell to tissue. *Journal of physics D: Applied physics* **50,** 113002 (2017).

53. Rappel, W.-J. & Edelstein-Keshet, L. Mechanisms of cell polarization. *Current opinion in systems biology* **3,** 43–53 (2017).

54. Coulthard, M. G., Morgan, M., Woodruff, T. M., Arumugam, T. V., Taylor, S. M., Carpenter, T. C., Lackmann, M. & Boyd, A. W. Eph/ephrin signaling in injury and inflammation. *The American journal of pathology* **181,** 1493–1503 (2012).

55. Parsons, J. T., Horwitz, A. R. & Schwartz, M. A. Cell adhesion: integrating cytoskeletal dynamics and cellular tension. *Nature reviews Molecular cell biology* **11,** 633–643 (2010).

56. Ridley, A. J., Schwartz, M. A., Burridge, K., Firtel, R. A., Ginsberg, M. H., Borisy, G., Parsons, J. T. & Horwitz, A. R. Cell migration: integrating signals from front to back. *Science* **302,** 1704–1709 (2003).

57. DeMali, K. A. & Burridge, K. Coupling membrane protrusion and cell adhesion. *Journal of cell science* **116,** 2389–2397 (2003).

58. Dufour, S., Mège, R.-M. & Thiery, J. P. α-catenin, vinculin, and f-actin in strengthening e-cadherin cell–cell adhesions and mechanosensing. *Cell adhesion & migration* **7,** 345–350 (2013).

59. Fily, Y. & Marchetti, M. C. Athermal phase separation of self-propelled particles with no alignment. *Phys. Rev. Lett.* **108,** 235702 (2012).

60. Levine, H., Rappel, W.-J. & Cohen, I. Self-organization in systems of self-propelled particles. *Phys. Rev. E* **63,** 017101 (2000).

61. Copenhagen, K., Malet-Engra, G., Yu, W., Scita, G., Gov, N. & Gopinathan, A. Frustration-induced phases in migrating cell clusters. *Science advances* **4,** eaar8483 (2018).

62. Sanoria, M., Malet-Engra, G., Scita, G., Gov, N. & Gopinathan, A. Chemotaxis-driven instabilities govern size, shape and migration efficiency of multicellular clusters. *PRX Life* **4,** 013004 (2026).

63. Dunn, G. & Brown, A. A unified approach to analysing cell motility. *Journal of Cell Science* **1987,** 81–102 (1987).

64. Fodor, É., Nardini, C., Cates, M. E., Tailleur, J., Visco, P. & van Wijland, F. How far from equilibrium is active matter? *Phys. Rev. Lett.* **117,** 038103 (2016).

65. Henkes, S., Kostanjevec, K., Collinson, J. M., Sknepnek, R. & Bertin, E. Dense active matter model of motion patterns in confluent cell monolayers. *Nature communications* **11,** 1405 (2020).

66. Wang, W. & Camley, B. A. Controlling tissue size by active fracture. *Phys. Rev. E* **113,** 034405 (2026).

67. Bi, D., Lopez, J., Schwarz, J. M. & Manning, M. L. A density-independent rigidity transition in biological tissues. *Nature Physics* **11,** 1074–1079 (2015).

68. Fletcher, A. G., Osterfield, M., Baker, R. E. & Shvartsman, S. Y. Vertex models of epithelial morphogenesis. *Biophysical journal* **106,** 2291–2304 (2014).

69. Lawson-Keister, E. & Manning, M. L. Jamming and arrest of cell motion in biological tissues. *Current Opinion in Cell Biology* **72,** 146–155 (2021).

70. Sussman, D. M. & Merkel, M. No unjamming transition in a Voronoi model of biological tissue. *Soft matter* **14,** 3397–3403 (2018).

71. Camley, B. A., Zhang, Y., Zhao, Y., Li, B., Ben-Jacob, E., Levine, H. & Rappel, W.-J. Polarity mechanisms such as contact inhibition of locomotion regulate persistent rotational motion of mammalian cells on micropatterns. *Proceedings of the National Academy of Sciences* **111,** 14770–14775 (2014).

72. Palmieri, B., Bresler, Y., Wirtz, D. & Grant, M. Multiple scale model for cell migration in monolayers: elastic mismatch between cells enhances motility. *Scientific reports* **5,** 11745 (2015).

73. Loewe, B., Chiang, M., Marenduzzo, D. & Marchetti, M. C. Solid-liquid transition of deformable and overlapping active particles. *Phys. Rev. Lett.* **125,** 038003 (2020).

74. Perez Ipiña, E., d'Alessandro, J., Ladoux, B. & Camley, B. A. Deposited footprints let cells switch between confined, oscillatory, and exploratory migration. *Proceedings of the National Academy of Sciences* **121,** e2318248121 (2024).

75. Zadeh, P. & Camley, B. A. Inferring nonlinear dynamics of cell migration. *PRX Life* **2,** 043020 (2024).

76. Chiang, M., Hopkins, A., Loewe, B., Marenduzzo, D. & Marchetti, M. C. Multiphase field model of cells on a substrate: from three dimensional to two dimensional. *Phys. Rev. E* **110,** 044403 (2024).

77. Cahn, J. W. & Hilliard, J. E. Free energy of a nonuniform system. i. interfacial free energy. *The Journal of chemical physics* **28,** 258–267 (1958).

78. Negro, G., Carenza, L. N., Gonnella, G., Mackay, F., Morozov, A. & Marenduzzo, D. Yield-stress transition in suspensions of deformable droplets. *Science Advances* **9,** eadf8106 (2023).

79. Graner, F. & Sawada, Y. Can surface adhesion drive cell rearrangement? Part II: a geometrical model. *Journal of theoretical biology* **164,** 477–506 (1993).

80. Teomy, E., Kessler, D. A. & Levine, H. Confluent and nonconfluent phases in a model of cell tissue. *Phys. Rev. E* **98,** 042418 (2018).

81. Huang, J., Levine, H. & Bi, D. Bridging the gap between collective motility and epithelial-mesenchymal transitions through the active finite Voronoi model. *Soft Matter* **19,** 9389–9398 (2023).

82. Boromand, A., Signoriello, A., Ye, F., O'Hern, C. S. & Shattuck, M. D. Jamming of deformable polygons. *Physical Review Letters* **121,** 248003 (2018).

83. Lv, J.-Q., Chen, P.-C., Chen, Y.-P., Liu, H.-Y., Wang, S.-D., Bai, J., Lv, C.-L., Li, Y., Shao, Y., Feng, X.-Q. & Li, B. Active hole formation in epithelioid tissues. *Nature Physics* **20,** 1313–1323 (2024).

84. Sandersius, S. A. & Newman, T. J. Modeling cell rheology with the subcellular element model. *Physical biology* **5,** 015002 (2008).

85. Ananthakrishnan, R. & Ehrlicher, A. The forces behind cell movement. *International journal of biological sciences* **3,** 303 (2007).

86. Noordstra, I., Díez Hermoso, M., Schimmel, L., Bonfim-Melo, A., Currin-Ross, D., *et al.* An e-cadherin-actin clutch translates the mechanical force of cortical flow for cell-cell contact to inhibit epithelial cell locomotion. *Developmental Cell* (2023).

87. Klein, R. Eph/ephrin signalling during development. *Development* **139,** 4105–4109 (2012).

88. Vicsek, T., Czirók, A., Ben-Jacob, E., Cohen, I. & Shochet, O. Novel type of phase transition in a system of self-driven particles. *Phys. Rev. Lett.* **75,** 1226–1229 (1995).

89. SenGupta, S., Parent, C. A. & Bear, J. E. The principles of directed cell migration. *Nature Reviews Molecular Cell Biology* **22,** 529–547 (2021).

90. Hu, B., Chen, W., Rappel, W.-J. & Levine, H. Physical limits on cellular sensing of spatial gradients. *Phys. Rev. Lett.* **105,** 048104 (2010).

91. Rogers, K. W. & Schier, A. F. Morphogen gradients: from generation to interpretation. *Annual review of cell and developmental biology* **27,** 377–407 (2011).

92. Feinerman, O., Germain, R. N. & Altan-Bonnet, G. Quantitative challenges in understanding ligand discrimination by $\alpha\beta$ t cells. *Molecular immunology* **45,** 619–631 (2008).

93. Bialek, W. & Setayeshgar, S. Physical limits to biochemical signaling. *Proceedings of the National Academy of Sciences* **102,** 10040–10045 (2005).

94. Ten Wolde, P. R., Becker, N. B., Ouldridge, T. E. & Mugler, A. Fundamental limits to cellular sensing. *Journal of Statistical Physics* **162,** 1395–1424 (2016).

95. Tkačik, G. & Wolde, P. R. t. Information processing in biochemical networks. *Annual review of biophysics* **54,** 249–274 (2025).

96. Hu, B., Chen, W., Rappel, W.-J. & Levine, H. How geometry and internal bias affect the accuracy of eukaryotic gradient sensing. *Phys. Rev. E* **83,** 021917 (2011).

97. Wasnik, V. H., Lipp, P. & Kruse, K. Accuracy of position determination in Ca^{2+} signaling. *Phys. Rev. E* **100,** 022401 (2019).

98. Mora, T. & Wingreen, N. S. Limits of sensing temporal concentration changes by single cells. *Phys. Rev. Lett.* **104,** 248101 (2010).

99. Rode, J., Novak, M. & Friedrich, B. M. Information theory of chemotactic agents using both spatial and temporal gradient sensing. *PRX Life* **2,** 023012 (2024).

100. Mou, D. & Cao, Y. Optimal cell shape for accurate chemical gradient sensing in eukaryote chemotaxis. *Physical Review Research* **7,** 033001 (2025).

101. Endres, R. G. & Wingreen, N. S. Maximum likelihood and the single receptor. *Phys. Rev. Lett.* **103,** 158101 (2009).

102. Endres, R. G. & Wingreen, N. S. Precise adaptation in bacterial chemotaxis through "assistance neighborhoods". *Proceedings of the National Academy of Sciences* **103,** 13040–13044 (2006).

103. Srinivasan, V., Wang, W. & Camley, B. A. Perfect adaptation in eukaryotic gradient sensing using cooperative allosteric binding. *Phys. Rev. E* **113,** 044414 (2026).

104. Mora, T. Physical limit to concentration sensing amid spurious ligands. *Phys. Rev. Lett.* **115,** 038102 (2015).

105. Singh, V. & Nemenman, I. Simple biochemical networks allow accurate sensing of multiple ligands with a single receptor. *PLOS Computational Biology* **13,** e1005490 (2017).

106. Noren, N. K. & Pasquale, E. B. Eph receptor–ephrin bidirectional signals that target ras and rho proteins. *Cellular signalling* **16,** 655–666 (2004).

107. Pasquale, E. B. Eph-ephrin bidirectional signaling in physiology and disease. *Cell* **133,** 38–52 (2008).

108. Arvanitis, D. & Davy, A. Eph/ephrin signaling: networks. *Genes & development* **22,** 416–429 (2008).

109. Wykosky, J., Palma, E., Gibo, D., Ringler, S., Turner, C. & Debinski, W. Soluble monomeric ephrina1 is released from tumor cells and is a functional ligand for the epha2 receptor. *Oncogene* **27,** 7260–7273 (2008).

110. Gong, J., Körner, R., Gaitanos, L. & Klein, R. Exosomes mediate cell contact–independent ephrin-eph signaling during axon guidance. *Journal of Cell Biology* **214,** 35–44 (2016).

111. Minami, M., Koyama, T., Wakayama, Y., Fukuhara, S. & Mochizuki, N. Ephrina/epha signal facilitates insulin-like growth factor-i–induced myogenic differentiation through suppression of the ras/extracellular signal–regulated kinase 1/2 cascade in myoblast cell lines. *Molecular biology of the cell* **22,** 3508–3519 (2011).

112. Wang, W., Law, R. A., Perez Ipiña, E., Konstantopoulos, K. & Camley, B. A. Confinement, jamming, and adhesion in cancer cells dissociating from a collectively invading strand. *PRX Life* **3,** 013012 (2025).

113. Chuai, M., Hughes, D. & Weijer, C. J. Collective epithelial and mesenchymal cell migration during gastrulation. *Current Genomics* **13,** 267–277 (2012).

114. Zhang, G., Mueller, R., Doostmohammadi, A. & Yeomans, J. M. Active inter-cellular forces in collective cell motility. *Journal of The Royal Society Interface* **17,** 20200312 (2020).

115. Kaplan, E. L. & Meier, P. Nonparametric estimation from incomplete observations. *Journal of the American statistical association* **53,** 457–481 (1958).

116. Nonomura, M. Study on multicellular systems using a phase field model. *PloS one* **7,** e33501 (2012).

117. Löber, J., Ziebert, F. & Aranson, I. S. Collisions of deformable cells lead to collective migration. *Scientific Reports* **5,** 1–7 (2015).

118. Mueller, R., Yeomans, J. M. & Doostmohammadi, A. Emergence of Active Nematic Behavior in Monolayers of Isotropic Cells. *Phys. Rev. Lett.* **122,** 048004 (2019).

119. Bitbol, A.-F. & Fournier, J.-B. Forces exerted by a correlated fluid on embedded inclusions. *Phys. Rev. E* **83,** 061107 (2011).

120. Shao, D., Rappel, W.-J. & Levine, H. Computational model for cell morphodynamics. *Physical review letters* **105,** 108104 (2010).

121. Brown, F. L. H. Elastic modeling of biomembranes and lipid bilayers. *Annu. Rev. Phys. Chem.* **59,** 685–712 (2008).

122. Zadeh, P. & Camley, B. A. Picking winners in cell-cell collisions: wetting, speed, and contact. *Phys. Rev. E* **106,** 054413 (2022).

123. Kulawiak, D. A., Camley, B. A. & Rappel, W.-J. Modeling Contact Inhibition of Locomotion of Colliding Cells Migrating on Micropatterned Substrates. *PLOS Computational Biology* **12,** e1005239 (2016).

124. Vilchez Mercedes, S. A., Bocci, F., Levine, H., Onuchic, J. N., Jolly, M. K. & Wong, P. K. Decoding leader cells in collective cancer invasion. *Nature Reviews Cancer* **21,** 592–604 (2021).

125. Sepúlveda, N., Petitjean, L., Cochet, O., Grasland-Mongrain, E., Silberzan, P. & Hakim, V. Collective Cell Motion in an Epithelial Sheet Can Be Quantitatively Described by a Stochastic Interacting Particle Model. *PLOS Computational Biology* **9,** e1002944 (2013).

126. Yang, Y. & Levine, H. Leader-cell-driven epithelial sheet fingering. *Physical Biology* **17,** 046003 (2020).

127. Leggett, S. E., Brennan, M. C., Martinez, S., Tien, J. & Nelson, C. M. Relatively rare populations of invasive cells drive progression of heterogeneous tumors. *Cellular and Molecular Bioengineering,* 1–18 (2024).

128. Cheung, K. J., Gabrielson, E., Werb, Z. & Ewald, A. J. Collective invasion in breast cancer requires a conserved basal epithelial program. *Cell* **155,** 1639–1651 (2013).

129. Hwang, P. Y., Brenot, A., King, A. C., Longmore, G. D. & George, S. C. Randomly Distributed K14+ Breast Tumor Cells Polarize to the Leading Edge and Guide Collective Migration in

Response to Chemical and Mechanical Environmental Cues. *Cancer Research* **79,** 1899–1912 (2019).

130. Zhang, J., Goliwas, K. F., Wang, W., Taufalele, P. V., Bordeleau, F. & Reinhart-King, C. A. Energetic regulation of coordinated leader-follower dynamics during collective invasion of breast cancer cells. *Proceedings of the National Academy of Sciences* **116,** 7867–7872 (2019).

131. Wisniewski, E. O., Mistriotis, P., Bera, K., Law, R. A., Zhang, J., Nikolic, M., Weiger, M., Parlani, M., Tuntithavornwat, S., Afthinos, A., *et al.* Dorsoventral polarity directs cell responses to migration track geometries. *Science advances* **6,** eaba6505 (2020).

132. Ilina, O., Gritsenko, P. G., Syga, S., Lippoldt, J., La Porta, C. A., Chepizhko, O., Grosser, S., Vullings, M., Bakker, G.-J., Starruß, J., *et al.* Cell–cell adhesion and 3d matrix confinement determine jamming transitions in breast cancer invasion. *Nature cell biology* **22,** 1103–1115 (2020).

133. Bi, D., Lopez, J. H., Schwarz, J. M. & Manning, M. L. Energy barriers and cell migration in densely packed tissues. *Soft Matter* **10,** 1885–1890 (2014).

134. Woillez, E., Kafri, Y. & Gov, N. S. Active trap model. *Physical Review Letters* **124,** 118002 (2020).

135. Duque, J., Bonfanti, A., Fouchard, J., Baldauf, L., Azenha, S. R., Ferber, E., Harris, A., Barriga, E. H., Kabla, A. J. & Charras, G. Rupture strength of living cell monolayers. *Nature materials* **23,** 1563–1574 (2024).

136. Chen, Y., Gao, Q., Li, J., Mao, F., Tang, R. & Jiang, H. Activation of topological defects induces a brittle-to-ductile transition in epithelial monolayers. *Phys. Rev. Lett.* **128,** 018101 (2022).

137. Gupta, S., Patteson, A. E. & Schwarz, J. M. The role of vimentin-nuclear interactions in persistent cell motility through confined spaces. *New J. Phys.* **23,** 093042 (2021).

138. Van Kampen, N. G. *Stochastic processes in physics and chemistry* (Elsevier, 1992).

139. Camley, B. A. Collective gradient sensing and chemotaxis: modeling and recent developments. *Journal of Physics: Condensed Matter* **30,** 223001 (2018).

140. Camley, B. A. & Rappel, W.-J. Velocity alignment leads to high persistence in confined cells. *Physical Review E* **89,** 062705 (2014).

141. Mitchel, J. A., Das, A., O'Sullivan, M. J., Stancil, I. T., DeCamp, S. J., Koehler, S., Ocaña, O. H., Butler, J. P., Fredberg, J. J., Nieto, M. A., *et al.* In primary airway epithelial cells, the unjamming transition is distinct from the epithelial-to-mesenchymal transition. *Nature communications* **11,** 5053 (2020).

142. Malinverno, C., Corallino, S., Giavazzi, F., Bergert, M., Li, Q., Leoni, M., Disanza, A., Frittoli, E., Oldani, A., Martini, E., *et al.* Endocytic reawakening of motility in jammed epithelia. *Nature materials* **16,** 587–596 (2017).

143. Mongera, A., Rowghanian, P., Gustafson, H. J., Shelton, E., Kealhofer, D. A., Carn, E. K., Serwane, F., Lucio, A. A., Giammona, J. & Campàs, O. A fluid-to-solid jamming transition underlies vertebrate body axis elongation. *Nature* **561,** 401–405 (2018).

144. Blauth, E., Kubitschke, H., Gottheil, P., Grosser, S. & Käs, J. A. Jamming in embryogenesis and cancer progression. *Frontiers in Physics* **9,** 666709 (2021).

145. Devany, J., Sussman, D. M., Yamamoto, T., Manning, M. L. & Gardel, M. L. Cell cycle–dependent active stress drives epithelia remodeling. *Proceedings of the National Academy of Sciences* **118,** e1917853118 (2021).

146. Czajkowski, M., Sussman, D. M., Marchetti, M. C. & Manning, M. L. Glassy dynamics in models of confluent tissue with mitosis and apoptosis. *Soft matter* **15,** 9133–9149 (2019).

147. Grosser, S., Lippoldt, J., Oswald, L., Merkel, M., Sussman, D. M., Renner, F., Gottheil, P., Morawetz, E. W., Fuhs, T., Xie, X., *et al.* Cell and nucleus shape as an indicator of tissue fluidity in carcinoma. *Physical Review X* **11,** 011033 (2021).

148. Gottheil, P., Lippoldt, J., Grosser, S., Renner, F., Saibah, M., Tschodu, D., Poßögel, A.-K., Wegscheider, A.-S., Ulm, B., Friedrichs, K., *et al.* State of cell unjamming correlates with distant metastasis in cancer patients. *Physical Review X* **13,** 031003 (2023).

149. Manning, M. L., Foty, R. A., Steinberg, M. S. & Schoetz, E.-M. Coaction of intercellular adhesion and cortical tension specifies tissue surface tension. *Proceedings of the National Academy of Sciences* **107,** 12517–12522 (2010).

150. Weng, S., Huebner, R. J. & Wallingford, J. B. Convergent extension requires adhesion-dependent biomechanical integration of cell crawling and junction contraction. *Cell Reports* **39** (2022).

151. McEvoy, E., Sneh, T., Moeendarbary, E., Javanmardi, Y., Efimova, N., Yang, C., Marino-Bravante, G. E., Chen, X., Escribano, J., Spill, F., *et al.* Feedback between mechanosensitive signaling and active forces governs endothelial junction integrity. *Nature communications* **13,** 7089 (2022).

152. Sadati, M., Qazvini, N. T., Krishnan, R., Park, C. Y. & Fredberg, J. J. Collective migration and cell jamming. *Differentiation* **86,** 121–125 (2013).

153. Tambe, D. T., Corey Hardin, C., Angelini, T. E., Rajendran, K., Park, C. Y., Serra-Picamal, X., Zhou, E. H., Zaman, M. H., Butler, J. P., Weitz, D. A., *et al.* Collective cell guidance by cooperative intercellular forces. *Nature materials* **10,** 469–475 (2011).

154. Russo, G. C., Crawford, A. J., Clark, D., Cui, J., Carney, R., Karl, M. N., Su, B., Starich, B., Lih, T.-S., Kamat, P., *et al.* E-cadherin interacts with egfr resulting in hyper-activation of erk in multiple models of breast cancer. *Oncogene* **43,** 1445–1462 (2024).

155. Padmanaban, V., Krol, I., Suhail, Y., Szczerba, B. M., Aceto, N., Bader, J. S. & Ewald, A. J. E-cadherin is required for metastasis in multiple models of breast cancer. *Nature* **573,** 439–444 (2019).

156. Garcia, S., Hannezo, E., Elgeti, J., Joanny, J.-F., Silberzan, P. & Gov, N. S. Physics of active jamming during collective cellular motion in a monolayer. *Proceedings of the National Academy of Sciences* **112,** 15314–15319 (2015).

157. Lord, S. J., Velle, K. B., Mullins, R. D. & Fritz-Laylin, L. K. Superplots: communicating reproducibility and variability in cell biology. *Journal of Cell Biology* **219** (2020).

158. Wang, W. & Camley, B. A. Limits on the accuracy of contact inhibition of locomotion. *Phys. Rev. E* **109,** 054408 (2024).

159. Kabla, A. J. Collective cell migration: leadership, invasion and segregation. *Journal of The Royal Society Interface* **9,** 3268–3278 (2012).

160. Treado, J. D., Roddy, A. B., Théroux-Rancourt, G., Zhang, L., Ambrose, C., Brodersen, C. R., Shattuck, M. D. & O'Hern, C. S. Localized growth and remodelling drives spongy mesophyll morphogenesis. *Journal of the Royal Society Interface* **19,** 20220602 (2022).

161. Davidson-Pilon, C. Lifelines: survival analysis in python. *Journal of Open Source Software* **4,** 1317 (2019).

162. Greenwood, M. The natural duration of cancer. *Reports on public health and medical subjects* **33,** 1–26 (1926).

163. Mukherjee, M. & Levine, H. Cluster size distribution of cells disseminating from a primary tumor. *PLOS Computational Biology* **17,** e1009011 (2021).

164. Nanda, P., Barrere, J., LaBar, T. & Murray, A. W. A dynamic network model predicts the phenotypes of multicellular clusters from cellular properties. *Current Biology* **34,** 2672–2683 (2024).

165. Desai, R. A., Gopal, S. B., Chen, S. & Chen, C. S. Contact inhibition of locomotion probabilities drive solitary versus collective cell migration. *Journal of The Royal Society Interface* **10,** 20130717 (2013).

166. Jain, S., Cachoux, V. M., Narayana, G. H., de Beco, S., D'alessandro, J., Cellerin, V., Chen, T., Heuzé, M. L., Marcq, P., Mège, R.-M., *et al.* The role of single-cell mechanical behaviour and polarity in driving collective cell migration. *Nature physics* **16,** 802–809 (2020).

167. Gardiner, C. *Stochastic Methods: A Handbook for the Natural and Social Sciences* (Springer, 2009).

168. Hänggi, P., Talkner, P. & Borkovec, M. Reaction-rate theory: fifty years after kramers. *Rev. Mod. Phys.* **62,** 251–341 (1990).

169. Hänggi, P. & Jung, P. Colored noise in dynamical systems. *Advances in chemical physics* **89,** 239–326 (1994).

170. Wexler, D., Gov, N., Rasmussen, K. Ø. & Bel, G. Dynamics and escape of active particles in a harmonic trap. *Phys. Rev. Res.* **2,** 013003 (2020).

171. Krapivsky, P. L., Redner, S. & Ben-Naim, E. *A kinetic view of statistical physics* See Chapter 6 (Fragmentation), Section 6.3 (Reversible polymerization), p. 184 (Cambridge University Press, 2010).

172. Puliafito, A., Hufnagel, L., Neveu, P., Streichan, S., Sigal, A., Fygenson, D. K. & Shraiman, B. I. Collective and single cell behavior in epithelial contact inhibition. *Proceedings of the National Academy of Sciences* **109,** 739–744 (2012).

173. Vennettilli, M., González, L., Hilgert, N. & Mugler, A. Autologous chemotaxis at high cell density. *Phys. Rev. E* **106,** 024413 (2022).

174. Trepat, X., Wasserman, M. R., Angelini, T. E., Millet, E., Weitz, D. A., Butler, J. P. & Fredberg, J. J. Physical forces during collective cell migration. *Nature physics* **5,** 426–430 (2009).

175. Öcal, K. & Stumpf, M. P. H. Cell size distributions in lineages. *Phys. Rev. Res.* **7,** 013302 (2025).

176. Amir, A. Cell size regulation in bacteria. *Phys. Rev. Lett.* **112,** 208102 (2014).

177. Jia, C., Singh, A. & Grima, R. Characterizing non-exponential growth and bimodal cell size distributions in fission yeast: an analytical approach. *PLOS Computational Biology* **18,** e1009793 (2022).

178. Taheri-Araghi, S., Bradde, S., Sauls, J. T., Hill, N. S., Levin, P. A., Paulsson, J., Vergassola, M. & Jun, S. Cell-size control and homeostasis in bacteria. *Current biology* **25,** 385–391 (2015).

179. Selmeczi, D., Mosler, S., Hagedorn, P. H., Larsen, N. B. & Flyvbjerg, H. Cell motility as persistent random motion: theories from experiments. *Biophysical journal* **89,** 912–931 (2005).

180. Kalbfleisch, J. D. & Prentice, R. L. in *The Statistical Analysis of Failure Time Data* 1–30 (John Wiley & Sons, 2002).

181. Ross, S. M. *Stochastic processes* (John Wiley & Sons, 1995).

182. Norris, J. R. *Markov Chains* (Cambridge University Press, 1997).

183. Soumya, S., Gupta, A., Cugno, A., Deseri, L., Dayal, K., Das, D., Sen, S. & Inamdar, M. M. Coherent motion of monolayer sheets under confinement and its pathological implications. *PLoS computational biology* **11,** e1004670 (2015).

184. Zhang, Y., Bastounis, E. E. & Copos, C. Emergence of multiple collective motility modes in a physical model of cell chains. *npj Systems Biology and Applications* **11,** 1–14 (2025).

185. Kaiyrbekov, K., Endresen, K., Sullivan, K., Zheng, Z., Chen, Y., Serra, F. & Camley, B. A. Migration and division in cell monolayers on substrates with topological defects. *Proceedings of the National Academy of Sciences* **120,** e2301197120 (2023).

186. Woillez, E., Kafri, Y. & Lecomte, V. Nonlocal stationary probability distributions and escape rates for an active ornstein–uhlenbeck particle. *Journal of Statistical Mechanics: Theory and Experiment* **2020,** 063204 (2020).

187. Martin, D., O'Byrne, J., Cates, M. E., Fodor, É., Nardini, C., Tailleur, J. & van Wijland, F. Statistical mechanics of active ornstein-uhlenbeck particles. *Phys. Rev. E* **103,** 032607 (2021).

188. Kloeden, P. E. & Platen, E. *Stochastic differential equations* (Springer, 1992).

189. Wang, W. & Camley, B. A. Divergence of detachment forces in the finite Voronoi model. *arXiv preprint arXiv:2604.15481* (2026).

190. Harper, K. L., Sosa, M. S., Entenberg, D., Hosseini, H., Cheung, J. F., Nobre, R., Avivar-Valderas, A., Nagi, C., Girnius, N., Davis, R. J., *et al.* Mechanism of early dissemination and metastasis in her2+ mammary cancer. *Nature* **540,** 588–592 (2016).

191. Bock, M., Tyagi, A. K., Kreft, J.-U. & Alt, W. Generalized Voronoi tessellation as a model of two-dimensional cell tissue dynamics. *Bulletin of mathematical biology* **72,** 1696–1731 (2010).

192. Schaller, G. & Meyer-Hermann, M. Multicellular tumor spheroid in an off-lattice Voronoi-Delaunay cell model. *Phys. Rev. E* **71,** 051910 (2005).

193. Sweet, C. R., Chatterjee, S., Xu, Z., Bisordi, K., Rosen, E. D. & Alber, M. Modelling platelet–blood flow interaction using the subcellular element langevin method. *Journal of The Royal Society Interface* **8,** 1760–1771 (2011).

194. Sandersius, S., Weijer, C. J. & Newman, T. J. Emergent cell and tissue dynamics from subcellular modeling of active biomechanical processes. *Physical biology* **8,** 045007 (2011).

195. Lawson-Keister, E., Zhang, T., Nazari, F., Fagotto, F. & Manning, M. L. Differences in boundary behavior in the 3d vertex and Voronoi models. *PLoS computational biology* **20,** e1011724 (2024).

196. Yue, H., Packard, C. R. & Sussman, D. M. Scale-dependent sharpening of interfacial fluctuations in shape-based models of dense cellular sheets. *Soft Matter* **20,** 9444–9453 (2024).

197. Spahn, P. & Reuter, R. A vertex model of Drosophila ventral furrow formation. *PloS one* **8,** e75051 (2013).

198. Smeets, B., Cuvelier, M., Pešek, J. & Ramon, H. The effect of cortical elasticity and active tension on cell adhesion mechanics. *Biophysical journal* **116,** 930–937 (2019).

199. Vangheel, J., Ramon, H. & Smeets, B. Rigidity transitions in a three-dimensional active foam model of cell monolayers with frictional contact interactions. *Phys. Rev. Res.* **8,** 013022 (2026).

200. Chu, Y.-S., Dufour, S., Thiery, J. P., Perez, E. & Pincet, F. Johnson-Kendall-Roberts theory applied to living cells. *Phys. Rev. Lett.* **94,** 028102 (2005).

201. Byers, S. W., Sommers, C. L., Hoxter, B., Mercurio, A. M. & Tozeren, A. Role of E-cadherin in the response of tumor cell aggregates to lymphatic, venous and arterial flow: measurement of cell-cell adhesion strength. *Journal of cell science* **108,** 2053–2064 (1995).

202. Rozema, D., Fagotto-Kaufmann, C., Ruppel, A., Lasko, P. & Fagotto, F. Remodeling of cadherin contacts in embryonic mesenchymal tissues during differential cell migration. *Developmental Cell* **60,** 2915–2930 (2025).

203. Eckert, J., Matylitskaya, V., Kasemann, S., Partel, S. & Schmidt, T. Cell–cell separation device: a new approach to measuring intercellular detachment forces. *Review of Scientific Instruments* **96,** 095002 (2025).

204. Scarpa, E., Roycroft, A., Theveneau, E., Terriac, E., Piel, M. & Mayor, R. A novel method to study contact inhibition of locomotion using micropatterned substrates. *Biology open* **2,** 901–906 (2013).

205. Brückner, D. B., Arlt, N., Fink, A., Ronceray, P., Rädler, J. O. & Broedersz, C. P. Learning the dynamics of cell–cell interactions in confined cell migration. *Proceedings of the National Academy of Sciences* **118,** e2016602118 (2021).

206. Li, D. & Wang, Y.-l. Coordination of cell migration mediated by site-dependent cell–cell contact. *Proceedings of the National Academy of Sciences* **115,** 10678–10683 (2018).

207. Zapata-Mercado, E., Biener, G., McKenzie, D. M., Wimley, W. C., Pasquale, E. B., Raicu, V. & Hristova, K. The efficacy of receptor tyrosine kinase EphA2 autophosphorylation increases with EphA2 oligomer size. *Journal of Biological Chemistry* **298,** 102370 (2022).

208. Lisabeth, E. M., Falivelli, G. & Pasquale, E. B. Eph receptor signaling and ephrins. *Cold Spring Harbor perspectives in biology* **5,** a009159 (2013).

209. Roycroft, A. & Mayor, R. Molecular basis of contact inhibition of locomotion. *Cellular and Molecular Life Sciences* **73,** 1119–1130 (2016).

210. Cayuso, J., Xu, Q. & Wilkinson, D. G. Mechanisms of boundary formation by eph receptor and ephrin signaling. *Developmental biology* **401,** 122–131 (2015).

211. Singh, V. & Nemenman, I. Universal properties of concentration sensing in large ligand-receptor networks. *Phys. Rev. Lett.* **124,** 028101 (2020).

212. Carballo-Pacheco, M., Desponds, J., Gavrilchenko, T., Mayer, A., Prizak, R., Reddy, G., Nemenman, I. & Mora, T. Receptor crosstalk improves concentration sensing of multiple ligands. *Phys. Rev. E* **99,** 022423 (2019).

213. Chakrabortty, T. & Varma, M. M. Theoretical limit of concentration sensing of single receptor artificial biosensors. *arXiv preprint arXiv:1907.05349* (2019).

214. Hopkins, A. & Camley, B. A. Chemotaxis in uncertain environments: hedging bets with multiple receptor types. *Phys. Rev. Res.* **2,** 043146 (2020).

215. Chylek, L. A., Holowka, D. A., Baird, B. A. & Hlavacek, W. S. An interaction library for the fcεri signaling network. *Frontiers in immunology* **5,** 172 (2014).

216. Lalanne, J.-B. & François, P. Chemodetection in fluctuating environments: receptor coupling, buffering, and antagonism. *Proceedings of the National Academy of Sciences* **112,** 1898–1903 (2015).

217. Karl, K., Paul, M. D., Pasquale, E. B. & Hristova, K. Ligand bias in receptor tyrosine kinase signaling. *Journal of Biological Chemistry* **295,** 18494–18507 (2020).

218. Gomez-Soler, M., Gehring, M. P., Lechtenberg, B. C., Zapata-Mercado, E., Ruelos, A., Matsumoto, M. W., Hristova, K. & Pasquale, E. B. Ligands with different dimeric configurations potently activate the epha2 receptor and reveal its potential for biased signaling. *iScience* **25** (2022).

219. Kay, S. M. *Fundamentals of statistical signal processing: estimation theory* (Prentice-Hall, Inc., 1993).

220. Li, B. & Babu, G. J. *A graduate course on statistical inference* (Springer, 2019).

221. Navratil, J. & Klusacek, D. *On linear DETs* in *2007 IEEE International Conference on Acoustics, Speech and Signal Processing-ICASSP'07* **4** (2007), IV–229.

222. Fawcett, T. An introduction to roc analysis. *Pattern recognition letters* **27,** 861–874 (2006).

223. Hu, J., Lipowsky, R. & Weikl, T. R. Binding constants of membrane-anchored receptors and ligands depend strongly on the nanoscale roughness of membranes. *Proceedings of the National Academy of Sciences* **110,** 15283–15288 (2013).

224. DiNapoli, K. T., Robinson, D. N. & Iglesias, P. A. A mesoscale mechanical model of cellular interactions. *Biophysical Journal* **120,** 4905–4917 (2021).

225. Adams, C. L., Nelson, W. J. & Smith, S. J. Quantitative analysis of cadherin-catenin-actin reorganization during development of cell-cell adhesion. *The Journal of cell biology* **135,** 1899–1911 (1996).

226. Janeš, J. A., Monzel, C., Schmidt, D., Merkel, R., Seifert, U., Sengupta, K. & Smith, A.-S. First-principle coarse-graining framework for scale-free bell-like association and dissociation rates in thermal and active systems. *Phys. Rev. X* **12,** 031030 (2022).

227. Govern, C. C. & ten Wolde, P. R. Fundamental limits on sensing chemical concentrations with linear biochemical networks. *Phys. Rev. Lett.* **109,** 218103 (2012).

228. Macdonald, J. L. & Pike, L. J. Heterogeneity in egf-binding affinities arises from negative cooperativity in an aggregating system. *Proceedings of the National Academy of Sciences* **105,** 112–117 (2008).

229. Singh, D. R., Kanvinde, P., King, C., Pasquale, E. B. & Hristova, K. The EphA2 receptor is activated through induction of distinct, ligand-dependent oligomeric structures. *Commun Biol* **1,** 1–12 (2018).

230. Skoge, M., Meir, Y. & Wingreen, N. S. Dynamics of cooperativity in chemical sensing among cell-surface receptors. *Phys. Rev. Lett.* **107,** 178101 (2011).

231. Bihr, T., Fenz, S., Sackmann, E., Merkel, R., Seifert, U., Sengupta, K. & Smith, A.-S. Association rates of membrane-coupled cell adhesion molecules. *Biophysical Journal* **107,** L33–L36 (2014).

232. Zarnitsyna, V. & Zhu, C. T cell triggering: insights from 2D kinetics analysis of molecular interactions. *Phys. Biol.* **9,** 045005 (2012).

233. Bicknell, B. A., Dayan, P. & Goodhill, G. J. The limits of chemosensation vary across dimensions. *Nat Commun* **6,** 7468 (2015).

234. Walther, G. R., Marée, A. F., Edelstein-Keshet, L. & Grieneisen, V. A. Deterministic versus stochastic cell polarisation through wave-pinning. *Bulletin of Mathematical Biology* **74,** 2570–2599 (2012).

235. Copos, C. & Mogilner, A. A hybrid stochastic–deterministic mechanochemical model of cell polarization. *Molecular Biology of the Cell* **31,** 1637–1649 (2020).

236. Ipiña, E. P., D'Alessandro, J., Ladoux, B. & Camley, B. A. Secreted footprints let cells switch between confined, oscillatory, and exploratory migration. *bioRxiv* (2023).

237. Kashyap, A., Wang, W. & Camley, B. A. Trade-offs in concentration sensing in dynamic environments. *Biophysical Journal* **123,** 1184–1194 (2024).

238. Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., *et al.* SciPy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods* **17,** 261–272 (2020).

239. Xu, Q., Lin, W.-C., Petit, R. S. & Groves, J. T. EphA2 receptor activation by monomeric ephrin-a1 on supported membranes. *Biophys J* **101,** 2731–2739 (2011).

240. Powell, M. J. An efficient method for finding the minimum of a function of several variables without calculating derivatives. *The computer journal* **7,** 155–162 (1964).

241. Wirth, D., Paul, M. D., Pasquale, E. B. & Hristova, K. Direct quantification of ligand-induced lipid and protein microdomains with distinctive signaling properties. *ChemSystemsChem* **4,** e202200011 (2022).

242. Davidescu, M. R., Romanczuk, P., Gregor, T. & Couzin, I. D. Growth produces coordination trade-offs in Trichoplax adhaerens, an animal lacking a central nervous system. *Proceedings of the National Academy of Sciences* **120,** e2206163120 (2023).

243. Leria, M., Daroueche, M.-m., Requin, M., Hill, R., Le Bivic, A., Clément, R. & Pasini, A. Fast calcium-dependent reorientation of motile cilia basal bodies in the simple metazoan, Trichoplax. *bioRxiv,* 2025–08 (2025).